Generalized fading models: Approximations, Asymptotics and Applications

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THESIS CERTIFICATE

This is to certify that the thesis titled **Generalized fading models: Approximations, Asymptotics and Applications**, submitted by **Muralikrishnan S**, to the Indian Institute of Technology Madras, for the award of the degree of **Doctor of Philosophy**, is a bona fide record of the research work done by him under our supervision. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

KEYWORDS: Fading; Generalized fading channels; MIMO; Optimum combining; Asymptotics; $\kappa - \mu$ fading; $\eta - \mu$ fading

In this thesis, we first explore the commonly used moment matching for deriving approximations for generalized fading channels in three different applications - a) We calculate secure outage probability and strictly positive secrecy capacity expressions over $\kappa - \mu$ shadowed fading channels. b) We derive outage probability (OP) and ergodic rate expressions for the uplink of multi-user massive multi-input multi-output (MIMO) systems with low-resolution analog-to-digital converters. c) We derive approximate OP and rate expressions for systems employing maximum ratio combining (MRC) when both the desired signal and the interfering signals undergo $\eta - \mu$ fading. Secondly, we determine the approximate symbol error rate and OP expressions when Optimum Combining is employed and when both the user and interferers experience Rician fading. We highlight the complicated nature of these expressions even when the eigenvalue statistics of non-central Wishart distribution is available. For both complex $\kappa - \mu$ and $\eta - \mu$ distribution, which do not have any matrix distribution, we propose a Wishart distributed approximation, by minimizing KL divergence. We explore the utility of the matrix approximations in some 5G-relevant applications. Finally, we assume that the user signal and the interferer signals experience $\kappa - \mu$ shadowed fading and determine the limiting distribution of the maximum and minimum over such random variables. Furthermore, we analyze the utility of the derived asymptotic results in some applications.

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ABBREVIATIONS

ADC	Analog-to-Digital Converter
AQNM	Additive Quantization Noise model
AWGN	Additive White Gaussian Noise
BS	Base Station
CCI	Co-channel Interference
CSI	Channel State Information
CUE	Cellular Users
D2D	Device-to-Device
DUE	D2D users
EVT	Extreme Value Theory
FAS	Full Array Selection
i.i.d.	independent and identically distributed
i.n.i.d.	independent and non-identically distributed
KL	Kullback-Leibler
MIMO	Multiple-Input Multi-Output
MSC	Maximum Sum Capacity
MU-MIMO	Multi-User MIMO
MRC	Maximum Ratio Combining
LOS	Line of Sight
NLOS	Non-line of Sight
OC	Optimum Combining
OP	Outage Probability
pdf	Probability Density Function
QoS	Quality of Service
RV	Random Variable
SER	Symbol Error Rate
SCV	Squared Coefficient of Variation
SIMO	Single-Input Multiple-Output

SINR	Signal-to-Interference plus Noise Ratio
SIQNR	Signal-to-Interference plus Quantization Noise Ratio
SIR	Signal-to-Interference Ratio
SOI	Signal of Interest
SOP	Secure Outage probability
SPSC	Strictly Positive Secrecy Capacity
SNR	Signal-to-Noise ratio
V2I	Vehicle-to-Interface Links
V2V	Vehicle-to-Vehicle Links
QAM	Quadrature Amplitude Modulation
ZF	Zero Forcing

NOTATION

$I_{(.)}$	Indicator function
${}_1F_1(.)$	Confluent hypergeometric function
$\Gamma(.,.)$	Upper incomplete gamma function
$\Gamma(.)$	Gamma function
$U(.)$	Kummer confluent hypergeometric function
${}_2F_1(.)$	Gauss hypergeometric function
$P(.)$	Probability of an event
$\mathbb{E}(.)$	Expectation of a random variable
$\mathcal{G}(.,.)$	Gamma random variable
κ	The ratio of the powers of dominant and scattered components in a κ-μ shadowed random variable
μ	The number of clusters in a κ-μ shadowed random variable
m	The shape of the Nakagami-m faded shadowing component in a κ-μ shadowed random variable
$\mathcal{CN}(.,.)$	Complex normal distribution
$\|.\|$	Magnitude function
$\|\|.\|\|$	Norm of a vector
diag(X)	Matrix formed by diagonal elements of a matrix X
$Var(.)$	Variance
$Cov(.)$	Covariance
$exp(.)$	Exponential function
$(.)^H$	Transpose of a matrix
det$(.)$	Determinant of a matrix
tr$(.)$	Trace of a matrix
$\mathcal{CW}(.)$	Complex Wishart random matrix
$\otimes$	Kronecker product
$\Psi(.)$	Digamma function
$etr(\mathbf{X})$	Exponential raised to trace of the matrix $\mathbf{X}$
sup	Supremum function
max	Maximum function
min	Minimum function
$\mathbb{R}$	Set of all Real numbers
$\mathbb{Z}$	Set of all Integers

CHAPTER 1

Introduction

Wireless communication has been one of the most transformative technologies of the past few decades. From the First generation (1G) technology in the early 80s to the fifth generation (5G) technology at present, wireless devices have become ubiquitous in every aspect of human life. What started as voice-only service in 1G has now transformed into a technology that will enable the Internet of Things (IoT), autonomous driving, and smart healthcare system in 5G. Hence, there have been many transformations in the field of wireless network communication when it comes to core functionality. With the demands on wireless networks increasing every day, the need for proper network performance measurement is essential than ever before. To evaluate the network performance, it is first necessary to statistically characterize the wireless fading channels.

1.1 The need for generalized fading channels

The most commonly used fading channels are Rayleigh and Rician fading. These fading distributions were derived assuming a homogeneous scattering field, resulting from randomly distributed scatterers. With such an assumption on the scattering field, the central-limit theorem leads to complex Gaussian processes, with in-phase and quadrature Gaussian-distributed variables having equal standard deviations. But this assumption is not always valid because the surfaces can be spatially correlated (Braun and Dersch, 1991). Also, in a lot of cases, the experimental data does not fit by any of the commonly used distributions (Stein, 1987; Yacoub, 2007; Cotton, 2015). To address this gap in the literature and to better fit experimental data, there has been a focus on generalized fading models, namely $\kappa - \mu$ and $\eta - \mu$ models, introduced in the seminal work (Yacoub, 2007). These distributions model the small scale variations in the fading channel in the line of sight and non-line of sight conditions, respectively. Furthermore, these generalized fading distributions include the well-studied Rayleigh, Rician, Nakagami, one-sided Gaussian distributions as special cases.

The fading model for the κ-μ distribution considers a signal composed of clusters of multipath waves, propagating in a non-homogeneous environment (Yacoub, 2007). Within any one cluster, the phases of the scattered waves are random and have similar delay times, with delay-time spreads of different clusters being relatively large. The clusters of multipath waves are assumed to have scattered waves with identical powers, but within each cluster, a dominant component is found, which presents an arbitrary power. κ is the ratio of the total power of LOS components to the total power of scattering components, and μ is the number of scattering clusters. Rician Fading model is a particular case of κ-μ fading with $\mu = 1$, and Nakagami fading model is another particular case with $\kappa = 0$.

$\eta - \mu$ distribution represents the small-scale variation of the fading signal in a non-line-of-sight condition. The distribution has two formats for characterization. In format-1, $0 < \eta < \infty$ is the scattered-wave power ration between the in-phase and quadrature components of each cluster of multipath. In format-2, $-1 < \eta < 1$ is the correlation coefficient between the scattered-wave in-phase and quadrature components of each cluster of multipath. A further generalization of the shadowed Rician fading is the $\kappa - \mu$ shadowed fading studied in both (Paris, 2014) and (Cotton, 2015). Also, $\kappa - \mu$ shadowed fading unifies the $\kappa - \mu$ and $\eta - \mu$ fading models (Moreno-Pozas *et al.*, 2016) and has a wide variety of applications ranging from land-mobile satellite systems to device to device communication (Cotton, 2015).

Though generalized fading channels have been well studied for a decade, which we will discuss in the next chapter, there are some challenges that one might encounter when using these fading models. The common three challenges are:

- Difficulty in implementing and analyzing the performance metric expressions
- Non-availability of matrix models
- Non-availability of an asymptotic distribution.

In the following sections, we will explore why one needs to overcome these challenges before one can meaningfully analyze the generalized fading channels.

1.2 Performance metrics in generalized fading channels

The outage probability (OP), coverage probability, ergodic rate, and other metrics of these generalized fading channels are analyzed by works such as (Morales-Jimenez *et al.*, 2012; Paris, 2013; Ermolova and Tirkkonen, 2014; Kumar and Kalyani, 2015; Parthasarathy and Ganti, 2017; Kumar and Kalyani, 2017*b*). A common feature among all these works is the complicated nature of the resulting expressions of the performance metrics. For example, the recent work (Kumar and Kalyani, 2017*b*) derives the Cumulative Distribution Function (CDF) of the Signal-to-Interference ratio (SIR). The work assumes that the source and interferers undergo $\kappa-\mu$ shadowed fading with parameters $(\kappa, \mu, m, \bar{x})$ and $\{(\kappa_i, \mu_i, m_i, \bar{y}_i); i = 1, \cdots, N\}$ respectively, where N is the number of interferers and $\bar{x}, \{\bar{y}_i; i = 1, \cdots, N\}$ are the expectations of the corresponding RVs. Then the CDF of the SIR, γ, is given by

$$\begin{aligned} F_\gamma(z) =& 1 - K_1 \sum_{p=0}^{\infty} \frac{(m)_p \left(1-\frac{\theta}{\lambda}\right)^p \Gamma\left[\sum_{i=1}^{N} \mu_i + \mu + p\right]}{(\mu)_p p!} F_D^{(2N)}(1-p-\mu, \\ & \mu_1 - m_1, \cdots, \mu_N - m_N, m_1, \cdots, m_N; 1 + \sum_{i=1}^{N} \mu_i; \\ & \frac{\theta}{\theta + z\theta_1}, \cdots, \frac{\theta}{\theta + z\theta_N}, \frac{\theta}{\theta + z\lambda_1}, \cdots, \frac{\theta}{\theta + z\lambda_N}), \end{aligned} \tag{1.1}$$

where $K_1 = \dfrac{\prod_{i=1}^{N} \left(\left(\frac{\theta}{\theta+z\theta_i}\right)^{\mu_i - m_i} \left(\frac{\theta}{\theta+z\lambda_i}\right)^{m_i} \right) \theta^m}{\Gamma\left[\sum_{i=1}^{N} \mu_i + 1\right] \Gamma[\mu] \lambda^m}$, $(m)_p = \dfrac{\Gamma[m+p]}{\Gamma[m]}$ is the Pochhammer symbol and

$$\theta = \frac{\bar{x}}{\mu_(1+\kappa)}, \ \theta_i = \frac{\bar{y}_i}{\mu_i(1+\kappa_i)}, \ \lambda = \frac{(\mu\kappa + m)\bar{x}}{\mu_(1+\kappa_)}, \lambda_i = \frac{(\mu_i\kappa_i + m_i)\bar{y}_i}{\mu_i(1+\kappa_i)m_i}. \tag{1.2}$$

in terms of an infinite summation of the Lauricella function of the fourth kind. The Lauricella function of the fourth kind, $(F_D^{(2N)}(.))$, has the following series expansion (Exton, 1976):

$$F_D^{(N)}(a, b_1, \cdots, b_N; c; x_1, \cdots, x_N) = \sum_{p_1, \cdots, p_N = 0}^{\infty} \frac{(a)_{p_1 + \cdots, p_N}}{(c)_{p_1 + \cdots + p_N}} \prod_{i=1}^{N} (b_i)_{p_i} \frac{x_i^{p_i}}{p_i!}. \tag{1.3}$$

Note that the Lauricella function involves an N-fold infinite summation, which is challenging to implement even in popular software such as MATLAB, Mathematica. Similarly, the work (Kumar and Kalyani, 2015) has derived the average rate expression when the signal of interest (SoI) undergoes $\kappa - \mu$ fading with fading parameters κ and μ, and the N co-channel interference (CCI) terms undergo $\eta - \mu$ fading with parameters η_i and μ_i for $i = 1, ..., N$. The expression for rate, R_r, is an infinite summation truncated to first M terms. This expressions is also in terms of infinite summations of the Lauricella function, and is given by

$$\begin{aligned} R_r = & \sum_{m=0}^{M} \frac{\Gamma\left(\sum_{i=1}^{N} 2\mu_i + \mu + m\right)(\mu\kappa)^m}{\Gamma(m+\mu)\, m! exp(\mu\kappa)\Gamma\left(\sum_{i=1}^{N} 2\mu_i + 1\right)} \\ & \sum_{i_1,...,i_{2N}=0}^{\infty} \frac{(1-\mu-m)_{i_1+...+i_{2N}} (\mu_1)_{i_1}(\mu_1)_{i_2}...(\mu_N)_{i_{2N}-1}(\mu_N)_{i_{2N}}}{\left(\sum_{i=1}^{N} 2\mu_i + 1\right)_{\sum_{j=1}^{2N} i_j} i_1!...i_{2N}!\left(\sum_{i=1}^{N} 2\mu_i + \sum_{j=1}^{2N} i_j\right)} \\ & F_D^{(2N)}\Big[1, \mu_1 + i_1, ..., \mu_N + i_{2N}; \sum_{i=1}^{N} 2\mu_i + \sum_{j=1}^{2N} i_j + 1 + 1; \\ & 1 - \frac{r^\alpha \mu(1+\kappa)}{d_1^\alpha \mu_1 (1+\eta_1^{-1})}, ..., 1 - \frac{r^\alpha \mu(1+\kappa)}{d_N^\alpha \mu_N (1+\eta_N^{-1})}\Big] \end{aligned} \tag{1.4}$$

Note that, in both these examples, we can observe that the expressions are not only difficult to implement but also not painful to analyze. Hence, there is a strong motivation to derive approximate expressions that are not only easy to implement but also more straightforward to explain in the context of generalized fading.

1.3 Matrix models for generalized fading channels

Most of the works mentioned in the previous section characterize the performance metrics in a Single-Input-Single-Output (SISO) or a Single-Input-Multiple-Output (SIMO) systems. In Single-Input-Single-Output (SISO) systems, a single random variable (RV) characterizes the channel gain. However, for Multiple-Input-Multiple-Output (MIMO) systems, the channel is in the form of a matrix. Hence, the characterization of random matrices plays an indispensable role in studying MIMO channel metrics. An $N_R \times N_T$ channel gain matrix $\mathbf{H}$ models the MIMO system. Here, N_R is the number of receive antennas, and N_T is the number of transmit antennas. Various performance metrics, such as capacity and rate, require the eigenvalue statistics of the Gram matrix $\mathbf{HH}^H$ (or

$\mathbf{HH}^H$).

When the elements of $\mathbf{H}$ are independent and identically distributed (i.i.d.) circular symmetric complex Gaussian with zero or non-zero mean (for Rayleigh or Rician fading MIMO channels), the Grammian $\mathbf{HH}^H$ can be characterized by Wishart matrices - central or non-central, respectively (James, 1964). These random matrix models and the eigenvalue statistics have been used widely for deriving capacity expressions in the case of Rayleigh faded MIMO channels (Telatar, 1999), and also Rician faded MIMO channels (Kang and Alouini, 2006). For the case of Nakagami-m fading, eigenvalue statistics are derived for a 2×2 case (Fraidenraich *et al.*, 2008). The joint eigenvalue density function is given by

$$p(\lambda_1, \lambda_2) = Ke^{-\frac{m(\lambda_1+\lambda_2)}{\Omega}}(\lambda_1 - \lambda_2)^2 F(\lambda_1, \lambda_2) \tag{1.5}$$

where K is a constant, m is the Nakagami-m parameter,

$$\begin{aligned} F(\lambda_1, \lambda_2) = \frac{\pi^4}{\Gamma(4m-2)} \sum_{i_1=0}^{m-1}\sum_{i_2=0}^{m-1}\sum_{k_1=0}^{i_1}\sum_{k_2=0}^{i_2} f(i_1, i_2, k_1, k_2) \\ (\lambda_1\lambda_2)^{\frac{1}{2}(i_1+i_2-2k_1-2k_2)+m-1}(\lambda_1-\lambda_2)^{-i_1-i_2+2(k_1+k_2+m-1)}, \end{aligned} \tag{1.6}$$

and

$$\begin{aligned} f(i_1, i_2, m_1, m_2) = & \frac{\binom{i_1}{k_1}\binom{i_2}{k_2}\binom{m-1}{i_1}\binom{m-1}{i_2}(-1)^{-i_1-2i_2+3m+1}2^{-2(k_1+k_2+1)}}{\Gamma(-\frac{i_1}{2}-\frac{i_2}{2}+m)} \\ & \frac{\Gamma(-\frac{i_1}{2}-\frac{i_2}{2}+m-\frac{1}{2})\Gamma(-\frac{i_1}{2}-\frac{i_2}{2}+k_1+k_2+m)}{\Gamma(-\frac{i_1}{2}-\frac{i_2}{2}+k_1+k_2+m+\frac{1}{2})} \\ & \Gamma(-\frac{i_1}{2}+\frac{i_2}{2}+k_1-k_2+2m-1)\Gamma(\frac{i_1}{2}-\frac{i_2}{2}-k_1+k_2+2m-1) \end{aligned} \tag{1.7}$$

Observe that even for a 2×2 matrix $\mathbf{H}$, the eigenvalue distribution is complicated and challenging to obtain inferences. Given the complicated pdf structure of complex variable $\kappa - \mu$ and $\eta - \mu$ fading distributions (Dias *et al.*, 2008; da Costa and Yacoub, 2007), it is challenging to develop the matrix distribution and the eigenvalue statistics for $\mathbf{HH}^H$, even when the elements of $\mathbf{H}$ are assumed to be i.i.d. Therefore, determining at least an approximate matrix model will address a significant gap in the literature of generalized fading models. Note that such an approximation will be useful in characterizing performance metrics such as ergodic capacity and OP, for which expressions

do not exist.

1.4 Asymptotic performance for generalized fading

MIMO systems have been widely accepted as a key to meet the increasing demand for wireless throughput in 5G networks (Marzetta, 2015). With the deployment of massive MIMO, one can expect transmitters/receivers with hundreds of antennas available for communication simultaneously. Larsson et al. in (Larsson *et al.*, 2014) show that the uplink spectral efficiency and the radiated power efficiency shall increase by 100 times with massive MIMO technology (with 100 antennas) and appropriate signal processing techniques. There are several examples in literature in which massive MIMO scenarios have more than 100 antennas. Works such as (Vieira *et al.*, 2014; Guan *et al.*, 2017) present various simulation results for realizing massive MIMO in a practical test-bed. Similarly, works such as (Gao *et al.*, 2013, 2015; Björnson *et al.*, 2016; Hanif *et al.*, 2018; Gao and Kaiser, 2016) present analysis of massive MIMO systems with over 100 antennas.

One of the impediments to a dense deployment of cellular networks, especially in MIMO systems, is the CCI, which is caused by the sharing of common system resources by multiple users and by frequency reuse among adjacent cells. Therefore, the effect of CCI on the quality of the wireless link has to be studied extensively before cell-planning and employing interference mitigation techniques. A vast amount of attention and research in literature is devoted to the study of SIR. Given the fact that massive MIMO is a promising technology for future cellular scenarios, the evaluation of the maximum SIR statistics over all the available antennas will be a useful metric for various performance analysis and other quality of service (QoS) provisioning applications. Very recently, (Gao *et al.*, 2018) discussed bounds on the rate of full antenna selection (FAS) architecture in a massive MIMO system, using statistics of the maximum SIR in a Rayleigh fading scenario.

Similarly, the maximum SIR is a critical performance metric in multi-user shared networks, where user scheduling is based on the channel conditions of the users. For example, the authors of (Sibomana and Zepernick, 2016) derive analytical expressions of the ergodic capacity for max-signal-to-interference-plus-noise-ratio (Max-SINR) schedul-

ing system in a cognitive radio network. Also, using the kth order statistics of the user SIR, (Al-Badarneh *et al.*, 2018) analyzes the asymptotic performance of a generalized multi-user diversity scheme of an underlay cognitive radio system in a Nakagami fading channel.

The CDF of the maximum of independent random RVs is the product of CDF of individual RVs. Hence, in the case of L i.i.d. RV's, the CDF of the maximum is given by the Lth power of the common CDF. In several cases, given that the CDF of a single RV can itself involve complicated functions, the CDF of order statistics like maximum and minimum, even over i.i.d. RVs will be more complicated. Also, providing a meaningful analysis for performance metrics like outage probability, the ergodic rate becomes intractable, if not impossible. In such cases, we can use extreme value theory (EVT) and propose a systematic approach to characterize the asymptotic maximum or minimum SIR in terms of simple probability distribution functions (PDF) or CDFs that are amenable to analysis. For example, works like (Jindal and q. Luo, 2006; Park and Love, 2008, 2009) study the capacity limits of Rayleigh faded multi-cast channels using EVT, without which the capacity limits would have been intractable to analyze. EVT has also been used effectively for studying the asymptotic behaviour of performance metrics in opportunistic scheduling. For example, the limiting distribution of spectral efficiency for multi-hop relaying techniques employing opportunistic scheduling is analyzed by works such as (Oyman, 2007; Oyman and Win, 2008; Oyman, 2010) using EVT. The authors of the work (Al-Ahmadi, 2012) investigate the ergodic capacity of opportunistic scheduling for a gamma-gamma composite fading channel. The asymptotic distributions of metrics such as ergodic capacity, mutual information, end-to-end signal to noise ratio (SNR), ergodic secrecy rate (ESR) in a multi-relay setup are discussed in works such as (Kountouris and Andrews, 2009; Xue and Abreu, 2010; Xia and Aissa, 2014; Biswas *et al.*, 2016; Xu *et al.*, 2016; Sibomana and Zepernick, 2016; Al-Badarneh *et al.*, 2018). The asymptotic PDF of the maximum of i.i.d. sums of independent and non-identically distributed (i.n.i.d.) gamma RVs is shown to be a Gumbel PDF in (Kalyani and Karthik, 2012). SIR-based asymptotic throughput analysis for opportunistic scheduling of MIMO downlink systems for Rayleigh fading channels is performed in (Pun *et al.*, 2011). Here, using EVT, the limiting distribution is found to be Frechet distribution. To the best of our knowledge, there is currently no work that gives results similar to (Pun *et al.*, 2011), even for Rician or Nakagami fading channels.

Also, the recent work (Gao *et al.*, 2018), derives the statistical upper channel capacity bounds for FAS systems using EVT in the large-scale limit only for Rayleigh fading channels.

For the case of $\kappa - \mu$ shadowed fading, we have already noticed that the recent work (Kumar and Kalyani, 2017*b*) derives the CDF of SIR in terms of an infinite summation of the Lauricella function of the fourth kind. This Lauricella function itself involves N-fold infinite summation (Here, N denotes the number of interferers). Now, determining the CDF of maximum over L such i.i.d. SIR realizations involve raising the CDF to power L, making mathematical analyses like computing rate very difficult. Even the evaluation of the exact CDF of the maximum of two SIR RVs having two i.n.i.d interferers takes more than an hour to compute in Mathematica with the series expansion given by (Kumar and Kalyani, 2017*b*, Eqn. (8)). Further, the evaluation of the exact CDF of the maximum of four SIR RVs with each SIR RV having four i.n.i.d interferers in a $\kappa - \mu$ shadowed fading environment times out in Mathematica. We can observe a similar problem for the case of the minimum also. Therefore, a limiting distribution for the maximum or minimum of SIR RVs, which is not only easy to compute but is also amenable to mathematical analysis, will have significant utility.

1.5 Our contributions

In this thesis, we explore some possible ways to overcome these challenges. The main contributions of the thesis are three-fold:

1. We first explore the commonly used moment matching for deriving approximations for generalized fading channels in the following three different applications:

    (a) We derive Secure outage probability (SOP), and strictly positive secrecy capacity (SPSC) expressions for single-input multiple-output systems over $\kappa - \mu$ shadowed fading channels. We also derive a novel approximation for the SPSC is derived in terms of a single Gauss hypergeometric function.

    (b) We consider the uplink of multi-user massive MIMO systems, in which the base station (BS) antennas have low-resolution analog-to-digital con-

verter (ADC). We assume that the channels undergo independent and non-identically distributed Nakagami-m fading. Employing moment matching, we approximate the signal-to-interference-plus-quantization-noise ratio at the output of maximum ratio combiner (MRC) by a gamma RV. Using this approximate probability density function (pdf), tight approximations for metrics, such as OP and rate, are derived.

(c) We then derive approximate OP and rate expressions for systems employing MRC, when both the desired signal and the interfering signals undergo $\eta - \mu$ fading, with the interferers having unequal power. The approximations are in terms of the Gauss hypergeometric function. Also, the utility of the applications is established in a vehicular application.

2. We then derive approximate symbol error rate (SER) and OP expressions when Optimum Combining (OC) is employed and when both the user and interferers undergo Rician fading. We will highlight the complicated nature of these expressions even when the eigenvalue statistics of non-central Wishart distribution is available. Hence, there is a motivation to develop an approximate matrix distribution for generalized fading distributions like $\kappa - \mu$ and $\eta - \mu$ distribution. For both complex $\kappa - \mu$ and $\eta - \mu$ distribution, we propose a Wishart distributed approximation of $\mathbf{H}\mathbf{H}^H$, such that the approximation has the least KL divergence from the original matrix distribution. For complex $\kappa - \mu$, we also propose another simple Wishart approximation of $\mathbf{H}\mathbf{H}^H$, such that the first moment is matched with the original matrix distribution of $\mathbf{H}\mathbf{H}^H$. Also, the degree of freedom is fixed to be the number of columns of the matrix $\mathbf{H}$. We explore the utility of the approximations in some applications.

3. We assume that the user signal and the interferer signals undergo i.n.i.d. $\kappa - \mu$ shadowed fading and determine the limiting distribution of the maximum/minimum of L such i.i.d. SIR RVs. We also analyze the utility of the derived asymptotic results in some applications. **This is a collaborative work**.

The most ideal situation would be to use $\kappa - \mu$ shadowed fading, across all the chapters. However, the phase distribution of $\kappa - \mu$ shadowed fading was not available in the open literature at the time of writing the thesis. Also, the works such as Atapattu *et al.* (2011); Zhengdao Wang and Giannakis (2003); Sharma Banjade *et al.* (2015) have also unified some fading distributions. But the phase statistics of these unified distributions are also not available. Without a phase distribution or an envelope-phase joint pdf, it is not possible to derive the approximate matrix distribution in Chapter 7, because the matrix $\mathbf{H}$ under consideration has complex elements. Also, expressions for MRC receivers used in Chapter 4 and 5 require the envelope-phase joint pdf. Therefore, we have used a variety of generalized fading channels across chapters depending on the availability of the phase statistics.

1.6 Thesis Overview

The rest of the thesis is organized as follows: In chapter 2, we provide a brief overview of the commonly used fading models. In chapter 3, we derive approximate SPSC expressions for single-input multiple-output systems over $\kappa-\mu$ shadowed fading channels. In chapter 4, we consider the uplink of multi-user massive MIMO systems, in which the BS antennas have low-resolution ADCs. In chapter 5, we derive approximate OP and rate expressions for systems employing MRC, when both the desired signal and the interfering signals undergo $\eta-\mu$ fading. In chapter 6, we derive approximate SER and OP expressions when OC is employed in Rician fading. In chapter 7, we propose a Wishart distributed approximation for the generalized fading matrix model and derive the ergodic capacity, OP, of such channels. In chapter 8, we derive the asymptotic results for generalized fading channels and discuss the utility of the results in some applications. In chapter 9, we conclude the work and give possible future directions.

CHAPTER 2

Common fading models

In this chapter, we discuss some of the commonly used fading models and their envelope distributions. Towards the end, we list popular literature in the context of generalized fading channels.

2.1 Rayleigh Fading Model

Rayleigh fading model, the most commonly used fading model, assumes that a large number of statistically independent scattered paths with random amplitudes contribute to a single tap of the channel (Tse and Viswanath, 2005). The complex channel gain is assumed to be a circularly symmetric complex Gaussian random variable, i.e., $c \sim \mathcal{CN}(0, \sigma^2)$. The power of the channel, $|c|^2$ is an exponential random variable with mean σ^2. For an $n_1 \times n_2$ matrix $\mathbf{C}$ with n_1 i.i.d rows taken from a multivariate complex Gaussian distribution with zero mean and $n_2 \times n_2$ covariance matrix Σ, the $n_1 \times n_1$ gram matrix, $\mathbf{R} = \mathbf{C}\mathbf{C}^H$ follows a complex Wishart distribution with n_2 degrees of freedom (James, 1964), i.e., $\mathbf{R} \sim \mathcal{CW}_{n_1}(n_2, \mathbf{\Sigma})$.

2.2 Rician Fading Model

Rician fading model is an extension to the Rayleigh fading model considering a line-of-sight (LOS) component along with scattering. The complex channel gain is modelled as a zero-mean complex Gaussian with an added LOS term as (Tse and Viswanath, 2005),

$$c = \sqrt{\Omega}\left(\sqrt{\frac{\kappa}{\kappa+1}}exp(j\theta) + \sqrt{\frac{1}{\kappa+1}}\mathcal{CN}(0,1)\right), \tag{2.1}$$

where Ω denotes the average received power and κ is the Rician factor which is the ratio of energy in LOS path to the energy in scattered paths. The magnitude of such

a random variable follows a Rician distribution. $\kappa = 0$ reduces to the Rayleigh case. Statistical model model for the gram matrix, $\mathbf{R} = \mathbf{C}\mathbf{C}^H$, can be given as

$$\mathbf{R} = (\mathbf{H} + \mathbf{M})(\mathbf{H} + \mathbf{M})^H, \tag{2.2}$$

where $\mathbf{H}$ is an $n_1 \times n_2$ matrix $\mathbf{C}$ with n_1 i.i.d rows taken from a multivariate complex Gaussian distribution with zero mean and $n_2 \times n_2$ covariance matrix $\mathbf{\Sigma}$. The $n_1 \times n_1$ gram matrix , $\mathbf{R}$, follows a con-central complex Wishart distribution with n_2 degrees of freedom (James, 1964). We follow the notation, $\mathbf{R} \sim \mathcal{CW}_{n_1}(n_2, \mathbf{\Sigma}, \mathbf{\Omega})$, where $\mathbf{\Omega} = \mathbf{\Sigma^{-1}}\mathbf{M}\mathbf{M^H}$.

2.3 Nakagami-m Fading Model

Nakagami-m physical model is based on the concept of "clustering" or "bunching" of scatterers (Nakagami, 1960). The channel consists of m clusters of scatterers located randomly within the confines of the channel. Each cluster is assumed to have a sufficiently large number of scatterers so that the contribution from each cluster is a complex Gaussian. Thus, the total received power can be modeled as

$$|c|^2 = \sum_{i=1}^{m} |h_i|^2, \tag{2.3}$$

where h_i are i.i.d complex gaussian random variables. The distribution of the power is given by a gamma distribution as

$$f_{|c|^2}(x) = \frac{x^{m-1}}{\Gamma(m)\left(\frac{\Omega}{m}\right)^m} exp\left(-\frac{m}{\Omega}x\right); x > 0, m > .5 \tag{2.4}$$

where Ω denotes the average power. Rayleigh Fading model can be considered as a special case of Nakagami-m fading with $m = 1$.

2.4 κ-μ Fading Model

The fading model for the κ-μ distribution (Yacoub, 2007) considers a signal composed of clusters of multipath waves, propagating in a non-homogeneous environment. Within

any one cluster, the phases of the scattered waves are random and have similar delay times, with delay-time spreads of different clusters being relatively large. The clusters of multipath waves are assumed to have scattered waves with identical powers, but within each cluster, a dominant component is found, which presents an arbitrary power. $\kappa - \mu$ power variable can be defined as,

$$R^2 = \sum_{i=1}^{\mu}(X_i + p_i)^2 + \sum_{i=1}^{\mu}(Y_i + q_i)^2, \tag{2.5}$$

where X_i and Y_i are mutually independent Gaussian processes with zero mean and σ^2 variance. Define $p^2 = \sum_{i=1}^{\mu} p_i^2$ and $q^2 = \sum_{i=1}^{\mu} q_i^2$. Let us define $\kappa = \frac{p^2+q^2}{2\mu\sigma^2}$. The envelope of $\kappa - \mu$ distribution is given by (Dias *et al.*, 2008),

$$f_R(r) = \frac{2\mu(1+\kappa)^{\frac{\mu+1}{2}}}{\kappa^{\frac{\mu-1}{2}} exp(\mu\kappa)\hat{r}}\left(\frac{r}{\hat{r}}\right)^{\mu} exp[-\mu(1+\kappa)\left(\frac{r}{\hat{r}}\right)^2] I_{\mu-1}[2\mu\sqrt{\kappa(1+\kappa)}\left(\frac{r}{\hat{r}}\right)], \tag{2.6}$$

where $\hat{r} = \mathbb{E}[r]$, $I(.)$ is the Bessel function and $\mathbb{E}$ denotes expectation. κ is defined as the ratio of total power of LOS components to the total power of scattering components, where μ is the number of scattering clusters. κ parameter can be interpreted as the ratio between the total power of the dominant components and the total power of the scattered waves. Rician Fading model can be considered as a special case of κ-μ fading with $\mu = 1$ and Nakagami fading model is the special case with $\kappa = 0$.

2.5 η-μ Fading model

$\eta - \mu$ distribution is used to better represent the small-scale variation of the fading signal in a non-line-of-sight condition. The power variable can be written as,

$$R^2 = \sum_{i=1}^{\mu}(X_i)^2 + \sum_{i=1}^{\mu}(Y_i)^2, \tag{2.7}$$

where X_i and Y_i are mutually independent Gaussian processes with zero mean and variances σ_X^2 and σ_Y^2 respectively. The distribution is defined in two formats. In format-1, $0 < \eta < \infty$ is the scattered-wave power ration between the in-phase and quadrature components of each cluster of multipath. There are two other parameters $h = \frac{2+\eta^{-1}+\eta}{4}$ and $\tilde{H} = \frac{\eta^{-1}-\eta}{4}$. The in-phase and quadrature components within each

cluster are assumed to be independent from each other and to have different powers and $\eta = \sigma_X^2/\sigma_Y^2$. In format-2, $-1 < \eta < 1$ is the correlation coefficient between the scattered-wave in-phase and quadrature components of each cluster of multipath. In this case, $h = \frac{1}{1-\eta^2}$ and $H = \frac{\eta}{1-\eta^2}$. The in-phase and quadrature components within each cluster are assumed to have identical powers σ^2 and to be correlated with each other and $\eta = \mathbb{E}(X_i Y_i)/\sigma^2$. The envelope of $\eta - \mu$ distribution is given by (Dias *et al.*, 2008),

$$f_R(r) = \frac{4\sqrt{\pi}\mu^{\mu+0.5}h^{\mu}}{\Gamma(\mu)\tilde{H}^{\mu-0.5}}\big(\frac{r}{\hat{r}}\big)^{2\mu} exp\Bigg[-2\mu h\big(\frac{r}{\hat{r}}\big)^2\Bigg] I_{\mu-1/2}[2\mu\tilde{H}\big(\frac{r}{\hat{r}}\big)^2], \tag{2.8}$$

where $\hat{r} = \mathbb{E}[r]$ and $I()$ is the Bessel function.

2.6 $\kappa - \mu$ shadowed fading

To investigate shadowing of the dominant component, a shadowed Rice model with random LOS component is introduced in (Abdi *et al.*, 2003). A further generalization of the shadowed Rician fading is the $\kappa - \mu$ shadowed fading, which has been studied in both (Paris, 2014) and (Cotton, 2015). Also, $\kappa - \mu$ shadowed fading has been shown to unify the $\kappa - \mu$ and $\eta - \mu$ fading models (Moreno-Pozas *et al.*, 2016) and to have a wide variety of applications ranging from land-mobile satellite systems to device to device communication (Cotton, 2015). A $\kappa - \mu$ shadowed RV X with parameters $(\kappa, \mu, m, \bar{x})$ has the following pdf (Paris, 2014):

$$f_X(x) = \frac{x^{\mu-1}}{\theta^{\mu-m}\lambda^m\Gamma[\mu]}e^{-\frac{x}{\theta}}\ {}_1F_1\left(m, \mu, \frac{x}{\theta} - \frac{x}{\lambda}\right),\ x \geq 0 \tag{2.9}$$

where ${}_1F_1(.)$ is the confluent hypergeometric function, $\Gamma[.]$ is the gamma function, $\theta = \dfrac{\bar{x}}{\mu(1+\kappa)}$, $\lambda = \dfrac{(\mu\kappa + m)\bar{x}}{\mu(1+\kappa)m}$ and $\bar{x} = \mathbb{E}[X]$.

2.7 Some important works on generalized fading channels

Performance metrics for $\kappa - \mu$ shadowed fading have been studied extensively in works like (García-Corrales *et al.*, 2014; Zhang *et al.*, 2015*a*; Chen *et al.*, 2016; Li *et al.*, 2017*a*; Zhang *et al.*, 2017*a*; Chandrasekaran and Kalyani, 2015; Thomas *et al.*, 2016). For example, the exact capacity and effective capacity expressions for $\kappa - \mu$ shadowed fading channel have been derived in (García-Corrales *et al.*, 2014) and (Zhang *et al.*, 2015*a*) respectively. Expressions for the effective rate of MISO systems over $\kappa - \mu$ shadowed fading models have been derived in (Li *et al.*, 2017*a*). However, all the above works either do not consider the impact of CCI or consider only Rayleigh faded interferers. There are works like (Morales-Jimenez *et al.*, 2012; Paris, 2013; Ermolova and Tirkkonen, 2014; Kumar and Kalyani, 2015; Kumar *et al.*, 2015*a*; Zhang *et al.*, 2017*a*), which consider CCI in a generalized fading setting and characterize the SIR. For example, outage probability expression for $\eta - \mu$ signal of interest (SOI) and Rayleigh faded interferers is derived in terms of confluent Lauricella function in (Morales-Jimenez *et al.*, 2012). Outage probability expressions, when SOI experiences $\eta - \mu$ or $\kappa - \mu$ fading and the interfering signals are subject to $\eta - \mu$ fading, have been derived in (Paris, 2013). This was further extended to cases where CCI can be either $\eta - \mu$ or $\kappa - \mu$ fading in the presence of white Gaussian noise in (Ermolova and Tirkkonen, 2014). Expressions for coverage probability and rate are derived in terms of Lauricella's function of the fourth kind in (Kumar and Kalyani, 2015) when SOI and interferers experience $\kappa - \mu$ and $\eta - \mu$ fading respectively. Approximate outage probability and rate expressions are derived in terms of the Appell function in (Kumar *et al.*, 2015*a*), when the user channel and the interferers experience $\kappa - \mu$ and $\eta - \mu$ fading respectively.

Though new, $\kappa - \mu$ shadowed fading has its fair share in the literature that characterize SIR. In (Parthasarathy and Ganti, 2017) coverage probability expressions are derived when the base stations are modeled as Poisson point process (PPP) and the channels experience $\kappa - \mu$ shadowed fading. Expressions for error vector magnitude (EVM) are derived in (Parthasarathy *et al.*, 2018) for an interference-limited system when both the desired channel and interferers experience i.n.i.d. $\kappa - \mu$ shadowed fad-

ing. Approximate outage probability and capacity expressions are derived for $\kappa - \mu$ shadowed fading channels in (Kumar, 2015). Exact outage and rate expressions in the presence of CCI has been studied in (Kumar and Kalyani, 2017*b*) only recently.

CHAPTER 3

Secrecy capacity of $\kappa - \mu$ shadowed fading channels

3.1 Introduction

A wiretap channel, in which a user (Alice) intends to communicate with the receiver (Bob) in the presence of an eavesdropper (Eve), was first proposed and analyzed in (Wyner, 1975). The impact of fading on key secrecy performance metrics, such as secure outage probability (SOP) and strictly positive secrecy capacity (SPSC), was analyzed from a wireless perspective in (Bloch *et al.*, 2008). (Liu, 2013) and (Sarkar *et al.*, 2009) extended the study to the cases of Rician, and Nakagami faded channels, respectively.

Expressions for SOP and SPSC for the case of $\kappa - \mu$ channels, with a passive eavesdropper[1], were derived by the works (Bhargav *et al.*, 2016) and (Iwata *et al.*, 2017). The treatise (Moualeu and Hamouda, 2017) extended these results to the single-input multiple-output (SIMO) case. Since these works derive expressions for SPSC in terms of infinite summations, they are challenging to analyze.

3.1.1 Our contributions

In this chapter, we derive an expression for SOP when both the user and the passive-eavesdropper channel are subject to $\kappa - \mu$ shadowed fading for the SIMO case. We also derive an expression for SPSC in terms of infinite summations. This result generalizes all the existing results from (Bhargav *et al.*, 2016; Iwata *et al.*, 2017; Moualeu and Hamouda, 2017). Furthermore, we also derive a simple approximate expression for SPSC in terms of the Gauss hypergeometric function. This approximation for SPSC is useful for the particular case of $\kappa - \mu$ fading, which otherwise has to make use of the

M. Srinivasan and S. Kalyani, "Secrecy Capacity of $\kappa - \mu$ Shadowed Fading Channels," in IEEE Communications Letters, vol. 22, no. 8, pp. 1728-1731, Aug. 2018.

[1]A passive eavesdropper detects the information by merely listening to the transmission in the broadcast medium

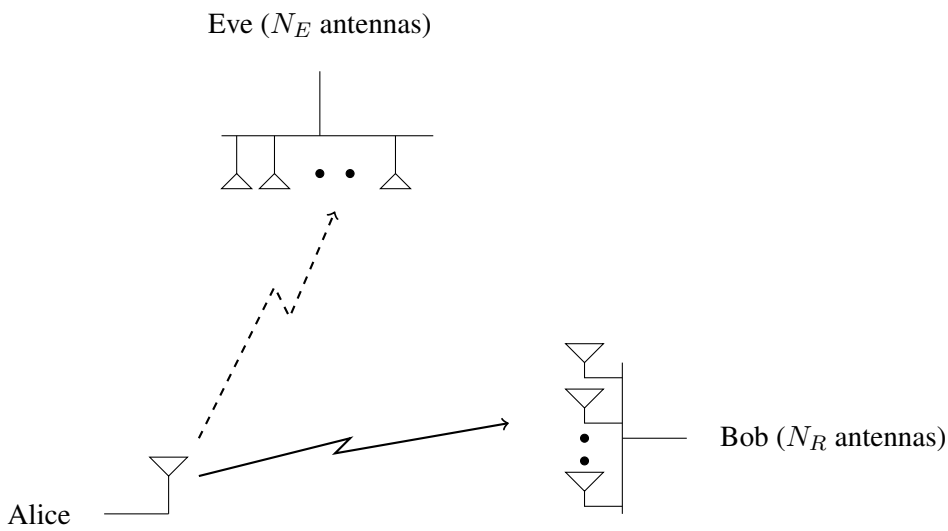

Figure 3.1: System model

expressions of (Bhargav *et al.*, 2016; Iwata *et al.*, 2017; Moualeu and Hamouda, 2017), all of which involve infinite summations. Based on this approximate expression, we study the impact of κ, μ, and the shadowing parameter m.

3.2 System model

The classic three-node model in which the legitimate transmitter (Alice) communicates with a legitimate receiver/user (Bob) in the presence of an eavesdropper (Eve) is considered (Refer Fig. 3.1). The receiver and eavesdropper have N_R and N_E antennas respectively, whereas the transmitter is assumed to have only one transmit antenna.

All the channels are subject to $\kappa - \mu$ shadowed fading, whose probability density function (pdf) is given by (Paris, 2014, Eq 4). The $N_R \times 1$ received signal $\mathbf{y}_b$ at Bob is given by $\mathbf{y}_b = \mathbf{c}_b u + \mathbf{z}_b$. Here, the powers of the elements of $\mathbf{c}_b = [c_b[1], ..., c_b[N_R]]^T$ are independent and identically distributed (i.i.d.) $\kappa - \mu$ shadowed random variables and $\mathbf{z}_b$ are additive white Gaussian noise (AWGN) variables with variance σ_b^2. The $N_E \times 1$ received signal $\mathbf{y}_e$ at Eve is given by $\mathbf{y}_e = \mathbf{c}_e u + \mathbf{z}_e$. The powers of elements of $\mathbf{c}_e = [c_e[1], ..., c_e[N_E]]^T$ are also i.i.d. $\kappa - \mu$ shadowed random variables and $\mathbf{z}_e$ are AWGN variables with variance σ_e^2. u denotes the desired user symbol belonging to unit energy M-QAM constellation.

The instantaneous signal-to-noise ratio (SNR) at each of the antennas i is given by $\frac{|\mathbf{c}_b[i]|^2}{\sigma_b^2}$ at Bob and by $\frac{|\mathbf{c}_e[i]|^2}{\sigma_e^2}$ at Eve. For the case of maximum ratio combining (MRC)

at the receivers, the total instantaneous SNR γ_b at Bob is given by $\sum_{i=1}^{N_R} \frac{|\mathbf{c}_b[i]|^2}{\sigma_b^2}$ and the average SNR $\bar{\gamma}_b = \sum_{i=1}^{N_R} \frac{\mathbb{E}[|\mathbf{c}_b[i]|^2]}{\sigma_b^2}$. Hence the pdf of γ_b is given by (Paris, 2014, Eq 4)

$$f_{\gamma_b}(\gamma_b) = \frac{(N_R\mu_b)^{N_R\mu_b}\,(N_R m_b)^{N_R m_b}\,(1+\kappa_b)^{N_R\mu_b}}{\Gamma(N_R\mu_b)(\bar{\gamma}_b)^{N_R\mu_b}} \frac{\gamma_b^{N_R\mu_b-1} e^{-\frac{N_R\mu_b(1+\kappa_b)\gamma_b}{\bar{\gamma}_b}}}{(N_R\mu_b\kappa_b + N_R m_b)^{N_R m_b}} \times {}_1F_1\left(N_R m_b, N_R\mu_b, \frac{N_R^2\mu_b^2\kappa_b(1+\kappa_b)\gamma_b}{(N_R\mu_b\kappa_b + N_R m_b)\bar{\gamma}_b}\right). \tag{3.1}$$

Similarly, the total instantaneous SNR γ_e at Eve is given by $\sum_{i=1}^{N_E} \frac{|\mathbf{c}_e[i]|^2}{\sigma_e^2}$ and the average SNR $\bar{\gamma}_e = \sum_{i=1}^{N_E} \frac{\mathbb{E}[|\mathbf{c}_e[i]|^2]}{\sigma_e^2}$. Hence, the pdf of γ_e is similar to (3.1), but with N_R, κ_b, μ_b, m_b and $\bar{\gamma}_b$ replaced by N_E, κ_e, μ_e, m_e and $\bar{\gamma}_e$ respectively.

3.3 SOP and SPSC Expressions

The secrecy capacity C_S is defined in (Bhargav *et al.*, 2016) as

$$C_S = \Big[\log_2(1+\gamma_b) - \log_2(1+\gamma_e)\Big] I_{(\gamma_b > \gamma_e)}. \tag{3.2}$$

SOP, denoted by P_{out}, is the probability that the secrecy capacity falls below a target secrecy rate R_S (Iwata *et al.*, 2017). Therefore,

$$\begin{aligned} P_{out}(R_S) &= P(C_S < R_S) = P\left(\gamma_b < (1+\gamma_e)2^{R_S} - 1\right) \\ &= \int_0^\infty f_{\gamma_e}(\gamma_e) \int_0^{(1+\gamma_e)2^{R_S}-1} f_{\gamma_b}(\gamma_b) d\gamma_b d\gamma_e. \end{aligned} \tag{3.3}$$

Substituting the pdfs from (3.1) in the above equation and expanding ${}_1F_1(.)$ using (Abramowitz and Stegun, 1972, Eq 13.1.2), we obtain

$$P_{out}(R_S) = C \sum_{p=0}^{\infty}\sum_{q=0}^{\infty} B_{pq} \int_0^\infty e^{-y\gamma_e}\gamma_e^{b_q-1} \int_0^{(1+\gamma_e)2^{R_S}-1} e^{-x\gamma_b}\gamma_b^{a_p-1} d\gamma_b d\gamma_e, \tag{3.4}$$

where $x = \frac{N_R\mu_b(1+\kappa_b)}{\bar{\gamma}_b}$, $y = \frac{N_E\mu_e(1+\kappa_e)}{\bar{\gamma}_e}$, $a_p = N_R\mu_b + p$, $b_q = N_E\mu_e + q$, $X = \frac{N_R\mu_b\kappa_b}{N_R\mu_b\kappa_b + N_R m_b}$, $Y = \frac{N_E\mu_e\kappa_e}{N_E\mu_e\kappa_e + N_E m_e}$, $C = \frac{x^{N_R\mu_b}(1-X)^{N_R m_b}}{\Gamma(N_R\mu_b)} \frac{y^{N_E\mu_e}(1-Y)^{N_E m_e}}{\Gamma(N_E\mu_e)}$ and $B_{pq} = \frac{(N_R m_b)_p}{(N_R\mu_b)_p p!}\frac{(N_E m_e)_q}{(N_E\mu_e)_q q!}(xX)^p\,(yY)^q$. We now solve the integrals using the identity (Grad-

shteyn and Ryzhik, 2007, Eq. 3.351) and obtain

$$P_{out}(R_S) = C\sum_{p=0}^{\infty}\sum_{q=0}^{\infty} B_{pq}\int_0^{\infty} e^{-y\gamma_e}\gamma_e^{b_q-1}\frac{(a_p-1)!}{x^{a_p}}\Big[1- e^{-\left((1+\gamma_e)2^{R_S}-1\right)x}\sum_{k=0}^{a_p-1}\frac{\left((1+\gamma_e)\,2^{R_S}-1\right)^k}{k!x^{-k}}\Big]d\gamma_e. \tag{3.5}$$

Further, we use the integration identity $\int_0^{\infty} e^{-px}x^{q-1}(1+ax)^{-v}dx = a^{-q}\Gamma(q)U(q, q+1-v, p/a)$ from (Gradshteyn and Ryzhik, 2007), to obtain the final SOP expression as

$$P_{out}(R_S) = 1 - C\sum_{p=0}^{\infty}\sum_{q=0}^{\infty} B_{pq}\frac{\Gamma(a_p)}{x^{a_p}}\sum_{k=0}^{a_p-1}\frac{e^{-(2^{R_S}-1)x}\Gamma(b_q)}{k!(x(2^{R_S}-1))^{-k}} \times\left(\frac{2^{R_S}}{2^{R_S}-1}\right)^{-b_q} U\left(b_q, b_q+1+k, \frac{y+2^{R_S}x}{\frac{2^{R_S}}{2^{R_S}-1}}\right). \tag{3.6}$$

We can now obtain SPSC, denoted by P_0, from the identity $P_0 = 1 - P_{out}(0)$, i.e., by substituting $R_S = 0$ in (3.5), as

$$P_0 = C\sum_{p=0}^{\infty}\sum_{q=0}^{\infty} B_{pq}\int_0^{\infty} e^{-y\gamma_e}\gamma_e^{b_q-1}\frac{\Gamma(a_p, \gamma_e x)}{x^{a_p}}d\gamma_e.$$

Using the identity $\Gamma(a, z) = \Gamma(a) - \frac{z^a}{a}\,{}_1F_1(a, a+1, -z)$ from (Wolfram Research, 2020*e*), we first obtain the incomplete gamma function in terms of ${}_1F_1$ function. We then solve the integral using the identity (Gradshteyn and Ryzhik, 2007, Eq. 7.522.9) and rearrange the terms to obtain

$$P_0 = D\sum_{p=0}^{\infty}\sum_{q=0}^{\infty} A_{pq}G_{pq}, \tag{3.7}$$

where $D = (1-X)^{N_R m_b}(1-Y)^{N_E m_e}$, $A_{pq} = \frac{(N_R m_b)_p}{p!}\frac{(N_E m_e)_q}{q!}X^pY^q$ and

$$G_{pq} = \left(\frac{y}{x}\right)^{b_q}\frac{\Gamma(b_q+a_p)}{\Gamma(a_p)\Gamma(b_q+1)}\,{}_2F_1\left(b_q, b_q+a_p, b_q+1, -\frac{y}{x}\right).$$

By first substituting $N_R = N_E = 1$, then applying limits $m_b \to \infty$ and $m_e \to \infty$ to the terms D and A_{pq} and finally using the identity (Abramowitz and Stegun, 1972, Eq. 15.3.5), we can obtain the expression for SPSC for $\kappa-\mu$ faded channels for the SISO case and is found to match with (Bhargav *et al.*, 2016, Eq. 21). Also, by substituting $\mu = 2\mu'$, $\kappa = \frac{1-\eta}{2\eta}$ and $m = \mu'$ (Moreno-Pozas *et al.*, 2016), we can obtain SPSC

expressions for $\eta - \mu$ distribution, which has not been derived in literature, to the best of our knowledge. Also, by appropriately substituting parameters, we can even obtain expressions for SPSC, when the eavesdropper channel is $\kappa - \mu$ faded and the user channel is $\eta - \mu$ faded and vice-versa. Note that the expression given by (3.7) is typically evaluated by truncating the infinite summations. The impact of truncating these infinite summations is discussed in Appendix A.

3.3.1 Approximation

Using moment matching, we determine a novel approximation, which does not involve infinite summations, for the SPSC. We approximate $\kappa - \mu$ shadowed random variable by a gamma variate $\mathcal{G}(.,.)$ as done in (Kumar, 2015). Approximately, $\gamma_b \sim \mathcal{G}(\alpha_b, \lambda_b) = \lambda_b \mathcal{G}(\alpha_b, 1)$, where $\alpha_b = \frac{N_R m_b N_R \mu_b (1+\kappa_b)^2}{N_R m_b + N_R \mu_b \kappa_b^2 + 2 N_R m_b \kappa_b}$, and $\lambda_b = \frac{\bar{\gamma}_b}{\alpha_b}$. Similarly, $\gamma_e \sim \mathcal{G}(\alpha_e, \lambda_e) = \lambda_e \mathcal{G}(\alpha_e, 1)$, where $\alpha_e = \frac{N_E m_e N_E \mu_e (1+\kappa_e)^2}{N_E m_e + N_E \mu_e \kappa_e^2 + 2 N_E m_e \kappa_e}$, and $\lambda_e = \frac{\bar{\gamma}_e}{\alpha_e}$. Hence, SPSC is given by

$$P(\gamma_b > \gamma_e) \approx P\left(\frac{\mathcal{G}(\alpha_b, \lambda_b)}{\mathcal{G}(\alpha_e, \lambda_e)} > 1\right) = P\left(\frac{\mathcal{G}(\alpha_e, 1)}{\mathcal{G}(\alpha_b, 1)} < \frac{\lambda_b}{\lambda_e}\right), \tag{3.8}$$

which is the cumulative distribution function (CDF) of the ratio of two gamma distribution evaluated at $\frac{\lambda_b}{\lambda_e}$ or $\frac{\bar{\gamma}_b \alpha_e}{\bar{\gamma}_e \alpha_b}$. This distribution is also called the beta-prime distribution or the beta distribution of the second kind (Dubey, 1970). The CDF of a beta-prime distributed random variable, say Z, with parameters α and β is given in (Cordeiro and Lemonte, 2012, Eq 2) as $P(Z < z) = \frac{(\beta)_\alpha z^\alpha \, {}_2F_1(\alpha+\beta, \alpha, \alpha+1, -z)}{\Gamma(\alpha+1)}$ for $z \geq 0$. Therefore, SPSC is given by

$$P_0 \approx \frac{(\alpha_b)_{\alpha_e} \bar{\gamma}_b^{\alpha_e} \alpha_e^{\alpha_e} \, {}_2F_1\left(\alpha_e + \alpha_b, \alpha_e, \alpha_e + 1, \frac{-\bar{\gamma}_b \alpha_e}{\bar{\gamma}_e \alpha_b}\right)}{\alpha_b^{\alpha_e} \bar{\gamma}_e^{\alpha_e} \Gamma(\alpha_e + 1)}. \tag{3.9}$$

This approximation involves only a simple Gauss hypergeometric function and is easy to analyze when compared to (3.7).

3.3.2 Inferences

We can make the following inferences from (3.9):

I1) Since the approximate SPSC given by (3.9) is a CDF, $P_0(z+\delta) > P_0(z)$ for $\delta > 0$. Therefore, an increase in $\bar{\gamma}_b$ or a decrease in $\bar{\gamma}_e$ leads to an increase in SPSC.

I2) From (Shaked and Shanthikumar, 2007, Theorem 1.A.12) and from the pdf of beta-prime distributed random variables given in (Dubey, 1970) for two different sets of parameters namely (α_e, α_b) and $(\alpha_e, \alpha_b + \delta)$, one can prove that

$$\mathcal{S}^{-}\left(\frac{\Gamma(\alpha_e+\alpha_b)}{\Gamma(\alpha_e)\Gamma(\alpha_b)} z^{\alpha_e-1}(1+z)^{-\alpha_e-\alpha_b} - \frac{\Gamma(\alpha_e+\alpha_b+\delta)}{\Gamma(\alpha_e)\Gamma(\alpha_b+\delta)} z^{\alpha_e-1}(1+z)^{-\alpha_e-\alpha_b-\delta}\right) = 1,$$

where $\mathcal{S}^{-}$ denotes the number of sign changes and $\delta > 0$. The sign change from $-$ to $+$ occurs at $z = \left(\frac{(\alpha_b+\delta)_{\alpha_e}}{(\alpha_b)_{\alpha_e}}\right)^{\frac{1}{\delta}} - 1$. Using the definition of stochastic ordering from (Shaked and Shanthikumar, 2007, 1.A.1), we can show that P_0 given by (3.9), evaluated using the parameter $\alpha_b + \delta$ is greater than the P_0 evaluated using the parameter α_b, for a constant $\frac{\bar{\gamma}_b \alpha_e}{\bar{\gamma}_e \alpha_b}$. The opposite is true for parameters $\alpha_e + \delta$ and α_e, for a constant $\frac{\bar{\gamma}_b \alpha_e}{\bar{\gamma}_e \alpha_b}$.

I3) An increase in N_R increases α_b, but decreases $\bar{\gamma}_b$ proportionately, thereby keeping $\frac{\bar{\gamma}_b \alpha_e}{\bar{\gamma}_e \alpha_b}$ constant. This leads to an increase in the SPSC due to inference I2. Similarly, an increase in N_E increases α_e keeping $\frac{\bar{\gamma}_b \alpha_e}{\bar{\gamma}_e \alpha_b}$ constant. Therefore, the SPSC decreases with an increase in N_E.

I4) An increase in μ_b or m_b increases α_b. But an increase in SPSC is not always guaranteed because $\frac{\bar{\gamma}_b \alpha_e}{\bar{\gamma}_e \alpha_b}$ decreases. On the other hand, if we proportionately increase $\bar{\gamma}_b$ or proportionately decrease $\bar{\gamma}_e$ to maintain a constant $\frac{\bar{\gamma}_b \alpha_e}{\bar{\gamma}_e \alpha_b}$, then SPSC increases. A similar inference can be made for a decrease in μ_e or m_e.

3.4 Numerical Results

P_{out} (SOP) and P_0 (SPSC) expressions given by (3.6) and (3.7) respectively are verified using Monte-Carlo simulations. While evaluating the expressions, all the infinite

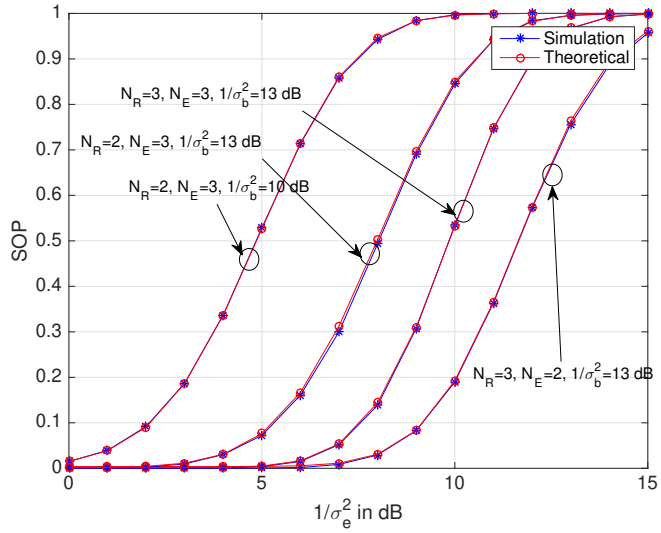

Figure 3.2: SOP vs $\frac{1}{\sigma_e^2}$ for $\kappa_b = 3, \kappa_e = 4, \mu_b = 5, \mu_e = 5, m_b = 3, m_e = 6, R_S = 1$

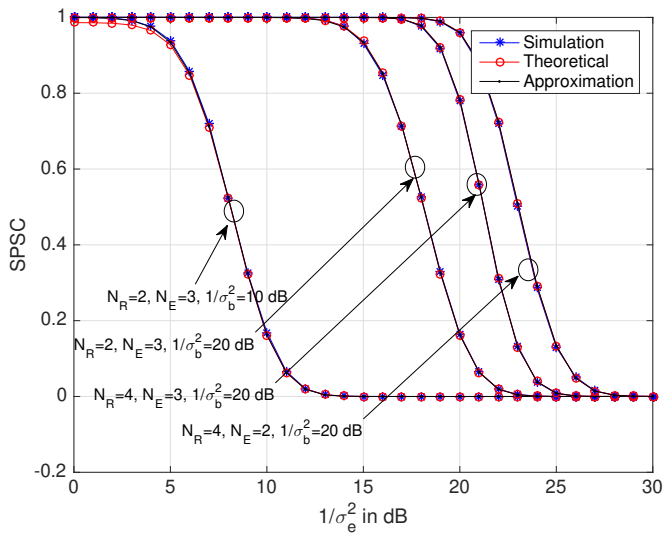

Figure 3.3: SPSC vs $\frac{1}{\sigma_e^2}$ for $\kappa_b = 3, \kappa_e = 4, \mu_b = 4, \mu_e = 5, m_b = 3, m_e = 6$

summations are truncated at $p = 100$ and $q = 100$ leading to a truncation error of order 10^{-3}. For Monte-Carlo simulations, we generate the $\kappa - \mu$ shadowed power random variables $|c_b[i]|^2$ for $i = 1, ..., N_R$ and $|c_e[i]|^2$ for $i = 1, ..., N_E$ and determine the SNRs γ_b and γ_e respectively. The probability that the secrecy capacity C_S (given by (3.2)) is lesser than a target secrecy rate R_S gives the required SOP. Similarly, the probability that the secrecy capacity is positive gives the required SPSC. From Fig. 3.2 and 3.3, we can observe that our expressions for the SOP (3.6) and the SPSC (3.7) matches the simulations for all values of N_R and N_E. Also, the approximation given by (3.9) for the SPSC is close to the simulated values.

As observed from Fig. 3.3, an increase in the number of user-antennas N_R increases

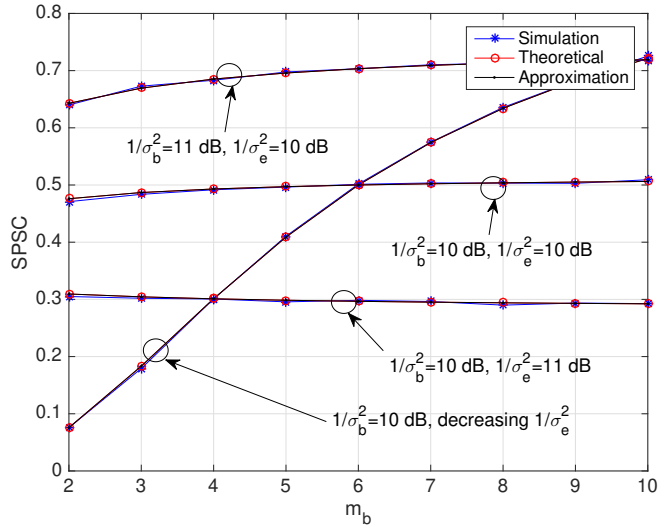

Figure 3.4: SPSC vs m_b for $N_R = 2, N_E = 2, \kappa_b = 3, \kappa_e = 3, \mu_b = 5, \mu_e = 5, m_e = 6$

the number of independent multi-path and hence leads to an increase in the SNR γ_b. Therefore, SPSC increases as N_R increases as corroborated by Fig. 3.3 and I3. The opposite happens as the number of eavesdropper-antennas N_E, increases. Also, SPSC increases with a decrease in the noise variance σ_b^2 at the user, or with an increase in noise variance σ_e^2 at the eavesdropper. This is as suggested by I1. On similar lines, SOP decreases with an increase in N_R or σ_b^2, as corroborated by Fig. 3.2. On the other hand, the SOP increases with increases with an increase in N_E or σ_e^2.

From Fig. 3.4, we can observe that an increase in m_b does not automatically guarantee an increase in SPSC at all times. Hence, for $\sigma_b^2 > \sigma_e^2$ and $\sigma_b^2 < \sigma_e^2$, there is a difference in how the SPSC trends with the increase in m_b. On the other hand, if the increase in m_b is accompanied by a decrease in $\bar{\gamma}_e$, then SPSC increases. This is also corroborated by I4. A figure with opposite trends can be obtained for an increase in m_e.

3.5 Conclusion

In this chapter, a classic three-node model, where a transmitter communicates with a receiver in the presence of an eavesdropper, was considered. The receiver and eavesdropper were assumed to have multiple antennas. SOP and SPSC expressions were derived for the case when all the channels were subject to $\kappa - \mu$ shadowed fading. Extensive Monte-Carlo simulations were performed and the expressions were found to match the simulation results for all cases. An approximation involving a single Gauss

hypergeometric function was also derived for SPSC. From the approximate expression, a number of inferences on the behavior of SPSC were drawn. One of the interesting observations was that an increase in the user's shadowing parameter does not guarantee an increase in SPSC for all user SNR values.

CHAPTER 4

Analysis of Massive MIMO with low resolution ADC in Nakagami-m fading

4.1 Introduction

Massive multi-input-multi-output (MIMO) system has been widely accepted as a key technology to meet the increasing demand for wireless throughput in both mobile and fixed scenarios and has been widely investigated in (Marzetta, 2015; Björnson *et al.*, 2016; Hanif *et al.*, 2018; Gao and Kaiser, 2016). However, one drawback with massive MIMO systems is that, since a large number of antennas are required, there is a substantial increase in the hardware cost and power consumption. Using high-speed high-resolution analog-to-digital converter (ADC) for all the antennas increases the power consumption of massive MIMO systems severely and this is considered as the bottleneck to realize massive MIMO in practice. To solve the power consumption problem, typically low-resolution ADCs (e.g., 1-3 bits) are employed at the radio frequency (RF) chains (Fan *et al.*, 2015).

Hence, it is imperative to study the performance of MIMO systems in conjunction with a quantizer. Works like (Singh *et al.*, 2009; Bai *et al.*, 2013; Orhan *et al.*, 2015) have analyzed such quantized MIMO systems. In (Singh *et al.*, 2009), the exact non-linear nature of quantizer is studied. Due to the complicated nature of the exact quantization error, in (Bai *et al.*, 2013; Orhan *et al.*, 2015) the quantization is modeled as additive and independent noise. This additive quantization noise model (AQNM) is further used in (Fan *et al.*, 2015) to study the impact of ADC resolution on the uplink rate for the case of Rayleigh fading channels. Some asymptotic results are also discussed in (Fan *et al.*, 2015). This was extended to the case of Rician fading in (Zhang *et al.*, 2016). In (Tan *et al.*, 2016), a mixed ADC architecture model is considered. In this architecture, a few antennas are equipped with costly full-resolution ADC and the

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rest with less expensive low-resolution ADC. This was further extended to the case of Rician fading channels in (Zhang *et al.*, 2017*b*).

Though the well-cited works like (Fan *et al.*, 2015; Zhang *et al.*, 2016; Tan *et al.*, 2016; Zhang *et al.*, 2017*b*) derive the expressions for uplink rate, in all these works, the average rate is obtained by approximating the expectation of logarithm of signal-to-interference plus quantization noise ratio (SIQNR). This was initially proposed in (Zhang *et al.*, 2014*b*) and then used in (Fan *et al.*, 2015; Zhang *et al.*, 2016; Tan *et al.*, 2016; Zhang *et al.*, 2017*b*). The limitation of this approximation is that it is very specific to obtaining rate expressions and cannot be adopted to obtaining approximate probability density function (pdf)/ cumulative distribution function (CDF) of SIQNR. Very recently, the authors of (Ding and Jing, 2018) have derived an approximate outage probability (OP) expression using the fact that if the squared coefficient of variation (SCV) of a random variable (RV) tends to zero, the RV approaches a deterministic value equal to its mean. However, the approximate expressions in (Ding and Jing, 2018) are shown to be tight only for a very large number of antennas at the base station (BS) and for Rayleigh fading channels. Hence, this approach for deriving the approximation for OP cannot be used if we assume any other type of channel fading, for example, Nakagami-m fading.

Though there are a few works like (Uddin *et al.*, 2014), which does performance analysis of massive MIMO systems with Nakagami-m fading channels, the impact of low resolution ADCs is not studied in these works. If low resolution ADCs are present, the quantization noise effect should also be taken into account. The quantization noise will have a covariance which depends on the fading channels, which makes the analysis completely different from those of conventional systems(Fan *et al.*, 2015).

4.1.1 Our contributions

In this chapter, we consider the channels to be independent and non-identically distributed (i.n.i.d.) Nakagami-m fading and derive a simple approximate pdf for the SIQNR, using moment matching. This pdf can be further used to obtain metrics like OP, rate, etc., for MIMO systems, where the quantization is modelled using AQNM. The derived approximations are simpler and tighter even for smaller values of number

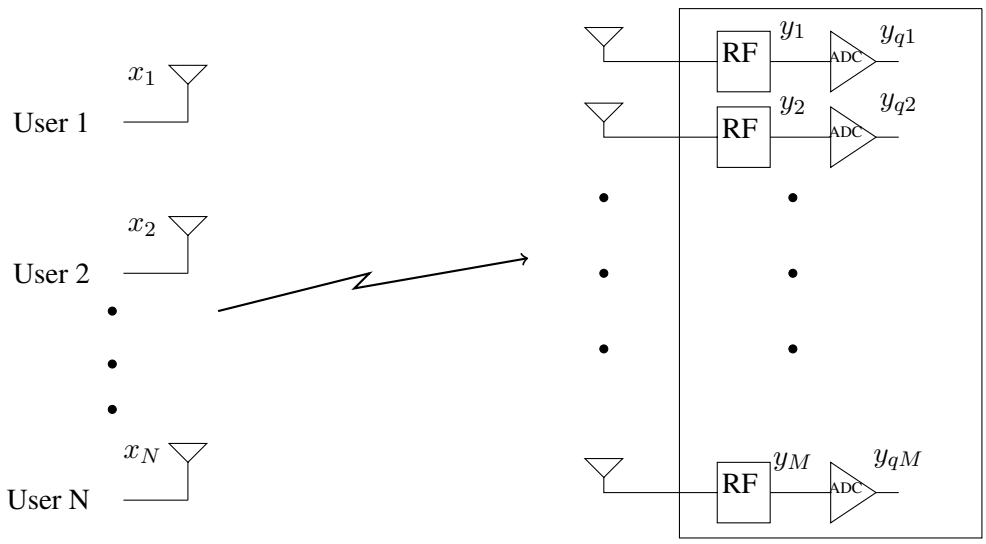

Figure 4.1: System model

of antennas when compared to the expressions for Rayleigh fading users in (Ding and Jing, 2018). Moreover, our approach can be used for deriving OP expressions for other types of fading channels also.

4.2 System model

The uplink of a multi-user MIMO (MU-MIMO) system formed by a BS with an array of M antennas serving N single-antenna user terminals is considered. All the user terminals are served in the same time-frequency resource. The schematic is shown in Fig. 4.1. The received $M \times 1$ vector $\mathbf{y}$ at BS can be expressed as

$$\mathbf{y} = \mathbf{G}\mathbf{x} + \mathbf{n}, \tag{4.1}$$

where $\mathbf{G}$ represents the $M \times N$ channel matrix between BS and users, $\mathbf{x}$ denotes unit-power $N \times 1$ symbols transmitted and $\mathbf{n}$ denotes the complex Gaussian noise i.e., $\mathbf{n} \sim \mathcal{CN}(0, \mathbf{I})$. The lnth element of the matrix $\mathbf{G}$ is given by the Nakagami-m fading complex channel coefficient g_{ln}, i.e., $g_{ln} = x_{ln} + jy_{ln}$ with density pdf (Yacoub *et al.*, 2005)

$$f(x_{ln}, y_{ln}) = \frac{m_n^{m_n} |x_{ln}|^{m_n-1} |y_{ln}|^{m_n-1} e^{-\frac{m_n}{\beta}\left(x_{ln}^2+y_{ln}^2\right)}}{\beta^{m_n}\Gamma^2\left(m_n/2\right)},$$

where β_n is the power coefficient and m_n is the number of clusters for user n. Note that, we consider i.n.i.d. Nakagami-m variables, i.e., β_n and m_n are different for different users. For AQNM, the output at the quantizer is given by (Fan *et al.*, 2015)

$$\mathbf{y}_q = \alpha\mathbf{y} + \mathbf{n}_q = \alpha\mathbf{G}\mathbf{x} + \alpha\mathbf{n} + \mathbf{n}_q, \tag{4.2}$$

with $\alpha = 1-\rho$, where ρ is the inverse of the signal-to-quantization-noise ratio, and $\mathbf{n}_q$ is the additive Gaussian quantization noise vector that is uncorrelated with $\mathbf{y}$. The relation between the number of quantization bins denoted by b and ρ is given in (Fan *et al.*, 2015, Table I) for $b \leq 5$ and is approximated as $\rho = \frac{\pi\sqrt{3}}{2}2^{-2b}$ for $b > 5$. For a fixed channel realization $\mathbf{G}$ and an identity input covariance matrix, $\mathbf{R}_{\mathbf{n_q n_q}}$ is the covariance of $\mathbf{n}_q$ and is given by (Fan *et al.*, 2015, Eq. 5)

$$\mathbf{R}_{\mathbf{n_q n_q}} = \alpha(1-\alpha)\text{diag}\left(\mathbf{G}\mathbf{G}^H + \mathbf{I}\right). \tag{4.3}$$

We assume that a maximum ratio combiner (MRC) is implemented at the BS similar to (Fan *et al.*, 2015; Zhang *et al.*, 2016; Tan *et al.*, 2016; Zhang *et al.*, 2017*b*). At the output of the MRC receiver, the received signal is given by

$$\mathbf{r} = \mathbf{G}^H\mathbf{y}_q. \tag{4.4}$$

By substituting (4.2) in (4.4), we obtain

$$\mathbf{r} = \alpha\mathbf{G}^H\mathbf{G}\mathbf{x} + \alpha\mathbf{G}^H\mathbf{n} + \mathbf{G}^H\mathbf{n}_q. \tag{4.5}$$

The nth element of $\mathbf{r}$ is given by

$$r_n = \alpha\mathbf{g}_n^H\mathbf{g}_n x_n + \alpha\sum_{i=1, i\neq n}^{N}\mathbf{g}_n^H\mathbf{g}_i x_i + \alpha\mathbf{g}_n^H\mathbf{n} + \mathbf{g}_n^H\mathbf{n}_q, \tag{4.6}$$

where $\mathbf{g}_n$ is the nth column of $\mathbf{G}$. Hence, the SIQNR of the nth user is given by

$$\gamma_n = \frac{X_n}{Y_n} = \frac{\alpha^2||\mathbf{g}_n||^4}{\alpha^2\left(\sum_{i=1,i\neq n}^{N}|\mathbf{g}_n^H\mathbf{g}_i|^2 + ||\mathbf{g}_n||^2\right) + \alpha(1-\alpha)Z_n}, \tag{4.7}$$

where $Z_n = \mathbf{g}_n^H diag(\mathbf{G}\mathbf{G}^H + \mathbf{I})\mathbf{g}_n$. The term Z_n can also be alternately, written as $\sum_{m=1}^{M}|g_{mn}|^2 + \sum_{m=1}^{M}|g_{mn}|^4 + \sum_{m=1}^{M}\sum_{i\neq n}|g_{mn}|^2|g_{mi}|^2$ as in (Fan *et al.*, 2015, Eq.

22).

4.3 The Approximation

To determine the exact pdf of the SIQNR is intractable, if not impossible. This is because of the term Z_n present in the denominator of (4.7). If this term had not been present, the equation reduces to finding signal-to-interference and noise ratio (SINR) at the output of the MRC for complex Nakagami-m fading channel. In order to circumvent this difficulty and to find the average rate expressions, i.e., $\mathbb{E}\left(\log_2\left(1+\frac{X_n}{Y_n}\right)\right)$, works like (Fan *et al.*, 2015) have approximated the rate expressions by $\log_2\left(1+\frac{\mathbb{E}(X_n)}{\mathbb{E}(Y_n)}\right)$. However, this method cannot be applied for obtaining the pdf/CDF of the SIQNR. To get an approximate expression for the pdf of SIQNR, we can approximate the RV γ_n by a known RV like gamma distribution using moment matching.[1] But these moment expressions for γ_n are again mathematically difficult to determine. It is precisely because of this difficulty that all prior works like (Fan *et al.*, 2015) determined even the rate expressions by approximations. To solve this problem, we propose to determine the approximate first and second moments of γ_n given by $\mathbb{E}[\gamma_n] = \mathbb{E}\left[\frac{X_n}{Y_n}\right]$ and $\mathbb{E}[\gamma_n^2] = \mathbb{E}\left[\left(\frac{X_n}{Y_n}\right)^2\right]$ respectively, by expanding these functions using Taylor series. In other words, we apply the second order bi-variate Taylor series expansion of the functions $f(X_n, Y_n) = \frac{X_n}{Y_n}$ and $g(X_n, Y_n) = \frac{X_n^2}{Y_n^2}$ around $\mathbb{E}[X_n]$ and $\mathbb{E}[Y_n]$ and determine the expectation to get these approximations. Hence,

$$\mathbb{E}[\gamma_n] \approx \frac{\mathbb{E}[X_n]}{\mathbb{E}[Y_n]} - \frac{Cov(X_n, Y_n)}{\mathbb{E}[Y_n]^2} + \frac{Var(Y_n)\mathbb{E}[X_n]}{\mathbb{E}[Y_n]^3} \tag{4.8}$$

$$\mathbb{E}[\gamma_n^2] \approx \frac{\mathbb{E}[X_n^2]}{\mathbb{E}[Y_n]^2} - \frac{4\mathbb{E}(X_n)Cov(X_n, Y_n)}{\mathbb{E}(Y_n)^3} + \frac{3\mathbb{E}(X_n)^2 Var(Y_n)}{\mathbb{E}(Y_n)^4}. \tag{4.9}$$

$\mathbb{E}[X_n], \mathbb{E}[Y_n], Var[X_n], Var[Y_n]$ and $Cov(X_n, Y_n) = \mathbb{E}[X_n Y_n] - \mathbb{E}[X_n]\mathbb{E}[Y_n]$ are determined in the Appendix and these expressions are substituted in (4.8) and (4.9) to obtain the first two moments. The pdf of the two moment matched gamma RV $\tilde{\gamma}_n$ is given by

[1] Since SIQNR is positive and only the first two moments of SIQNR are matched, the approximating distribution should be a two parameter distribution and be non-negative. Gamma distribution matches these criteria,has a simple pdf and CDF and has been extensively used in matching moments of non-negative RVs (Kumar *et al.*, 2015*a*; Al-Ahmadi and Yanikomeroglu, 2010).

(Kumar *et al.*, 2015*a*)

$$f(\tilde{\gamma}_n) = \frac{\tilde{\gamma}_n^{a_n-1} e^{-\frac{x}{b_n}}}{\Gamma(a_n) b_n^{a_n}}, \tag{4.10}$$

where $a_n = \mathbb{E}[\tilde{\gamma}_n]^2/(\mathbb{E}[\tilde{\gamma}_n^2] - \mathbb{E}[\tilde{\gamma}_n]^2)$ and $b_n = \mathbb{E}[\tilde{\gamma}_n]/a_n$ are the shape and scale parameters respectively. The corresponding CDF/OP at a threshold T is given by (Kumar *et al.*, 2015*a*, Eq. 5)

$$P(\tilde{\gamma}_n < T) = \frac{T^{a_n}}{b_n^{a_n}\Gamma(a_n+1)} \, {}_1F_1\left(a_n, a_n+1, \frac{-T}{b_n}\right), \tag{4.11}$$

where ${}_1F_1$ is the confluent hyper-geometric function. [2] Note, our approach of deriving the pdf expressions through moment matching can be extended to other types of fading channels, since the approximation depends only on the moment expressions. Further, the OP expressions in (Ding and Jing, 2018) is suitable only for very large M, as the results are obtained assuming $M \to \infty$. Whereas, we have made no such assumption when we derived our approximations. Since we have approximated the SIQNR to a gamma RV, our OP approximations are much simpler when compared to (Ding and Jing, 2018, Eq 15). Also, from our pdf, we can obtain a simple rate expression for user n by first writing the logarithm in terms of Meijer-G function using the identity $ln(1+x) = G_{2,2}^{1,2}\left(\begin{smallmatrix}1,1\\1,0\end{smallmatrix}\middle| x\right)$ (Wolfram Research, 2020*f*) and then solving the integral using the identity (Gradshteyn and Ryzhik, 2007, Eq 7.813.1), as,

$$Rate \approx \mathbb{E}[\log_2(1+\tilde{\gamma}_n)] = \frac{G_{3,2}^{1,3}\left(\begin{smallmatrix}1-a_n,1,1\\1,0\end{smallmatrix}\middle| b_n\right)}{ln2\Gamma(a_n)}. \tag{4.12}$$

From the pdf, we can also derive expressions for metrics like symbol error rate, error vector magnitude, etc. The moment-matching method can be extended to the cases of imperfect channel state information (CSI), multi-cell massive MIMO, since these cases involve only additional interference terms leading to calculation of more algebraic terms in the expectation expressions.

[2]The authors of (Ding and Jing, 2018) have derived approximate OP using the SCV of the various components of SIQNR. The pdf of the term $\sum_{i=1,i\neq n}^{N} |\mathbf{g}_n^H \mathbf{g}_i|^2$ is then determined and the OP is hence determined. But the disadvantage with this method is that it is intractable to determine the pdf of this term, when $\mathbf{g}_i$ have a Nakagami-m pdf.

4.4 Numerical results

The derived OP and rate approximate expressions, (4.11) and (4.12) respectively, are verified using Monte-Carlo simulations. For simulations, we generate the Nakagami-m RVs from the complex pdf given in (Yacoub *et al.*, 2005). We then use these variables to determine the SIQNR γ_n in (4.7). It can be observed from Fig. 4.2 that the simulation results for Nakagami-m fading channel matches our approximation for various values of α and M. Recall that, as α decreases, resolution of ADC decreases, which is our regime of interest. Note from Fig. 4.2 that, our approximation is tighter as the resolution decreases. This is because, as α decreases, the quantization term Z_n dominates the interference term in the denominator. In such scenarios, the ratio becomes much closer to a gamma distribution and moment-matching with gamma distribution is tighter. Moreover, increase in the ADC resolution (given by a decrease in α) or an increase in M decreases the OP.

Also, as M gets halved, we can observe that the threshold has to be halved (i.e., reduced by 3dB) to maintain the same OP. This can be explained from our approximate expressions. At large M, $\mathbb{E}[X_n]$ and $\mathbb{E}(X_n^2)$ are proportional to M^2 and M^4. Similarly, at large M, one can observe that $\mathbb{E}[Y_n]$, $\mathbb{E}[Y_n^2]$ and $\mathbb{E}[X_nY_n]$ are proportional to M, M^2 and M^3 respectively. Hence, from (4.8) and (4.9), $\mathbb{E}[\gamma_n]$ and $\mathbb{E}[\tilde{\gamma}_n^2]$ are proportional to M and M^2, at large M. From the expressions of a_n and b_n, a_n is constant and b_n is proportional to M. Hence, if the number of antenna is halved, OP can be maintained constant only if the threshold T is also halved. Similarly, from Fig. 4.3, for Rayleigh fading parameters, our approximation is tighter when compared to the existing approximation in (Ding and Jing, 2018), even for larger M and N. The authors in (Ding and Jing, 2018) have derived approximate results for very large M. On the other hand, we have imposed no such condition in our approximation.

A similar trend can be observed in Fig. 4.4 for the rate expressions also. For Rayleigh fading channels, our approximations are tighter when compared to the rate approximations derived in (Fan *et al.*, 2015). This is so because, the approximation first proposed in (Zhang *et al.*, 2014*b*, Appendix I) will be more and more accurate as the number of RVs of X_n and Y_n increase. In other words $\mathbb{E}[\frac{X}{Y}] \approx \frac{\mathbb{E}[X]}{\mathbb{E}[Y]}$ as the number of terms in Y increases and Y tends to a deterministic value equal to its mean by law of large numbers. However, our approximation involves a second order Taylor

approximation containing the second order correction term $\frac{Cov(X_n,Y_n)}{\mathbb{E}[Y_n]^2} - \frac{Var(Y_n)\mathbb{E}[X_n]}{\mathbb{E}[Y_n]^3}$ and hence a tighter approximation even when N or M is smaller is achieved. In the case of Nakagami-m fading, we can observe that with an increase in α from 0.6366 to 0.8825, the SIQNR increases leading to an increase in the sum rate of the users. But as α gets further increased, the increase in rate is subdued, as seen for $\alpha = 0.9975$. This is because, as α gets closer to unity, the moment expressions (4.8) and (4.9) become invariant to changes in α.

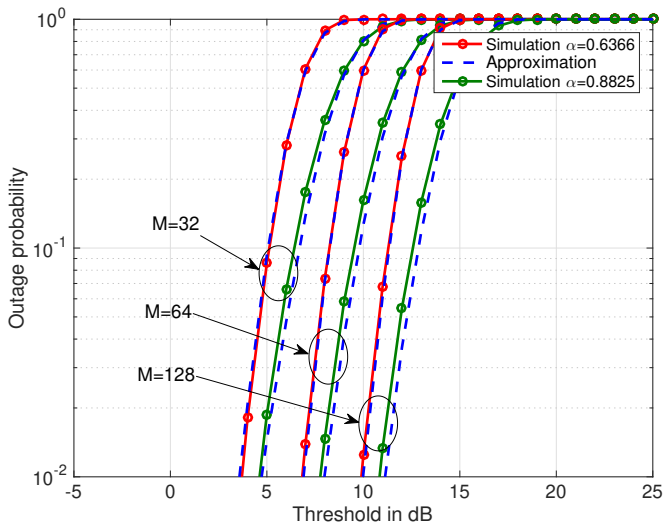

Figure 4.2: Outage probability vs Target SIR for Nakagami-m fading with $N = 6, \beta_n = 2, m_n = 2, \beta_i = m_i = 1, 3, 1, 1, 1$

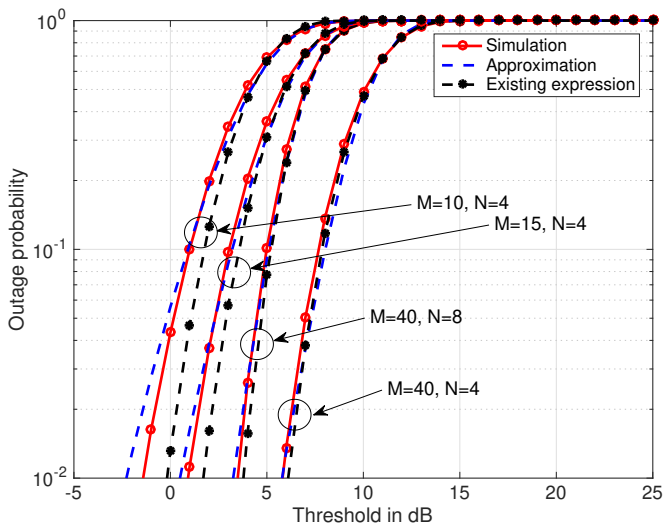

Figure 4.3: Outage probability vs Target SIR for Rayleigh fading with $\rho = 0.03454, \beta_i = 1, m_i = 1$

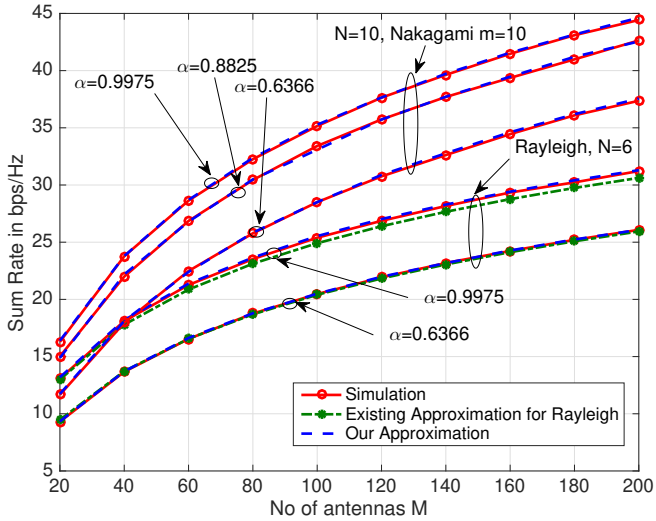

Figure 4.4: Sum Rate vs No of antennas M for $\beta_i = 1$ for all $i = 1$ to N

4.5 Conclusion

The uplink of Massive-MIMO with low resolution ADC architecture was considered in this chapter. All the channels were assumed to undergo i.n.i.d. Nakagami-m fading. By means of moment matching and second order bi-variate Taylor series expansion, the SIQNR was approximated by a gamma RV. Using this approximate pdf, tight approximations for metrics like OP and rate were derived. Extensive Monte-Carlo simulations were performed and the approximation matches the simulation results for all parameters. Also, from the approximate expressions, it was observed that at large M, if the number of antennas is halved, OP can be maintained constant only if the threshold T is also halved.

CHAPTER 5

Analysis of MRC with $\eta - \mu$ co-channel interference

5.1 Introduction

Outage probability (OP) of Maximum ratio combining (MRC) has been studied in the presence of co-channel interference (CCI) for Rayleigh channels by (Shah and Haimovich, 2000; Zhang and Beaulieu, 2006) and Rician channels by (Chayawan and Aalo, 2002). Outage probability analysis of MRC has been studied for Nakagami faded channels by (Romero-Jerez *et al.*, 2007) with the assumption that the interferer fading is the same in all diversity branches and by (Wang *et al.*, 2006) with no such assumption.

There has been some work in literature, for example, (Moraes *et al.*, 2012; Peppas, 2013; Ansari *et al.*, 2013; Morales-Jimenez *et al.*, 2012; Paris, 2013; Ermolova and Tirkkonen, 2014; Chen *et al.*, 2016), which analyze the performance metrics of generalized faded signals. In (Moraes *et al.*, 2012) equal gain combiner is considered for the case of generalized fading. The performance evaluation of dual-hop decode-and-forward (DF) cooperative systems operating over $\eta - \mu$ fading channels is considered in (Peppas, 2013). In (Morales-Jimenez *et al.*, 2012; Chen *et al.*, 2016), generalized fading channels are assumed only for the signal of interest (SOI), whereas interferers are assumed to be Rayleigh faded. In (Ansari *et al.*, 2013), the SOI is assumed to be $\eta - \mu$ distributed, but CCI is not considered. Extending the analysis in (Morales-Jimenez *et al.*, 2012; Chen *et al.*, 2016) for the case of general $\eta - \mu$ interferers seems mathematically intractable, if not impossible.

5.1.1 Our contributions

In this chapter, we determine approximate expressions for OP and rate when both SOI and the CCI are $\eta - \mu$ distributed, with the interference fading being different across diversity branches. We assume that the noise power is negligible when compared to the

M. Srinivasan and S. Kalyani, "Analysis of MRC With $\eta - \mu$ Co-Channel Interference," in IEEE Transactions on Vehicular Technology, vol. 69, no. 1, pp. 738-745, Jan. 2020.

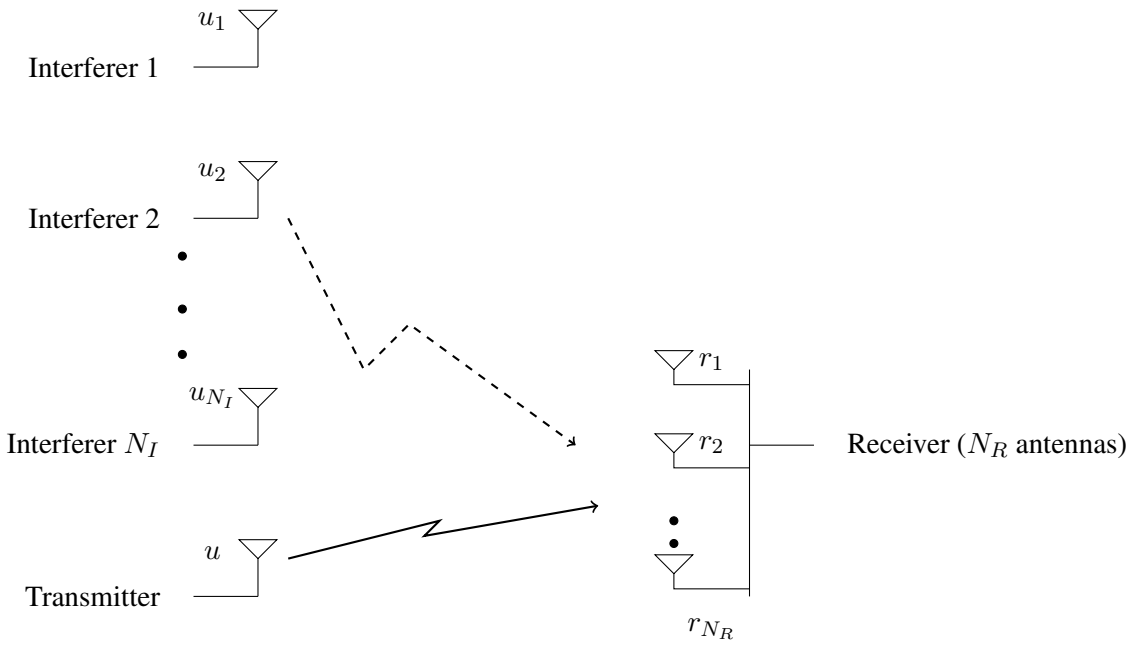

Figure 5.1: System model

interference power, as is the case in most practical wireless systems. The interferers are assumed to be uncorrelated and have unequal powers.

Our expressions also have an important vehicular application. The resource allocation and power control problem for device-to-device (D2D) based vehicular communication underlay cellular networks has been studied from multiple perspectives in literature (Wang *et al.*, 2017*b*, 2018; Liang *et al.*, 2017; Liang *et al.*, 2018). Most of these work consider the problem of resource allocation and power control in a scenario where each cellular user shares spectrum with D2Ds in Rayleigh or Rician fading scenarios. Also, the above works consider single antenna terminals for both transmission and reception. In the current cellular scenario, one can typically expect the base station (BS) to be equipped with multiple antennas and employ a combining technique like MRC. Therefore, in this chapter, we demonstrate the utility of our approximate expressions in solving the joint power control and resource allocation problem of D2D based vehicular communication sharing spectrum with cellular users for a BS employing MRC in a $\eta - \mu$ faded scenario.

5.2 System model

Let the number of receive antennas be N_R and let the number of interferers be N_I. The $N_R \times 1$ user signal $\mathbf{c} = [c[1], ..., c[N_R]]^T$ are assumed to be $\eta - \mu$ random variables (da Costa and Yacoub, 2007) and the N_I interferers denoted $\mathbf{c}_i$ are also independent $\eta - \mu$ random variables with complex probability density function (pdf) given by (da Costa and Yacoub, 2007)

$$f(x, y) = \frac{\mu^{2\mu}|xy|^{2\mu-1}}{\Omega_X^{\mu}\Omega_Y^{\mu}\Gamma^2(\mu)} exp[-\mu\Big(\frac{x^2}{\Omega_X} + \frac{y^2}{\Omega_Y}\Big)], \tag{5.1}$$

where Ω is the power parameter given by $\Omega = 2\sigma^2\mu$. Here, σ^2 is the power of the Gaussian variable in each cluster and 2μ is the number of clusters. Note, $\Omega_X = (1 - \eta)\Omega/2$, $\Omega_Y = (1 + \eta)\Omega/2$ and $-1 \leq \eta \leq 1$ (Format 2). Henceforth, the subscript int and $user$ will be used to differentiate interferer $\eta - \mu$ parameters from user $\eta - \mu$ parameters. u denotes the desired user symbol and u_i the ith interferer symbol, with both belonging to unit energy QAM constellation. The $N_R \times 1$ received vector in a receive diversity system is given by

$$\mathbf{r} = \mathbf{c}u + \sum_{i=1}^{N_I} \sqrt{E_i}\mathbf{c}_i u_i, \tag{5.2}$$

where E_i is the mean energy of each of the interfering signals. The schematic is given in Fig. 5.1. In MRC, the received vector $\mathbf{r}$ is weighted by $\mathbf{c}^H$ to produce the following output at the combiner:

$$z = \mathbf{c}^H\mathbf{r} = \mathbf{c}^H\mathbf{c}u + \sum_{i=1}^{N_I} \sqrt{E_i}\mathbf{c}^H\mathbf{c}_i u_i. \tag{5.3}$$

Hence the SIR at the output of the combiner is given by (Shah and Haimovich, 2000)

$$\gamma = \frac{|\mathbf{c}^H\mathbf{c}|^2}{\sum_{i=1}^{N_I} E_i|\mathbf{c}^H\mathbf{c}_i|^2} = \frac{|\mathbf{c}^H\mathbf{c}|^2}{\sum_{i=1}^{N_I} E_i\mathbf{c}^H\mathbf{c}_i\mathbf{c}_i^H\mathbf{c}} = \frac{|\mathbf{c}^H\mathbf{c}|^2}{\mathbf{c}^H\mathbf{H}\mathbf{H}^H\mathbf{c}}, \tag{5.4}$$

where the $N_R \times N_I$ matrix $\mathbf{H} = [\mathbf{c}_1, \mathbf{c}_2, ..., \mathbf{c}_{N_I}]$ denotes the interferer fading coefficients.

If the interferers were Rayleigh faded, then $\mathbf{H}\mathbf{H}^H$ is given by a Wishart distributed

matrix (James, 1964). For any $\mathbf{c}$ and Wishart distributed $\mathbf{H}\mathbf{H}^H$,

$$c^H\mathbf{H}\mathbf{H}^H\mathbf{c} \sim \mathbf{c}^H\mathbf{c}\mathcal{G}(N_I),$$

where $\mathcal{G}(x)$ is a Gamma variate with shape parameter x. Such a simplification is used in studying Rayleigh faded interferers in (Shah and Haimovich, 2000; Zhang and Beaulieu, 2006).

However, a matrix distribution for $\mathbf{H}\mathbf{H}^H$ is not available in open literature, when the elements of $\mathbf{H}$ are assumed to be $\eta - \mu$ faded complex random variables.[1] However, some random matrix models have been developed in (Fraidenraich *et al.*, 2008) and (Gholizadeh *et al.*, 2015), for the Grammian $\mathbf{H}\mathbf{H}^H$, when $\mathbf{H}$ is assumed to be of dimension 2×2 and the elements of $\mathbf{H}$ are Nakagami-m distributed. Given that, extending these results even for Nakagami-m matrices with larger dimension using this method is cumbersome, extending the results to any general $\eta - \mu$ matrices is mathematically intractable. Hence, inspired from the results of (Shah and Haimovich, 2000), we use the idea of moment matching to approximate the denominator of (5.4) by a gamma random variable and simplify the SIR expression. Though we give approximate results, these results can serve as a good start for deriving any future results in generalized NLOS fading scenario or for deriving results for more advanced interference cancellation like optimum combining. Also, the results are applicable for the more practical case of different interferer powers.

The recent work (Paris, 2013) also derives outage probability of MRC for $\eta - \mu$ faded interferers. However, in (Paris, 2013), the SIR is given by $\gamma = \frac{\mathbf{c}^H\mathbf{c}}{\sum_{i=1}^{N_I}|Y_i|^2}$, where $|Y_i|^2$ denotes N_I independent $\eta - \mu$ faded interferer power coefficients. Also note the distinct difference in the denominator of γ, when one compares (Paris, 2013) with the conventional MRC model used in this chapter and in (Shah and Haimovich, 2000; Zhang and Beaulieu, 2006; Romero-Jerez *et al.*, 2007; Morales-Jimenez *et al.*, 2012; Chen *et al.*, 2016). In other words, the denominator of (5.4) is $\mathbf{c}^H\mathbf{H}\mathbf{H}^H\mathbf{c}$, while the denominator of (Paris, 2013) is $\sum_{i=1}^{N_I}|Y_i|^2$. The difference is due to the fact that the MRC model in (Paris, 2013) adds up the total interference power, i.e., $\sum_{i=1}^{N_I}|Y_i|^2$, at each of the diversity branches, without taking into account the multiplicative effect of the user signal $\mathbf{c}$ on each of the interference signal.

[1]This is the reason why determining the distribution of the largest eigenvalue for the case of MIMO-MRC is also intractable.

5.3 Approximation

Since, obtaining the exact pdf of γ seems mathematically intractable, we go for a simple approximation to obtain the pdf. Consider the expression for SIR given by (5.4). We propose to approximate each of the term in the denominator $E_i|\mathbf{c}^H\mathbf{c}_i|^2$ by x_i, which is defined by the product of $|\mathbf{c}^H\mathbf{c}|$ and a gamma random variable z_i (independent of $|\mathbf{c}^H\mathbf{c}|$), for $i = 1, ..., N_I$, by matching the first and the second moment. In other words, we match the first and the second moment of $E_i|\mathbf{c}^H\mathbf{c}_i|^2$ and $x_i = |\mathbf{c}^H\mathbf{c}|z_i$.[2]

The gamma random variable $z_i \sim \mathcal{G}(a_i, b_i)$ is given by its pdf

$$f_{z_i}(z_i) = \frac{1}{\Gamma(a_i)b_i^{a_i}} {z_i}^{a_i-1} e^{-\frac{z_i}{b_i}}$$

, with $z_i \in (0, \infty)$, $a_i > 0$ being the shape parameter and $b_i > 0$ being the scale parameter. The first moment of the term $|\mathbf{c}^H\mathbf{c}_i|^2$ is given by (Yacoub, 2007),

$$\mathbb{E}[E_i|\mathbf{c}^H\mathbf{c}_i|^2] = E_i N_R(\Omega_{Xuser} + \Omega_{Yuser})(\Omega_{Xint} + \Omega_{Yint}) \tag{5.5}$$

and the second moment can be derived by expanding $|\mathbf{c}^H\mathbf{c}_i|^2$, taking its square and taking expectation of each term explicitly, to obtain,

$$\begin{aligned}
\mathbb{E}[(E_i|\mathbf{c}^H\mathbf{c}_i|^2)^2] = E_i^2 \Bigg[& (\Omega_{Xuser}^2 + \Omega_{Yuser}^2)(\Omega_{Xint}^2 + \Omega_{Yint}^2) \\
& \Big(3N_R(N_R-1) + N_R \frac{\mu_{user}+1}{\mu_{user}} \frac{\mu_{int}+1}{\mu_{int}}\Big) \\
& + (\Omega_{Xuser}^2 + \Omega_{Yuser}^2)(\Omega_{Xint}\Omega_{Yint})\Big(2N_R(N_R-1) + 2N_R \frac{\mu_{user}+1}{\mu_{user}}\Big) \\
& + (\Omega_{Xint}^2 + \Omega_{Yint}^2)(\Omega_{Xuser}\Omega_{Yuser})\Big(2N_R(N_R-1) + 2N_R \frac{\mu_{int}+1}{\mu_{int}}\Big) \\
& + (\Omega_{Xint}\Omega_{Yint}\Omega_{Xuser}\Omega_{Yuser})\Big(4N_R^2 + 8N_R(N_R-1)\Big)\Bigg],
\end{aligned} \tag{5.6}$$

where $\mathbb{E}[.]$ denotes expectation[3]. The first moment of gamma random variable is

[2]This method is inspired from the case of Rayleigh faded inteferers (Shah and Haimovich, 2000), where each of the term in the denominator is exactly a product of $|\mathbf{c}^H\mathbf{c}|$ and a gamma random variable.

[3]A similar expression can be found for i.n.i.d. $\eta - \mu$ interferers, but not included here.

given by $\mathbb{E}[z_i] = a_i b_i$. Hence, matching the first moment is equivalent to

$$\mathbb{E}[x_i] = E[|\mathbf{c}^H\mathbf{c}|]a_i b_i, \tag{5.7}$$

where $E[|\mathbf{c}^H\mathbf{c}|] = N_R(\Omega_{Xuser} + \Omega_{Yuser})$. The second moment is given by $\mathbb{E}[z_i^2] = a_i b_i^2 + a_i^2 b_i^2$. Hence, matching the second moment is equivalent to,

$$\mathbb{E}[x_i^2] = \mathbb{E}[|\mathbf{c}^H\mathbf{c}|^2](a_i b_i^2 + a_i^2 b_i^2), \tag{5.8}$$

where $\mathbb{E}[|\mathbf{c}^H\mathbf{c}|^2] = N_R\Big[(\frac{\mu_{user}+1}{\mu_{user}} + N_R - 1)(\Omega_{Xuser}^2 + \Omega_{Yuser}^2) + 2N_R\Omega_{Xuser}\Omega_{Yuser}\Big]$. Note that $\mathbb{E}[|\mathbf{c}^H\mathbf{c}|^2]$ can be obtained by expanding the square and explicitly calculating the expectation of each term. From (5.7) and (5.8), we obtain, $b_i = \frac{\frac{\mathbb{E}[(E_i|\mathbf{c}^H\mathbf{c}_i|^2)^2]}{\mathbb{E}[|\mathbf{c}^H\mathbf{c}|^2]} - \left(\frac{\mathbb{E}[E_i|\mathbf{c}^H\mathbf{c}_i|^2]}{\mathbb{E}[|\mathbf{c}^H\mathbf{c}|]}\right)^2}{\frac{\mathbb{E}[E_i|\mathbf{c}^H\mathbf{c}_i|^2]}{\mathbb{E}[|\mathbf{c}^H\mathbf{c}|]}}$ and $a_i = \frac{\frac{\mathbb{E}[E_i|\mathbf{c}^H\mathbf{c}_i|^2]}{\mathbb{E}[|\mathbf{c}^H\mathbf{c}|]}}{b_i}$. Now, the SIR at the output of the combiner is given by,

$$\gamma \approx \frac{|\mathbf{c}^H\mathbf{c}|^2}{\sum_{i=1}^{N_I}|\mathbf{c}^H\mathbf{c}|z_i} = \frac{|\mathbf{c}^H\mathbf{c}|^2}{|\mathbf{c}^H\mathbf{c}|\sum_{i=1}^{N_I} z_i} = \frac{|\mathbf{c}^H\mathbf{c}|}{\sum_{i=1}^{N_I} z_i}. \tag{5.9}$$

The denominator is now a sum of N_I gamma random variables z_i. Also, from (Heath *et al.*, 2013), it is again known that a sum of gamma variables can be well approximated by a single gamma random variable by matching the first two moments Hence approximately, $\sum_{i=1}^{N_I} z_i \sim \mathcal{G}(a, b)$, where

$$a = \frac{\left(\sum_{i=1}^{N_I} a_i b_i\right)^2}{\sum_{i=1}^{N_I} a_i b_i^2} \tag{5.10}$$

and

$$b = \frac{\sum_{i=1}^{N_I} a_i b_i^2}{\sum_{i=1}^{N_I} a_i b_i}. \tag{5.11}$$

Note that for $E_1 = E_2 = ... = E_{N_I}$, the above approximation of sum of gamma random variables by another gamma random variable becomes exact.

The SIR at the output of the combiner is given by $\gamma \approx \frac{y}{x}$, where y is the sum of N_R $\eta - \mu$ power variables denoted by y_i $i = 1, .., N_R$ and $x \sim \mathcal{G}(a, b)$. The $\eta - \mu$ random variable in the numerator can also be approximated by a Gamma random variable $y' \sim \mathcal{G}(p, q) = q\mathcal{G}(p, 1)$ using moment matching. Equating with the moments of y', we obtain $p = \frac{2\mu_{user}N_R}{1+\eta_{user}^2}$ and $q = \frac{N_R(\Omega_{X_{user}}+\Omega_{Y_{user}})}{p}$. Hence $\gamma \approx \frac{y'}{x} = \frac{q}{b}\frac{\mathcal{G}(p,1)}{\mathcal{G}(a,1)}$.

The ratio of two gamma distribution $z = \frac{\mathcal{G}(p,1)}{\mathcal{G}(a,1)}$ is called the beta-prime distribution or beta distribution of the second kind with pdf given by $\frac{\Gamma(a+p)}{\Gamma(a)\Gamma(p)} z^{p-1}(1+z)^{-a-p}$ for $z > 0$ (Dubey, 1970). Hence the pdf of SIR can also be given as .

$$f_\gamma(\gamma) \approx \frac{\Gamma(a+p)}{\Gamma(a)\Gamma(p)}(\frac{b}{q})^p \gamma^{p-1}(1+\frac{b}{q}\gamma)^{-a-p} \quad \gamma > 0.$$

The CDF of a beta-prime distributed random variable, say Z, with parameters m and n, is given in (Cordeiro and Lemonte, 2012, Eq 2), as $P(Z < z) = \frac{(n)_m z^m \, {}_2F_1(m+n,m,m+1,-z)}{\Gamma(m+1)}$ for $z \geq 0$. Therefore, outage is given by,

$$P_{out} \approx \frac{(a)_p(\frac{b}{q}\gamma_0)^p \, {}_2F_1(p+a,p,p+1,\frac{-b}{q}\gamma_0)}{\Gamma(p+1)}, \tag{5.12}$$

Similarly using the pdf of beta-prime distribution, one can obtain the approximate rate, at high SIR, as

$$\begin{aligned} R &\approx \int_0^\infty \log_2(x)\frac{\Gamma(a+p)}{\Gamma(a)\Gamma(p)}(\frac{b}{q})^p x^{p-1}(1+\frac{b}{q}x)^{-a-p}dx \\ &= \frac{\psi(p)-\psi(a)-ln(b/q)}{ln2}, \end{aligned} \tag{5.13}$$

where $\psi(.)$ is the di-gamma function (Abramowitz and Stegun, 1972, 6.3.1). The second equality is obtained by using the integral identity (Gradshteyn and Ryzhik, 2007, 4.292.14).

5.4 Inferences and Numerical Results

The derived approximation involves only a simple Gauss hypergeometric function. We can make the following inferences from (5.12):

I1) Since the approximate outage probability given by (5.12) is a CDF evaluated at $\frac{b}{q}\gamma_0$, $P_0(\frac{b}{q}\gamma_0+\delta) > P_0(\frac{b}{q}\gamma_0)$ for $\delta > 0$. In other words, decrease in user power $\Omega_{X_{user}} + \Omega_{Y_{user}}$, which corresponds to a decrease in q or increase in target SIR γ_0 increases outage probability.

I2) From (Shaked and Shanthikumar, 2007, Theorem 1.A.12) and the pdf of beta-prime distributed random variables given in (Dubey, 1970), for two different sets

of parameters namely (p, a) and $(p, a+\delta)$, one can prove that

$$\mathcal{S}^{-}\Big(\frac{\Gamma(a+p)}{\Gamma(a)\Gamma(p)}z^{p-1}(1+z)^{-a-p} - \frac{\Gamma(a+p+\delta)}{\Gamma(p)\Gamma(a+\delta)}z^{p-1}(1+z)^{-a-p-\delta}\Big) = 1,$$

where $\mathcal{S}^{-}$ denotes the number of sign changes and $\delta > 0$. The sign change from $-$ to $+$ occurs at $z = \left(\frac{(a+\delta)_p}{(a)_p}\right)^{\frac{1}{\delta}} - 1$. This implies that, using the definition of stochastic ordering from (Shaked and Shanthikumar, 2007, 1.A.1), we can show that P_{out} given by (5.12), evaluated using the parameter $a+\delta$ is greater than P_{out} evaluated using the parameter a, for a constant $\frac{b}{q}\gamma_0$. Similarly, the opposite is true for parameters $p+\delta$ and p, for a constant $\frac{b}{q}\gamma_0$.

I3) With a decrease in μ_{int}, outage probability decreases (coverage increases) only when there is a proportionate increase in the target SIR γ_0. This is so because μ_{int} leads to an increase in $\mathbb{E}[(E_i|\mathbf{c}^H\mathbf{c}_i|^2)^2]$ given by (5.6). This in turn decreases a given by (5.10) and increases b given by (5.11). This implies that apart from a decrease in a, the CDF evaluation point $\frac{b}{q}\gamma_0$ also increases. To really guarantee a decrease in the outage probability according to I2, the target SIR γ_0 has to be increased to maintain a constant $\frac{b}{q}\gamma_0$.

I4) A similar inference can be obtained for an increase in $\mathbb{E}[(E_i|\mathbf{c}^H\mathbf{c}_i|^2)^2]$, due to an increase in η_{int}. A change in η_{user} and μ_{user} leads to a change in all the parameters, namely a, p, b and q of the outage expression. Hence, it is intractable to form similar inferences using I2 for a change in η_{user} and μ_{user}. These inferences have been numerically verified, but not shown here due to space constraints.

The derived outage probability approximate expression (5.12) is verified using Monte-Carlo simulations. For Monte-Carlo simulation, we generate the $\eta-\mu$ random variables $\mathbf{c}$ and $\mathbf{c}_i$ from the complex pdf given in (da Costa and Yacoub, 2007). We then use these variables in (5.4) to determine the SIR γ and plot the CDF of γ to obtain the outage probability. From Fig. 5.2, we can observe that both our approximations are tight for all values of N_R and N_I. An increase in N_R results in an increase in the number of

η_{int}	μ_{int}	η_{user}	μ_{user}	KL divergence
0.1	2	0.1	2	0.0057
0.1	2	0.1	4	0.0076
0.1	2	0.9	4	0.0057
0.1	4	0.9	4	0.0108
0.9	4	0.9	4	0.0186

Table 5.1: KL divergence for the approximations for $N_R = 3$, $N_I = 2$ and $E_I = -1, -1$ dB

independent multi-paths and hence leads to increase in SIR. Therefore outage probability decreases as N_R increases, as corroborated by Fig. 5.2 and Fig. 5.3. Increase in μ_{user} also decreases the outage probability as observed in Fig. 5.2. This can again be explained by an increase in independent multi-path due to an increase in μ. Similarly, increase in the magnitude of η_{user} contributes to increase in the correlation between the real and imaginary components in each cluster of $\eta - \mu$ fading. This leads to a decrease in the number of independent multi-paths and hence leads to increase in SIR. In other words, there is a reduction in diversity, with an increase in correlation among the multi-paths. Hence, an increase in the outage probability can be observed as in Fig. 5.2. Similarly, a decrease in interference power leads to a decrease in outage probability as in Fig. 5.4. To determine the tightness of the approximations, KL-divergence is calculated empirically between the actual SIR and the approximate SIRs and the same is given in Table 5.1. We can observe that the approximation becomes tighter with the interferers becoming close to Rayleigh, i.e., μ_{int} approaching unity and η_{int} approaching 0. This is along expected lines because the approximations are exact for Rayleigh faded interferers. Also, the approximations are tighter for larger η_{user} and smaller μ_{user}.

5.5 Resource allocation and power control for D2D based vehicular communication

We now demonstrate the utility of our approximate expressions in solving the joint power control and resource allocation problem of D2D based vehicular communication sharing spectrum with cellular users for a BS employing MRC in a $\eta - \mu$ faded sce-

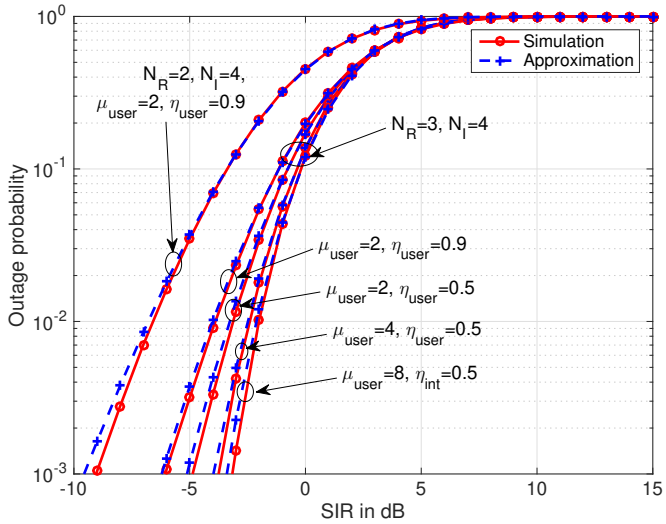

Figure 5.2: Outage probability vs Target SIR for $\eta_{int} = 0.3, \mu_{int} = 2, E_I = -1, -3, -5, -7$dB

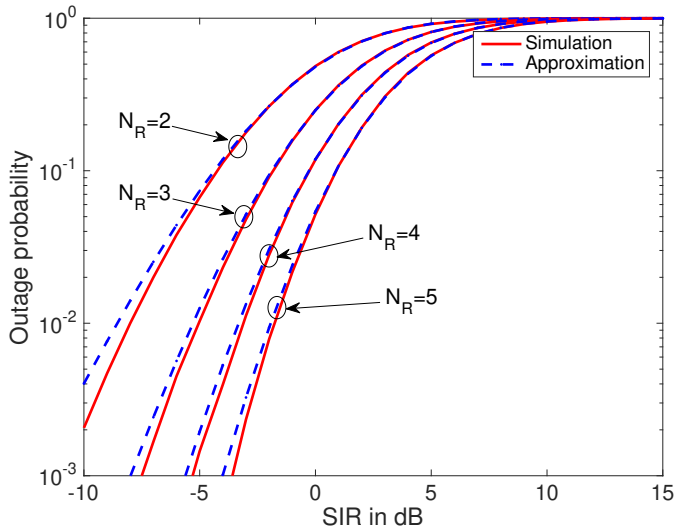

Figure 5.3: Outage probability vs Target SIR for $N_R = 3, N_I = 4, \eta_{int} = 0.5, \mu_{int} = 1, \eta_{user} = -0.5, \mu_{user} = 1$

nario. Device-to-device (D2D) communication enables a pair of closely located mobile users to establish a direct link for their user-plane traffic without going through the entire network infrastructure, while reusing the spectrum allocated for traditional cellular communication. In D2D underlying cellular networks, strong interference may occur between cellular and D2D links sharing the same spectrum. To mitigate the interference, many resource allocation schemes have been proposed in (Wang *et al.*, 2017*b*, 2018; Liang *et al.*, 2017). All these works consider K pairs of D2D users reusing the spectrum of M cellular users. All these works assume that the number of D2D users is less and hence each portion of the spectrum is accessed by at most one D2D users. This

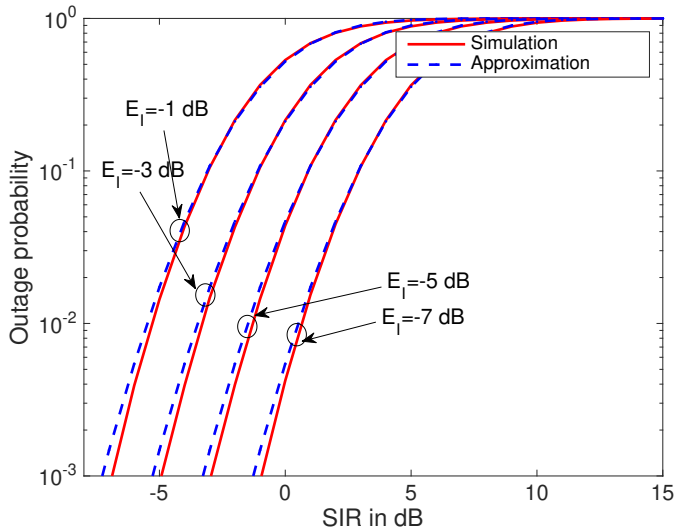

Figure 5.4: Outage probability vs Target SIR for $N_I = 3, E_I = -2$dB, $\eta_{int} = 0.5, \mu_{int} = 3, \eta_{user} = -0.5, \mu_{user} = 2$

was extended to the case of more D2D users accessing the same portion of the spectrum in (Liang *et al.*, 2018). Here the D2D users are clustered into disjoint spectrum-sharing clusters using graph partitioning algorithms to mitigate their mutual interference. The spectrum allocation is problem is then solved for each of the cluster. In all the existing works (Wang *et al.*, 2017*b*; Liang *et al.*, 2017; Wang *et al.*, 2018; Liang *et al.*, 2018), only one antenna at BS is considered. It makes practical sense to consider the BS to be equipped with more than one antenna. Since the BS is equipped with more than one antenna, the interference at the BS can be mitigated by MRC. Similar to (Wang *et al.*, 2018), we consider a D2D-enabled communications network, where M single-antenna cellular users (CUE) perform high-capacity uplink cellular communications with the base station (BS) with N_R antennas. The entire schematic is shown in Fig. 5.5.

Consider a D2D-based vehicular communications network similar to (Wang *et al.*, 2018). There are M vehicle to interface links (V2I) or (CUEs) and K vehicle-to-vehicle links (V2V) (D2D pairs or DUEs). The V2I links are denoted by the set $\mathcal{M} = \{1, \cdots, M\}$ and the V2Vs are denoted by the set $\mathcal{K} = \{1, \cdots, K\}$. We assume that each of the V2I links occupy an orthogonal spectrum. Also, the number of V2V links tends to be larger than that of the V2I links, i.e., $K \gg M$. This is the model considered in (Liang *et al.*, 2018). Therefore, spectrum reuse among V2V links becomes imperative. Further, we assume that the BS is assumed to have N_R antennas. Also, due to the rapidly varying channel with respect to the V2V links we assume that

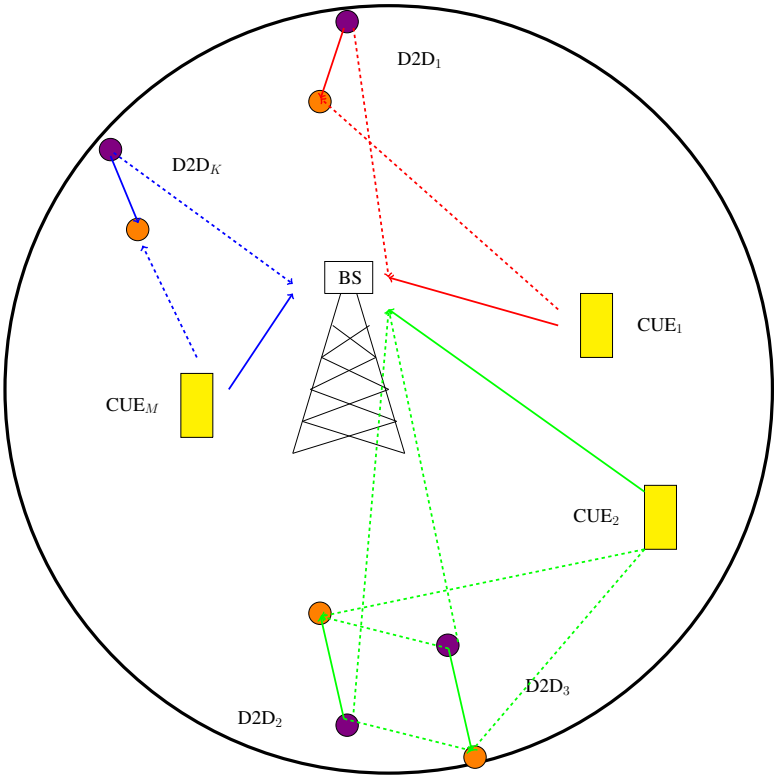

Figure 5.5: This figure shows a general D2D underlay cellular network, with M CUEs, K D2D pairs and $M << K$. Different colours are used for different spectrum. The dashed lines indicate the interferer channels. BS has only the statistical information on the fast fading parameters of the D2Ds

the BS does not have the instantaneous channel estimates of the V2V links. However, the statistical nature and the slow fading parameters of the V2Vs are available at the BS. If the BS station has the perfect CSI of the V2I links, MRC as discussed in previous sections, is employed at the BS to combine the multi antenna signals. If the perfect CSI is not available, there is a Gaussian additive error due to estimation for which the expressions for MRC can be obtained.

The channel power gain, $g^i_{m,B}$, from the transmitter of the mth V2I link to the ith antenna at the BS is

$$g^i_{m,B} = \sqrt{\alpha_{m,B}} h^i_{m,B}, \tag{5.14}$$

where $h^i_{m,B}$ is the small-scale fading component, assumed to be $\eta - \mu$ faded and independent across the links and antennas. $\alpha_{m,B}$ denotes the large-scale fading effects, i.e., including path loss and shadowing. We also assume that the large scale fading effects to be invariant of the antenna index i. Similarly, we can define the kth V2V channel to be g_k. The interfering channel from the k'th V2V transmitter to the kth V2V receiver

is $g_{k',k}$. The interfering channel from the mth V2I transmitter to the kth V2V receiver is $g_{m,k}$, and the interfering channel from the kth V2V transmitter to the ith antenna is $g^i_{k,B}$.

Now, assuming an interference-limited scenario, SIRs of the mth V2I link at the BS after MRC and the kth V2V link at the V2V receiver can be expressed as

$$\gamma^c_m = \frac{P^c_m |\mathbf{g}_{m,B}|^2}{\sum\limits_k \rho_{m,k} P^d_k |\mathbf{g}^H_{m,B} \mathbf{g}_{k,B}|^2} \tag{5.15}$$

and

$$\gamma^d_k = \frac{P^d_k |g_k|^2}{\sum\limits_m \rho_{m,k} P^c_m |g_{m,k}|^2 + \sum\limits_{k' \neq k} \rho_{m,k'} P^d_{k'} |g_{k',k}|^2}, \tag{5.16}$$

respectively. Here, P^c_m and P^d_k denote transmit powers of the mth V2I transmitter and the kth V2V transmitter respectively. $\rho_{m,k} \in \{0, 1\}$ is the spectrum allocation indicator. $\rho_{m,k} = 1$ indicates that the mth V2I link is sharing the spectrum with the kth V2V link and $\rho^c_{m,k} = 0$ otherwise.

The spectrum and power allocation problem is formulated as:

$$\max_{\substack{\{\rho_{m,k}\} \\ \{P^c_m, P^d_k\}}} \sum_m \mathbb{E}\left[\log_2(1 + \gamma^c_m)\right] \tag{5.17}$$

$$\text{s.t.} \quad \Pr\left\{\gamma^d_k \leq \gamma^d_0\right\} \leq p_0, \forall k \tag{5.17a}$$

$$\sum_m \rho_{m,k} = 1, \forall k \tag{5.17b}$$

$$0 \leq P^c_m \leq P^c_{\max}, \forall m \tag{5.17c}$$

$$0 \leq P^d_k \leq P^d_{\max}, \forall k \tag{5.17d}$$

$$\rho_{m,k} \in \{0, 1\}, \forall m, k, f, \tag{5.17e}$$

where γ^d_0 in (5.17a) is the minimum SIR needed to guarantee reliability of the V2V link. p_0 in (5.17a) is the maximum tolerable outage probability. $P^c_{\max}$ and $P^d_{\max}$ are the maximum transmit powers of the V2I and V2V transmitters respectively. Constraint (5.17a) represents the minimum reliability requirement for the K V2V links. Constraint (5.17b) only one V2I link shares spectrum with each V2V link. Constraints (5.17c) and (5.17d) ensure that the transmit powers of V2I and V2V links have maximum limits.

While this spectrum allocation problem has been discussed for a BS with single antenna and Rayleigh faded fast fading channel coefficients in (Liang *et al.*, 2017; Liang *et al.*, 2018; Wang *et al.*, 2017*b*), we, for the first time, demonstrate a sub-optimal solution for the spectrum allocation problem for a BS with multiple antennas where MRC is applied and the coefficients undergo $\eta - \mu$ fading. The above spectrum and power allocation problem is complicated due to presence non-linear objective function and reliability constraints. The following sub-optimal solution is hence given in three steps and can be centrally implemented at the BS. First, since the V2Vs reuse spectrum, they have to be clustered such that the intra-cluster interference is minimized. Second, for a given ADC resolution profile and V2V clustering, the power allocation problem for every pair of V2I and V2V-cluster is solved. Finally, the resource matching is performed such that the V2I ergodic sum rate is maximized.

5.5.1 V2V Clustering

Next we divide the V2V links into different clusters such that the intra-cluster mutual interference is minimized. To divide the V2V links into clusters, we use the Max N cut algorithm given in (Liang *et al.*, 2018). Since, only the slow fading parameters of the V2Vs are available at the BS, the algorithm forms a cluster of V2Vs based on these slow fading parameters. Each V2V link is modeled as a vertex. If any two links are mutually interfering, i.e, they belong to the same cluster, the corresponding vertices are joined by an edge, such that the edge weight $w_{k',k} = \alpha_{k',k}$, where $\alpha_{k',k}$ is the large-scale fading CSI of the interference channel from the k'th V2V transmitter to the kth V2V receiver. Now we have to partition the K vertices into N sets (clusters), $C_1, \cdots, C_N$, where $N \ll K$, such that the the intra-cluster interference across all clusters, i.e., $\sum_n \left(\sum_{k',k \in C_n} w_{k',k} \right)$ is minimized. This is equivalent to the MAX N-CUT problem in graph theory.

5.5.2 Power Allocation

For any pair of V2I-V2V cluster, the V2I capacity is now maximized such that the outage condition of all V2V links in that cluster is satisfied. The power optimization problem for the (m, n) V2I V2V-cluster pair, where m denotes the V2I index and n

Table 5.2: Heuristic Algorithm for MAX N-CUT (Liang *et al.*, 2018)

Algorithm 1 Heuristic Algorithm for V2V Partitioning

1: Arbitrarily assign one V2V link to each of the N clusters.
2: **for** $k \in \mathcal{K}$ **and** not already in any cluster **do**
3: **for** $n = 1 : N$ **do**
4: Compute the increased intra-cluster interference using $\sum_{k' \in C_n} (w_{k,k'} + w_{k',k})$.
5: **end for**
6: Assign the kth V2V link to the n^*th cluster with $n^* = \arg\min \sum_{k' \in C_n} (w_{k,k'} + w_{k',k})$.
7: **end for**
8: Return the V2V clustering result.

denotes the V2V cluster index, is therefore formulated as

$$\max_{P_m^c, \mathbf{P}_k^d} \mathbb{E}\left[\log_2\left(1 + \frac{P_m^c |\mathbf{g}_{m,B}|^2}{\sum_{k \in C_n} \rho_{m,k} P_k^d |\mathbf{g}_{m,B}^H \mathbf{g}_{k,B}|^2}\right)\right], \tag{5.18}$$

$$\text{s.t. Pr}\left\{\frac{P_k^d |g_k|^2}{P_m^c |g_{m,k}|^2 + \sum_{k' \neq k \in C_n} P_{k'}^d |g_{k',k}|^2} \leq \gamma_0^d\right\} \leq p_0, \tag{5.18a}$$

$$0 \leq P_m^c \leq P_{\text{max}}^c \tag{5.18b}$$

$$0 \leq P_k^d \leq P_{\text{max}}^d, \forall k \in C_n. \tag{5.18c}$$

From (Liang *et al.*, 2018), for generalized fast fading distributions with unit mean power, i.e., $\mathbb{E}[|g_k|2] = 1$, the left hand side of (5.18a) can be upper bounded as

$$\begin{aligned} &\text{Pr}\left\{\frac{P_k^d |g_k|^2}{P_m^c |g_{m,k}|^2 + \sum_{k' \neq k \in C_n} P_{k'}^d |g_{k',k}|^2} \leq \gamma_0^d\right\} \\ &\leq F_{|g_k|^2}\left(\gamma_0^d \frac{P_m^c \alpha_{m,k} + \sum_{k' \neq k \in C_n} P_{k'}^d \alpha_{k',k}}{P_k^d \alpha_k}\right) \end{aligned} \tag{5.19}$$

when the fading cumulative distribution function $F_{|g_k|^2}$ of the kth V2V link, is assumed to be concave on $\mathbf{R}^+$. But, for $\eta - \mu$ distributed RV, $F_{|g_k|^2}$ is not concave on $\mathbf{R}^+$ for all η and μ parameters. But it shown in (Papandriopoulos *et al.*, 2006) that even if the $F_{|g_k|^2}$ is not concave, the upper bound is a tight approximation for the outage constraints. Using

this approximation the transformed optimization problem is

$$\max_{P_m^c, \mathbf{P}_k^d} \mathbb{E}\left[\log_2\left(1+\frac{P_m^c|\mathbf{g}_{m,B}|^2}{\sum\limits_{k\in C_n}\rho_{m,k}P_k^d|\mathbf{g}_{m,B}^H\mathbf{g}_{k,B}|^2}\right)\right], \tag{5.20}$$

$$\text{s.t. } \frac{P_k^d\alpha_k}{P_m^c|g_{m,k}|^2+\sum\limits_{k'\neq k\in C_n}P_{k'}^d|g_{k',k}|^2} \geq \frac{\gamma_0^d}{F_{|g_k|^2}^{-1}(p_0)}, \forall k\in C_n \tag{5.20a}$$

$$0 \leq P_m^c \leq P_{\text{max}}^c \tag{5.20b}$$

$$0 \leq P_k^d \leq P_{\text{max}}^d, \forall k \in C_n. \tag{5.20c}$$

Note that (5.20a) are linear inequality constraints in P_m^c and P_k^d, $\forall k \in C_n$. Since the objective (5.20) is monotonically decreasing with P_k^d, $\forall k \in C_n$, using the same argument as in (Liang *et al.*, 2018), it can be shown that the objective is maximized only if the outage constraints are satisfied with equality. Therefore, similar to (Liang *et al.*, 2018), the optimal solution to 5.20 is given by

$$P_m^{c^*} = \min\left\{P_{\text{max}}^c, \left\{\frac{P_{\text{max}}^d}{\bar{\gamma}_0\boldsymbol{\phi}_i^H\boldsymbol{\alpha}_m}\right\}_{i=1}^{N_{c_n}}\right\}, \tag{5.21}$$

and

$$\mathbf{P}_n^{d^*} = \boldsymbol{\Phi}^{-1}\bar{\gamma}_0\left(P_m^{c^*}\boldsymbol{\alpha}_m\right), \tag{5.22}$$

where $\mathbf{P}_n^d \in \mathbb{R}^{N_{c_n}\times 1}$ denotes the transmit powers of all N_{c_n} V2V links in the nth cluster, $\bar{\gamma}_0 = \frac{\gamma_0^d}{F_{|g_k|^2}^{-1}(p_0)}$, $\boldsymbol{\alpha}_m = (\alpha_{m,1}, \cdots, \alpha_{m,N_{c_n}})^T \in \mathbb{C}^{N_{c_n}\times 1}$, $\mathbf{1}$ is an all-one vector, $\boldsymbol{\Phi} \in \mathbb{C}^{N_{c_n}\times N_{c_n}}$ is given by

$$\boldsymbol{\Phi}_{i,j} = \begin{cases} \alpha_i, & \text{if } i=j, \\ -\bar{\gamma}_0\alpha_{j,i}, & \text{otherwise}, \end{cases} \tag{5.23}$$

and $\boldsymbol{\phi}_i^H$ is the ith row of the inverse of $\boldsymbol{\Phi}$, i.e., $\boldsymbol{\Phi}^{-1}$.

5.5.3 Spectrum Allocation

With the optimal power allocations $(P_m^{c^*}, \mathbf{P}_n^{d^*})$, under each given channel allocation, we aim to maximize the cellular rates by searching over all possible channel allocation

schemes. We first determine the ergodic capacity of mth V2I when it shares spectrum with nth V2V cluster is $C_{m,n}$ for a power allocation $(P_m^{c^*}, \mathbf{P}_n^{d^*})$. Since, MRC is used at the BS, the rate $C_{m,k}(P_m^{c^*}, \mathbf{P}_n^{d^*}) = \mathbb{E}[\log_2(1 + \gamma_m^{c\,*})]$, where $\gamma_m^{c\,*}$ is the SIR at the output of MRC for the power allocation $(P_m^{c^*}, \mathbf{P}_n^{d^*})$. This rate can be easily determined by means of the rate expressions derived in the previous sections. Without the rate expressions, it wouldn't have been possible to consider an interference mitigation scheme like MRC at the BS. Therefore, the spectrum allocation problem becomes

$$\max_{\rho_{m,n}} \sum_{m \in \mathcal{M}} C_{m,k},$$
$$\sum_{\forall m \in \mathcal{M}} \rho_{m,n} \leq 1, \; \rho_{m,n} \in \{0,1\}, \; \forall m \in \mathcal{M},$$
$$\sum_{\forall n \in \mathcal{N}} \rho_{m,n} \leq 1, \; \rho_{m,n} \in \{0,1\}, \; \forall n \in \mathcal{N}.$$

This is a maximum weight bipartite matching problem and can be solved by the Hungarian method as in (Liang *et al.*, 2017; Wang *et al.*, 2017*b*, 2018).

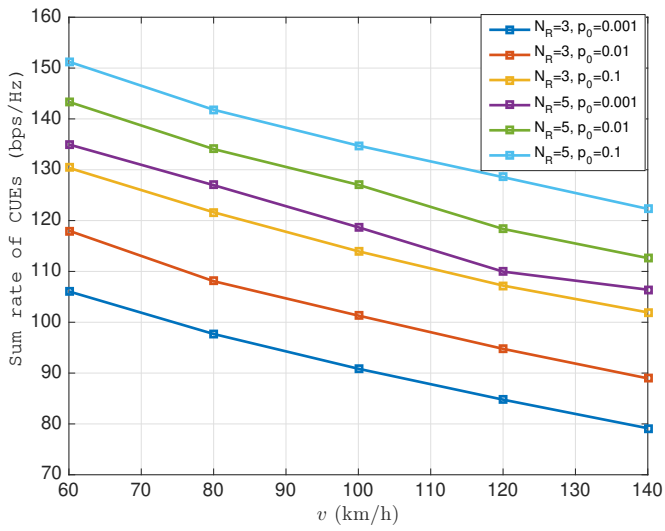

Figure 5.6: Sum rate of V2I vs speed of vehicle for $\eta_{V2I} = 0$, $\mu_{V2I} = 0.5$, $\eta_{V2V} = 0.5$, $\mu_{V2V} = 2$

We now present simulation results to validate the proposed resource allocation algorithm. We now use the same simulation parameters as in (Liang *et al.*, 2017; Liang *et al.*, 2018). We model a multi-lane freeway that passes through a single cell where the BS is located at its center. The vehicles are dropped on the roads according to spatial Poisson process and the vehicle density is determined by the vehicle speed as in (Liang *et al.*, 2018). The M V2I and K V2V links are randomly chosen among the vehicles.

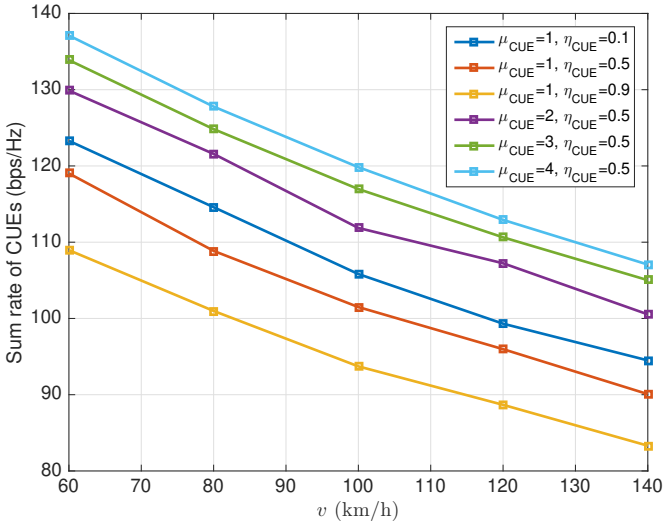

Figure 5.7: Sum rate of V2I vs speed of vehicle for $N_R = 3$, $\eta_{V2V} = 0.5$, $\mu_{V2V} = 4$, $p_0 = 0.001$

The simulation parameters are listed in Table IX in (Liang *et al.*, 2018) and the channel models for V2I and V2V links are listed in Table X in (Liang *et al.*, 2018). The number of V2V clusters, N, is set equal to the number of V2I links, M. Sum rate of the V2Is for various vehicle speeds are provided in Fig. 5.6 and 5.7. With decreasing p_0, we can observe that the reliability guarantee at V2Vs become stricter. This implies that the power allocated to the V2Vs become larger and therefore, the rate of the V2Is decreases. This can be compensated by increasing the number of antennas at N_R or by increasing the number of clusters of the fading parameters at the V2I μ_{V2I} or by decreasing η_{V2I}.

5.6 Conclusion

In this chapter, we considered receiver diversity system in the presence of multiple unequal power uncorrelated $\eta-\mu$ interferers. Closed-form approximate outage probability and rate expressions for the MRC system were derived for the case when the desired user and the interfering signals are subject to $\eta-\mu$ fading. Extensive Monte-Carlo simulations were performed and the approximation matches the simulation results for all $\eta-\mu$ parameters. The effect of the variation of η and μ on the outage probability was also analyzed using tools from stochastic ordering. Also, the utility of the expressions were studied by means of a vehicular application.

CHAPTER 6

Analysis of Optimal Combining in Rician Fading with Co-channel Interference

6.1 Introduction

Among the various diversity combining schemes, optimum combining (OC) proposed in (Winters, 1984) maximizes the signal to interference plus noise ratio (SINR). Performance of OC receivers has been extensively studied for various cases when both the desired and interfering signals are subjected to Rayleigh fading (McKay *et al.*, 2009; Ali *et al.*, 2010; Rahimzadeh and Beaulieu, 2010; Chung, 2012; Shah and Haimovich, 1998; Kwak and Lee, 2002; Chiani *et al.*, 2003*c*; Mallik *et al.*, 2004; Menon and Kalyani, 2015; Yue *et al.*, 2006; Zhang and Cui, 2004; Afana *et al.*, 2015). Many practical scenarios exist such as indoor propagation, micro-cellular channels, satellite channels, inter-vehicular communications, etc, where both the desired and interfering signals may have line-of-sight (LoS) paths. Symbol error rate (SER) expressions for OC have been derived, when either the desired signals or the interfering signals undergo Rician fading, while the other undergoes Rayleigh fading (McKay *et al.*, 2009).

Rician fading has found applications even in recent times in the study of the performance of distributed multiple input multiple output (MIMO) with zero forcing (ZF) receivers over correlated Rician fading channels (Li *et al.*, 2017*b*), in deriving expressions for achievable rates of MIMO relay systems with ZF processing over Rician fading systems (Wang *et al.*, 2017*a*) and in the study of the performance of co-operative relaying systems with non-orthogonal multiple access (Jiao *et al.*, 2017). Existing 4G and emerging 5G systems are both interference-limited. Hence, receiver techniques like OC and MRC will play a key role in the performance analysis of these systems (Ali *et al.*, 2017). However, SER and/or outage probability (OP) expressions when OC is employed and when both the user and interferers undergo Rician fading have not been

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studied. Further, characterization of OC receivers, which takes into account practical scenarios such as unequal interference power and correlation among interferers is not present in open literature.

6.1.1 Our contributions

We address the gap in the literature with the following contributions through this chapter:

- We derive exact expressions for the Laplace transform of SINR at the output of OC considering Rician faded users and a) mixture of Rayleigh and Rician faded interferers, b) only Rician faded interferers and c) only Rayleigh faded interferers. A simple approximation, which avoids determinant evaluation, is also derived for Rayleigh faded interferers in an interference dominated scenario.

- We also derive exact Laplace transform expressions of SINR for unequal power interferers and correlated interferers, which occur due to correlated channel fading, shadowing and from spatial distribution of transmitters (Zhong *et al.*, 2014; Kumar and Kalyani, 2017*a*, 2016). An extension to $\kappa - \mu$ faded users is also proposed.

- Using these Laplace transform expressions, we derive approximate SER expressions that are functions of a double infinite series, which are truncated to finite series with arbitrarily small truncation error. The series terms are functions of Tricomi hypergeometric functions, which has been used extensively in analyzing throughput and the rate of wireless systems over various fading channels (Li *et al.*, 2017*a*; Zhang *et al.*, 2014*a*; You *et al.*, 2016; Zhang *et al.*, 2015*b*).

- We also derive an expression for the moments of the SINR η. The first two moments are then matched with those of a beta-prime random variable to obtain approximate outage probability and rate expressions. Inferences on the impact of the fading parameters are analytically studied by using stochastic ordering tools on the outage and SER expressions. All our results are compared with corre-

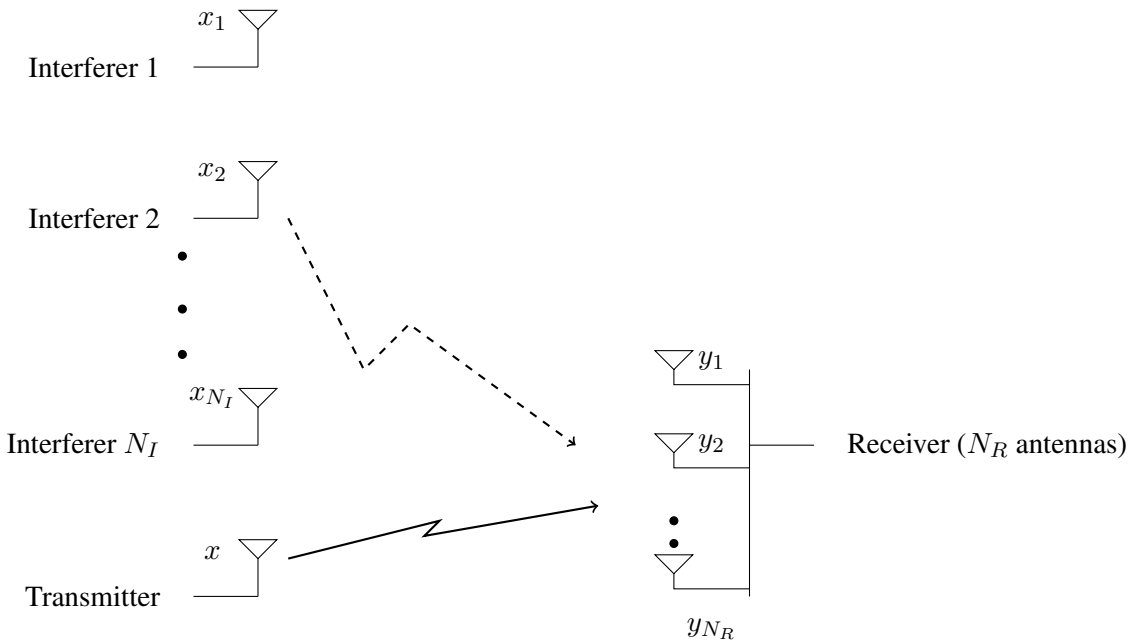


Figure 6.1: System model

sponding Monte-Carlo simulations and a close match is observed. We also give an application of OC in vehicular technology networks.

6.2 System Model

Let N_R denote the number of receive antennas, N_I denote the number of interferers, $\mathbf{c}$ denote the $N_R \times 1$ channel from the transmitter to the user, $\mathbf{c}_i$ denote the $N_R \times 1$ channel from the ith interferer to the user, x denotes the desired user symbol belonging to unit-energy quadrature amplitude modulation (QAM) constellation and x_i denote the ith interferer symbol also belonging a unit energy QAM constellation. The schematic is given in Fig. 6.1. The received vector is given by

$$\mathbf{y} = \mathbf{c}x + \sum_{i=1}^{N_I} \mathbf{c}_i x_i + \mathbf{n}, \tag{6.1}$$

where $\mathbf{n}$ is the $N_R \times 1$ additive white complex Gaussian noise vector, with a power of σ^2 per dimension i.e., $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I}_{N_R})$. The interferer channels are modeled as i.i.d. Rician i.e., $\mathbf{c}_i \sim \mathcal{CN}(\sqrt{a'}\mathbf{m}'_\mathbf{i}, b'\mathbf{I}_{N_R})$, where $a' = \frac{\kappa_i}{\kappa_i+1}$, $b' = \frac{1}{\kappa_i+1}$, κ_i is the ratio of the power of the line of sight component to the scattering component of the interferer

signals and $\mathbf{m}'_{\mathbf{i}}$ is an $N_R \times 1$ arbitrary vector with elements of unit magnitude. The user channel is also assumed to be i.i.d Rician i.e., $\mathbf{c} \sim \mathcal{CN}(\sqrt{a}\mathbf{m}, b\mathbf{I}_{N_R})$, where $a = \frac{\kappa_s}{\kappa_s+1}$, $b = \frac{1}{\kappa_s+1}$. Note that, the Rician parameter κ_s is the ratio of the power of the line of sight component to that of the scattering component and $\mathbf{m}$ is an $N_R \times 1$ mean vector with elements of unit magnitude and uniform phase. Let $E''_I = N_I \times E'_I$ denote the total energy of the interfering signals, where E'_I is the mean energy of each of the interfering signals. The covariance matrix of the interference term plus the noise term is given by

$$\mathbf{R} = E'_I \mathbf{C}'\mathbf{C}'^H + \sigma^2\mathbf{I} = E_I \mathbf{C}\mathbf{C}^H + \sigma^2\mathbf{I}, \tag{6.2}$$

where $\mathbf{C}' = [\mathbf{c}_1, ..., \mathbf{c}_{N_I}]$, $\mathbf{C}' \sim \mathcal{CN}(\sqrt{a'}\mathbf{M}', b'\mathbf{I}_{N_R} \otimes \mathbf{I}_{N_I})$. $\mathbf{M}'$ is an arbitrary deterministic matrix obtained by stacking $\mathbf{m}'_{\mathbf{i}}$s, such that, $\mathbf{M}' = [\mathbf{m}'_{\mathbf{1}}, \mathbf{m}'_{\mathbf{2}}, \mathbf{m}'_{\mathbf{3}}, ..., \mathbf{m}'_{\mathbf{N_I}}]$ and tr$(\mathbf{M}'\mathbf{M}'^H) = N_R N_I$. Here, $\mathbf{C} \sim \mathcal{CN}(\mathbf{M}, \mathbf{I}_{N_R} \otimes \mathbf{I}_{N_I})$, $E_I = E'_I \times b'$ and $\mathbf{M} = \sqrt{a'}/\sqrt{b'}\mathbf{M}' = \sqrt{\kappa_i}\mathbf{M}'$. The received SINR for the OC is given by (Chiani *et al.*, 2003*c*)

$$\eta = E_D \mathbf{c}^H \mathbf{R}^{-1} \mathbf{c}, \tag{6.3}$$

where E_D is the mean energy of the user signal. In the next two sections, we will detail the procedure to obtain the Laplace transform expressions.

6.3 Laplace transform for equal power uncorrelated interferers

A general expression for the Laplace transform $M_\eta(s)$ of SINR η can now be obtained from Theorem 1 of (McKay *et al.*, 2009). We further simplify this expression for the specific case of Rician distribution. Let $n_2 = \max(N_R, N_I)$ and $n_1 = \min(N_R, N_I)$[1].

$$M_\eta(s) = (-1)^{N_R}(\sigma^2/E_I)^{(N_R-n_1)}\mathbb{E}_{\mathbf{\Lambda}_R}\left[\left(\prod_{i=1}^{n_1}\frac{\sigma^2/E_I + \lambda_i}{\lambda_i^{(N_R-n_1)}}\right)\frac{\det(\mathbf{J})}{V_{n_1}(\mathbf{\Lambda}_R)}\right], \tag{6.4}$$

where $V_{n_1}(\mathbf{\Lambda}_R)$ is the determinant of the Vandermonde matrix formed by eigenval-

[1]According to definition, moment generating function (mgf) should ideally exist in an interval around 0. But in all works including (McKay *et al.*, 2009), $M_\eta(s)$ exists only for $s < 0$. So we believe that calling $M_\eta(s)$ Laplace transform is a more appropriate, as Laplace transform can be one-sided unlike mgf.

ues of non-central Wishart matrix $\mathbf{C}\mathbf{C}^H$. $\mathbf{J}$ is an $n_1 \times n_1$ matrix with elements,

$$\mathbf{J}_{i,j} = \begin{cases} h_1(s,\lambda_i) - \sum_{t=1}^{N_R-n_1} h_t(s,0)\lambda_i^{t-1}, \; j=1, \\ \lambda_i^{N_R-j}, \; j=2,...,n_1, \end{cases} \tag{6.5}$$

and

$$h_t(s,x) = \frac{{}_1F_1(t; N_R; \frac{aN_R s}{xE_I/E_D+\sigma^2/E_D-bs})}{(bsE_D/E_I - \sigma^2/E_I - x)^t}, \tag{6.6}$$

with the series expansion of ${}_1F_1(.)$ given by ${}_1F_1(a;b;z) = \sum_{z=0}^{\infty} \frac{(a)_k z^k}{(b)_k k!}$. Laplace transform is derived in (McKay *et al.*, 2009) for two cases: a) Rician signal with Rayleigh interferers b) Rayleigh signal with Rician interferers. In the former case, (6.4) is used along with the eigenvalue distribution of central Wishart matrix to arrive at a closed form expression for the Laplace transform. In the latter case, the fact that user signal $\mathbf{c}$ exhibits Rayleigh fading and hence invariant under unitary transformation is exploited to derive a closed form expression for the Laplace transform. To the best of our knowledge, there is no open literature that proves that Rician distribution is invariant under unitary transformation. Therefore, for the case of Rician signals with Rician interferers, we propose to evaluate the expectation in (6.4), by using the eigenvalue distribution of non-central Wishart matrix and subsequently simplify it by using properties of hypergeometric functions. The joint probability density function (pdf) of ordered eigenvalues $(\lambda_1 > \lambda_2 > ... > \lambda_{n_1})$ of non-central Wishart matrix is given by (Jin *et al.*, 2008),

$$f(\lambda_1,...,\lambda_s) = c_1 \det(\boldsymbol{\Upsilon}) \prod_{i<j}^{n_1}(\lambda_i - \lambda_j) \prod_{k=1}^{n_1} \lambda_k^{n_2-n_1} e^{-\lambda_k}, \tag{6.7}$$

where $\boldsymbol{\Upsilon}$ is a $n_1 \times n_1$ matrix whose $(i,j)^{th}$ entry $\forall i = 1,...,n_1$ is given by,

$$\boldsymbol{\Upsilon}_{i,j} = \begin{cases} {}_0F_1(n_2-n_1+1; w_j\lambda_i), & j=1,...,L, \\ \lambda_i^{n_1-j}\frac{(n_2-n_1)!}{(n_2-j)!}, & j=L+1,...,n_1, \end{cases}$$

and

$$c_1 = \frac{e^{-\mathrm{tr}(\boldsymbol{\Omega})}((n_2-n_1)!)^{-n_1}}{\prod_{i=1}^{n_1-L}(n_1-L-i)! \prod_{i=1}^{L} w_i^{n_1-L} \prod_{i<j}^{L}(w_i-w_j)}.$$

Note that w_is are the ordered L non-zero eigenvalues of the non-centrality matrix, $\boldsymbol{\Omega} = \mathbf{M}^H\mathbf{M}$ and the series expansion of the hypergeometric function ${}_0F_1(.)$ is given

by ${}_0F_1(b;z) = \sum_{k=0}^{\infty} \frac{z^k}{(b)_k k!}$. Substituting (6.7) in (6.4),

$$M_\eta(s) = c_1(-1)^{N_R}(\sigma^2/E_I)^{(N_R-n_1)} \int_0^\infty \left[\left(\prod_{i=1}^{n_1} \frac{\sigma^2/E_I + \lambda_i}{\lambda_i^{(N_R-n_1)}} \right) \frac{\det(\mathbf{J})}{V_{n_1}(\mathbf{\Lambda}_R)} \right] \det(\mathbf{\Upsilon})$$
$$\times \prod_{i<j}^{n_1} (\lambda_i - \lambda_j) \prod_{k=1}^{n_1} \lambda_k^{n_2-n_1} e^{-\lambda_k} d\lambda_1 ... d\lambda_{n_1}$$
$$= (-1)^{N_R}(\sigma^2/E_I)^{(N_R-n_1)} \int_0^\infty c_1 \left(\prod_{i=1}^{n_1} (\sigma^2/E_I + \lambda_i) \times e^{-\lambda_i} \right)$$
$$\det(\mathbf{J}) \det(\mathbf{\Upsilon})\, d\lambda_1 ... d\lambda_{n_1}. \tag{6.8}$$

Using Theorem 2 in Appendix of (Chiani *et al.*, 2003*b*), we can simplify Laplace transform in (6.8) to obtain,

$$M_\eta(s) = c \det(\mathbf{N}), \tag{6.9}$$

where

$$c = \frac{e^{-\mathrm{tr}(\mathbf{\Omega})}((n_2-n_1)!)^{-n_1}(-1)^{N_R}(\sigma^2/E_I)^{(N_R-n_1)}}{\prod_{i=1}^{n_1-L}(n_1-L-i)! \prod_{i=1}^{L} w_i^{n_1-L} \prod_{i<j}^{L}(w_i - w_j)},$$

$$\mathbf{N}_{i,j} = \begin{cases} \int_0^\infty (\frac{\sigma^2}{E_I} + x) e^{-x} x^{n_2-N_R}\, {}_0F_1(n_2-n_1+1; w_i x) \\ \qquad [h_1(s,x) - \sum_{t=1}^{N_R-n_1} h_t(s,0) x^{t-1}] dx,\ j=1,\ i=1,...,L, \\ \int_0^\infty (\frac{\sigma^2}{E_I} + x) e^{-x} x^{n_2-N_R} x^{n_1-i} \frac{(n_2-n_1)!}{(n_2-i)!} [h_1(s,x) - \sum_{t=1}^{N_R-n_1} h_t(s,0) x^{t-1}] dx, \\ \qquad j=1,\ i=L+1,...,n_1, \\ \int_0^\infty (\frac{\sigma^2}{E_I} + x) e^{-x} x^{n_2-N_R}\, {}_0F_1(n_2-n_1+1; w_i x) x^{N_R-j} dx, \\ \qquad j=2,...,n_1\ i=1,...,L, \\ \int_0^\infty (\frac{\sigma^2}{E_I} + x) e^{-x} x^{n_2-N_R} x^{n_1-i} \frac{(n_2-n_1)!}{(n_2-i)!} x^{N_R-j} dx, \\ \qquad j=2,...,n_1\ i=L+1,...,n_1. \end{cases}$$

Further simplification of $\mathbf{N}_{i,j}$ is given in Appendix C. The final expression for the entries of $\mathbf{N}_{i,j}$ is given in (6.10) and this can be substituted in (6.9) to obtain the final Laplace transform of SINR. In (6.10), $p = \sigma^2/E_I$, $q = n_2 - n_1 + 1$, $u = aN_R s E_D/E_I$, $v = bsE_D/E_I - \sigma^2/E_I$. Also, note that the expression for $N_{i,j}$, for $L = n_1$ and $L = 0$, which correspond to all Rician faded interferers and all Rayleigh faded interferers respectively, get significantly simplified. When we do a Laplace expansion of the determinant $\det(\mathbf{N}_{L=0})$ along the first column and substitute $\zeta_t(k) = \frac{\sigma^2}{E_I}\Gamma(n_1+n_2-N_R+t-k+1)+\Gamma(n_1+n_2-N_R+t-k+2)$, we observe that the expression for $M_\eta(s)$

$$
\mathbf{N}_{i,j} = \begin{cases}
\sum_{k=0}^{T_2} \frac{w_i^k}{q_k k!} \Bigg[\sum_{l=0}^{T_1} \frac{u^l}{(N_R)_l} \Bigg[-\Gamma(k+n_2-N_R+2)U(l+1, l-k-n_2+N_R, -v) \\
-p\Gamma(k+n_2-N_R+1)U(l+1, -k-n_2+N_R+l+1, -v) \Bigg] \Bigg] \\
- \sum_{t=1}^{N_R-n_1} \frac{{}_1F_1(t;N_R;u/v)}{v^t} \times \Bigg[p\Gamma(t+n_2-N_R)\, {}_1F_1(t+n_2-N_R; q; w_i) \\
\quad + \Gamma(t+n_2-N_R+1) {}_1F_1(t+n_2-N_R+1; q; w_i) \Bigg], \\
\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad j=1,\ i=1,...,L, \\
\frac{(n_2-n_1)!}{(n_2-i)!} \Bigg[\sum_{l=0}^{T_1} \frac{u^l}{(N_R)_l} \Bigg[-\Gamma(n_2-N_R+n_1-i+2) \\
U(l+1, l-n_2+N_R-n_1+i, -v) \\
-p\Gamma(n_2-N_R+n_1-i+1)U(l+1, -n_2+N_R-n_1+i+l+1, -v) \Bigg] \\
- \sum_{t=1}^{N_R-n_1} \frac{{}_1F_1(t;N_R;u/v)}{v^t} \Bigg[p\Gamma(t+n_2+n_1-N_R-i) \\
\quad + \Gamma(t+n_2+n_1-N_R-i+1) \Bigg] \Bigg], \\
\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad j=1,\ i=L+1,...,n_1, \\
p\, {}_1F_1(n_2-j+1; q; w_i)\Gamma(n_2-j+1) + \Gamma(n_2-j+2) {}_1F_1(n_2-j+2; q; w_i), \\
\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad j=2,...,n_1,\ i=1,...,L. \\
\frac{(n_2-n_1)!}{(n_2-i)!} [p\Gamma(n_2+n_1-i-j+1) + \Gamma(n_2+n_1-i-j+2)], \\
\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad j=2,...,n_1,\ i=L+1,...,n_1.
\end{cases} \tag{6.10}
$$

for the Rayleigh interferers case is the same as the one obtained in (McKay *et al.*, 2009, Eq. 13).

Interference-limited scenario

Recently, there has been a lot of interest in characterizing the performance of cellular networks/wireless system in an interference-limited scenario. Throughput and rate have been studied in (Paris, 2013; Kumar and Kalyani, 2015; Kumar *et al.*, 2015*a*; Zhou *et al.*, 2015; Chayawan and Aalo, 2002) and references therein, assuming the noise can be neglected (i.e., $\sigma^2 = 0$), in an interference-limited scenario. Motivated by these works, we now show that the Laplace expansion can be substantially simplified in the case of Rayleigh interferers. Note that an interference-limited scenario is possible only for $N_I > N_R$. For $N_R > N_I$ and $\sigma^2 = 0$, the receive antennas can cancel every interfering signal. If the number of non-zero eigenvalues $L = 0$ as is the case for Rayleigh

fading, then $M_\eta(s) = c \det(\mathbf{N}_{\sigma^2=0,L=0})$, where $c = \frac{((n_2-n_1)!)^{-n_1}}{\prod_{i=1}^{n_1}(n_1-i)!}(-1)^{N_R}\left(\frac{\sigma^2}{E_I}\right)^{(N_R-n_1)}$. Note that in c, we do not neglect σ^2. The expression can be further simplified as shown in Appendix D to obtain,

$$M_\eta(s) = \frac{(-1)^{N_R+1}(\sigma^2/E_I)^{(N_R-n_1)}n_2!}{(n_2-n_1)!\prod_{i=1}^{n_1}(n_1-i)!}\sum_{i=1}^{n_1}(-1)^{i+1}A(i)\frac{\prod_{j=1}^{n_1}(j-1)!}{(n_1-i)!(i-1)!}, \quad (6.11)$$

where $A(i)$ is given in Appendix D below (D.2).

6.4 Laplace Transform for correlated interferers and unequal power interferers

In the previous section, Laplace transform expressions are derived for the case of equal power uncorrelated interferers. But, in practice, the interferers can have different power and/or can be correlated. In a practical cellular system, there can be one or more of the following: a) receiver side correlation, b) interferer correlation, c) unequal power interferers.

The general non-central Wishart matrix $\mathbf{W}$ is written as $\mathbf{W} = \mathbf{C}'\mathbf{C}'^H$, where $\mathbf{C}' \sim \mathcal{CN}(\mathbf{M}, \mathbf{\Sigma}\otimes\mathbf{\Psi})$. Here, the $N_R \times N_R$ matrix $\mathbf{\Sigma}$ denotes the receive correlation and the $N_I \times N_I$ matrix $\mathbf{\Psi}$ denotes the transmit correlation or interferer correlation in our case. Suppose, we consider only receive side correlation and assume that the interferer correlation is not present, i.e., $\mathbf{\Psi}$ is an identity matrix. This reduces to the non-central Wishart matrix denoted by $\mathbf{W} \sim \mathcal{CW}(N_I, \mathbf{\Sigma}, \mathbf{\Sigma}^{-1}\mathbf{M}\mathbf{M}^H)$. This case, where $\mathbf{\Psi}$ is assumed to be an identity matrix, is widely discussed in the literature. The eigenvalue distribution of this case, i.e., a non-central Wishart matrix with a covariance matrix $\mathbf{\Sigma}$ which is not an identity matrix, is analyzed in (Ratnarajah *et al.*, 2005) in terms of zonal polynomials. However, using this eigenvalue distribution to obtain the Laplace transform expression of η becomes mathematically intractable. Hence, considering receive correlation is beyond the scope of this work.

On the other hand, the cases of $\mathbf{\Psi}$ being a diagonal matrix, i.e., unequal power interferers or $\mathbf{\Psi}$ being a full matrix, i.e., correlated interferers, have barely received attention in statistic literature. There do not even exist matrix variate and eigenvalue

distribution results for this case. However, we do provide results for this case by considering the problem as two sub-problems a) for $N_R \geq N_I$ exact results are provided, b) for $N_R < N_I$ approximate results are provided. In short, in this section, we derive Laplace transform expressions for $\mathbf{C}' \sim \mathcal{CN}(\mathbf{M}, \mathbf{I}_{N_R} \otimes \mathbf{\Psi})$. Note, $\mathbf{W} = \mathbf{C}'\mathbf{C}'^H$ can be decomposed into $\mathbf{W} = \mathbf{C}\mathbf{\Psi}\mathbf{C}^H$, such that $\mathbf{C} \sim \mathcal{CN}(\mathbf{M}\mathbf{\Psi}^{-\frac{1}{2}}, \mathbf{I}_{N_R} \otimes \mathbf{I}_{N_I})$ (McKay and Collings, 2005). Let us first consider the case of correlated interferers. The covariance matrix of the interference term plus the noise term is given by

$$\mathbf{R} = \mathbf{C}\mathbf{\Psi}\mathbf{C}^H + \sigma^2\mathbf{I}. \tag{6.12}$$

The received SINR for the OC is given by (6.3). We will consider this problem as two cases: a) $N_R \geq N_I$, b) $N_R < N_I$.

6.4.1 $N_R \geq N_I$

Similar to (6.4), the general expression for the Laplace transform $M_\eta(s)$ of SINR η is given by,

$$M_\eta(s) = (-1)^{N_R}(\sigma^2)^{(N_R-n_1)}\mathbb{E}_{\mathbf{\Lambda}_R}\left[\left(\prod_{i=1}^{n_1}\frac{\sigma^2+\lambda_i}{\lambda_i^{(N_R-n_1)}}\right)\frac{\det(\mathbf{J})}{V_{n_1}(\mathbf{\Lambda}_R)}\right], \tag{6.13}$$

where $V_{n_1}(\mathbf{\Lambda}_R)$ is the determinant of the Vandermonde matrix formed by eigenvalues of non-central Wishart matrix $\mathbf{C}\mathbf{\Psi}\mathbf{C}^H$. $\mathbf{J}$ is an $n_1 \times n_1$ matrix with elements given by (6.5) and

$$h_t(s,x) = \frac{{}_1F_1(t; N_R; \frac{aN_R s}{x/E_D+\sigma^2/E_D-bs})}{(bsE_D-\sigma^2-x)^t}. \tag{6.14}$$

Recall that, in the case of equal power uncorrelated interferers, we simplified the expression in (6.4) using the eigenvalue distribution of the non-central Wishart matrix. But for the case of correlated interferers, there exists no matrix variate distribution formula in the open literature and deriving one requires integration over the Stiefel manifold (Ratnarajah *et al.*, 2005). Also, there exists no eigenvalue distribution for this case. Hence, initially, we consider the case of Rayleigh-faded interferers, i.e., $\mathbf{M} = \mathbf{0}$.

Rayleigh faded correlated interferers

We exploit the property that $\mathbf{W} = \mathbf{C}\mathbf{\Psi}\mathbf{C}^H$ has the same non-zero eigenvalues as that of $\mathbf{\Psi}^{\frac{1}{2}}\mathbf{C}^H\mathbf{C}\mathbf{\Psi}^{\frac{1}{2}}$, where $\mathbf{C}^H\mathbf{C}$ is also a Wishart matrix. The eigenvalue distribution of $\mathbf{\Psi}^{\frac{1}{2}}\mathbf{C}^H\mathbf{C}\mathbf{\Psi}^{\frac{1}{2}}$, is given by (Couillet and Debbah, 2011),

$$f(\lambda_1, ..., \lambda_s) = c_1 \det(\mathbf{\Upsilon}) \prod_{i<j}^{N_I} (\lambda_i - \lambda_j) \prod_{k=1}^{N_I} \lambda_k^{N_R - N_I}, \tag{6.15}$$

where $\mathbf{\Upsilon}$ is a $N_I \times N_I$ matrix whose $(i,j)^{th}$ entry $\forall i, j = 1, ..., N_I$ is given by, $\mathbf{\Upsilon}_{i,j} = e^{-\frac{\lambda_i}{r_j}}$ and $c_1 = (-1)^{\frac{1}{2}N_I(N_I-1)} \frac{\det(\mathbf{\Psi})^{-N_R}}{\prod_{i<j}^{N_I}(\frac{1}{r_i} - \frac{1}{r_j}) \prod_{k=1}^{N_I}(N_R - k)!}$. Also, note that r_is are the ordered N_I distinct non-zero eigenvalues of $\mathbf{\Psi}$. Using (6.15) to simplify (6.13), we obtain the Laplace transform of η as

$$\begin{aligned} M_\eta(s) &= (-1)^{N_R} (\sigma^2)^{(N_R - N_I)} \\ &\times \int_0^\infty c_1 \prod_{i=1}^{n_1} (\sigma^2 + \lambda_i) \det(\mathbf{J}) \det(\mathbf{\Upsilon}) \, d\lambda_1 ... d\lambda_{N_I}. \end{aligned} \tag{6.16}$$

As in the case of equal power interferers, we use Theorem 2 in the Appendix of (Chiani *et al.*, 2003*b*) and the identity $\int_0^\infty e^{-px} x^{s-1} dx = p^{-s}\Gamma(s)$ from (Gradshteyn and Ryzhik, 2007). We can solve the integral for $j = 1$, by expanding the ${}_1F_1$ hypergeometric series and interchanging the integration and summation. The approach followed in Appendix C can be followed here. The Laplace transform after simplification becomes,

$$M_\eta(s) = c \det(\mathbf{N}), \tag{6.17}$$

where $c = \frac{(-1)^{N_R}(\sigma^2)^{(N_R - N_I)}(-1)^{\frac{1}{2}N_I(N_I-1)} \det(\mathbf{\Psi})^{-N_R}}{\prod_{i<j}^{N_I}(\frac{1}{r_i} - \frac{1}{r_j}) \prod_{k=1}^{N_I}(N_R - k)!}$ and N is given by

$$\mathbf{N}_{i,j} = \begin{cases} \sum_{l=0}^{\infty} \frac{(aN_R s E_D)^l}{(N_R)_l} (\sigma^2 |bsE_D - \sigma^2|^{-l} U(1, 1-l, r_i |bsE_D - \sigma^2|) \\ \quad + |bsE_D - \sigma^2|^{-l+1} U(2, 2-l, r_i |bsE_D - \sigma^2|)) \\ \quad - \sum_{t=1}^{N_R - N_I} \frac{{}_1F_1(t; N_R; \frac{aN_R s}{\sigma^2/E_D - bs})}{(bsE_D - \sigma^2)^t} (\sigma^2 r_i^t \Gamma(t) + r_i^{t+1}\Gamma(t+1)], \\ \quad j = 1, \, i = 1, ..., N_I, \\ \sigma^2 r_i^{N_R - j + 1} \Gamma(N_R - j + 1) + r_i^{N_R - j + 2} \Gamma(N_R - j + 2), \\ \quad j = 2, ..., N_I \, i = 1, ..., N_I, \end{cases} \tag{6.18}$$

where r_i, $1 \leq i \leq N_I$ are the eigenvalues of $\mathbf{\Psi}$. We can truncate the converging infinite series for $j = 1$ at a finite value, with an arbitrarily small truncation error. The convergence proof is similar to the one given in Appendix C. Note that this is an exact Laplace transform expression and is novel for the case of Rayleigh faded correlated interferers with Rician faded users. Earlier works like (McKay *et al.*, 2009), considers only equal power uncorrelated interferers, while recent works like (Menon and Kalyani, 2015) consider only Rayleigh faded user. An approximation which works for $\sigma^2 \approx 0$ is also derived in Appendix F. The expression is as follows:

$$M_\eta(s) \approx c \sum_{i=1}^{N_I} (-1)^{i+1} B(i) r_i^{-N_R+N_I-2} \det\left(V^i(\mathbf{r})\right), \tag{6.19}$$

where $c = \frac{(-1)^{N_R}(\sigma^2)^{(N_R-N_I)}(-1)^{\frac{1}{2}N_I(N_I-1)}\det(\mathbf{\Psi})^{-N_R}}{\prod_{i<j}^{N_I}(\frac{1}{r_i}-\frac{1}{r_j})\prod_{k=1}^{N_I}(N_R-k)!}$, $A(i)$ and $\det\left(V^i(\mathbf{r})\right)$ are given in Appendix F.

Rician faded correlated interferers

For the case of Rician faded interferers, the above approach is not possible even for $N_R \geq N_I$. This is because we have to use the zonal-polynomial based eigenvalue distribution from (Ratnarajah *et al.*, 2005) to simplify the Laplace transform, which is not mathematically tractable. Nevertheless, we arrive at a mathematically tractable solution, wherein we propose to approximate the non-central Wishart matrix by a central Wishart matrix through moment matching and then use the derived expression given in (6.17).

The expected value of any matrix of the form $\mathbf{A}^H\mathbf{A}$, if $\mathbf{A} \sim \mathcal{CN}(\mathbf{M}, \mathbf{I}_{N_I} \otimes \mathbf{I}_{N_R})$, with N_R degrees of freedom, is given by $\mathbb{E}[\mathbf{A}^H\mathbf{A}] = N_R + \mathbf{M}^H\mathbf{M}$. This implies that for $\mathbf{W} = \mathbf{\Psi}^{\frac{1}{2}}\mathbf{C}^H\mathbf{C}\mathbf{\Psi}^{\frac{1}{2}}$, the first moment is given by, $\mathbb{E}[\mathbf{W}] = N_R\mathbf{\Psi} + \mathbf{\Psi}^{1/2}\mathbf{M}^H\mathbf{M}\mathbf{\Psi}^{1/2}$. The first moment of any central Wishart matrix $\mathbf{W}_1 \sim \mathcal{CW}(N_R, \mathbf{\Phi})$ with the same degree of freedom N_R, is given by $\mathbb{E}[\mathbf{W}_1] = N_R\mathbf{\Phi}$. When the first moments of the $\mathbf{W}$ and $\mathbf{W}_1$ are equated, we obtain $\mathbf{\Phi} = \mathbf{\Psi} + \frac{1}{N_R}\mathbf{\Psi}^{1/2}\mathbf{M}^H\mathbf{M}\mathbf{\Psi}^{1/2}$. We have now obtained a central Wishart approximation of the non-central Wishart matrix. Hence, $\mathbf{W}$ can be approximated as a central Wishart $\mathcal{CW}(N_R, \mathbf{\Psi} + \frac{1}{N_R}\mathbf{\Psi}^{1/2}\mathbf{M}^H\mathbf{M}\mathbf{\Psi}^{1/2})$. A similar approximation is performed in (Xu *et al.*, 2009). Now that we have a central Wishart matrix, the expressions derived for the case of Rayleigh faded interferers hold, but with

the matrix $\mathbf{\Psi}$ in (6.17) replaced by $\mathbf{\Psi} + \frac{1}{N_R}\mathbf{\Psi}^{1/2}\mathbf{M}^H\mathbf{M}\mathbf{\Psi}^{1/2}$.

Unequal power interferers

All the above analysis holds for a general $\mathbf{\Psi}$. For the case of unequal power interferers, $\mathbf{\Psi}$ is just a diagonal matrix, with the interferer powers occupying the diagonal. Hence, the Laplace transform expressions (6.17) can be used for unequal power Rayleigh-faded and Rician-faded interferers respectively.

6.4.2 Rayleigh faded correlated interferers for $N_I > N_R$

For the case of Rayleigh-faded correlated interferers, for $N_I > N_R$, the covariance matrix of the interference term plus the noise term is given by $\mathbf{R} = \mathbf{C}\mathbf{\Psi}\mathbf{C}^H + \sigma^2\mathbf{I}$. Here, $\mathbf{C} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_{N_R} \otimes \mathbf{I}_{N_I})$. From (Ohlson and Koski, 2012), the distribution of $\mathbf{W} = \mathbf{C}\mathbf{\Psi}\mathbf{C}^H$ is the same as that of $\sum_{i=1}^{N_I} \lambda_i \mathbf{W}_i$, where λ_i are the eigenvalues of $\mathbf{\Psi}$ and $\mathbf{W}_i \sim \mathcal{CW}(1, \mathbf{I}_{N_R})$. Though this method works for $N_R \geq N_I$, we can use the analysis given in the previous subsection for determining the Laplace expansion.

From (Kumar *et al.*, 2015*b*), the sum of central Wishart matrices can be approximated by another central Wishart matrix. In our case, from (Kumar *et al.*, 2015*b*), $\mathbf{W} \approx \mathbf{S}\frac{\sum_{i=1}^{N_I} \lambda_i}{p_s}$, where $\mathbf{S} \sim \mathcal{CW}(p_s, \mathbf{I}_{N_R})$ and $p_s = \left[\frac{(\sum_{i=1}^{N_I} \lambda_i)^2}{\sum_{i=1}^{N_I} \lambda_i^2}\right]$ rounded to the nearest integer. Note that, this has reduced to a case of a Wishart matrix with an identity covariance matrix. Hence, the Laplace transform expression derived for the case of equal power Rayleigh interferers, i.e., expressions corresponding to $L = 0$ given in Section III, can now be used. Also, the determinant simplification that has been derived in the case of equal power Rayleigh faded interferers holds for this case.

6.5 SER Expressions

Using the standard assumption that the contribution of the interference and the noise at the output of optimum combiner, for a fixed η, can be well-approximated to be Gaussian as in (Poor and Verdu, 1997) and (Giorgetti and Chiani, 2005) and references therein, the probability of symbol error for an M-ary square QAM constellation is given by

(Proakis and Salehi, 2007),

$$\mathcal{P}_e \approx k_1 Q(\sqrt{k_2\eta}) - k_3 Q(\sqrt{k_2\eta})^2, \tag{6.20}$$

where $k_1 = 4\big(1 - \frac{1}{\sqrt{M}}\big)$, $k_2 = \frac{3}{M-1}$, $k_3 = \frac{k_1^2}{4}$ and the Q-function is given by $Q(x) = \frac{1}{2\pi}\int_x^\infty e^{-u^2/2}du$. The assumption that the contribution of the interference and the noise at the output of OC, for a fixed η, is Gaussian, is valid even when the number of interferers N_I is small (Giorgetti and Chiani, 2005) and such a system model assumption is made in a number of papers (Chiani *et al.*, 2003*c*), (Chiani *et al.*, 2001), (Chiani *et al.*, 2002) to derive the SER expression. Using the approximation $Q(x) \approx \frac{1}{12}e^{-\frac{1}{2}x^2} + \frac{1}{4}e^{-\frac{2}{3}x^2}$, one can write $\mathcal{P}_e$ as (Chiani *et al.*, 2003*a*), $\mathcal{P}_e \approx \sum_{l=1}^{5} a_l e^{-b_l\eta}$, where a_1=$\frac{k_1}{12}$, a_2=$\frac{k_1}{4}$, a_3=$\frac{-k_3}{144}$, a_4=$\frac{-k_3}{16}$, a_5=$\frac{-k_3}{24}$, b_1=$\frac{k_2}{2}$, b_2=$\frac{2k_2}{3}$, b_3=k_2, b_4=$\frac{4k_2}{3}$ and b_5=$\frac{7k_2}{6}$. The average SER obtained by averaging $\mathcal{P}_e$ over all channel realizations is,

$$SER \approx \mathbb{E}_\eta[\mathcal{P}_e] = \mathbb{E}_\eta[\sum_{l=1}^{5} a_l e^{-b_l\eta}] = \sum_{l=1}^{5} a_l M_\eta(s)|_{s=-b_l}, \tag{6.21}$$

where $M_\eta(s)$ is the Laplace transform of η and $\mathbb{E}_\eta$ denotes expectation over SINR η. Now, the SER approximations can be directly obtained by substituting the Laplace transform expressions in the above equation. Wherever the Laplace transform is simplified to circumvent the determinant expansion, we get simplified SER expressions. For example, for an interference-limited scenario ($N_I > N_R$ and $\sigma^2 = 0$), by substituting $n_1 = N_R$ and $n_2 = N_I$, the SER becomes,

$$SER \approx \sum_{l=1}^{5} a_l \frac{(-1)^{N_R+1} N_I!}{(N_I - N_R)! \prod_{i=1}^{N_R}(N_R - i)!} \sum_{i=1}^{N_R} (-1)^{i+1} A(i) \frac{\prod_{j=1}^{N_R}(j-1)!}{(N_R - i)!(i-1)!}|_{s=-b_l}. \tag{6.22}$$

Ours is the first work to obtain SER expression in an interference-limited scenario, for Rayleigh faded interferers in a closed form. All existing works, so far, require an explicit evaluation of the determinant. Further, the expression derived also gives an approximation of SER, for $N_I > N_R$ for very low noise values $\sigma^2 \approx 0$. If we substitute $n_1 = N_I$ and $n_2 = N_R$ in (6.9) and ignore the σ^2 term inside the determinant, we also

obtain an approximation for the SER as,

$$SER \approx \sum_{l=1}^{5} a_l \frac{(-1)^{N_R+1}(\sigma^2/E_I)^{(N_R-N_I)} N_R!}{(N_R-N_I)! \prod_{i=1}^{N_I}(N_I-i)!} \sum_{i=1}^{N_I} (-1)^{i+1} A(i) \frac{\prod_{j=1}^{N_I}(j-1)!}{(N_I-i)!(i-1)!} |_{s=-b_l}. \tag{6.23}$$

Note that the dependence of c term on σ^2 is not present for $N_I > N_R$. On the other hand, the σ^2 term exists in c term for $N_R > N_I$. Note that SER expressions obtained for Rayleigh interferers in (McKay *et al.*, 2009) involve not only an explicit evaluation of determinants but also numerical integration, while results here require neither. For the case of Rayleigh/Rician faded correlated interferers and $N_R > N_I$, we can substitute (6.17) in (6.21) to get the SER approximations. For this case also, we have a Laplace expansion in (6.19) that doesn't involve determinant evaluation. This approximation works very well when σ^2 is actually small or when interferer powers are large compared to σ^2. This can be further substituted in (6.21), to obtain the approximate expression for SER as,

$$\begin{aligned} SER \approx & \sum_{l=1}^{5} \frac{a_l (-1)^{N_R} (\sigma^2)^{(N_R-N_I)} (-1)^{\frac{1}{2}N_I(N_I-1)} \det(\mathbf{\Psi})^{-N_R}}{\prod_{i<j}^{N_I}(\frac{1}{r_i}-\frac{1}{r_j}) \prod_{k=1}^{N_I}(N_R-k)!} \\ & \times \sum_{i=1}^{N_I} (-1)^{i+1} B(i)_{s=-b_l} r_i^{-N_R+N_I-2} \det\left(V^i(\mathbf{r})\right). \end{aligned} \tag{6.24}$$

In case of correlated or unequal power Rayleigh faded interferers and $N_I > N_R$, we have discussed an approximate expression for Laplace transform in Section III.B, which can be used to determine the SER expressions. In case, of correlated or unequal power Rician faded interferers for $N_I > N_R$, it is mathematically intractable to give an Laplace transform expression and hence derive an SER expression. Nevertheless, the existing SER expressions derived for the case of equal power uncorrelated interferers can be used as an upper bound. If we consider all the interferers to have the same power as that of the maximum-power interferer, our expression gives an upper bound for the actual SER, i.e., our expressions give the worst case SER. Similarly, the expressions for the uncorrelated case gives the worst case SER, i.e., a good upper bound on for the actual SER of correlated interferers. This is because, correlated interferers cause partial interference alignment (Menon and Kalyani, 2015) and hence the receive antennas can cancel the interferers better, leading to a lower SER when compared to the uncorrelated case.

6.6 Outage and Rate approximations

Apart from SER, outage probability and rate are the other performance metrics that are useful in characterizing the performance of any wireless system . In the preceding section, we used the exponential approximation to determine the approximate SER at the output of OC. But, no such straightforward method exists for determining expressions for outage probability and rate. This is so because expressions for the pdf of the SINR or signal to interference ratio (SIR) are mathematically intractable to derive. However, we, in this section, detail a moment-matching method to determine approximate expressions for outage probability and rate. To the best of our knowledge, ours is the first work to obtain even approximate expressions for outage probability and rate. For doing this, we will first determine the exact moments of the SINR.

6.6.1 Moments of Rician-Rician

The lth moment of SINR for Rician faded user and Rician faded interferers is given by,

$$\mu_l^{Ric-Ric} = \frac{d^l}{ds^l} M_\eta(s)|_{s=0} = \alpha_l^{Ric} c \sum_{k=1}^{n_1} (-1)^k \det(Y_k)\, d_l$$

where $c = \frac{e^{-\mathrm{tr}(\mathbf{\Omega})}((n_2-n_1)!)^{-n_1}}{\prod_{i<j}^{n_1}(w_i-w_j)}(-1)^{N_R}(\sigma^2/E_I)^{(N_R-n_1)}$, d_l is given by (E.2), Y_k is the matrix formed by omitting the kth row and 1st column of the matrix $Y_{i,j} = \frac{\sigma^2}{E_I}\, {}_1F_1(n_2 - j+1; n_2-n_1+1; w_k)\Gamma(n_2-j+1) + \Gamma(n_2-j+2){}_1F_1(n_2-j+2; n_2-n_1+1; w_i)$ and $\alpha_l^{Ric} = b^l \sum_{k=0}^{l} \binom{l}{k} \frac{(aN_R/b)^k}{(N_R)_k}$. The derivation is given in Appendix E and is very similar to the one given in (McKay *et al.*, 2009). Though we have derived the moments for the uncorrelated case in the presence of noise, similar moment expressions can be obtained for an interference-limited scenario or for Rayleigh faded correlated/unequal power interferers.

6.6.2 Moment matched approximation

Inspired by the simplicity of the results in (Shah and Haimovich, 1998), we now use the idea of moment matching to approximate the SINR/SIR random variables for our

case too. If we had Rayleigh faded users and interferers as in the case of (Shah and Haimovich, 1998), the SIR would be distributed according to a beta-prime distribution, because the ratio of two gamma random variables follow a beta-prime distribution. We propose to match the first two moments of the SINR/SIR with the moments of the beta-prime distribution. As an example, let us consider the interference-limited scenario, i.e., $N_I > N_R$ and $\sigma^2 = 0$. The lth moment is given by

$$\begin{aligned}\mu_l = &\frac{e^{-\mathrm{tr}(\mathbf{\Omega})}((N_I - N_R)!)^{-N_R}}{\prod_{i<j}^{N_R}(w_i - w_j)}(-1)^{N_R}\sum_{k=1}^{N_R}(-1)^{k+1} \\ &\times (-1)b^l l!(\frac{E_D}{E_I})^l \sum_{m=0}^{l}\binom{l}{m}\frac{(\frac{aN_R}{b})^m\Gamma(N_I - N_R - l + 1)}{(N_R)_m} \\ &\times {}_1F_1(N_I - N_R - l + 1, N_I - N_R + 1, w_k)\det(Y_k)\,, \end{aligned} \tag{6.25}$$

where $Y_{i,j} = \Gamma(N_I - j + 2){}_1F_1(N_I - j + 2; N_I - N_R + 1; w_i)$ and Y_k is the matrix Y with k^{th} row and first column removed. The first two moments μ_1 and μ_2 of the SIR can be matched with the first two moments $\frac{\alpha}{\beta-1}$ and $\frac{\alpha(\alpha+1)}{(\beta-1)(\beta-2)}$, respectively, of a beta prime distribution with parameters α, β. In other words, $\frac{\alpha}{\beta-1} = \mu_1$ and $\frac{\alpha(\alpha+1)}{(\beta-1)(\beta-2)} = \mu_2$. This implies that

$$\beta = \frac{\mu_1^2 - \mu_1 - 2\mu_2}{\mu_1^2 - \mu_2} \tag{6.26}$$

and

$$\alpha = \mu_1(\beta - 1). \tag{6.27}$$

Since the first and second moments do not involve any infinite summations, the parameters α and β are obtained in closed form. These parameters can be substituted in the cumulative distribution function (CDF) of a beta-prime distributed random variable, say Z, with parameters α and β given by (Cordeiro and Lemonte, 2012, Eq 2)

$$P_0(Z < z) = \frac{(\beta)_\alpha z^\alpha\, {}_2F_1(\alpha + \beta, \alpha, \alpha + 1, -z)}{\Gamma(\alpha + 1)} \tag{6.28}$$

to obtain the closed form outage expression for OC. Here, ${}_2F_1(.)$ denotes the Gauss hypergeometric function (Abramowitz and Stegun, 1972, 15.1.1). Similarly using the pdf of beta-prime distribution given by $f(z) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}z^{\alpha-1}(1+z)^{-\alpha-\beta}$, one can obtain

the approximate rate as

$$
\begin{aligned}
R &\approx \int_0^\infty \log_2(1+x)\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(\alpha+x)^{-\alpha-\beta}dx \\
&= \frac{\psi(\alpha+\beta)-\psi(\beta)}{ln2},
\end{aligned}
\tag{6.29}
$$

where $\psi(.)$ is the di-gamma function (Abramowitz and Stegun, 1972, 6.3.1). The second equality is obtained by using the integral identity (Gradshteyn and Ryzhik, 2007, 4.292.14).

Here, we have considered the case of all interferers to be Rician faded. We can even consider the case of a mix of Rayleigh faded and Rician faded interferers, especially single rank non-centrality matrix $\mathbf{M}\mathbf{M}^H$ or the case of correlated interferers and do a similar moment matching to obtain the approximate outage probability and rate expressions.

6.6.3 Analysis using stochastic ordering

Now, we can study the impact of the Rician parameters on the approximate expressions using stochastic ordering. According to (Shaked and Shanthikumar, 2007, Theorem 1.A.12), for two random variables X and Y with pdf f and g, respectively, if $\mathcal{S}^-(g-f)=1$ and the sign sequence is $-,+$, then $X \leq_{st} Y$ or in other words $P(X \leq x) \geq P(Y \leq y)$, i.e., outage probability of X is greater than that of Y. Here $\mathcal{S}^-$ denotes the number of sign changes and $\leq_{st}$ denotes stochastic ordering (Shaked and Shanthikumar, 2007, 1.A.1). In case we assume f and g to be beta-prime with parameters $(\alpha, \beta+\delta)$ and (α, β), we can easily prove that the sign change from $-$ to $+$ occurs at $z = \left(\frac{(\beta+\delta)_\alpha}{(\beta)_\alpha}\right)^{\frac{1}{\delta}} - 1$, where $(x)_y$ denotes the Pochhammer symbol (Abramowitz and Stegun, 1972, 13.1.2). This implies that, P_0 given by (6.28), evaluated using parameter $\beta+\delta$ is greater than P_0 evaluated using the parameter β. Similarly, we can also prove P_0 evaluated using the parameter $\alpha+\delta$ is lesser than P_0 evaluated using the parameter α. This method is also adopted in (Srinivasan and Kalyani, 2018). Since, it is intractable to directly analyze the approximate SER expressions, we'll introduce a Laplace ordering result to connect the variations in outage probability with variations in SER. Let X and Y be

two non-negative random variables such that

$$\mathbb{E}[exp(sX)] \geq \mathbb{E}[exp(sY)], \quad \forall s < 0. \tag{6.30}$$

Then X is said to be smaller than Y in the Laplace transform order denoted by $X \leq_{Lt} Y$. According to (Shaked and Shanthikumar, 2007, Theorem 5.A.6), $X \leq_{St} Y$, then $X \leq_{Lt} Y$. In other words, $P(X \leq x) \geq P(Y \leq y)$ implies $E[exp(sX)] \geq E[exp(sY)]$. Recall that in the probability of error is given by (6.20), if we neglect the second term, which is usually small in magnitude, SER, by exponential approximation of Q-function, is given by

$$SER \approx \mathbb{E}_{\eta}[\sum_{l=1}^{2} a_l e^{-b_l \eta}], \tag{6.31}$$

where a_1=$\frac{k_1}{12}$, a_2=$\frac{k_1}{4}$, b_1=$\frac{k_2}{2}$ and b_2=$\frac{2k_2}{3}$. Now if $P(X \leq x) \geq P(Y \leq y)$, then by the preceding Laplace ordering result, SER for X is greater than that for Y. Though, these results actually pertain to the moment-matched η and not the actual η, we can still get some approximate trends in SER based on the variations in outage probability. We also can make the following inferences about variation of outage probability and SER with respect to fading parameters:

- I1) We can observe that an increase in E_I, decreases μ_2 and μ_1, which in turn increases β. Similarly, an increase in E_I decreases α. According to the previous stochastic ordering result, an increase in β and a decrease in α increases the outage probability. By the Laplace ordering result, the SER also increases.

- I2) It is also clear that an increase in Rician parameter of the user κ decreases μ_2, keeping μ_1 constant. This in turn increases α. Also, β increases very negligibly. Though, the stochastic ordering result cannot be used directly if both α and β increases, the increase in β is actually negligible. Hence, the outage probability decreases with increase in α due to an increase in κ. Also, by the Laplace ordering result, the SER decreases.

- I3) Also, an increase in N_R or a decrease in N_I decreases β and increases α. This in turn decreases the outage probability.

6.7 A short extension to $\kappa - \mu$ faded user and Rayleigh faded interferers

Recently there has been a lot of interest in studying the effect of a general fading model such as the $\kappa-\mu$ fading model for a LoS scenario. Moreover, $\kappa-\mu$ fading also includes Rician, Nakagami-m, Rayleigh and one-sided Gaussian as its special cases (Yacoub, 2007). Given, the fairly complicated nature of these general fading distributions and due to unavailability of a matrix model for these distributions, it is mathematically intractable to determine any expressions considering $\kappa - \mu$ fading in the interferers. However, it is possible to consider $\kappa - \mu$ fading in the user and Rayleigh fading in the interferers and determine exact OP and approximate SER expressions. Recall that the SINR for OC is given by (McKay *et al.*, 2009), $\eta = \frac{E_D}{E_I}\mathbf{c}^H\mathbf{R}^{-1}\mathbf{c}$, where $\mathbf{R} = \sum_{i=1}^{N_I}\mathbf{c}_i\mathbf{c}_i^H$, where the user $\mathbf{c}$ is $\kappa - \mu$ faded and the interferers $\mathbf{c}_i$ are Rayleigh faded. We consider only the case of $N_I > N_R$ and neglect noise in the analysis. Note that, if noise is neglected, $\mathbf{R}$ is a complex Wishart matrix, i.e., $\mathbf{R} \sim \mathcal{CW}_{N_R}(N_I, \mathbf{I}_{N_R})$. From (Nagar and Gupta, 2011), it is known that for any $\mathbf{R} \sim \mathcal{CW}_m(n, \mathbf{I}_m)$, such that $n \geq m$ $\mathbf{SRS}^H \sim \mathcal{CW}_m(n, \mathbf{I}_m)$ for any $m \times m$ unitary matrix $\mathbf{S}$. Let $\mathbf{V} = \mathbf{SRS}^H$ and $\mathbf{S}$ be a unitary matrix such that, $\mathbf{S}^H = (\mathbf{S}_1^H, (\mathbf{c}^H\mathbf{c})^{-\frac{1}{2}}\mathbf{c})$ where $\mathbf{S}_1$ is $m \times m-1$ matrix and $\mathbf{c} \in \mathcal{C}^m$. From (Nagar and Gupta, 2011), we obtain $\mathbf{c}^H\mathbf{R}^{-1}\mathbf{c} = (\mathbf{c}^H\mathbf{c})v^{mm}$, where the elements of $\mathbf{V}^{-1} = (v^{jk})$. Letting $\mathbf{V} = \mathbf{LL}^H$, where $\mathbf{L} = (l_{ij})$ is a complex lower triangular matrix with positive diagonal elements, we get, $v^{mm} = l_{mm}^{-2}$. Using this result for the case $m = N_R$,

$$\eta = \frac{E_D}{E_I}\mathbf{c}^H\mathbf{R}^{-1}\mathbf{c} = \frac{E_D}{E_I}(\mathbf{c}^H\mathbf{c})l_{N_RN_R}^{-2} = \frac{E_D}{E_I}\frac{y}{x}, \tag{6.32}$$

where $y = \mathbf{c}^H\mathbf{c}$ and $x^{-1} = l_{N_RN_R}^{-2}$ and are independent.The pdf of $\eta = \frac{(\mathbf{c}^H\mathbf{c})}{l_{N_RN_R}^2}\frac{E_D}{E_I} = \frac{y}{x}\frac{E_D}{E_I}$ is can now be obtained by solving the integral using using (C.1) as,

$$\begin{aligned} f(\eta) =& \frac{\frac{E_I}{E_D}^{N_R\mu}e^{-N_R\kappa\mu}}{\Gamma(N_R\mu)\Gamma(a)}(\mu(1+\kappa))^{N_R\mu}\eta^{N_R\mu-1}\Gamma(N_R\mu+a) \\ &\times (\mu(1+\kappa)\frac{E_I}{E_D}\eta+1)^{-N_R\mu-a}\,{}_1F_1(N_R\mu+a, N_R\mu, \frac{N_R\mu^2\kappa(1+\kappa)\frac{E_I}{E_D}\eta}{\mu(1+\kappa)\frac{E_I}{E_D}\eta+1}). \end{aligned} \tag{6.33}$$

Though there are existing results which derive the CDF of this ratio (Kumar *et al.*, 2015*a*; Ermolova and Tirkkonen, 2014), obtaining the pdf of SIR η by differentiating these CDF expressions is not straightforward. The outage probability for our case can be obtained by substituting $\beta = a$, $\lambda = 1$, $\mu = N_R\mu$, $r^a = \frac{E_I}{E_D}$ and $T = \frac{T}{N_R}$ in (Kumar *et al.*, 2015*a*, Eq.11), as,

$$P(\eta < T) = A\Psi_1(a + N_R\mu, 1; a + 1, N_R\mu; \frac{1}{\mu(1+\kappa)T\frac{E_I}{E_D}}, \frac{N_R\mu^2(1+\kappa)T\frac{E_I}{E_D}}{\mu(1+\kappa)T\frac{E_I}{E_D}}), \tag{6.34}$$

where $A = 1 - e^{-N_R\kappa\mu}\frac{\Gamma(N_R\mu+a)}{\Gamma(a+1)\Gamma(N_R\mu)}\frac{(\mu(1+\kappa)T\frac{E_I}{E_D})^{N_R\mu}}{(1+\mu(1+\kappa)T\frac{E_I}{E_D})^{a+N_R\mu}}$ and Ψ_1 is the confluent Appell function (Kumar *et al.*, 2015*a*).

We apply the following identities from (Abramowitz and Stegun, 1972; Wolfram Research, 2020*a*) sequentially to simplify (6.33): a) ${}_1F_1(p+a, p, z) = e^z\,{}_1F_1(-a, p, -z)$, b) ${}_1F_1(-a, p, -z) = \frac{a!}{(p)_a}L_a^{p-1}(-z)$, c) $L_a^{p-1}(-z) = \frac{\Gamma(a+p)}{a!}\sum_{k=0}^{a}\frac{(-a)_k(-z)^k}{\Gamma(k+p)k!}$ and d) $(-a)_k = \frac{a!}{(a-k)!}k!$ for $p \in \mathbb{R}^+$ and $a \in \mathbb{Z}^+$. Note, in our case $p = N_R\mu$ and $a \in \mathbb{Z}^+$. Finally, using the Taylor series expansion for the exponential term and the identity for Laplace expansion $M_\eta(s) = \int_0^\infty e^{-s\eta}f(\eta)d(\eta)$, we obtain

$$\begin{aligned} M_\eta(s) =& \frac{e^{-N_R\kappa\mu}\Gamma(N_R\mu + a)}{\Gamma(N_R\mu)\Gamma(a)}\sum_{k=0}^{a}\frac{a!(N_R\mu\kappa)^k}{(a-k)!k!(N_R\mu)_k} \\ &\times \int_0^\infty e^{-s\eta}\sum_{l=0}^{\infty}\frac{(\frac{E_I}{E_D}\mu(1+\kappa))^{N_R\mu+k+l}(N_R\mu\kappa)^l}{l!} \\ &\times \eta^{N_R\mu+k+l-1}(1 + \frac{\mu(1+\kappa)E_I\eta}{E_D})^{-N_R\mu-a-l-k}d\eta. \end{aligned}$$

The summation and integration can be interchanged by the direct application of Tonelli's theorem (Tao, 2011), since $f_l(\eta) = (\frac{E_I}{E_D}\mu(1+\kappa))^{N_R\mu+k+l}\frac{(N_R\mu\kappa)^{k+l}}{l!}\eta^{N_R\mu+k+l-1}(\mu(1+\kappa)\frac{E_I}{E_D}\eta + 1)^{-N_R\mu-a-l-k} > 0$. We now use the integration identity $\int_0^\infty e^{-px}x^{q-1}(1+ax)^{-v}dx = a^{-q}\Gamma(q)U(q, q+1-v, p/a)$ from (Gradshteyn and Ryzhik, 2007), to ob-

tain the Laplace expansion of η as,

$$\begin{aligned}
M_\eta(s) =& \frac{e^{-N_R\kappa\mu}\Gamma(N_R\mu+a)}{\Gamma(N_R\mu)\Gamma(a)}\sum_{k=0}^{a}\frac{a!}{(a-k)!k!(N_R\mu)_k}\sum_{l=0}^{\infty}(\frac{E_I}{E_D}\mu(1+\kappa))^b \\
& \frac{(N_R\mu\kappa)^{k+l}}{l!}(\frac{E_I}{E_D}\mu(1+\kappa))^{-b}\Gamma(b)U(b,b+1-b-a,\frac{s}{\mu(1+\kappa)\frac{E_I}{E_D}}) \\
=& \frac{e^{-N_R\kappa\mu}\Gamma(N_R\mu+a)}{\Gamma(N_R\mu)\Gamma(a)}\sum_{k=0}^{a}\frac{a!}{(a-k)!k!(N_R\mu)_k} \\
& \sum_{l=0}^{\infty}\frac{(N_R\mu\kappa)^{k+l}\Gamma(b)}{l!}U(b,1-a,\frac{s}{\mu(1+\kappa)\frac{E_I}{E_D}}). \qquad (6.35)
\end{aligned}$$

where $b = N_R\mu + k + l$. Substituting the expression for the Laplace expansion from (6.35) in (6.21), we obtain the average SER as

$$\begin{aligned}
SER = & \sum_{m=1}^{5} a_m \frac{e^{-N_R\kappa\mu}\Gamma(N_R\mu+a)}{\Gamma(N_R\mu)\Gamma(a)}\sum_{k=0}^{a}\frac{a!}{(a-k)!k!(N_R\mu)_k} \\
& \times\sum_{l=0}^{\infty}\frac{(N_R\mu\kappa)^{k+l}\Gamma(b)}{l!}U\left(b,1-a,\frac{b_m}{\mu(1+\kappa)\frac{E_I}{E_D}}\right), \qquad (6.36)
\end{aligned}$$

where $U(.)$ is the Tricomi Hypergeometric function (Zhang *et al.*, 2014*a*). Since $U(a+1,c+1,x) < \frac{-1}{c}U(a,c,x)$ for $a > 0 > c$ and $x > 0$ from (Baricz *et al.*, 2016), we obtain

$$U(b,1-a,\frac{b_m}{\mu(1+\kappa)\frac{E_I}{E_D}}) < \frac{U(N_R\mu+k,1-a-l,\frac{b_m}{\mu(1+\kappa)\frac{E_I}{E_D}})}{(N_I-N_R)_l}.$$

Also, since $U(m,n-k,x)$ decreases monotonically with k (Zhang *et al.*, 2014*a*), $\sum_{l=0}^{\infty}\frac{(N_R\mu\kappa)^{k+l}}{l!}\Gamma(b)U(b,1-a,\frac{b_m}{\mu(1+\kappa)\frac{E_I}{E_D}}) < \sum_{l=0}^{\infty}\frac{(N_R\mu\kappa)^{k+l}}{l!}\Gamma(b)\frac{U(N_R\mu+k,N_R-N_I,\frac{b_m}{\mu(1+\kappa)\frac{E_I}{E_D}})}{(N_I-N_R)_l}$ which is a converging series. Hence the infinite summation can be truncated to a finite series with arbitrarily low truncation error.

6.8 Numerical Results

The derived SER expressions are verified using Monte-Carlo simulations. The total interference power is denoted as E_I'', from which average interference power per interferer is obtained as $E_I' = E_I''/N_I$. The mean energy of the received signal, E_D, is taken

to be unity without loss of generality. SIR is given by E_D/E_I''. The determinant of $\mathbf{N}$ matrix whose entries are given by (6.10) is determined with the infinite summation truncated to $T_1 = T_2 = 70$. For a SER of 10^{-5}, with $T_1 = 100$ and $T_2 = 100$ terms, the numerical evaluation completes in 40 seconds[2]. Also, for Rayleigh faded interferers, the expressions have only a single infinite series, which takes a maximum of 2 seconds for evaluation. On the other hand, Monte-Carlo simulations take close to 500 seconds. By substituting the determinant of $\mathbf{N}$ matrix in (6.9), the Laplace transform is evaluated for $s = b_l, \forall l = 1\ to\ 5$. These Laplace transform values are substituted in (6.21) and theoretical SER is calculated for values of signal to noise ratio (SNR) in the range 5 dB to 25 dB. For the Monte-Carlo simulation, the deterministic matrix $\mathbf{M}'$ is first obtained with unit magnitude and uniform phase, satisfying the condition tr$(\mathbf{M}'\mathbf{M}'^H) = N_R N_I$. This matrix is fixed during a set of simulations.

For the case of equal power and uncorrelated interferers, a close match between the theoretical and simulated SER is observed in Fig. 6.2(a) and Fig. 6.2(b). The exponential approximation $Q(x) \approx \frac{1}{12}e^{-\frac{1}{2}x^2} + \frac{1}{4}e^{-\frac{2}{3}x^2}$, from (Chiani *et al.*, 2003*a*) provides a very tight upper bound for values of $x > 0.5$ and the bound becomes tighter as x increases. Since, for $N_I > N_R$, the average SINR is much lower when compared to the case $N_R > N_I$, we can observe a small mismatch between the theoretical and the simulated SER in Fig. 6.2(b). Also, the SER approximation plot in Fig. 6.2 is tight beyond 15 dB. For Rayleigh interferers, i.e., $\kappa_I = 0$, the SER approximation computed using (6.23), match with the simulation results at high SNR, for $N_R > N_I$ as seen from Fig. 6.2(a). We can also see from Fig. 6.2 that, when the interference power dominates the noise power, as is the case when $E_I'' = -1$ dB, the SER approximation is tight even at 15 dB SNR.

Similar Monte-Carlo simulations are performed for the case of correlated and/or unequal power interferers. The difference is that, random covariance matrix $\mathbf{R}$ of the interference terms plus the noise term is now calculated using (6.12). Also note that the determinant of $\mathbf{N}$ matrix whose entries are given by (6.18). From Fig. 6.3, we can observe that for correlated Rayleigh faded interferers, SER computed by means of (6.17) and (6.18) matches the simulated values. We can also see that the approximation for $\sigma^2 \approx 0$ computed using (6.24) gives good match with the simulated results for higher values of SNR. We consider exponential correlation between interferers (Menon

[2]in MATLAB R2015b run in an iMac with 2.8GHz with intel i5 core and 8 GB RAM in Sierra OS

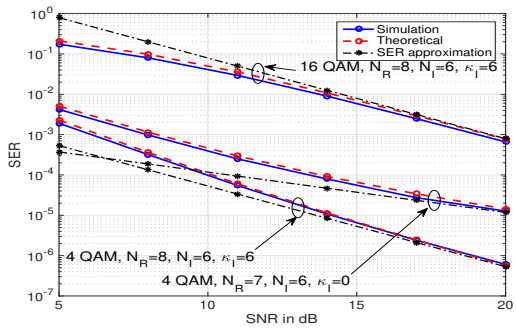
Figure	Inference
6.2(a)	With increase in N_R SER decreases.
6.2(b)	SER saturates with an increase in SNR for $N_I > N_R$
6.3	The SER approximation given by (6.24) is tight as interference power dominates the noise power, i.e., $E_I'' = -1$ dB
6.4(b)	The approximation in Section IV.B is tight if the interferer powers do not vary widely
6.5(a)	Outage probability decreases with an increase in N_R
6.5(b)	Outage probability decreases with a decrease in κ_I

Table 6.1: Inferences from simulations

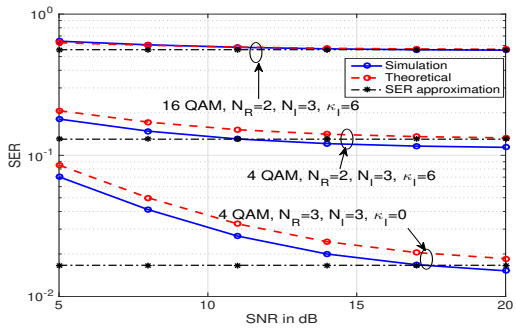
and Kalyani, 2015), i.e., $\mathbf{\Psi}(i,j) = \rho^{|i-j|}$ and $0 \leq \rho \leq 1$, where $\mathbf{\Psi}$ is the interferer covariance matrix. In Fig. 6.4(a), we studied the case of mix of Rayleigh and Rician faded unequal power interferers. Here also, the SER approximation given by (6.24) gives a good match to the simulated SER for high SNR values. For Rayleigh faded interferers with unequal power and $N_I > N_R$, the approximation in Section IV.B gives a fairly good match to the theoretical values as seen from Fig. 6.4(b), as long as the interferer powers do not vary widely.

It is known that OC trades off the effect of noise and interference at the receiver. In the absence of noise it maximizes the average SIR and specializes to a ZF receiver, which is the ideal receiver for interference cancellation. In the presence of noise, OC balances between noise cancellation and interference cancellation. This results in a sub-optimal average SIR. A better SIR translates to a lower SER and hence we observe a lower SER for both 4-QAM and 16 QAM, for $N_I > N_R$ and $\sigma^2 = 0$. This is captured by the fact that SER approximation that assumes $\sigma^2 = 0$ forms a lower bound for the SER values of $N_I > N_R$ with non-zero σ^2. This can be observed in Fig. 6.2(b). Also, note that the bound becomes tighter for large values of SNR. In this regime, OC mimics the performance of ZF. In the case of $N_R > N_I$, the approximation is ad-hoc and computationally less intensive due to the absence of determinant evaluation.

The variation of the outage probability is shown in Fig. 6.5(a) for changes in N_R and N_I. As discussed in I3, the outage probability decreases with an increase in N_R or a decrease in N_I. Also, as discussed in I1, the outage probability decreases with a decrease in E_I. In Fig. 6.5(b), κ_I, the Rician parameter of the interferers is varied. We can observe that the outage probability decreases with a decrease in κ_I. Finally, the Monte-Carlo simulations and approximate expressions for rate are shown in Fig. 6.6.



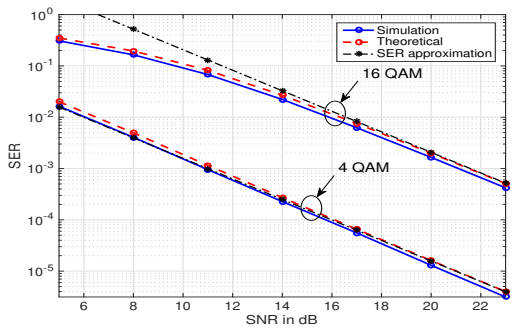
(a) $E_I'' = -1dB, \kappa_s = 3$



(b) $E_I'' = -1dB, \kappa_s = 3$

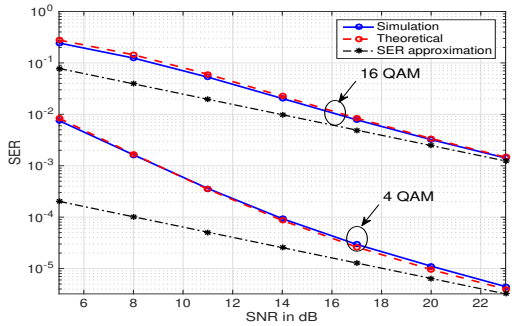
Figure 6.2: SER for $N_R \geq N_I$ and $N_R \leq N_I$

We can observe that the rate increases with an increase in N_R or SNR or a decrease in N_I. Finally, from Fig. 6.7 and 6.8, we can observe that the OP expressions (6.34) and SER expressions (6.36) for $\kappa - \mu$ fading users match the simulations. Inferences are provided in Table 6.1 for readability.

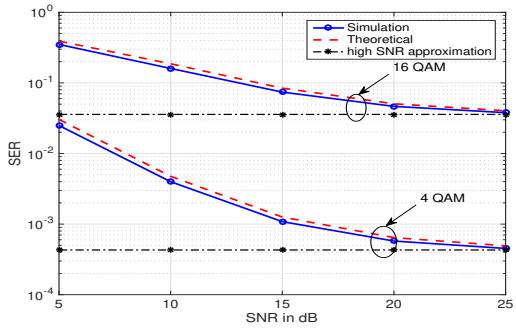


(a) correlated Rayleigh interferers $\kappa_s = 3, N_R = 4, N_I = 2, E_I'' = -1$ dB, $\rho = 0.6$

Figure 6.3: SER for correlated Rayleigh interferers

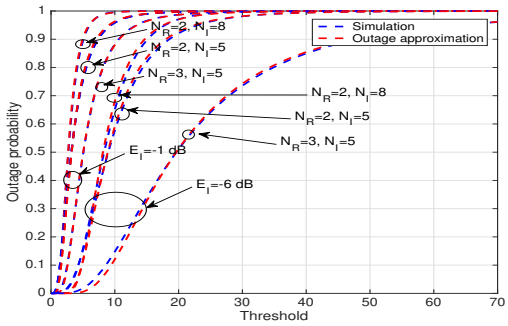


(a) mixture of two Rician and two Rayleigh interferers $\kappa_s = 2, \kappa_i = 5, N_R = 5, N_I = 4, E_I'' = -1, -1, 5, -2, -2.5$ dB

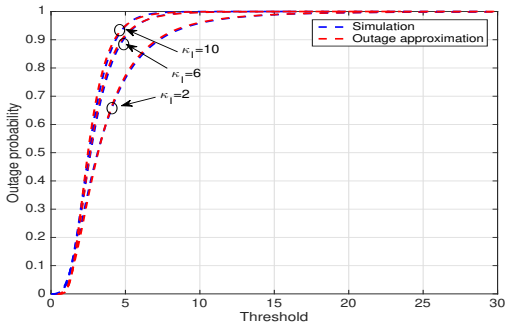


(b) unequal power Rayleigh interferers $\kappa_s = 2, N_R = 3, N_I = 5, E_I'' = -6, -9, -10, -10, -10$ dB

Figure 6.4: SER for unequal power Rayleigh/Rician interferers



(a) $E_I'' = -1$ dB, $\kappa_s = 6, \kappa_I = 3$



(b) $N_R = 2, N_I = 5, E_I'' = -1$ dB, $\kappa_s = 6$

Figure 6.5: Outage probability for Rician interferers

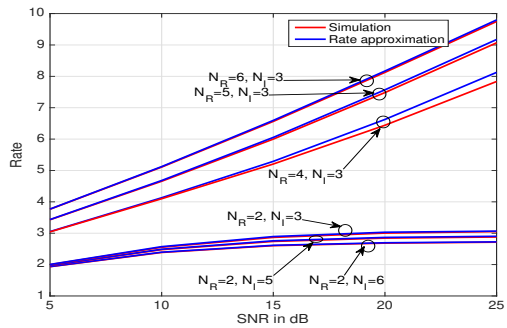


Figure 6.6: Rate vs SNR, $\kappa_s = 8, \kappa_I = 3, E_I = -4$ dB

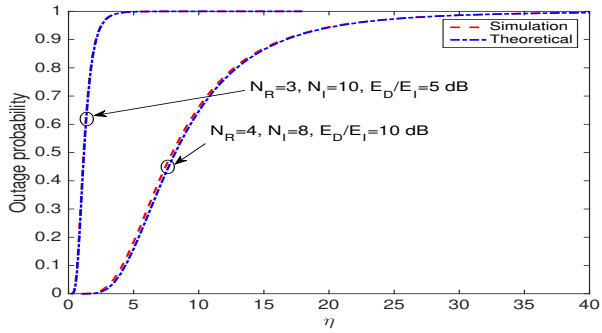


Figure 6.7: Outage probability for $\kappa - \mu$ faded users $\kappa = 1.5, \mu = 4.0$

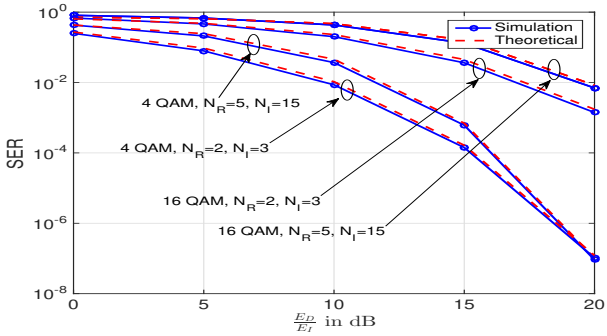


Figure 6.8: SER vs $\frac{E_D}{E_I}$ for $\kappa = 4.0, \mu = 5.3$

6.9 Applications

Device-to-device (D2D) communication enables a pair of closely located mobile users to establish a direct link for their user-plane traffic without going through the entire network infrastructure, while reusing the spectrum allocated for traditional cellular communication. Due to the existence of a LoS component, it is appropriate to model D2D links by Rician fading. In fact, in (Wang *et al.*, 2018) and references therein, several performance metrics were analyzed in Rician fading D2D systems. In D2D underlying cellular networks, strong interference may occur between cellular and D2D links sharing the same spectrum. To mitigate the interference, many resource allocation schemes

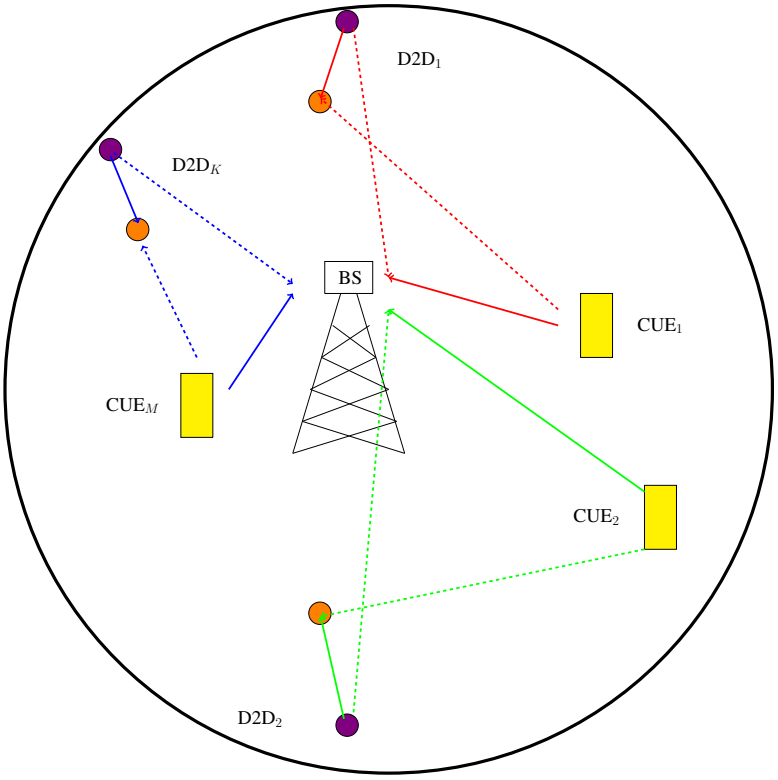


Figure 6.9: This figure shows a general D2D underlay cellular network, with M CUEs, K D2D pairs. Different colours are used for different spectrum. The dashed lines indicate the interferer channels. BS has only the statistical information on the fast fading parameters of the D2Ds

have been proposed in (Wang *et al.*, 2017*b*, 2018; Liang *et al.*, 2017). Similar to (Wang *et al.*, 2018), we consider a D2D-enabled communications network, where M single-antenna cellular users (CUE) perform high-capacity uplink cellular communications with the base station (BS) with N_R antennas. In (Wang *et al.*, 2017*b*; Liang *et al.*, 2017; Wang *et al.*, 2018), only one antenna at BS is considered. But it makes practical sense to consider the BS to be equipped with more than one antenna. Also, consider K pairs of single-antenna users doing local data exchange in the form of D2D communications. These are denoted as DUEs or D2D users. The K DUEs can reuse the M sub-channels in the underlay mode. Since the BS is equipped with more than one antenna, the interference at the BS can be cancelled by means of OC. The entire schematic is shown in Fig. 6.9.

Let $\mathbf{h}_{m,B}$ denote the $N_R \times 1$ Rician channel (with Rician factor $\kappa_{m,B}$) from the mth CUE to the BS. h_k denotes the channel fast fading coefficient between the kth DUE pair. Due to the presence of strong LoS component between the D2D receiver (Rx) and transmitter (Tx), h_k is assumed to be Rician with Rician factor κ as in (Wang *et al.*, 2018). $\mathbf{h}_{k,B}$ denote the $N_R \times 1$ interferer channel fast fading coefficients from the kth

DUE Tx to BS. $h_{m,k}$ denotes the channel fast fading coefficient (with Rician factor $\kappa_{m,k}$) from the mth CUE to the the Rx of the kth DUE. This alone is assumed to be Rayleigh faded as in (Wang *et al.*, 2018). Also, let $s_{m,B}$, s_k, $s_{k,B}$ and $s_{m,k}$ denote the corresponding slow fading coefficients. Since we consider a high-speed scenario where fast-fadings are difficult to estimate and only slow-fadings are available, we can assume the slow-fadings to be known constants and the fast-fadings to be random variables. Denote the CUE set as $\mathcal{M} = \{1, ..., M\}$ and the DUE set as $\mathcal{K} = \{1, ..., K\}$. The SINR at the output of OC at the BS is

$$\gamma_m^c = P_m^c s_{m,B} \mathbf{h}_{m,B}^H (\sum_{k \in \mathcal{K}} \rho_{m,k} P_k^d s_{k,B} \mathbf{h}_{k,B} \mathbf{h}_{k,B}^H + \sigma^2 \mathbf{I})^{-1} \mathbf{h}_{m,B}$$

and the SINR at the the kth D2D Rx is

$$\gamma_k^d = \frac{P_k^d s_k |h_k|^2}{\sum_{m \in \mathcal{M}} \rho_{m,k} P_m^c s_{m,k} |h_{m,k}|^2 + \sigma^2}, \tag{6.37}$$

where P_m^c and P_k^d denote transmit powers of the mth CUE and the kth DUE, respectively, σ^2 is the noise power and $\rho_{m,k}$ is the spectrum allocation indicator with $\rho_{m,k} = 1$ indicating the kth DUE reuses the spectrum of the mth CUE and $\rho_{m,k} = 0$ otherwise. The ergodic capacity of the mth CUE is given by $C_m = \mathbb{E}[\log_2(1 + \gamma_m^c)]$. We maximize the sum ergodic capacity of M CUEs while guaranteeing the minimum reliability for each DUE. In addition, we set a minimum capacity requirement for each CUE. The reliability of DUEs is guaranteed through controlling the probability of outage events, where its received SINR γ_k^d is below a predetermined threshold γ_0^d . Hence the radio resource allocation problem in vehicular networks is formulated as (Wang *et al.*, 2017*b*, 2018)

$$\begin{aligned}
&\max_{\rho_{m,k} P_m^c, P_k^d} \sum_{m \in \mathcal{M}} \mathbb{E}[\log_2(1 + \gamma_m^c)], \\
&\text{s.t. } \mathbb{E}[\log_2(1 + \gamma_m^c)] \geq r_0^c, \ \forall m \in \mathcal{M}, \\
&Pr\{\gamma_k^d \leq \gamma_0^d\} \leq p_0, \forall k \in \mathcal{K}, \\
&0 \leq P_m^c \leq P_{max}^c, \forall m \in \mathcal{M} \text{ and } 0 \leq P_k^d \leq P_{max}^d, \forall k \in \mathcal{K}, \\
&\sum_{\forall m \in \mathcal{M}} \rho_{m,k} \leq 1, \ \rho_{m,k} \in \{0, 1\}, \ \forall m \in \mathcal{M}, \\
&\sum_{\forall k \in \mathcal{K}} \rho_{m,k} \leq 1, \ \rho_{m,k} \in \{0, 1\}, \ \forall k \in \mathcal{K},
\end{aligned} \tag{6.38}$$

where r^c is the minimum capacity requirement of the rate intensive CUEs and γ_0^d is the minimum SINR needed by the DUEs to establish a reliable link. p_0 is the tolerable outage probability of the D2D links. P_{max}^c and P_{max}^d are the maximum transmit powers of the CUE and DUE, respectively. The last two constraints mathematically model the assumption that the spectrum of one CUE can only be shared with a single DUE and one DUE is only allowed to access the spectrum of a single CUE.

The power allocation problem can be written for kth DUE sharing the band of the mth CUE as

$$\begin{aligned}
&\max_{P_m^c, P_k^d} \mathbb{E}[\log_2(1+\gamma_m^c)], \\
&\text{s.t. } Pr\{\gamma_k^d \leq \gamma_0^d\} \leq p_0, \\
&0 \leq P_m^c \leq P_{max}^c \text{ and } 0 \leq P_k^d \leq P_{max}^d.
\end{aligned} \tag{6.39}$$

Since the D2D links are Rician and the link between the the CUE and Rx of the D2D link is Rayleigh as in (Wang *et al.*, 2018), the reliability constraint of the kth DUE, $Pr\{\gamma_k^d \leq \gamma_0^d\} \leq p_0$, is given by (Wang *et al.*, 2018, Eq 6) $P(\gamma_k^d \leq \gamma_0^d) = \frac{\theta}{\theta+1} e^{-\frac{\kappa}{\theta+1}} = f(\theta)$, where $\theta = \frac{\gamma_0^d P_m^c s_{m,k}}{P_k^d s_k}$. From (Wang *et al.*, 2018, Eq.9), we can obtain an upper bound of P_m^c as $P_m^c \leq \frac{s_k \theta_0}{s_{m,k}\gamma_0^d} P_k^d$, where $\theta_0 = f^{-1}(p_0)$ and $\beta = \frac{s_k \theta_0}{sm,k\gamma_0^d}$. Using the same reasoning as in (Wang *et al.*, 2018), the optimal power allocation is obtained as

$$(P_m^{c^*}, P_k^{d^*}) = \begin{cases} (P_{max}^c, \frac{P_{max}^c}{\beta}), \text{ if } \beta > \frac{P_{max}^c}{P_{max}^d}, \\ (P_{max}^d \beta, P_{max}^d), \text{ if } \beta < \frac{P_{max}^c}{P_{max}^d}. \end{cases} \tag{6.40}$$

With the optimal power allocations $(P_m^{c^*}, P_k^{d^*})$, under each given channel allocation, we aim to maximize the cellular rates by searching over all possible channel allocation schemes. We first determine the ergodic capacity of mth CUE when it shares spectrum with kth DUE is $C_{m,k}$ for a power allocation $(P_m^{c^*}, P_k^{d^*})$. Since, OC is used at the BS, the rate $C_{m,k}(P_m^{c^*}, P_k^{d^*}) = \mathbb{E}[\log_2(1+\gamma_m^c{}^*)]$, where $\gamma_m^c{}^*$ is the SINR at the output of OC for the power allocation $(P_m^{c^*}, P_k^{d^*})$. This rate can be easily determined by means of the rate expressions (6.29). Without the rate expressions, it wouldn't have been possible to consider an interference cancellation scheme like OC at the BS. To guarantee the minimum transmission rates for CUEs and DUEs, we have to exclude those CUE-DUE pairs unable to meet the rate requirements even with the optimal transmitting

powers. This condition is interpreted as $\rho_{m,k} = 0$ if the pairing of CUE m with DUE k violates the rate constraints. If for one such pair (m, k), $C_{m,k}(P_m^{c^*}, P_k^{d^*})$ is lesser than the minimum rate r_0^c, then the rate is replaced by $-\infty$. The exact procedure is as discussed in (Liang *et al.*, 2017; Wang *et al.*, 2017*b*). Hence,

$$C_{m,k}^* = \begin{cases} C_{m,k}(P_m^{c^*}, P_k^{d^*}), \text{ if } C_{m,k}(P_m^{c^*}, P_k^{d^*}) \geq r_0^c, \\ -\infty, \text{ otherwise} \end{cases} \tag{6.41}$$

The spectrum allocation problem becomes

$$\begin{aligned} &\max_{\rho_{m,k}} \sum_{m \in \mathcal{M}} C_{m,k}^*, \\ &\sum_{\forall m \in \mathcal{M}} \rho_{m,k} \leq 1, \ \rho_{m,k} \in \{0, 1\}, \ \forall m \in \mathcal{M}, \\ &\sum_{\forall k \in \mathcal{K}} \rho_{m,k} \leq 1, \ \rho_{m,k} \in \{0, 1\}, \ \forall k \in \mathcal{K}. \end{aligned}$$

This is a maximum weight bipartite matching problem and can be solved by the Hungarian method as in (Liang *et al.*, 2017; Wang *et al.*, 2017*b*, 2018). Simulation results are presented to evaluate the performance of joint power and channel allocation scheme. The simulation parameters are as follows: $M = 40$, $N = 20$, $r_0^c = 0.1$ bps/Hz, $\gamma_0 = 5$dB, $p_0 = 0.1$, $P_{max}^c = P_{max}^d = 25$ dBm and $\kappa_{m,B} = \kappa_{k,B} = 0.1$. BS antenna gain is 8dBi and the user antenna gain is 3 dBi. The noise power is $\sigma^2 = -114$ dBm. The loss exponents of the links to the BS and user antenna are 3.76 and 2.27, respectively and the shadow fading standard deviations are 8 dB and 3dB, respectively. It can

κ	Sum rate of CUE for $N_R = 3$	Sum rate of CUE for $N_R = 2$
0	815.6465	780.0624
0.5	816.7338	781.1835
1	818.1748	782.7159
2	819.3933	783.8709
3	819.4748	783.9478

Table 6.2: Sum rate of CUE in bps/Hz

be observed from the Table 6.2 that the sum rate in bps/Hz increases with increase in N_R and the Rician factor κ between the D2D pairs. Also, the utility of our expressions lies in the fact that it can be used in any application, where the user and interferers undergo Rician fading and OC is employed such as this application and Vehicular Ad-hoc

Network (VANET).

6.10 Conclusions

Approximate SER, outage probability and rate expressions have been derived for OC for the case of Rician faded users and a) Rician faded interferers, b) mixture of Rician and Rayleigh faded interferers and c) Rayleigh faded interferers when the interferers are correlated/uncorrelated and have equal or unequal powers. SER is also derived for an interference-limited scenario and the expressions obtained are significantly simpler than the existing expressions. The Monte-Carlo simulation closely match the derived results. An application where our results have significant utility is also discussed.

CHAPTER 7

Approximate Random Matrix Models for Generalized Fading MIMO Channels

7.1 Introduction

The need for high data rates has been one of the driving factors for the evolution of the wireless systems from Single Input Single Output (SISO) systems to Multiple Input Multiple Output (MIMO) systems. MIMO systems are being used increasingly in modern wireless standards and it is imperative to study the channel capacity and other quality of service (QoS) metrics of such systems. But metrics likes capacity depend not only on channel fading statistics and but also on whether the statistics are known at the receiver and the transmitter. To capture the fading statistics, channel gain is characterized by a single random variable (RV) in SISO systems. But for MIMO systems, the channel is in the form of a matrix, hence the characterization of random matrices plays an indispensable role in studying MIMO channel metrics.

The MIMO system is modelled by an $N_R \times N_T$ channel gain matrix $\mathbf{H}$, where N_R is the number of receive antennas and N_T is the number of transmit antennas. Various performance metrics such as capacity, rate, etc. require the eigenvalue statistics of the Gram matrix $\mathbf{H}\mathbf{H}^H$ (or $\mathbf{H}\mathbf{H}^H$). When the elements of $\mathbf{H}$ are i.i.d. circular symmetric complex Gaussian with zero or non-zero mean, i.e., when Rayleigh or Rician faded MIMO channels are considered, the Grammian $\mathbf{H}\mathbf{H}^H$ can be characterized by Wishart matrices - central or non-central respectively (James, 1964). These random matrix models have been used widely for deriving capacity expressions in the case of Rayleigh faded MIMO channels (Telatar, 1999; Chiani *et al.*, 2003*b*; Goldsmith *et al.*, 2003) and also Rician faded MIMO channels (Kang and Alouini, 2006; McKay and Collings, 2005). Recently, there has been focus on generalized fading models namely $\kappa - \mu$ and $\eta - \mu$ models, which were introduced in the seminal work (Yacoub, 2007).

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These distributions model the small scale variations in the fading channel in the line of sight and non-line of sight conditions respectively. Further, these generalized fading distributions include the well-studied Rayleigh, Rician, Nakagami, one-sided Gaussian distributions as special cases. $\kappa - \mu$ and $\eta - \mu$ fading distributions have been widely used in capacity and outage probability analysis for SISO and MISO systems.

Average channel capacity of single branch $\kappa - \mu$ and $\eta - \mu$ faded receivers is studied by the authors of (Costa and Yacoub, 2007). The outage probability (OP), coverage probability and rate of these generalized fading channels are analyzed by works such as (Morales-Jimenez and Paris, 2010; Paris, 2013; Ermolova and Tirkkonen, 2014; Kumar and Kalyani, 2015; Kumar *et al.*, 2015*a*) and the references therein. OP of MRC in $\kappa-\mu$ fading channels in the presence of co-channel interference (CCI) is studied by (Morales-Jimenez *et al.*, 2012) and for $\eta - \mu$ fading channels by (Srinivasan and Kalyani, 2020). For example, OP expression for $\eta - \mu$ signal of interest (SOI) and Rayleigh faded interferers is derived in terms of confluent Lauricella function in (Morales-Jimenez and Paris, 2010). OP expressions, when SOI experiences $\eta-\mu$ or $\kappa-\mu$ fading and the interfering signals are subject to $\eta - \mu$ fading, have been derived in (Paris, 2013). This was further extended to cases where CCI can be either $\eta - \mu$ or $\kappa - \mu$ fading in (Ermolova and Tirkkonen, 2014). Expressions for coverage probability and rate are derived in terms of Lauricella's function of the fourth kind in (Kumar and Kalyani, 2015), when SOI and CCI experience $\kappa - \mu$ and $\eta - \mu$ fading respectively. Approximate OP and rate expressions are derived in terms of the Appell function in (Kumar *et al.*, 2015*a*), when the user channel and the interferers experience $\kappa - \mu$ and $\eta - \mu$ fading respectively. OP analysis of $\kappa - \mu$ fading is performed for optimum combining in (Srinivasan and Kalyani, 2019*b*). Secrecy capacity analysis is carried out in (Bhargav *et al.*, 2016; Srinivasan and Kalyani, 2018) and effective throughput in MISO systems is determined in (Zhang *et al.*, 2014*a*, 2013). Analysis of decode and forward relay system for generalized fading models is performed in (Li and Chen, 2010; Fikadu *et al.*, 2016; Kumar and Dhaka, 2016). Asymptotic analysis of generalized fading channels using extreme value theory is performed in (Subhash *et al.*, 2019).

On the other hand, the capacity of MIMO systems for these generalized fading channels has been less analyzed for want of a random matrix model that characterizes the channel matrix. Nevertheless, some random matrix models have been developed for Nakagami and Rician-shadowed fading channels. A random matrix model has been

developed for Nakagami-q fading in (Kumar and Pandey, 2010) and the pdf of eigenvalues of the Gram $\mathbf{HH}^H$ is obtained in terms of a Pfaffian. In (Gholizadeh *et al.*, 2015), the ergodic capacity of MIMO correlated Nakagami-m fading channel has been derived using the concept of a copula. But the work presents an analysis only for 2×2 MIMO channel and determining the capacity of MIMO channels with a larger number of receive and transmit antennas using this method is cumbersome. Recently a MIMO capacity upper bound was derived in (Vergara and Barbin, 2012) for the $\kappa - \mu$ and $\eta - \mu$ fading channels. In (Alfano *et al.*, 2010; Moreno-Pozas and Martos-Naya, 2016) a MIMO model has been developed for Rician-shadowed fading as a unification model for MIMO-Rayleigh and MIMO-Rician fading models. But, to the best of our knowledge, no work has presented even an approximate matrix model for the $\kappa - \mu$ and $\eta - \mu$ fading channels.

Given the complicated pdf structure of complex variable $\kappa - \mu$ and $\eta - \mu$ fading distributions (Dias *et al.*, 2008; da Costa and Yacoub, 2007), it is challenging to develop the matrix distribution and the eigenvalue statistics for $\mathbf{HH}^H$, even when the elements of $\mathbf{H}$ are assumed to be i.i.d. complex $\kappa - \mu$ or $\eta - \mu$ random variables. Hence, in this chapter, we develop an approximate matrix model for $\mathbf{HH}^H$ (or $\mathbf{H}^H\mathbf{H}$) in terms of a Wishart distribution, which is a very well-studied matrix distribution (James, 1964). There is some prior literature which focuses on approximation of random matrices with Wishart matrices. Approximating any matrix distributions by central Wishart by means of Taylor expansion is studied in (Kollo and von Rosen, 1995), but the approximation requires the knowledge of not only one or more cumulants and moments of the random matrix but also the derivatives of central Wishart matrix. Finding the cumulants of the random matrix requires the knowledge of the characteristic function, which is not available in our case. Also, the approximation of non-central Wishart matrix by a central Wishart by means of Laguerre polynomial expansion is given in (Tan and Gupta, 1982) and by means of the moment generating functions in (H.S. and J.J.J., 1962). But, the Laguerre polynomial expansion and the moment generating functions are difficult to derive for matrix variate $\kappa - \mu$ and $\eta - \mu$ fading distributions.

7.1.1 Our contributions

Our contribution in this chapter, are as follows:

a) For both complex $\kappa - \mu$ and $\eta - \mu$ distribution[1], we propose a Wishart distributed approximation of $\mathbf{H}\mathbf{H}^H$, such that the approximation has the least KL divergence from the original matrix distribution.

b) For complex $\kappa - \mu$, we also propose another simple Wishart approximation of $\mathbf{H}\mathbf{H}^H$, such that the approximation has its first moment matched with the original matrix distribution of $\mathbf{H}\mathbf{H}^H$ and the degree of freedom is constrained to be the number of columns of the matrix $\mathbf{H}$. This method is similar to the approximation of Rician channels in (H.S. and J.J.J., 1962).

The proposed approximation is discussed in Section 7.3. Though the approximations are derived for $\kappa - \mu$ and $\eta - \mu$ fading, the same methodology can be extended to other generalized fading models like $\kappa - \mu$ shadowed fading (Moreno-Pozas *et al.*, 2016; Cotton, 2015), $\alpha - \kappa - \mu$ or $\alpha - \eta - \mu$ fading (Fraidenraich and Yacoub, 2006), or $\alpha - \kappa - \mu$ shadowed fading (Ramirez-Espinosa *et al.*, 2019), if the joint complex envelope-phase distributions are available. Even if the exact MIMO matrix models are determined in future, the Wishart approximations derived in the chapter will still remain significantly simpler. In Section 7.4, the utility of the approximation is shown via three applications namely:

a) Determining zero-forcing (ZF) ergodic rate expressions of single-cell uplink MIMO systems in $\kappa - \mu$ or $\eta - \mu$ fading channels for the scenario in which only low-resolution ADCs are deployed at the receiver RF chains.

b) Determining the capacity of MIMO systems with i.i.d. $\kappa - \mu$ or $\eta - \mu$ channel gains.

c) Determining OP expressions of MIMO-maximum ratio combining (MIMO-MRC).

To the best of our knowledge, ours is the first work to derive even an approximate capacity expression in the presence of $\kappa - \mu/\eta - \mu$ MIMO channels. Also, no prior work has given OP expressions for a receiver diversity system employing MIMO-MRC with $\kappa - \mu$ or $\eta - \mu$ fading channels. To show the utility of our expressions in a recent

[1]We use the distribution of the in-phase and quadrature components

5G technology, we have derived rate expressions for ZF receiver in massive MIMO scenario. The derived approximations are compared with Monte-Carlo simulations and a close match is found between the theoretical results and simulation results. While we have only shown the utility of the approximation in three applications namely capacity computation and outage probability computation, the approximation can be used in any application which deals with random $\kappa - \mu/\eta - \mu$ matrix models.

7.2 Generalized fading models

In this section, we revisit the complex variable pdfs of commonly used generalized fading models.

7.2.1 $\eta - \mu$ model

The $\eta - \mu$ is a fading distribution that represents small scale fading effects in non-line of sight condition. The elements h_{ij} of $\mathbf{H}$ are independent and identical $\eta - \mu$ distributed random variables with density (da Costa and Yacoub, 2007),

$$f_{xy}(x_{ij}, y_{ij}) = \frac{\mu^{2\mu}|x_{ij}y_{ij}|^{2\mu-1}}{\Omega_X^{\mu}\Omega_Y^{\mu}\Gamma^2(\mu)} exp(-\mu\Big(\frac{x_{ij}^2}{\Omega_X} + \frac{y_{ij}^2}{\Omega_Y}\Big)) \tag{7.1}$$

where Ω is the power parameter given by $\Omega = 2\sigma^2\mu$, σ^2 is the power of the Gaussian variable in each cluster, μ is the number of clusters. Note, $\Omega_X = (1-\eta)\Omega/2$, $\Omega_Y = (1+\eta)\Omega/2$, $-1 \leq \eta \leq 1$.

7.2.2 $\kappa - \mu$ model

In $\kappa - \mu$ fading model, the signal is divided into different clusters of waves. The number of clusters is μ and in each of the clusters, there is a deterministic LOS component with arbitrary power and scattered waves with identical powers. Note, κ is the ratio between the total power of the dominant components and the total power of the scattered waves. Suppose the elements $h_{i,j} = x_{ij} + jy_{ij}$ of $\mathbf{H}$ are i.i.d. $\kappa - \mu$ random variables, where x_{ij} and y_{ij} are the real and imaginary components respectively, then the joint distribution

is given by (Dias *et al.*, 2008),

$$f_{xy}(x_{ij}, y_{ij}) = \frac{|x_{ij}y_{ij}|^{\mu/2}}{4\sigma^4|pq|^{\mu/2-1}} exp(-\frac{(x_{ij}-p)^2+(y_{ij}-q)^2}{2\sigma^2})$$
$$sech(\frac{px_{ij}}{\sigma^2})sech(\frac{qy_{ij}}{\sigma^2})I_{\frac{\mu}{2}-1}(\frac{|px_{ij}|}{\sigma^2})I_{\frac{\mu}{2}-1}(\frac{|qy_{ij}|}{\sigma^2}). \quad (7.2)$$

Here $p^2 = \sum_{i=1}^{\mu} p_i^2$ and $q^2 = \sum_{i=1}^{\mu} q_i^2$, where p_i and q_i are the LOS components of in-phase and quadrature components respectively of multipath waves of each cluster. $\kappa = \frac{p^2+q^2}{2\mu\sigma^2}$, where σ^2 is the power of the scattered waves.

7.2.3 Special case: Nakagami-m model

If the elements h_{ij} of $\mathbf{H}$ are independent and identically distributed Nakagami-m variables with density,

$$f_{xy}(x_{ij}, y_{ij}) = \frac{m^m|x_{ij}|^{m-1}|y_{ij}|^{m-1}}{\Omega^m\Gamma^2(m/2)} exp(-\frac{m}{\Omega}(x_{ij}^2+y_{ij}^2)).$$

We can obtain a Nakagami random variable by substituting $\eta = 0$ i.e., $\Omega_X = \Omega_Y = \Omega/2$ and $m = 2\mu$ in $\eta - \mu$ random variable given by (7.1). It can also be obtained by substituting $\kappa = 0$ i.e., $p = q = 0$ and $\Omega = 2\mu\sigma^2$ in $\kappa - \mu$ random variable given by (7.2).

7.3 Proposed Matrix Approximations

Though the statistical characterization of MIMO fading channels is of interest for generalized fading models, it is very extremely challenging to derive the eigenvalue distribution for these models. Hence, it is essential to develop at least an approximate matrix model to study metrics like capacity. In the following subsection, we explain why it is challenging to develop the matrix distribution and the eigenvalue statistics for $\mathbf{HH^H}$, when the elements of $\mathbf{H}$ are assumed to be i.i.d. complex $\kappa - \mu$ or $\eta - \mu$ random variables. In the subsequent subsections, we describe the proposed approximation in detail.

7.3.1 Why is deriving the exact eigenvalue statistic intractable?

The eigenvalue statistics of any Grammian $\mathbf{W} = \mathbf{H}\mathbf{H}^H$ can be obtained by first decomposing $\mathbf{H}$ as $\mathbf{H} = \mathbf{L}\mathbf{Q}$. Here, $\mathbf{L}$ is a complex triangular matrix with real positive diagonal elements and $\mathbf{Q}$ is a complex unitary matrix. Hence, the distribution $p(\mathbf{H})$ is transformed to $p(\mathbf{L}, \mathbf{Q})$. Next, integrating over $\mathbf{Q}$, we obtain the distribution $p(\mathbf{L})$. Then, we perform the transformation $\mathbf{W} = \mathbf{L}\mathbf{L}^H$ and obtain the distribution of the Grammian $p(\mathbf{W})$. Finally, we perform eigenvalue decomposition $\mathbf{W} = \mathbf{S}\mathbf{\Lambda}\mathbf{S}^H$ and integrate over $\mathbf{S}$ to obtain the joint eigenvalue distribution of $\mathbf{W}$. This method used in (Fraidenraich *et al.*, 2008) to determine the eigenvalue statistics of 2×2 $\mathbf{W} = \mathbf{H}\mathbf{H}^H$, where the elements $h_{ij} = x_{ij} + jy_{ij}$ are i.i.d. complex Nakagami-m random variables with uniform phase.

Consider that the elements $h_{ij} = x_{ij} + jy_{ij}$ are i.i.d. complex $\eta - \mu$ random variables with pdf (7.1). The joint distribution is simply the product of the pdf of the individual elements. Therefore, the joint distribution of elements of a 2×2 matrix $\mathbf{H}$ for the complex $\eta - \mu$ case is given by

$$p(\mathbf{H}) = Kexp(-\frac{\mu tr(\mathbf{H}\mathbf{H}^H)}{\Omega_X}) \prod_{i,j=1}^{2} |x_{ij}y_{ij}|^{2\mu-1} exp(-\mu y_{ij}^2 \Big(\frac{1}{\Omega_X} - \frac{1}{\Omega_Y}\Big)), \tag{7.3}$$

where $K = (\frac{\mu^{2\mu}}{\Omega_X^{\mu}\Omega_Y^{\mu}\Gamma^2(\mu)})^4$. Using LQ decomposition, the matrix $\mathbf{H}$ can be decomposed as $\mathbf{H} = \mathbf{L}\mathbf{Q}$, where the matrix $\mathbf{Q}$ is given by (Fraidenraich *et al.*, 2008)

$$\mathbf{Q} = \begin{pmatrix} e^{j\phi_1}cos\theta & e^{j\phi_2}sin\theta \\ -e^{j(\phi_3-\phi_2)}sin\theta & e^{j(\phi_3-\phi_1)}cos\theta \end{pmatrix} \tag{7.4}$$

with $0 \leq \phi_1, \phi_2, \phi_3 \leq 2\pi$, $0 \leq \theta \leq \pi/2$. The matrix $\mathbf{L}$ is given by

$$\mathbf{L} = \begin{pmatrix} l_{11} & 0 \\ l_{21R} + jl_{21I} & l_{22} \end{pmatrix} \tag{7.5}$$

where l_{11} and l_{22} are real. From the Jacobian of this transformation given by $|J| = l_{11}^3 l_{22} sin\theta cos\theta$, the joint pdf of $\mathbf{L}$ and $\mathbf{Q}$ is obtained. In the case of Rayleigh distribution, $p(\mathbf{H})$ doesn't depend on $\mathbf{Q}$, hence there is no integration involved over elements of $\mathbf{Q}$. Integrating over the elements of $\mathbf{Q}$, namely ϕ_1, ϕ_2, ϕ_3 and θ, for the case of $\eta - \mu$

distribution becomes intractable. A similar case is seen for $\kappa - \mu$ fading also. This methodology leads to a simpler form only for Nakagami-m RV with uniform phase (Fraidenraich *et al.*, 2008). Even in the case of Nakagami-m RV with a non-uniform phase, the eigenvalue distribution obtained by the above methodology is not in closed form (Gholizadeh *et al.*, 2015). Summarizing, the above method cannot be adopted for complex $\eta - \mu/\kappa - \mu$ scenario.

7.3.2 Minimizing K-L divergence

An alternative to determining the exact matrix distribution is finding a Wishart distribution that has the least K-L divergence from the actual distribution. As seen earlier, Wishart distribution is a well-studied distribution and its properties can be readily applied (James, 1964). Let $\mathbf{H}$ be an $n_1 \times n_2$ random matrix with independent and identically distributed elements and $\mathbf{X} = \mathbf{H}\mathbf{H}^H$ be an $n_1 \times n_1$ random matrix. Say, the exact matrix distribution of $\mathbf{X}$ denoted by $p(\mathbf{X})$ is not known, as in the case of $\eta - \mu/\kappa - \mu$/Nakagami-m distributions. Let $q(\mathbf{X})$ be that Wishart distribution which minimizes the K-L divergence between $p(\mathbf{X})$ and all the complex Wishart distributions $\mathcal{CW}_{n_1}(n, \mathbf{\Sigma})$, i.e.,

$$\begin{aligned} q(\mathbf{X}) = \underset{q(\mathbf{X})}{argmin} KL(p(\mathbf{X})||q(\mathbf{X})) &= \underset{q(\mathbf{X})}{argmax} \int p(\mathbf{X})[ln(q(\mathbf{X})) - ln(p(\mathbf{X}))]d\mathbf{X} \\ &= \underset{q(\mathbf{X})}{argmax} \int p(\mathbf{X}) ln(q(\mathbf{X}))d\mathbf{X}. \end{aligned} \tag{7.6}$$

Note that, we assume an unknown degrees of freedom as n in this case, while in the Wishart approximation we had constrained the degree of freedom to n_2. The density of an $n_1 \times n_1$ complex Wishart matrix $\mathbf{X} \sim \mathcal{CW}_{n_1}(n, \mathbf{\Sigma})$ is given by (Couillet and Debbah, 2011),

$$q(\mathbf{X}) = \frac{1}{C\Gamma_{n_1}(n)(det\mathbf{\Sigma})^n} etr(-\mathbf{\Sigma}^{-1}\mathbf{X})(det\mathbf{X})^{n-n_1},$$

where $C\Gamma_{.}(.)$ is the complex multivariate gamma function (Couillet and Debbah, 2011). Substituting the density in (7.6), we obtain

$$\begin{aligned} q(\mathbf{X}) &= \underset{q(\mathbf{X})}{argmax} \int p(\mathbf{X}) \Big[- ln(C\Gamma_{n_1}(n)) - nln \det(\mathbf{\Sigma}) \\ &\qquad + Tr(-\mathbf{\Sigma}^{-1}\mathbf{X}) + (n-n_1) ln \det(\mathbf{X}) \Big] d\mathbf{X} \\ &= \underset{q(\mathbf{X})}{argmax} \Big[- ln(C\Gamma_{n_1}(n)) - nln \det(\mathbf{\Sigma}) \\ &\quad + Tr(-\mathbf{\Sigma}^{-1} E_p[\mathbf{X}]) + (n-n_1) E_p[ln \det(\mathbf{X})] \Big]. \end{aligned}$$

Denoting $\mathbf{Z} = E_p[\mathbf{X}]$ and $Y = \mathbb{E}_p[ln \det(\mathbf{X})]$, we get

$$q(\mathbf{X}) = \underset{q(\mathbf{X})}{argmax} \Big[- ln(C\Gamma_{n_1}(n)) - nln \det(\mathbf{\Sigma}) + Tr(-\mathbf{\Sigma}^{-1}\mathbf{Z}) + (n-n_1)Y \Big]. \tag{7.7}$$

To obtain the minimizing distribution, we can differentiate the above equation with respect to two variables namely, $\mathbf{\Sigma}$ and n. Differentiating equation (7.7) w.r.t. $\mathbf{\Sigma}$, we obtain

$$\frac{dq(\mathbf{X})}{d\mathbf{\Sigma}} = -n\mathbf{\Sigma}^{-1} + \mathbf{\Sigma}^{-1}\mathbf{Z}^T\mathbf{\Sigma}^{-1}.$$

When the above equation is equated to zero, we obtain

$$\mathbf{\Sigma} = \frac{1}{n}\mathbf{Z} = \frac{1}{n}\mathbb{E}_p[\mathbf{X}]. \tag{7.8}$$

Note that we obtain the same $\mathbf{\Sigma}$ when we equate the expectations of the matrix with respect to distributions $p(\mathbf{X})$ and $q(\mathbf{X})$, i.e., $\mathbb{E}_p[\mathbf{X}] = \mathbb{E}_q[\mathbf{X}]$. Now differentiating equation (7.7) w.r.t. n, we obtain

$$\frac{dq(\mathbf{X})}{dn} = -ln \det(\mathbf{\Sigma}) + Y - \sum_{i=1}^{n_1} \psi(n-i+1),$$

where $\psi(.)$ is the digamma function (Granstrom and Orguner, 2011). Equating the derivative to zero, we get

$$-ln \det(\mathbf{\Sigma}) + Y - \sum_{i=1}^{n_1} \psi(n-i+1) = 0.$$

By substituting $\mathbf{\Sigma} = \frac{1}{n}\mathbf{Z}^T$ from (7.8), we obtain,

$$n_1 ln(n) - ln\det(\mathbf{Z}) + Y - \sum_{i=1}^{n_1} \psi(n-i+1) = 0. \tag{7.9}$$

Matching the expectations $\mathbb{E}_p[ln\det(\mathbf{X})] = \mathbb{E}_q[ln\det(\mathbf{X})]$ also leads to (7.9).

Hence minimizing the K-L divergence has reduced to a simple case of matching the expectations $\mathbb{E}_p[\mathbf{X}]$ and $\mathbb{E}_p[ln\det(\mathbf{X})]$ with $\mathbb{E}_q[\mathbf{X}]$ and $\mathbb{E}_q[ln\det(\mathbf{X})]$ respectively. Though this is the most ideal approximation, solving (7.9) for n requires the knowledge of $Y = \mathbb{E}_p[ln\det(\mathbf{X})]$. Since finding the actual random matrix variate distribution of $\mathbf{X}$ is intractable, finding the exact expectation of the log-determinant $\mathbb{E}_p[ln\det(\mathbf{X})]$ analytically is not possible. Nevertheless, it is possible to approximate the log determinant $ln\det(\mathbf{X})$ (Boyd *et al.*, 2004, Page 644). One of the simplest approximations is a second order Taylor expansion of the $ln\det(\mathbf{X})$ at $\mathbb{E}_p[\mathbf{X}]$ given by,

$$\begin{aligned} ln\det(\mathbf{X}) \approx\; & ln\det(\mathbb{E}_p[\mathbf{X}]) + tr\left(\mathbb{E}_p[\mathbf{X}]^{-1}(\mathbf{X}-\mathbb{E}_p[\mathbf{X}])\right) \\ & - \frac{1}{2}tr\left(\mathbb{E}_p[\mathbf{X}]^{-1}(\mathbf{X}-\mathbb{E}_p[\mathbf{X}])\,\mathbb{E}_p[\mathbf{X}]^{-1}(\mathbf{X}-\mathbb{E}_p[\mathbf{X}])\right). \end{aligned} \tag{7.10}$$

Evaluating the expectations on both sides, we obtain

$$\mathbb{E}_p[ln\det(\mathbf{X})] \approx ln\det(\mathbb{E}_p[\mathbf{X}]) - \frac{1}{2}tr\mathbb{E}_p\left(\mathbb{E}_p[\mathbf{X}]^{-1}\mathbf{X}\mathbb{E}_p[\mathbf{X}]^{-1}\mathbf{X}\right) + \frac{1}{2}n_1. \tag{7.11}$$

The above approximation can now evaluated for $\eta-\mu$ and $\kappa-\mu$ fading.

$\eta-\mu$

Note that the approximation involves the mean of the random matrix given by $\mathbf{Z} = \mathbb{E}_p[\mathbf{X}] = \mathbb{E}_p[\mathbf{H}\mathbf{H}^H]$. The diagonal elements z_{ii} are means of sums of $\eta-\mu$ envelope square variables. i.e., $z_{ii} = \mathbb{E}[\sum_{j=1}^{n_2}[x_{ij}^2 + y_{ij}^2]]$. Hence by (Yacoub, 2007), $z_{ii} = n_2(\Omega_X + \Omega_Y)$. The off-diagonal elements are given by,

$$z_{ij} = \mathbb{E}[\sum_{k=1}^{n_2}[(x_{ik} + jy_{ik})(x_{kj} - jy_{kj})]].$$

Since x_{ik} and y_{ik} are i.i.d., we obtain $\forall i, j$ and $i \neq j$,

$$z_{ij} = \sum_{k=1}^{n_2} (\mathbb{E}[x_{ik}]^2 + \mathbb{E}[y_{kj}]^2) = 0.$$

The off-diagonal elements are zero, because the marginals of (7.1), given by $f_x(x_{ij})$ and $f_y(y_{ij})$, are odd functions. Therefore, $\mathbf{Z} = \mathbb{E}_p[\mathbf{X}] = n_2(\Omega_X + \Omega_Y)\mathbf{I}_{n_1}$. Also, $ln \det(\mathbb{E}_p[\mathbf{X}]) = n_1 ln\,(n_2(\Omega_X + \Omega_Y))$. Also, $tr\mathbb{E}_p\,(\mathbb{E}_p[\mathbf{X}]^{-1}\mathbf{X}\mathbb{E}_p[\mathbf{X}]^{-1}\mathbf{X})$ evaluates to

$$\begin{aligned} tr\mathbb{E}_p\left(\mathbb{E}_p[\mathbf{X}]^{-1}\mathbf{X}\mathbb{E}_p[\mathbf{X}]^{-1}\mathbf{X}\right) &= (n_2(\Omega_X + \Omega_Y))^{-2}\, tr\mathbb{E}_p\,(\mathbf{X}\mathbf{X}) \\ &= \frac{n_2 n_1}{n_2\,(\Omega_X + \Omega_Y)^{-2}}\Big((\frac{\mu+1}{\mu} + n_2 - 1)(\Omega_X^2 + \Omega_Y^2) \\ &\quad + 2n_2\Omega_X\Omega_Y + (n_1 - 1)(\Omega_X + \Omega_Y)^2\Big) \end{aligned}$$

Hence,

$$\begin{aligned} \mathbb{E}_p[ln \det(\mathbf{X})] \approx n_1 ln\,(n_2(\Omega_X + \Omega_Y)) - \frac{0.5 n_2 n_1}{n_2\,(\Omega_X + \Omega_Y)^{-2}}\Big((\frac{\mu+1}{\mu} + n_2 - 1) \\ (\Omega_X^2 + \Omega_Y^2) \quad + 2n_2\Omega_X\Omega_Y + (n_1 - 1)(\Omega_X + \Omega_Y)^2\Big) + \frac{1}{2}n_1 \end{aligned}$$

Once $\mathbb{E}_p[\mathbf{X}]$ and $\mathbb{E}_p[ln \det(\mathbf{X})]$ are obtained, the $\mathbf{\Sigma}$ and n of the minimizer-Wishart distribution are obtained by (7.8) and (7.9). For $\eta - \mu$ faded channels, $\mathbf{\Sigma}$ will be diagonal. To quantify how far is the actual $\mathbb{E}_p[ln \det(\mathbf{X})]$ from the approximation, we give the following table (Table 7.1) that provides the relative error in the approximation. We can observe that the relative error ((approximate log determinant -actual log determinant)/actual log determinant) is negligible. The error in the approximation decreases with increase in either N_R or N_T and increases with increase in either μ or η.

$\kappa - \mu$ model

The diagonal elements of $\mathbf{Z} = \mathbb{E}_p[\mathbf{X}] = \mathbb{E}_p[\mathbf{H}\mathbf{H}^H]$, i.e, z_{ii} are nothing but the mean of n_2 $\kappa - \mu$ envelope square variables. i.e., $z_{diag} = z_{ii} = \mathbb{E}[\sum_{j=1}^{n_2}[x_{ij}^2 + y_{ij}^2]]$. Hence,

$$z_{diag} = z_{ii} = 2\sigma^2 n_2(1 + \kappa)\mu, \tag{7.12}$$

N_R	N_T	μ	η	Error
3	2	1	0.1	0.11
4	2	1	0.1	0.04
5	2	1	0.1	0.0188
5	2	2	0.1	0.0195
5	2	4	0.1	0.02
2	3	1	0.1	0.12
2	4	1	0.1	0.04
2	5	1	0.1	0.018
2	5	1	0.5	0.0187
2	5	1	0.9	0.0305

Table 7.1: Relative error of approximation of log determinant

where $\kappa = \frac{p^2+q^2}{2\mu\sigma^2}$ (Yacoub, 2007). The off diagonal elements of z_{ij} are given by

$$\begin{aligned} z_{off} = z_{ij} &= \mathbb{E}[\sum_{k=1}^{n_2}[(x_{ik} + jy_{ik})(x_{kj} - jy_{kj})]] \\ &= \sum_{k=1}^{n_2} \mathbb{E}[x_{ik}x_{kj} + y_{ik}y_{kj} - jx_{ik}y_{kj} + jx_{kj}y_{ik}]. \end{aligned} \tag{7.13}$$

Since x_{ik} and y_{ik} are i.i.d., we obtain $\forall i, j$,

$$z_{off} = z_{ij} = \sum_{k=1}^{n_2}((\mathbb{E}[x_{ik}])^2 + (\mathbb{E}[y_{kj}])^2). \tag{7.14}$$

$\mathbb{E}[x_{ik}]$ and $\mathbb{E}[y_{ik}]$ are given by,

$$\mathbb{E}[x_{ik}] = \int_{-\infty}^{\infty} x \frac{|x|^{\mu/2}}{2\sigma^2|p|^{\mu/2-1}} exp(-\frac{(x-p)^2}{2\sigma^2}) sech(\frac{px}{\sigma^2}) I_{\frac{\mu}{2}-1}(\frac{|px|}{\sigma^2}) dx \tag{7.15}$$

$$\mathbb{E}[y_{ik}] = \int_{-\infty}^{\infty} y \frac{|y|^{\mu/2}}{2\sigma^2|q|^{\mu/2-1}} exp(-\frac{(y-q)^2}{2\sigma^2}) sech(\frac{qy}{\sigma^2}) I_{\frac{\mu}{2}-1}(\frac{|qy|}{\sigma^2}) dy \tag{7.16}$$

$\forall i, j$. The closed form expressions for the above integrals seem mathematically intractable. However an approximation for the integral is derived in Appendix G and given by (G.5) and (G.6). [2] Since all the $\kappa - \mu$ elements of the matrix $\mathbf{H}$ are i.i.d., the mean of all the off-diagonal elements are equal. Substituting the results from (G.5) and

[2]These closed form approximations are also compared with both numerical integration evaluation of (7.15) and (7.16) and Monte-Carlo simulation and an excellent match is observed with both.

(G.6) in (7.14), we obtain $\forall i, j$ and $i \neq j$,

$$z_{ij} \approx n_2 \Bigg[\Big[2pe^{-\frac{p^2}{2\sigma^2}} (\frac{4\sigma^2}{4\sigma^2 + 2p^2\pi})^{\mu/2+1} \frac{\Gamma(\mu/2+1)}{\Gamma(\mu/2)} \Psi_1(\mu/2+1, 1, 3/2, \mu/2, \frac{2p^2\pi}{2p^2\pi + 4\sigma^2}, \frac{2p^2}{4\sigma^2 + 2p^2\pi}) \Big]^2 + \Big[2qe^{-\frac{q^2}{2\sigma^2}} (\frac{4\sigma^2}{4\sigma^2 + 2q^2\pi})^{\mu/2+1} \frac{\Gamma(\mu/2+1)}{\Gamma(\mu/2)} \Psi_1(\mu/2+1, 1, 3/2, \mu/2, \frac{2q^2\pi}{2q^2\pi + 4\sigma^2}, \frac{2q^2}{4\sigma^2 + 2q^2\pi}) \Big]^2 \Bigg]. \tag{7.17}$$

Now, to evaluate (7.11), note that

$$ln \det (\mathbb{E}_p[X]) = (z_{diag} - z_{off}) + n_1 z_{off} \tag{7.18}$$

Similarly, we need to determine $tr\mathbb{E}_p (\mathbb{E}_p[\mathbf{X}]^{-1}\mathbf{X}\mathbb{E}_p[\mathbf{X}]^{-1}\mathbf{X})$. It is easy to derive the closed form expression for $\mathbf{X}$ with dimension 2×2, i.e., when either N_T or N_T can be fixed at 2. This is shown in Appendix H. Once $\mathbb{E}_p[\mathbf{X}]$ and $\mathbb{E}_p[ln|\mathbf{X}|]$ are obtained, the $\mathbf{\Sigma}$ and n of the minimizer-Wishart distribution are obtained by (7.8) and (7.9). Unlike $\eta - \mu$ fading, we do not obtain a diagonal covariance matrix $\mathbf{\Sigma}$ for $\kappa - \mu$ fading[3]. The derivation becomes algebraically complicated with increasing N_R or N_T, but not intractable. To circumvent even this algebraic complexity, we also come up with an alternative approximation, which retains matching the first moment.

7.3.3 1-Moment matching approximation for $\kappa - \mu$ fading

We approximate the density $p(\mathbf{X})$ by an $n_1 \times n_1$ Wishart matrix whose distribution is $q(\mathbf{X}) = \mathcal{CW}_{n_1}(n, \mathbf{\Sigma})$ with n degrees of freedom and covariance matrix $\mathbf{\Sigma}$, such that $\mathbb{E}_p[\mathbf{X}] = \mathbb{E}_q[\mathbf{X}]$, i.e., their first moments are matched

$$\mathbb{E}_q(\mathbf{X}) = n_2\mathbf{\Sigma} = \mathbb{E}_p[\mathbf{X}] \tag{7.19}$$

and the degrees of freedom is fixed to be equal to the number of columns of $\mathbf{H}$ i.e., $n = n_2$. This approximation is inspired from the popular paper (H.S. and J.J.J., 1962), where a non-central Wishart matrix $\mathcal{CW}_{n_1}(n_2, \mathbf{\Sigma}', \mathbf{M}\mathbf{M}^H)$ is approximated by a

[3]This is because the first moment of $\mathbf{H}\mathbf{H}^H$ has non-zero off-diagonal elements

Wishart matrix $\mathcal{CW}_{n_1}(n_2, \mathbf{\Sigma})$, where $\mathbf{\Sigma} = \mathbf{\Sigma}' + \frac{1}{n_2}\mathbf{M}\mathbf{M}^H$, by first moment matching. Note, n_2 denotes the number of transmitter antennas or the number of interferers in the MIMO channel matrix. For any Wishart distributed matrix $\mathbf{A} = \mathbf{B}\mathbf{B}^H \sim \mathcal{CW}_{n_1}(n_2, \mathbf{\Sigma})$, the degrees of freedom also denote the number of columns of complex Gaussian $\mathbf{B}$. Hence, we retain the same number n_2, even in approximation, given that there is no correlation in the transmitter side and all the elements of the matrix are i.i.d. The covariance matrix of the corresponding Wishart matrix is given by $\mathbf{\Sigma} = \frac{1}{n_2}(\mathbb{E}_p[\mathbf{X}])^T = \frac{1}{n_2}\mathbf{Z}^T$. Therefore, for $\eta - \mu$ fading, $\mathbf{\Sigma} = (\Omega_X + \Omega_Y)\mathbf{I}_{n_1}$. It is interesting to note that, the approximation doesn't depend on η. Further, it doesn't depend on μ if we normalize the powers. Hence, this approximation is not useful in characterizing the η and μ dependence of the $\eta - \mu$ MIMO matrix. On the other hand, for $\kappa - \mu$ fading, since $\mathbf{\Sigma} = \frac{\mathbf{Z}}{n_2}$, we have

$$\mathbf{\Sigma}_{ii} = z_{diag}/n_2 = 2\sigma^2(1+\kappa)\mu \text{ and } \mathbf{\Sigma}_{ij} = z_{off}/n_2 = z_{ij}/n_2,\ i \neq j. \tag{7.20}$$

Though this approximation does not minimize the KL divergence, it is much easier to evaluate in case of $\kappa - \mu$ fading.

7.3.4 What about more generalized fading models?

Recently there has been significant focus on more generalized fading models such as $\kappa - \mu$ shadowed fading (Moreno-Pozas *et al.*, 2016; Cotton, 2015), $\alpha - \kappa - \mu$ fading, $\alpha - \eta - \mu$ fading (Fraidenraich and Yacoub, 2006), $\alpha - \kappa - \mu$ shadowed fading (Ramirez-Espinosa *et al.*, 2019), etc. Once $\mathbb{E}_p[\mathbf{X}]$ and $\mathbb{E}_p[ln \det(\mathbf{X})]$ are obtained for these generalized fading models from the complex phase-envelope distribution, the $\mathbf{\Sigma}$ and n of the minimizer-Wishart distribution can be obtained by (7.8) and (7.9). For all these cases, there are currently no exact MIMO model available in literature. Even if they are determined in near future, the Wishart approximation will still be a simpler approximation and easily amenable to analysis. [4]

[4]For example, even the exact MIMO model derived in (Gholizadeh *et al.*, 2015) for the well-studied Nakagami-m fading channel with non-uniform phase distribution is not in closed form.

7.4 Applications and Numerical results

In this section, to demonstrate the utility of our work, we apply the above approximation in three very different applications. First, we determine the ZF ergodic rate expressions of massive MIMO employing ADCs. We then determine the MIMO channel capacity for $\kappa - \mu/\eta - \mu$ faded channel coefficients. We finally determine the outage probability of MIMO-MRC for $\kappa - \mu/\eta - \mu$ faded channel coefficients. Determining ergodic rate expressions involves determining the expectation of the inverse of $\mathbf{H}\mathbf{H}^H$. Finding ergodic MIMO channel capacity involves finding the expectation of log determinant of Gram matrix $\mathbf{H}\mathbf{H}^H$, where entries of $\mathbf{H}$ are $\kappa - \mu/\eta - \mu$ faded. On the other hand, finding OP expressions for MIMO-MRC involves characterizing the CDF of the maximum eigenvalue of $\mathbf{H}\mathbf{H}^H$.

7.4.1 Ergodic rate of ZF receiver in massive MIMO with low resolution ADCs

Massive multi-input-multi-output (MIMO) system has been widely accepted as a key technology to meet the increasing demand for wireless throughput in both mobile and fixed scenarios and has been widely investigated in (Marzetta, 2015; Björnson *et al.*, 2016; Hanif *et al.*, 2018; Gao and Kaiser, 2016). However, one drawback with massive MIMO systems is that, since a large number of antennas are required, there is a substantial increase in the hardware cost and power consumption. Using high-speed high-resolution analog-to-digital converter (ADC) for all the antennas increases the power consumption of massive MIMO systems severely and this is considered as the bottleneck to realize massive MIMO in practice. To solve the power consumption problem, typically low-resolution ADCs (e.g., 1-3 bits) are employed at the RF chains (Fan *et al.*, 2015).

Hence, it is imperative to study the performance of MIMO systems in conjunction with a quantizer. Works like (Singh *et al.*, 2009; Bai *et al.*, 2013; Orhan *et al.*, 2015) have analyzed such quantized MIMO systems. Due to the complicated nature of the exact quantization error, the quantization is modeled as additive and independent noise. This additive quantization noise model (AQNM) is further used in (Fan *et al.*, 2015) to study the impact of ADC resolution on the uplink rate for the case of Rayleigh fading

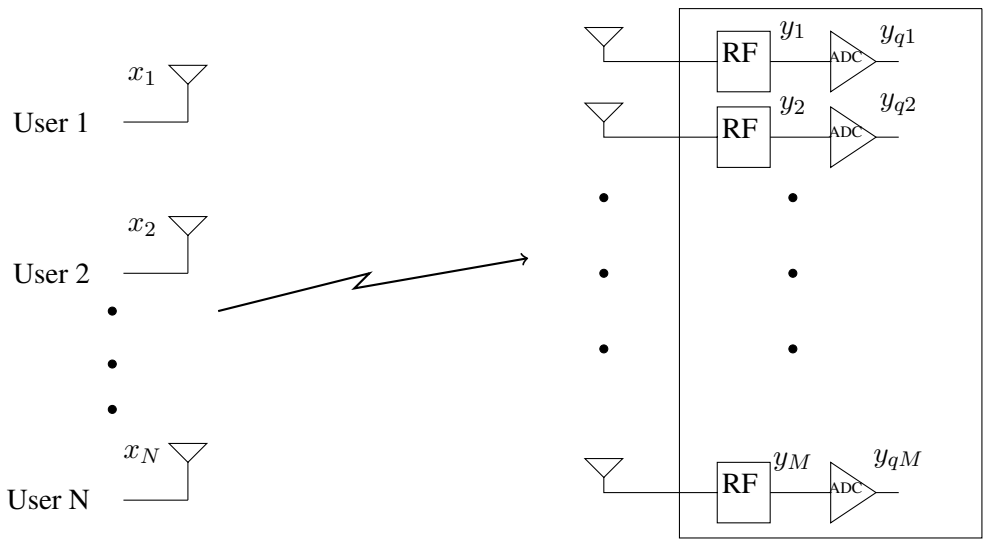


Figure 7.1: System model

channels. Some asymptotic results are also discussed in (Fan *et al.*, 2015). This was extended to the case of Rician fading in (Zhang *et al.*, 2016). In (Tan *et al.*, 2016), a mixed ADC architecture model is considered. In this architecture, a few antennas are equipped with costly full-resolution ADC and the rest with less expensive low-resolution ADC. This was further extended to the case of Rician fading channels in (Zhang *et al.*, 2017*b*) and Nakagami-m channels (Srinivasan and Kalyani, 2019*a*). Very recently, the authors of (Ding and Jing, 2018) have derived an approximate outage probability (OP) expression using the fact that if the squared coefficient of variation (SCV) of a random variable (RV) tends to zero, the RV approaches a deterministic value equal to its mean.

7.5 System model

All the above works consider MRC at the receiver. There are a few works like (Qiao *et al.*, 2016; Jacobsson *et al.*, 2017), which consider ZF at the receiver . It is shown that the zero-forcing (ZF) receiver outperforms the MRC receiver in rate when the resolution of ADCs, the numbers of users (UEs) and BS antennas are fixed. In this chapter, we derive approximations for ergodic rate when the channels are $\kappa - \mu$ or $\eta - \mu$ faded and ZF is employed at the BS. The uplink of a multi-user MIMO (MU-MIMO) system formed by a BS with an array of M antennas serving N single-antenna user terminals is considered. All the user terminals are served in the same time-frequency resource. The

schematic is shown in Fig. 7.1. The received $M \times 1$ vector $\mathbf{y}$ at BS can be expressed as

$$\mathbf{y} = \mathbf{G}\mathbf{x} + \mathbf{n}, \tag{7.21}$$

where $\mathbf{G}$ represents the $M \times N$ channel matrix between BS and users, $\mathbf{x}$ denotes unit-power $N \times 1$ symbols transmitted and $\mathbf{n}$ denotes the complex Gaussian noise i.e., $\mathbf{n} \sim \mathcal{CN}(0, \sigma^2\mathbf{I})$. The lnth element g_{ln} of the matrix $\mathbf{G}$ is $\eta - \mu$ or $\kappa - \mu$ fading complex coefficient. For AQNM, the output at the quantizer is given by (Fan *et al.*, 2015)

$$\mathbf{y}_q = \alpha\mathbf{y} + \mathbf{n}_q = \alpha\mathbf{G}\mathbf{x} + \alpha\mathbf{n} + \mathbf{n}_q, \tag{7.22}$$

with $\alpha = 1-\rho$, where ρ is the inverse of the signal-to-quantization-noise ratio, and $\mathbf{n}_q$ is the additive Gaussian quantization noise vector that is uncorrelated with $\mathbf{y}$. The relation between the number of quantization bins denoted by b and ρ is given in (Fan *et al.*, 2015, Table I) for $b \leq 5$ and is approximated as $\rho = \frac{\pi\sqrt{3}}{2}2^{-2b}$ for $b > 5$. For a fixed channel realization $\mathbf{G}$ and an identity input covariance matrix, $\mathbf{R}_{\mathbf{n_q n_q}}$ is the covariance of $\mathbf{n}_q$ and is given by (Fan *et al.*, 2015, Eq. 5)

$$\mathbf{R}_{\mathbf{n_q n_q}} = \alpha(1-\alpha)\text{diag}\left(\mathbf{G}\mathbf{G}^H + \sigma^2\mathbf{I}\right). \tag{7.23}$$

We assume that a zero-forcing receiver (ZF) is implemented at the BS. At the output of the ZF receiver, the received signal is given by

$$\mathbf{r} = \left(\mathbf{G}\left(\mathbf{G}^H\mathbf{G}\right)^{-1}\right)^H \mathbf{y}_q. \tag{7.24}$$

By substituting (7.22) in (7.24), we obtain

$$\mathbf{r} = \alpha\left(\mathbf{G}\left(\mathbf{G}^H\mathbf{G}\right)^{-1}\right)^H \mathbf{G}\mathbf{x} + \alpha\left(\mathbf{G}\left(\mathbf{G}^H\mathbf{G}\right)^{-1}\right)^H \mathbf{n} + \left(\mathbf{G}\left(\mathbf{G}^H\mathbf{G}\right)^{-1}\right)^H \mathbf{n}_q. \tag{7.25}$$

The nth element of $\mathbf{r}$ is given by

$$r_n = \alpha x_n + \alpha\tilde{\mathbf{g}}_\mathbf{n}^H\mathbf{n} + \tilde{\mathbf{g}}_\mathbf{n}^H\mathbf{n}_q, \tag{7.26}$$

where $\tilde{\mathbf{g}}_{\mathbf{n}} = \left[\mathbf{G}\left(\mathbf{G}^H\mathbf{G}\right)^{-1}\right]_{:,n}$ is the filter for nth user. Hence, the SIQNR of the nth user is given by

$$\gamma_n = \frac{X_n}{Y_n} = \frac{\alpha^2}{\alpha^2||\tilde{\mathbf{g}}_{\mathbf{n}}||^2 + \alpha(1-\alpha)Z_n}, \tag{7.27}$$

where $Z_n = \tilde{\mathbf{g}}_{\mathbf{n}}^H diag(\mathbf{G}\mathbf{G}^H + \sigma^2\mathbf{I})\tilde{\mathbf{g}}_{\mathbf{n}}$. Now using the popular approximation from (Fan *et al.*, 2015; Zhang *et al.*, 2016, 2017*b*), the approximate ergodic rate is given by

$$\begin{aligned}
\text{Rate} &= \mathbb{E}[\log_2(1+\gamma_n)] && (7.28)\\
&= E\left[\log_2\left(1 + \frac{\alpha^2}{\alpha^2||\tilde{\mathbf{g}}_{\mathbf{n}}||^2 + \alpha(1-\alpha)Z_n}\right)\right] && (7.29)\\
&\approx \log_2\left(1 + \frac{\alpha^2}{\mathbb{E}[\alpha^2||\tilde{\mathbf{g}}_{\mathbf{n}}||^2 + \alpha(1-\alpha)Z_n]}\right). && (7.30)
\end{aligned}$$

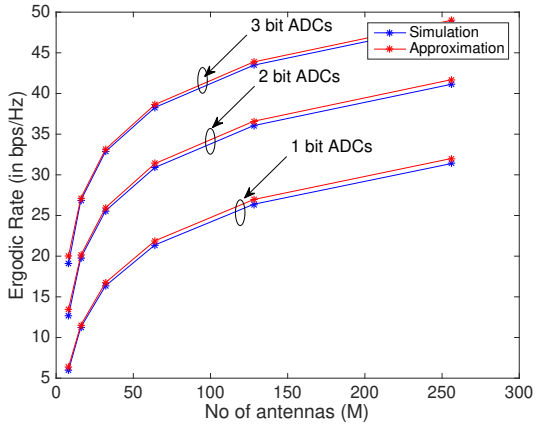
Note that, since $\tilde{\mathbf{g}}_{\mathbf{n}} = \left[\mathbf{G}\left(\mathbf{G}^H\mathbf{G}\right)^{-1}\right]_{:,n}$, $||\tilde{\mathbf{g}}_n||^2 = \left[\left(\mathbf{G}^H\mathbf{G}\right)^{-1}\right]_{n,n}$. For large M, from (Liu *et al.*, 2019), the following approximation identity holds $\tilde{\mathbf{g}}_n^H\mathbf{R}_{n_q n_q}\tilde{\mathbf{g}}_n \approx \tilde{\mathbf{g}}_n^H\mathbb{E}[\mathbf{R}_{n_q n_q}]\tilde{\mathbf{g}}_n$. Therefore,

$$\begin{aligned}
\mathbb{E}[Z_n] &\approx E\left[\tilde{\mathbf{g}}_n^H\mathbb{E}[\mathbf{R}_{n_q n_q}]\tilde{\mathbf{g}}_n\right]\\
&= E\left[\tilde{\mathbf{g}}_n^H\mathbb{E}[diag(\mathbf{G}\mathbf{G}^H + \sigma^2\mathbf{I})]\tilde{\mathbf{g}}_n\right]\\
&= E\left[||\tilde{\mathbf{g}}_n||^2\right]\left[E(\mathbf{G}\mathbf{G}^H)_{1,1} + \sigma^2\right], && (7.31)
\end{aligned}$$

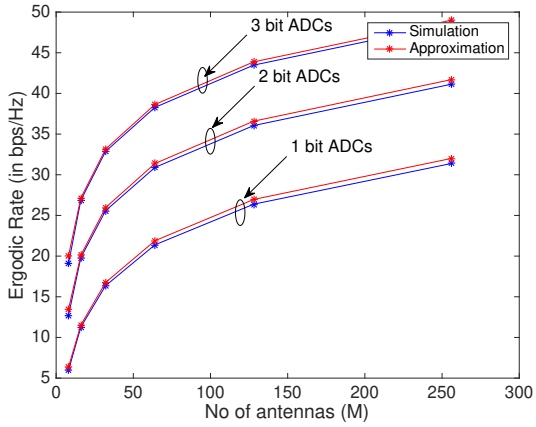
where $E(\mathbf{G}\mathbf{G}^H)_{1,1}$ is the expectation of the diagonal element of $\kappa-\mu$ or $\eta-\mu$ faded channel and is already determined in Section III.B. Therefore,

$$\text{Rate} \approx \log_2\left(1 + \frac{\alpha^2}{\mathbb{E}[\alpha^2||\tilde{\mathbf{g}}_{\mathbf{n}}||^2] + \alpha(1-\alpha)E\left[||\tilde{\mathbf{g}}_n||^2\right]\left[E(\mathbf{G}\mathbf{G}^H)_{1,1} + \sigma^2\right]}\right). \tag{7.32}$$

For $\eta-\mu$ channel, $E(\mathbf{G}\mathbf{G}^H)_{1,1} = K(\Omega_X + \Omega_Y)$ and for $\kappa-\mu$ channel, $E(\mathbf{G}\mathbf{G}^H)_{1,1} = K2\sigma^2(1+\kappa)\mu$. Once $\mathbb{E}[||\tilde{\mathbf{g}}_{\mathbf{n}}||^2]$ is determined, we can determine the expressions for rate. But it is intractable to exactly determine $\mathbb{E}[||\tilde{\mathbf{g}}_{\mathbf{n}}||^2]$ for $\kappa-\mu$ or $\eta-\mu$ faded elements of G. With our Wishart approximation, one can use Wishart properties to determine $\mathbb{E}[||\tilde{\mathbf{g}}_{\mathbf{n}}||^2]$. $\mathbf{G}^H\mathbf{G}$ is approximated by a Wishart matrix $\mathcal{CW}_K(m, \mathbf{\Sigma})$, where $\mathbf{\Sigma}$ and m are obtained by (7.8) and (7.9). Therefore, using the Wishart identity $E\left[\left[\left(\mathbf{G}^H\mathbf{G}\right)^{-1}\right]_{n,n}\right] \approx \frac{\left[\Sigma^{-1}\right]_{n,n}}{m-K}$, one can substitute $\mathbb{E}[||\tilde{\mathbf{g}}_{\mathbf{n}}||^2]$ in (7.32) to determine the approximate rate ex-



(a) Ergodic rate for $K = 5$, $\eta = 0.1$ and $\mu = 1$



(b) Ergodic rate for $K = 5$, $\kappa = 0.2$ and $\mu = 2$

Figure 7.2: Ergodic rate expressions for varying K and M

pressions.[5]

The derived ergodic rate expressions are verified using Monte-Carlo simulations. A close match is found between the theoretical and simulation results for all the cases as can be seen from the Fig. 7.2. For both the cases it can be seen that the ergodic rate increases with increase in the number of transmitting antennas M. This is because of the increase in diversity from the users to the BS. Also, the ergodic rate increases with better resolution of ADCs.

7.5.1 MIMO channel Capacity

We consider an $N_R \times N_T$ MIMO channel matrix $\mathbf{H}$, where N_R denotes the number of receive antennas and N_T denotes the number of transmit antennas. Let $\mathbf{x}$ be the $N_T \times 1$

[5]Note that, even after the approximation, it is difficult to determine the exact pdf of SIQNR due to the presence of quantization noise (Fan *et al.*, 2015; Tan *et al.*, 2016).

transmitted vector and $\mathbf{n}$ be the $N_R \times 1$ zero mean i.i.d. complex Gaussian noise vector. The $N_R \times 1$ received vector $\mathbf{y}$ is given by, $\mathbf{y} = \mathbf{Hx} + \mathbf{n}$. Assuming that the transmitter has no channel state information (CSI), the capacity of the MIMO channels when the transmitter has no CSI, is given by(Chiani *et al.*, 2003*b*),

$$C' = log_2 \, det(\mathbf{I} + \frac{\rho}{N_T}\mathbf{HH}^H), \tag{7.33}$$

where ρ is the average signal to noise ratio (SNR) per receiving antenna.[6]. Since $\mathbf{HH}^H$ and $\mathbf{H^H H}$ have the same non-zero eigenvalue statistics, from (Chiani *et al.*, 2003*b*),

$$C' = \sum_{i=1}^{n_1} log_2(1 + \frac{\rho}{N_T}\lambda_i),$$

where $n_1 = min(N_R, N_T)$ and $\lambda_1, ..., \lambda_{n_1}$ are the non-zero eigenvalues of $\mathbf{R}$, which is given by,

$$\mathbf{R} = \begin{cases} \mathbf{HH}^H & if \; N_R \leq N_T \\ \mathbf{H}^H\mathbf{H} & if \; N_R > N_T. \end{cases}$$

Hence, the mean value of C' is given by (Telatar, 1999),

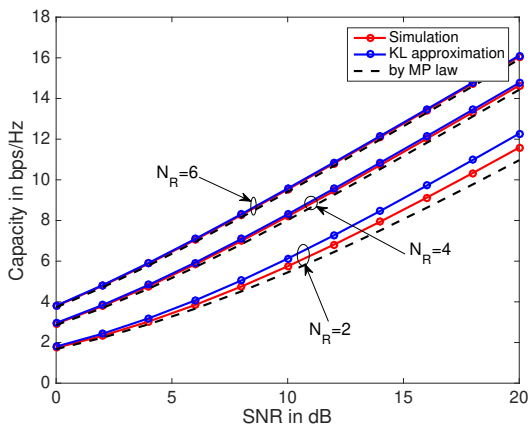
$$C = E_{\mathbf{\Lambda}}\Big[\sum_{i=1}^{n_1} log_2(1 + \frac{\rho}{N_T}\lambda_i)\Big].$$

We do not know the exact eigenvalue distribution of $\mathbf{R}$, when $\mathbf{H}$ comprises i.i.d. $\kappa - \mu$ or $\eta - \mu$ variables. Hence, we apply the Wishart approximation developed in the last section and then determine C.

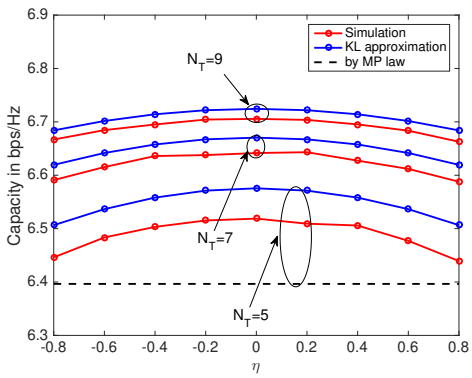
$\eta - \mu$

For $\eta - \mu$ faded channel, we approximate the $N_R \times N_R$ matrix $\mathbf{HH}^H$ by a Wishart matrix $\mathcal{CW}_{N_R}(n, b\mathbf{I}_{N_R})$, where b and n are obtained by (7.8) and (7.9). Once the Wishart approximation is obtained, we can draw expressions from the vast literature of expressions derived for Rayleigh faded MIMO channels. For example, the expressions given

[6]Though, for capacity, expectation of $\log_2 \det(\mathbf{I} + \frac{\rho}{N_T}\mathbf{HH}^H)$ can be matched instead of that of $log_2 \, det(\mathbf{HH}^H)$(as done in Section III), the resultant Wishart matrix will not be a KL divergence minimizer



(a) Capacity vs SNR for $N_T = 2$, $\mu = 2$ and $\eta = 0.1$



(b) Capacity vs SNR for $N_R = 2$ and $\mu = 1$

Figure 7.3: Capacity for varying η, μ, N_R and N_T

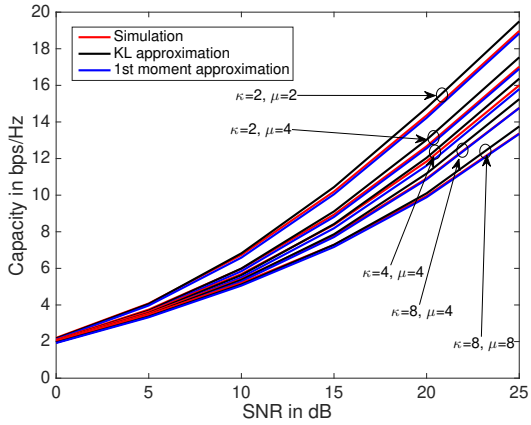
in (Telatar, 1999) can be used directly as follows:

$$C \approx \sum_{k=0}^{N_R-1} \frac{k!}{\Gamma(k+n-N_R+1)} \int_0^\infty log_2(1+\frac{\rho b}{N_T}\lambda)\lambda^{n-N_R}e^{-\lambda}(L_k^{n-N_R}(\lambda))^2 d\lambda \quad (7.34)$$

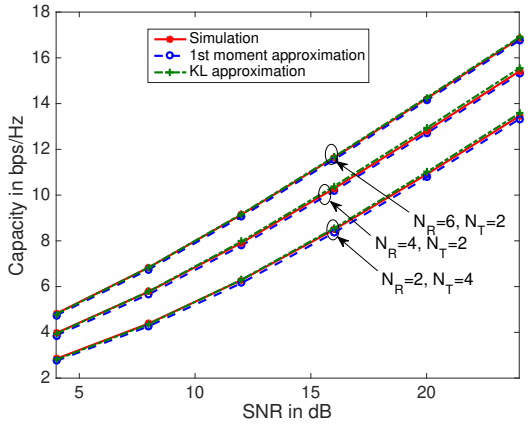
The above expression is valid for $N_R < N_T$. In case $N_T < N_R$, we approximate $\mathbf{H}^H\mathbf{H}$ instead of $\mathbf{H}\mathbf{H}^H$ since both have the same non-zero eigenvalues.

$\kappa - \mu$

Unlike $\eta - \mu$ fading, $\kappa - \mu$ fading approximation involves a non-diagonal covariance matrix with repeated eigenvalues. Though a lot of results exists for diagonal covariance matrix and a few results exists for non-diagonal covariance matrix, especially when the covariance matrix has repeated eigenvalues, the approximate expression for C is derived in Appendix I for the sake of completion and better readability of the manuscript. For



(a) Capacity vs SNR for $N_R = 3$ and $N_T = 4$



(b) Capacity vs SNR for $\kappa = 2$ and $\mu = 2$

Figure 7.4: Capacity for varying κ, μ, N_R and N_T

$N_T \geq N_R$, by substituting $n_1 = N_R$, $w_1 = 2\sigma^2(1+\kappa)\mu - y$ and $w_2 = 2\sigma^2(1+\kappa)\mu + (N_R - 1)y$ which are the eigenvalues of $\mathbf{\Sigma}^{-1}$ in (I.4), we can get the average capacity approximation. $\mathbf{\Sigma}$ and $n_2 = n$ of the Wishart matrix are obtained by (7.8) and (7.9). In case $N_T < N_R$, we approximate $\mathbf{H}^H\mathbf{H}$ instead of $\mathbf{H}\mathbf{H}^H$ since both have the same non-zero eigenvalues. For $N_T \leq N_R$, we substitute $n_1 = N_T$ in (I.4) to get the capacity approximation. Here, $w_1 = 2\sigma^2(1+\kappa)\mu - y$ and $w_2 = 2\sigma^2(1+\kappa)\mu + (N_T - 1)y$ are the eigenvalues of $\mathbf{\Sigma}$ with multiplicity $N_T - 1$ and 1 respectively.

The derived capacity expressions are verified using Monte-Carlo simulations. For each Monte-Carlo simulation, the $N_R \times N_R$ random matrix $\mathbf{H}\mathbf{H}^H$ is generated such that $\mathbf{H}$ has i.i.d. $\kappa - \mu$ or $\eta - \mu$ complex variables following the distribution that is given in (Dias *et al.*, 2008; da Costa and Yacoub, 2007). For a given SNR ρ and N_T, capacity is evaluated using (7.33). This procedure is repeated over many realizations of $\mathbf{H}\mathbf{H}^H$ and the mean is taken to obtain the average capacity. This procedure is repeated for

various values of κ/η, μ, N_R and N_T. A close match is found between the theoretical and simulation results for all the cases as can be seen from the Fig. 7.3- Fig. 7.4. For the case of $\eta - \mu$ distribution, from Fig. 7.3 (b) it can be seen that the average capacity increases with decrease in the magnitude of η. Also, the average capacity increases with increase in the number of transmitters N_T, but the increase is diminished with larger N_T. For any further increase in capacity one has to increase either N_R or the SNR, as shown in Fig. 7.3 (a). The results are also compared with those obtained by Marchenko-Pasteur (MP) law (Telatar, 1999, Eq. 13). Though, the asymptotics obtained from MP law works even for smaller values of N_R or N_T, the capacity expressions from MP law are not dependent on η or μ and cannot be used to plot the variation of capacity with changes in η or μ as evident from 7.3 (b). It can be observed from Fig. 7.4 (a), that capacity decreases with both κ and μ. In case of $\kappa - \mu$, $\mathbf{H}$ has a non-zero mean.

Asymptotic capacity

MP law has been traditionally derived only for $\mathbf{H}$ matrices with zero mean. To the best of our knowledge, there are only a few literature such as (Dumont *et al.*, 2005; Lebrun *et al.*, 2006) that apply MP law for random matrices with non-zero mean. The former derives an asymptotic capacity after a lengthy algebraic calcuation and Taylor expansion (Dumont *et al.*, 2005, Eq. 14). The latter derives only asymptotic bounds for the capacity expression and do not derive exact capacity (Lebrun *et al.*, 2006). Unlike $\eta - \mu$ random variable, it is difficult to obtain the asymptotic capacity for all SNR values, due to the presence of correlation matrix $\mathbf{\Sigma}$. Hence, we will derive approximate asymptotic capacity only at high SNR, based on the matrix approximation that we have obtained. At high SNR, the capacity is given by

$$\begin{aligned} C &\approx \mathbb{E}[log_2\, det(\frac{\rho}{N}\mathbf{H}\mathbf{H}^H)] = \mathbb{E}[log_2\, det(\frac{\rho}{N}\mathbf{\Sigma^{1/2}}\mathbf{H'}\mathbf{H'^H}\mathbf{\Sigma^{1/2}})] \\ &= \mathbb{E}[log_2\, det(\frac{\rho}{N}\mathbf{H'}\mathbf{H'}^H)] + log_2\, det(\mathbf{\Sigma}) \\ &= \mathbb{E}_\Lambda \sum_{i=1}^{N} log_2(\frac{\rho}{N}\lambda_i) + log_2(w_1^{N-1}w_2). \end{aligned}$$

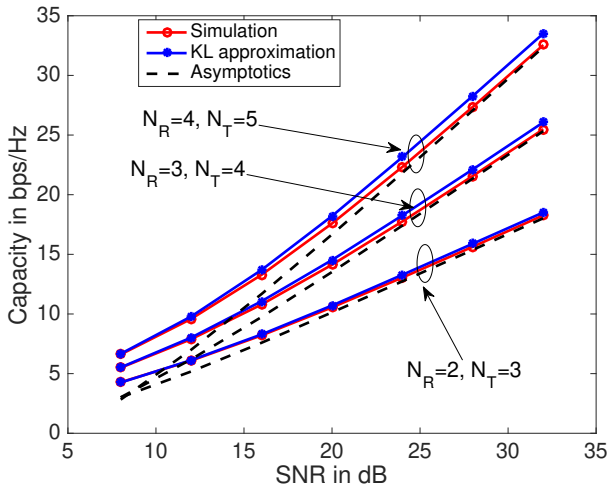


Figure 7.5: Capacity vs SNR for $\kappa = 1$ and $\mu = 3$

where $\mathbf{H}' \sim \mathcal{CN}_N(0, \mathbf{I})$. Therefore, the asymptotic capacity after applying the MP distribution is is given by,

$$\begin{aligned} C &\approx \mathbb{E}_\Lambda \sum_{i=1}^{N} log_2(\frac{\rho}{N}\lambda_i) + (N-1)log_2(w_1) + log_2(w_2) \\ &= N \int_a^b log_2(\rho\lambda)g(\lambda) + (N-1)log_2(w_1) + log_2(w_2), \end{aligned} \tag{7.35}$$

where $g(\lambda) = (2\pi c\lambda)^{-1}\sqrt{(b-\lambda)(\lambda-a)}$ and $a = (1-\sqrt{c})^2$, $b = (1+\sqrt{c})^2$ and $c = N_R/N_T$. If $N_R = N_T$, the equation simplifies to

$$\frac{1}{N}C \approx log_2(\frac{\rho}{e}) + log_2(w_1) \tag{7.36}$$

From the above equation it is clear that, at high SNR, capacity grows as a logarithmic function of SNR ρ. This can now be verified by Fig 7.5.

7.5.2 Outage probability of MIMO-MRC

We consider a wireless link equipped with N_T antenna at the transmitter and N_R antenna at the receiver. The received vector at the receiver can be modelled as

$$\mathbf{y} = \mathbf{H}\mathbf{w}s + \mathbf{n}, \tag{7.37}$$

where s is the transmitted signal of the desired user and $\mathbf{n}$ is the additive white Gaussian noise with power σ_n^2. $\mathbf{w}$ the weight vector at the transmitter with power Ω_D and $\mathbf{H}$ is the channel gain matrix. For MIMO-MRC, the maximum output SNR at the receiver is (Ming Kang and Alouini, 2003, Eq. 27)

$$\gamma = \frac{\Omega_D}{\sigma_n^2}\lambda_{max}, \tag{7.38}$$

where λ_{max} is the largest eigenvalue of $\mathbf{HH}^H$. The outage probability is the CDF of the output SNR evaluated at γ_{th}.

$\eta - \mu$

Since $\mathbf{HH}^H \sim \mathcal{CW}_{N_R}(n, b\mathbf{I}_{N_R})$, the OP is directly given by (Ming Kang and Alouini, 2003, Eq. 33)

$$P_{out} = \Pr(\gamma \leq \gamma_{th}) \tag{7.39}$$

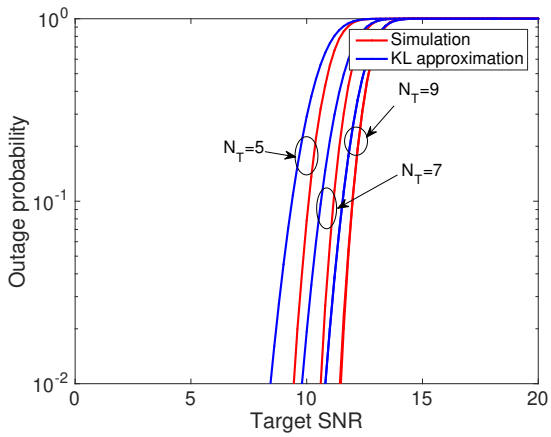
$$= \frac{\det\left(\mathbf{\Psi}_c(\frac{\sigma_n^2\gamma_{th}}{\Omega_D b}\right)}{\prod_{k=1}^{N_R}\Gamma(n-k+1)\Gamma(N_R-k+1)}, \tag{7.40}$$

where $\{\mathbf{\Psi}_c(x)\}_{i,j} = \gamma(n - N_R + i + j - 1, x)$ and $\gamma(.,.)$ is the incomplete gamma function. The above expression is valid for $N_R < N_T$. In case $N_T < N_R$, we approximate $\mathbf{H}^H\mathbf{H}$ instead of $\mathbf{HH}^H$ since both have the same non-zero eigenvalues.

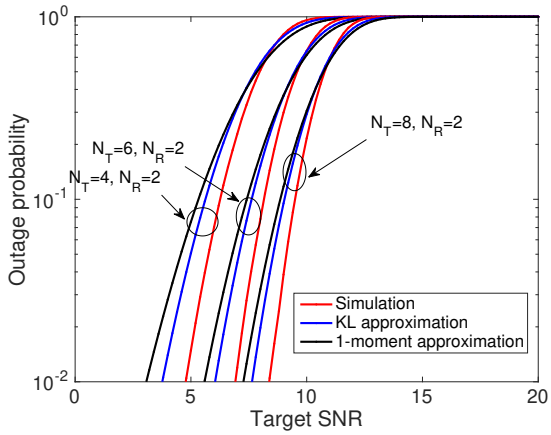
$\kappa - \mu$

The outage expressions for $\kappa - \mu$ fading in MRC is derived in Appendix I. For $N_T \geq N_R$, by substituting $n_1 = N_R$, $x = \frac{\gamma_{th}\sigma_n^2}{\Sigma_D}$, $w_1 = 2\sigma^2(1+\kappa)\mu - y$ and $w_2 = 2\sigma^2(1+\kappa)\mu + (N_R - 1)y$ which are the eigenvalues of $\mathbf{\Sigma}^{-1}$ in (I.6), we can get the outage probability approximation. In case $N_T < N_R$, we approximate $\mathbf{H}^H\mathbf{H}$ instead of $\mathbf{HH}^H$ since both have the same non-zero eigenvalues. For $N_T \leq N_R$, we substitute $n_1 = N_T$ in (I.6) to get the outage approximation. Here, $w_1 = 2\sigma^2(1+\kappa)\mu - y$ and $w_2 = 2\sigma^2(1+\kappa)\mu + (N_T - 1)y$ are the eigenvalues of $\mathbf{\Sigma}$ with multiplicity $N_T - 1$ and 1 respectively.

The derived capacity expressions are verified using Monte-Carlo simulations. The



(a) Outage probability for $N_R = 5$, $\eta = 0.3$ and $\mu = 3$



(b) Outage probability for $N_R = 2$, $\kappa = 0.2$ and $\mu = 2$

Figure 7.6: Outage probability of MRC for varying N_T

theoretical and simulation results for all the cases are within a 2dB difference as can be seen from the Fig. 7.6. For both the cases it can be seen that the outage decreases with increase in the number of transmitting antennas N_T. The KL approximation, even though provides the Wishart distribution with the least KL divergence from the actual distribution, doesn't approximate the tail of the distribution perfectly. This is reflected by means of the small gap between the simulation and the approximation for outage in the order of 10^{-1}. Nevertheless, the KL approximation performs better than the 1-moment approximation as seen from Fig. 7.6(b).

7.6 Conclusions and Future work

Approximate random matrix models have been derived for $\mathbf{H}\mathbf{H}^H$ when the elements of $\mathbf{H}$ are i.i.d $\kappa - \mu$ or $\eta - \mu$ random variables. The approximation is terms of the complex Wishart matrix, which has the least KL-divergence from the original matrix distribution. Further, for $\kappa - \mu$ distribution, we also derive a simple approximation that matches the first moment but fixes the degree of freedom. The utility of our result is shown by a) deriving approximate capacity expressions for $\kappa - \mu$ or $\eta - \mu$ MIMO models b) deriving approximate outage expressions of MIMO-MRC for $\kappa - \mu$ or $\eta - \mu$ channels c) deriving ergodic rate expressions for zero-forcing (ZF) receiver in an uplink single cell massive MIMO scenario with low resolution analog-to-digital converters (ADCs) in the antennas. For these applications, extensive Monte-Carlo simulations have been performed and an excellent match with the approximate expressions has been observed. Given the simplicity of the approximation, it can also be extended to other types of generalized fading matrix models.

CHAPTER 8

Asymptotic maximum and minimum order statistic for SIR in $\kappa - \mu$ shadowed fading

8.1 Introduction

For the case of $\kappa - \mu$ shadowed fading, the recent work (Kumar and Kalyani, 2017*b*) derives the CDF of SIR in terms of an infinite summation of the Lauricella function of the fourth kind. Since, determining the CDF of maximum over L such i.i.d. SIR realizations involve raising the CDF to power L, making mathematical analyses like computing rate is very difficult. We can observe a similar problem for the case of the minimum also. Therefore, a limiting distribution for the maximum or minimum of SIR RVs, which is not only easy to compute but is also amenable to mathematical analysis, will have significant utility.

8.1.1 Our contributions

The contributions in this chapter are as follows:

- Assuming that the user signal and the interferer signals undergo independent and non-identically distributed (i.n.i.d.) $\kappa - \mu$ shadowed fading, we prove, using tools from extreme value theory (EVT), that the limiting distribution of the maximum of L such independent and identically distributed (i.i.d.) signal-to-interference ratio (SIR) random variables (RV)s is a Frechet distribution. We then prove the convergence of moments of the maximum RV to those of the limiting distribution.

- We also prove that the minimum of L such i.i.d. SIR RVs is a Weibull distribution.

A. Subhash, M. Srinivasan and S. Kalyani, "Asymptotic Maximum Order Statistic for SIR in $\kappa - \mu$ Shadowed Fading," in IEEE Transactions on Communications, vol. 67, no. 9, pp. 6512-6526, Sept. 2019.

A Subhash, M Srinivasan, S Kalyani, L Hanzo, "Transmit Power Policy and Ergodic Multicast Rate Analysis of Cognitive Radio Networks in Generalized Fading", to appear in *IEEE Transactions on Communications* (Early Access)

- We also derive the rate of convergence of the actual maximum distribution to the asymptotic distribution. This sheds light on how well the limiting distribution approximates the actual distribution for finite values of L and N. In order to further demonstrate the practical validity of the work, we also study the empirical Kullback-Leibler (KL) divergence between the empirical maxima distribution and the derived asymptotic distribution. The KL divergence results indicate the quantitative closeness between the asymptotic results and the exact results, for finite L, whereas the rate of convergence results discusses the order of convergence.

- Further, we use results from stochastic ordering to analyze the variations in the asymptotic distribution of the maximum. This analysis will not be possible with the exact but complicated distribution of the maximum RV.

- Finally, we analyze the utility of the derived asymptotic results in the following applications:
    1. Analysis of asymptotic outage probability and asymptotic ergodic rate of the user in each time slot of a Maximum Sum Capacity (MSC) scheduling system.
    2. Derivation of upper bound on the rate in Full Array Selection (FAS) architectures for antenna selection in massive Multiple-Input-Multiple Output (MIMO) scenario.
    3. Derivation of optimal power allocation in cognitive radio scenario.

Also, the above-mentioned results hold for Rayleigh, Rician, Nakagami-m, $\kappa - \mu$ and $\eta - \mu$ faded user and interferer scenarios since all of these are special cases of the $\kappa - \mu$ shadowed fading model. Since we assume i.n.i.d. interferers, we also account for interferers having different path-loss or having unequal powers.

The rest of the chapter is organized in the following fashion. In Section 8.2, we find the asymptotic distribution of the maximum SIR using tools from EVT. We mention

the asymptotic distribution of minimum SIR also. We also give brief notes on the convergence of moments and the rate of convergence. Further, in Section 8.4, we give an analysis of the asymptotic distribution and analyze the convergence of the true maxima distribution to the asymptotic results derived in terms of the empirical KL divergence. Then, in Section 8.5, we present three applications of the derived results and their corresponding simulations. Finally, we conclude the work in Section 8.7.

8.2 EVT based maxima and minima of L i.i.d. SIR RVs

Let γ_{max}^{L} denote the maximum of L i.i.d. SIR RVs, where the source and the interferers are assumed to experience i.n.i.d. $\kappa - \mu$ shadowed fading, i.e., $\gamma_{max}^{L} = max\{\gamma_1, \cdots, \gamma_L\}$ and $\gamma_j \sim F_\gamma(z)$, $\forall\, j \in \{1, \cdots, L\}$. Similarly, $\gamma_{min}^{L} = min\{\gamma_1, \cdots, \gamma_L\}$. In this section, (a) the asymptotic distribution of γ_{max}^{L} represented as $F_{\gamma_{max}}(z)$ is derived. Using similar mathematics, we give the asymptotic minimum without derivation. (b) convergence of the moments of $F_{\gamma_{max}}(z)$ to the moments of the true maxima distribution $F_{\gamma_{max}^{L}}(z)$ is analyzed and (c) the rate of convergence of $F_{\gamma_{max}^{L}}(z)$ to $F_{\gamma_{max}}(z)$ is derived.

8.2.1 Maximum and Minimum of SIR RVs in $\kappa - \mu$ shadowed fading environment.

We first prove that the cumulative distribution function (CDF) of maximum of L i.i.d. SIR RVs converges to the CDF of a Frechet RV. For this, we make use of Fisher-Tippet theorem, which forms the corner-stone of EVT. The seminal theorem is as follows (De Haan and Ferreira, 2007):

Theorem 8.1. *Fisher-Tippet Theorem, Limit Laws for Maxima:*
Let $z_1, z_2, \cdots, z_L$ be a sequence of L i.i.d. RVs and $M_L = max\ \{z_1, z_2, \cdots, z_L\}$; if $\exists$ constants $a_L > 0$ and $b_L \in \mathbb{R}$ and some non-degenerate CDF G_β such that, as $L \to \infty$,

$$a_L^{-1}\left(M_L - b_L\right) \xrightarrow{D} G_\beta, \tag{8.1}$$

where $\xrightarrow{D}$ denotes convergence in distribution, then the CDF G_β is one of the three CDFs:

$$Frechet \ : \ \Lambda_1(z) := \begin{cases} 0, & z \leq 0, \\ exp(-z^{-\beta}), & z > 0, \end{cases}$$

$$Reversed\,Weibull \ : \ \Lambda_2(z) := \begin{cases} exp(-(-z)^{\beta}), & z \leq 0, \\ 1, & z > 0, \end{cases}$$

$$Gumbel \ : \ \Lambda_3(z) := exp(-exp(-z)), \quad z \in \mathbb{R}.$$

Proof. Please refer to page 6 in (De Haan and Ferreira, 2007) for the proof. □

To determine the limiting distribution from the above three, we have to first define the maximum domain of attraction (*MDA*).

Definition 1. Maximum Domain of Attraction (De Haan and Ferreira, 2007): The CDF F of i.i.d. RVs $z_1, \cdots, z_L$ belongs to the MDA of the extreme value distribution (EVD) G_β, if and only if $\exists$ constants $a_L > 0$ and $b_L \in \mathbb{R}$, such that (8.1) holds.

Theorem 8.2. *A CDF F belongs to the MDA of the Frechet distribution, if it satisfies the following relation from (De Haan and Ferreira, 2007):*

$$\lim_{t \to \infty} \frac{1 - F(tz)}{1 - F(t)} \ = \ z^{-\beta}. \tag{8.2}$$

Proof. Please refer to page 19 in (De Haan and Ferreira, 2007) for the proof. □

Now, if we show that the CDF $F_\gamma(z)$ satisfies the relation in (8.2), then from the definition of the MDA of an EVD, we can conclude that there exists a_L and b_L satisfying (8.1). A choice for the corresponding constants for the Frechet distribution is given in (De Haan and Ferreira, 2007) as $b_L = 0$ and $a_L = F^{-1}(1 - L^{-1})$.

Theorem 8.3. *The CDF $F_\gamma(z)$ is in the MDA of the Frechet distribution.*

Proof. Please refer to Appendix J for the detailed proof. □

Thus, we conclude that the CDF of γ_{max}^L converges to the CDF of a Frechet RV γ_{max} with shape parameter

$$\beta = \sum_{i=1}^{N} \mu_i \tag{8.3}$$

and scale parameter

$$a_L = F_\gamma^{-1}(1 - L^{-1}). \tag{8.4}$$

The asymptotic distribution of the CDF of γ_{max}^L is hence given by

$$F_{\gamma_{max}}(z) = \begin{cases} 0, & z \leq 0, \\ \exp\left(-\left(z/a_L\right)^{-\beta}\right), & z > 0. \end{cases} \tag{8.5}$$

The above expression is far easier to evaluate than the Lth power of (J.1) for large values of L.

Theorem 8.4. *The asymptotic distribution of $\gamma_{\min}^L = \min\{\gamma_1, \gamma_2, ..., \gamma_L\}$ is a Weibull distribution having the shape parameter $\upsilon = \mu$ and scale parameter $a_L = F_\gamma^{-1}\left(\frac{1}{L}\right)$, where $F_\gamma(z)$ is the common CDF of i.i.d. RVs γ_l. Let, $\gamma_{\min} = \lim_{L\to\infty} \gamma_{\min}^L$, then we have,*

$$F_{\gamma_{\min}}(z) = \begin{cases} 1 - exp(-(z/a_L)^\upsilon), & z \geq 0, \\ 0, & z < 0. \end{cases} \tag{8.6}$$

Proof. The proof is similar to that for maxima RV and is omitted here. □

8.2.2 Moment Convergence

We will examine the convergence of moments of γ_{max}^L to those of γ_{max}. This is useful in evaluating various average ergodic performance metrics with respect to the maximum SIR RV. We make use of the following results from (De Haan and Ferreira, 2007) to prove the convergence of moments.

Lemma 8.1. *If F, the CDF of a RV Z, belongs to the domain of attraction of G_β, then $\forall -\infty < z < \omega(F)$,*

$$\mathbb{E}[|Z|^\nu \, \mathbf{1}_{Z>z}] : \begin{cases} < \infty, & if\ 0 < \nu < \beta^+, \\ = \infty, & if\ \nu > \beta^+, \end{cases} \tag{8.7}$$

where $\beta^+ := max\{0, \beta\}$, $\omega(F) := sup\{z \in \mathbb{R} : F(z) < 1\}$ and $\mathbf{1}_{Z>z}$ is the indicator function for the event given by $Z > z$.

Proof. Please refer to (De Haan and Ferreira, 2007) for the proof. □

Theorem 8.5. *Let Z be an F distributed RV and F belongs to the domain of attraction of G_β, if $\mathbb{E}[Z^\nu]$ is finite for some $\nu < \beta^+$ then,*

$$\lim_{L\to\infty} \mathbb{E}\left[\left(\frac{M_L - b_L}{a_L}\right)^\nu\right] = \int_{-\infty}^{\infty} z^\nu dG_\beta(z), \tag{8.8}$$

where $\beta^+ := max\{0, \beta\}$.

Proof. Please refer to page 176 in (De Haan and Ferreira, 2007) for the proof. □

From Lemma 8.1, we observe that $\mathbb{E}[Z^\nu]$ is finite for all values of $0 < \nu < \beta^+$. So, according to Theorem 8.5, (8.8) holds for all ν in this range. Hence, we conclude that the νth moment of the RV γ_{max}^L converges to the νth moment of γ_{max}, for all $\nu < \sum_{i=1}^{N} \mu_i$.

8.3 Rate of convergence

8.3.1 Rate of convergence for maxima distribution

Note that, (8.1) guarantees the convergence of the distribution of γ_{max}^L to a Frechet distribution, but does not discuss the rate of convergence. In other words, it does not discuss how fast $\gamma_{max}^L \xrightarrow{D} \gamma_{max}$. The rate of convergence is not the same for all distributions in any domain of attraction. In fact, it is a function of the initial distribution parameters and depends on the equivalence of the tail of the initial distribution function to the tail of a generalized Pareto distribution (GPD) (De Haan and Ferreira, 2007). The closer the tail-behaviour of the initial distribution to the tail-behaviour of a GPD, faster is its rate of convergence. We now give the rate of convergence for our case through the following theorem.

Theorem 8.6. *The rate of convergence of $F_{\gamma_{max}^L}(z)$ to the Frechet distribution is $O\left(L^{-\left(\sum_{i=1}^{N}\mu_i\right)^{-1}} + L^{-1}\right)$.*

Proof. Please see Appendix. K for the detailed proof. □

This result is equivalent to stating that

$$\sup_{B\in\mathbb{B}}\left|\mathbb{P}\left(\left(\left(\frac{\gamma_{max}^{L}}{a}\right)/L^{\beta}\right)\in B\right)-\Lambda_1(B)\right|=O\left(\left(\frac{1}{L}\right)^{\delta}+\frac{1}{L}\right), \tag{8.9}$$

where $\delta = \left(\sum_{i=1}^{N}\mu_i\right)^{-1}$, $\mathbb{B}$ denotes the Borel σ algebra on $\mathbb{R}$, $\Lambda_1(.)$ is the asymptotic maxima distribution and a is a positive constant. Hence, we can see that the maximum deviation between the true distribution of the maximum and the asymptotic distribution of the maxima over all the points, decreases with an increase in L or $1/\beta$. (Note that, this is the rate of convergence at the point of maximum possible deviation over the entire support of the maximum distribution. We can expect faster convergence over some subsets of the support of the maxima distribution). Further, this result says that the rate of convergence is determined by the number of interferers N and the number of clusters μ_i (for $i = 1, \cdots, N$) in the interferers' fading distribution. The convergence rate decreases as the number of interferers increases or the number of clusters μ_i increases. Thus, the distributions of γ_{max}^{L} for interferers with fading environments having $\mu_i = 1$ (Rayleigh, Rician or shadowed Rician) converge faster to the asymptotic distribution of the maximum, given by (8.5) than those having $\mu_i > 1$ (Nakagami-m, κ-μ or η-μ). Also, note that the parameters κ, μ and m of the source and the parameters κ_i and m_i (for $i = 1, ..., N$) of the interferers do not affect the convergence rate.

8.3.2 KL divergence between asymptotic maximum distribution and the true maximum distribution

To get a quantitative idea of how the convergence of the true distribution of the maximum to the asymptotic distribution of the maximum varies for different values of L and β, we compute the empirical KL divergence between the maximum SIR samples and the samples from the corresponding Frechet distribution[1]. To calculate the empirical KL-divergence, we use the method discussed in (Wang *et al.*, 2005). Let $\{X_1, \cdots, X_n\}$ and $\{Y_1, \cdots, Y_n\}$ be i.i.d samples from the distributions P and Q respectively. Now,

[1]Since the exact CDF of the maximum SIR has a complicated structure, it is mathematically intractable to derive an expression for the KL divergence and hence we calculate the empirical KL divergence

we compute their histograms over the complete range of samples divided into equispaced bins. According to the Freedman-Diaconis rule[2] (Freedman and Diaconis, 1981), the number of bins is computed for both set of samples and the maximum of the two, given by W, is chosen to compute the histograms. If u_i and v_i represent the number of samples in the ith bin of histograms of P and Q respectively, then the corresponding empirical KL divergence is computed as (Wang *et al.*, 2005)

$$D_{KL}(P||Q) \approx \sum_{i=1}^{W} \frac{u_i}{n} \log\left(\frac{u_i}{v_i}\right). \tag{8.10}$$

The following tables give the empirical KL divergence of the asymptotic distribution of the maximum from the true distribution of the maximum for different values of N, L and β. The number of samples in each case is $n = 10^6$. Tables 8.1 (a)-(c) gives the KL divergence in Rayleigh fading scenario for different number of interferers. The smaller the KL divergence, the closer are two distributions. We can see that the KL divergence decreases as L increases for all the cases as expected. Similarly, for the same value of L, we can see that the KL divergence increases with N. This observation is also in agreement with the rate of convergence results derived which says that the rate decreases with an increase in β. For the case of Rayleigh fading, we have $\beta = N$.

L	(a) N=1	(b) N=2	(c) N=3	(d) $\beta = 2$	(e) $\beta = 3$	(f) $\beta = 4$
20	3.05683e-04	6.91740e-02	1.86644e-01	8.41624e-02	8.32787e-02	4.31212e-01
40	2.40126e-04	3.43137e-02	1.36535e-01	3.1705e-02	2.09614e-02	4.11388e-01
60	1.74081e-04	2.28956e-02	9.69221e-02	2.65476e-02	1.22219e-02	3.74502e-01
80	1.17382e-04	1.72509e-02	8.34451e-02	1.87795e-02	1.16313e-02	3.37304e-01
100	9.64249e-05	1.34699e-02	6.68859e-02	1.12030e-02	1.04119e-02	2.79680e-01

Table 8.1: Empirical KL divergence values for Rayleigh and $\kappa - \mu$ shadowed fading

Similarly, Tables 8.1 (d)-(f) give the empirical KL divergence for the case of $\kappa - \mu$ shadowed fading for different values of β. Here too, we can see that the KL divergence increases with a decrease in L or an increase in β. From the above tables, it is clear that for $L > 60$, the KL divergence values are small and hence our asymptotic distribution very well approximates the true distribution of the maximum, for maximum taken over sequences of length greater than 60.[3] This along with the rate of convergence analysis reaffirms the claim that our asymptotic results can be reliably used for the performance

[2] $Number\ of\ bins = \frac{Max(\{X_i\}) - Min(\{X_i\})}{2 \times IQR \times n^{-1/3}}$ where IQR is the interquartile range.

[3] Note that, similar behaviour of KL divergence is observed for different values of κ, μ and m.

analysis in all scenarios where we need the statistics of the maximum SIR.

8.3.3 Rate of convergence for minima distribution

In the case of minima distribution, we have the following theorem

Theorem 8.7. *The rate of convergence of* $F_{\gamma_{\min}^L}(z)$ *to the Weibull distribution is* $\mathcal{O}\left(L^{-\mu^{-1}} + L^{-1}\right)$ *where* $\gamma_{\min}^L = \min\{\gamma_1, \cdots, \gamma_L\}$.

From this result, we observe that the rate of convergence depends on the length of the sequence L and the source fading parameter μ. The simulated and theoretical distribution are expected to be closer for large values of L. Further, the convergence will be faster for smaller values of μ, the number of multi paths in the source to desired receiver link.

8.4 Analysis of extreme-value distribution

Now that we have derived the asymptotic distribution of the maximum of L i.i.d. SIR RVs where the source and the interferers are assumed to experience i.n.i.d. $\kappa - \mu$ shadowed fading, in this section, we analyze the impact of fading parameters κ, μ and m on the asymptotic distribution. For this, we give the following key lemma.

Lemma 8.2. *Consider two Frechet RVs* P *and* Q *with parameters* $\{a_{L1}, \beta\}$ *and* $\{a_{L2}, \beta\}$ *respectively.* P *is stochastically larger than* Q *if*

$$\mathbb{P}(P < z) < \mathbb{P}(Q < z),\ \forall z > 0. \tag{8.11}$$

In other words, $P >_{st} Q$ *if*

$$exp\left(-\left(\frac{z}{a_{L1}}\right)^{-\beta}\right) < exp\left(-\left(\frac{z}{a_{L2}}\right)^{-\beta}\right). \tag{8.12}$$

The above condition is achieved when $a_{L1} \geq a_{L2}$.

According to Lemma 8.2, the variations in the asymptotic CDF of the maximum SIR is governed by the variations in a_L where $a_L = F_\gamma^{-1}(1 - L^{-1})$. Hence, the variation in

the CDF of the maximum SIR with respect to the variations in the source fading environment can be studied by analyzing the variations in a_L. However, the relationship between various parameters and a_L is highly non linear, and therefore comprehending these variations with respect to changes in the fading parameters is very difficult. One way to circumvent this problem is to use moment matching as in (Kumar, 2015), and approximate each of the $\kappa - \mu$ shadowed RV as a gamma RV. The $\kappa - \mu$ shadowed RV corresponding to the users fading coefficients with parameters $(\kappa, \mu, m, \bar{x})$ can be approximated with a gamma RV with shape parameter $\psi_1 = \frac{m\mu(1+\kappa)^2}{m+\mu\kappa^2+2m\kappa}$ and scale parameter $\psi_2 = \frac{\bar{x}}{\psi_1}$. Similarly, each of the $\kappa - \mu$ shadowed interferer can be first approximated as a gamma RV and their sum can be further approximated by another gamma RV with parameters (ϕ_1, ϕ_2) using (Kumar, 2015, Eqn. (4)). Here, we have $F_\gamma(z) = \mathbb{P}(\gamma \leq z) \approx \mathbb{P}\left(\frac{\mathbf{\Gamma}(\psi_1, \psi_2)}{\mathbf{\Gamma}(\phi_1, \phi_2)} \leq z\right) = \mathbb{P}\left(\frac{\mathbf{\Gamma}(\psi_1, 1)}{\mathbf{\Gamma}(\phi_1, 1)} \leq z\frac{\phi_2}{\psi_2}\right)$, where $\mathbf{\Gamma}(.,.)$ represents a gamma distributed RV. This ratio of gamma RVs has a beta-prime CDF (Dubey, 1970) with parameters ψ_1 and ϕ_1 evaluated at $z\frac{\phi_2}{\psi_2}$. Now, the analysis in (Srinivasan and Kalyani, 2018) can be used to make inferences about the approximate variation in $F_\gamma(z)$, with respect to the changes in κ, μ and m. Based on the analysis, we give the following observations.

Observation 1 **: Scale parameter of the Frechet distribution a_L increases with increase in μ or m.**

Observe that, an increase in μ or m results in an increase in ψ_1. According to $I4$ in Section III of (Srinivasan and Kalyani, 2018), with an increase in ψ_1 along with a proportionate increase in $\bar{x}$, we can observe a decrease in $F_\gamma(z)$. Since CDF is an monotonically increasing function, to obtain the same CDF value of $1 - \frac{1}{L}$ even after an increase in μ or m, the CDF evaluation point, which in our case is a_L, has to increase.

Observation 2 **: Scale parameter of the Frechet distribution a_L increases with increase in κ if $m - \mu \geq 0$ and decreases otherwise .**

The derivative of ψ_1 with respect to κ is given by $\frac{2\kappa(1+\kappa)m\mu(m-\mu)}{(m+2\kappa m+\kappa^2\mu)^2}$. This shows that ψ_1 increases with an increase in κ if $m - \mu > 0$ and decreases otherwise. This in turn implies that the scale parameter $F_\gamma^{-1}(1 - L^{-1})$ increases with an increase in κ, if $m - \mu > 0$ and decreases otherwise. Hence, following the same reasoning given in *Observation 2*, we can infer that an increase in κ increases a_L, if $m - \mu > 0$, owing to the increase in ψ_1. Similarly, an increase in κ results in an decrease in a_L, if $m - \mu < 0$.

Thus *Observation 1*, *Observation 2* along with Lemma 8.2 gives inferences on the

variation of the asymptotic maximum distribution with respect to the changes in the source's fading environment. Further, Table I in (Moreno-Pozas *et al.*, 2016) summarizes the relation between $\kappa - \mu$ shadowed fading model and many common fading models like Rayleigh, Rician, Nakagami, etc. Using these results, we can analyze the variations in the maximum SIR for any specific fading environment as well.

8.5 Applications and Simulations

Our results can be used in any application which involves the maximum SIR statistic. Here, we present some example applications.

8.5.1 Asymptotic outage probability and ergodic rate of MSC system.

The limiting distribution of the maximum SIR RV is useful for the asymptotic performance analysis of channel-aware packet scheduling systems (Song and Li, 2006). Consider a time-slotted downlink channel shared among L users. With MSC scheduling, the channel is assigned to the user with the maximum SIR in each time slot. The authors of (Song and Li, 2006) have analyzed the performance of such a system in a noise-limited scenario under Rayleigh fading channel using EVT. We can readily generalize these results to a $\kappa - \mu$ shadowed fading environment using the results derived in the previous section.

Outage probability

The probability of outage of the user in each time slot of an MSC scheduling system, for a threshold γ_T, in an interference limited scenario is given by

$$\mathbb{P}(\gamma_{max}^{L} \leq \gamma_T) = \prod_{j=1}^{L} F_{\gamma_i}(z), \tag{8.13}$$

where γ_i is the SIR of the ith user. When all the users experience identical fading, the above probability is the same as $(F_\gamma(z))^L$ evaluated at γ_T. Note that, the evaluation of

the true distribution of the maximum given by $(F_\gamma(z))^L$ is not computationally tractable in a $\kappa-\mu$ shadowed fading environment. However, the asymptotic distribution function derived in the previous section can be used to compute this outage probability easily. In the following figures, we show the simulated and asymptotic CDF of maximum for a two interferer scenario.

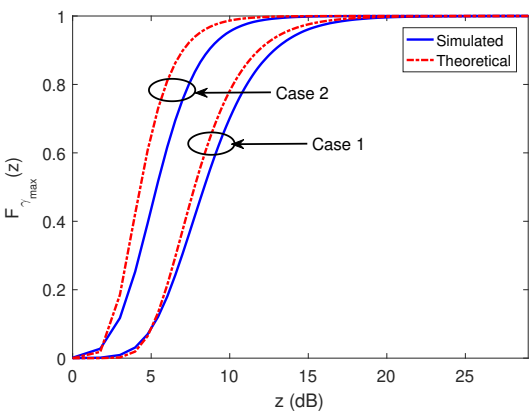


Figure 8.1: CDF of γ_{max}^L for $\kappa-\mu$ shadowed fading with i.n.i.d. interferers, L=64.

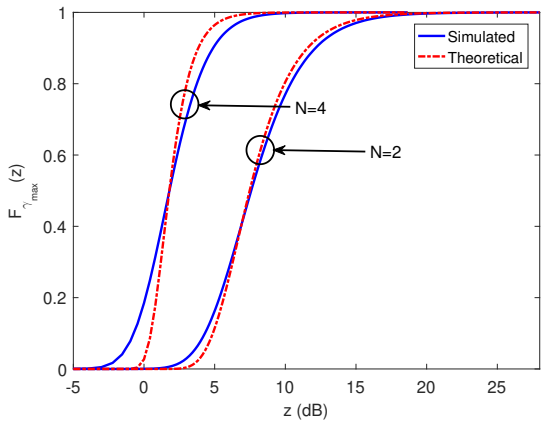


Figure 8.2: CDF of γ_{max}^L for Rayleigh fading with i.i.d. interferers, L=32.

Fig. 8.1, shows the CDFs for the case of $\kappa-\mu$ shadowed fading with i.n.i.d interferers. Here, case 1 corresponds to the scenario where $\kappa = 2, \mu = 2, m = 3$, $\kappa_i = \{2,2\}, \mu_i = \{2,1\}, m_i = \{3,2\}$ and case 2 corresponds to $\kappa = 2, \mu = 1, m = 2$ and $\kappa_i = \{2,2\}, \mu_i = \{1,1\}, m_i = \{2,1\}$. Fig. 8.2 show the case of Rayleigh fading channel for different values of N when $L = 32$.Fig. 8.3-8.6 validates Observation 1 and 2 in Section 8.4. An increase in μ or m increases the scale parameter a_L. From Lemma 8.2, an increase in the scale parameter for a constant shape parameter results in a shift of the CDF to the right. This results in a lesser probability of outage for the same threshold. For clarity, we show the results only for $L = 200$.

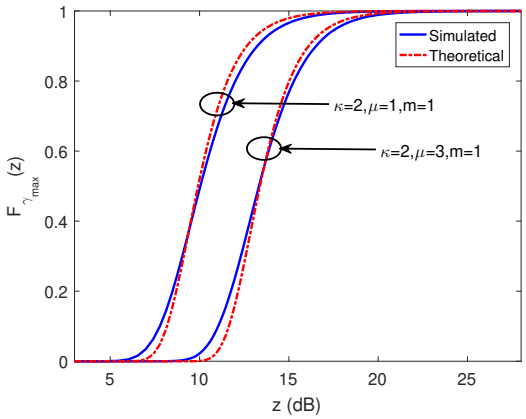


Figure 8.3: CDF for N=1, L=200, $\kappa_I = 2$, $\mu_I = 3$, $m = 1$.

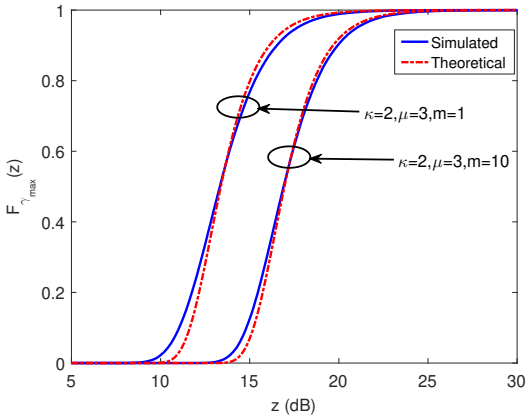


Figure 8.4: CDF for N=1, L=200, $\kappa_I = 2$, $\mu_I = 3$, $m = 1$.

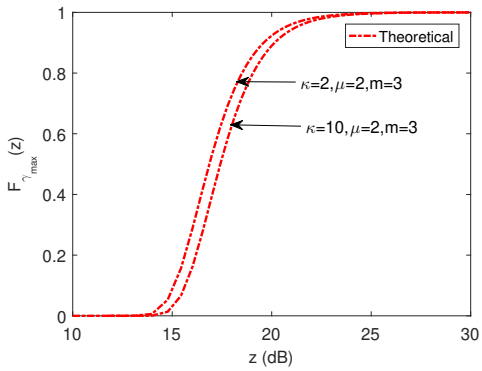


Figure 8.5: CDF for N=1, L=200, $\kappa_I = 2$, $\mu_I = 3$, $m = 1$.

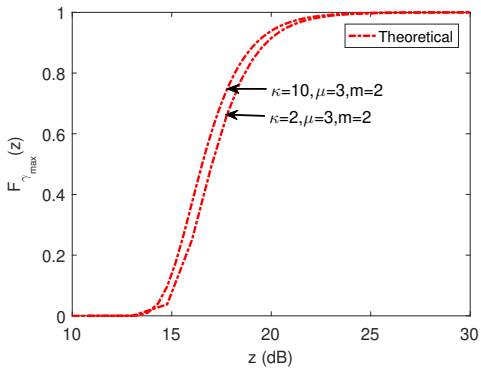


Figure 8.6: CDF for N=1, L=200, $\kappa_I = 2$, $\mu_I = 3$, $m = 1$.

Fig. 8.5 and 8.6 show the variation of CDF with respect to variation in κ. From *Observation 2* in Section 8.4, we know that, with an increase in κ the scale parameter a_L decreases, if $m - \mu$ is positive. This would result in a decrease in the outage probability. Fig. 8.5 shows such a scenario and the result agrees with the expected observation. Fig. 8.6 corresponds to a case where $\mu > m$. In this case, it can be observed that an increase in κ results in an increase in the outage probability. The change in outage probability with the change in κ is not very large. Hence, for clarity, in Figs. 8.5 and 8.6, we have given only the theoretical Frechet distribution curves.

Fig. 8.7 compares the simulated and theoretical values of the first moment of γ_{max}^L for different values of L and N. As discussed in Section 8.2.2, the first moment of the asymptotic distribution converge to the first moment of the original distribution of γ_{max}^L. From the results, it is clear that the simulated and theoretical values of expectation get closer as L increases and this convergence is faster for smaller values of N.

Ergodic rate

The asymptotic ergodic rate of the user in each time slot of the MSC scheduling system is given by

$$R_{max}^{L} = \mathbb{E}\left[\log_2(1+\gamma_{max}^{L})\right]. \tag{8.14}$$

where $\gamma_{max}^{L} = max\{\gamma_1, \cdots, \gamma_L\}$. Recall that γ_{max}^{L} converges in distribution to a Frechet RV γ_{max}, i.e., $\gamma_{max}^{L} \xrightarrow{\text{D}} \gamma_{max}$. We still have to prove that $\lim\limits_{L\to\infty} \mathbb{E}[R_{max}^{L}] = \mathbb{E}[R_{max}]$ where $R_{max} = log_2(1+\gamma_{max})$. To prove this, we first utilize continuous mapping theorem, which is given as follows (Billingsley, 2013):

Theorem 8.8. *Let $\{X_n\}_{n=1}^{\infty}$ be a sequence of random variables and X another random variable, all taking values in the same metric space $\mathcal{X}$. Let $\mathcal{Y}$ be another metric space and $f : \mathcal{X} \to \mathcal{Y}$ a measurable function and $C_f := \{x : \ f \ is \ continuous \ at \ x\}$. Suppose that $X_n \xrightarrow{D} X$ and $\mathbb{P}(X \in C_f) = 1$, then $f(X_n) \xrightarrow{D} f(X)$.*

Let $R_{max}^{L} = log_2(1+\gamma_{max}^{L})$. Since $f(x) = log_2(1+x)$ is a continuous function, using Theorem. 8.8, $R_{max}^{L} \xrightarrow{\text{D}} R_{max}$. Finally, we use monotone convergence theorem, which is given below (Billingsley, 2008).

Theorem 8.9. *Let $g_n \geq 0$ be a sequence of measurable functions such that $g_n(\omega) \to g(\omega)\ \forall\ \omega$ except maybe on a measure zero set and $g_n(\omega) \leq g_{n+1}(\omega),\ n \geq 1$. We then have*

$$\lim_{n\to\infty} \int g_n \, d\mu = \int g \, d\mu. \tag{8.15}$$

Here, we know that $\gamma_{max}^{L} \leq \gamma_{max}^{L+1},\ \forall\ L$ and hence $\mathbb{P}(\gamma_{max}^{L} \leq l) \geq \mathbb{P}(\gamma_{max}^{L+1} \leq l)$. Thus, $1 - F_{\gamma_{max}^{L}}(l) \leq 1 - F_{\gamma_{max}^{L+1}}(l)$. Logarithm is a monotonic function and hence $1 - F_{R_{max}^{L}}(l) \leq 1 - F_{R_{max}^{L+1}}(l)$. For a positive RV X, note that the expectation is given by

$$\mathbb{E}[X] = \int_0^{\infty} \mathbb{P}(X > x)\, dx = \int_0^{\infty} (1 - F_X(x))\, dx. \tag{8.16}$$

Thus, making use of Theorem. 8.9 we have $\lim\limits_{L\to\infty} \mathbb{E}[R_{max}^{L}] = \lim\limits_{L\to\infty} \int\limits_0^{\infty} \mathbb{P}(R_{max}^{L} > l)\, dl = \int\limits_0^{\infty} \lim\limits_{L\to\infty} \mathbb{P}(R_{max}^{L} > l)\, dl = \mathbb{E}[R_{max}]$. Hence, we have the required result. Given that we have proved the convergence of moments of R_{max}^{L} to the moments of R_{max}, for large L, the expectation in (8.14) can now be evaluated using the CDF of Frechet RV

given in (8.5), instead of using the true CDF of γ_{max}^{L}, which is difficult to evaluate. The asymptotic ergodic rate is thus given by $R = \int_0^\infty \log_2(1+z)\ f_{\gamma_{max}}(z)\ dz$, where $f_{\gamma_{max}}(z)$ is the asymptotic probability distribution functions (PDF) of the Frechet RV γ_{max}^{L}. Substituting the Frechet PDF, the previous expression can be rewritten as follows :

$$R = \beta(a_L)^\beta \int_0^\infty \log_2(1+z) z^{-\beta-1} e^{-\left(\frac{z}{a_L}\right)^{-\beta}}\ dz. \tag{8.17}$$

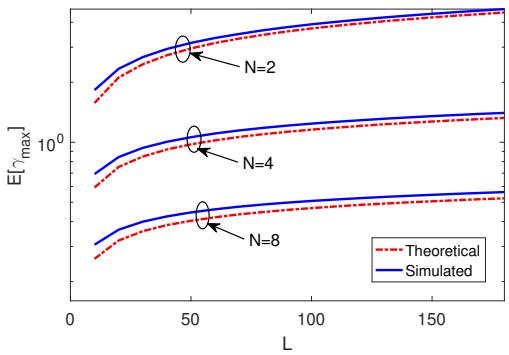


Figure 8.7: L vs $\mathbb{E}[\gamma_{max}^{L}]$ for $\kappa = 2$, $\bar{\mu} = 3$, $m = 2$, $\kappa_I = 2$, $\mu_I = 2$, $m_I = 3$.

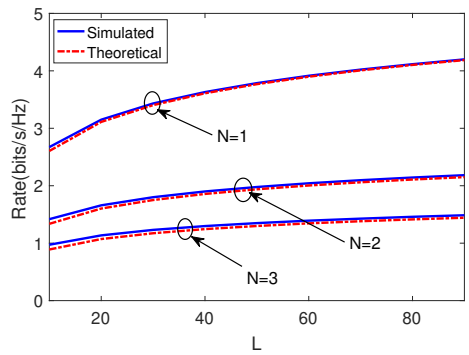


Figure 8.8: L vs ergodic rate for $\{\kappa = 2, \mu = 3, m = 2\}$, $\{\kappa_I = 2, \mu_I = 2, m_I = 3\}$.

From (8.17), we obtain the following observation :

Observation 3 : **Ergodic rate increases with increase in a_L, for a constant shape parameter β.**

Note that both β and z are non-negative and hence $\log_2(1+z)z^{-\beta-1}$ will also be non-negative for all values of β and z. Hence, with an increase in a_L, the rate increases. *Observation 1* and *Observation 2* along with *Observation 3* will facilitate obtaining inferences on the variations of the asymptotic data rate with respect to the $\kappa-\mu$ shadowed fading parameters.

Fig. 8.8 compares simulated values of the rate with the rate computed using (8.17), for different values of L and N. The results show that there is a good match between the simulated and theoretical values over a wide range of L and N. As expected, the rate increases with the number of antennas at the receiver. However, with an increase in the number of interferers SIR signals of smaller magnitude are available at the receivers and hence the rate reduces significantly.

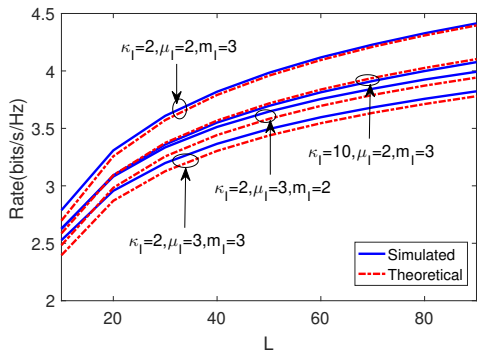


Figure 8.9: L vs ergodic rate for $\{\kappa = 2, \mu = 1, m = 1\}$, N =1.

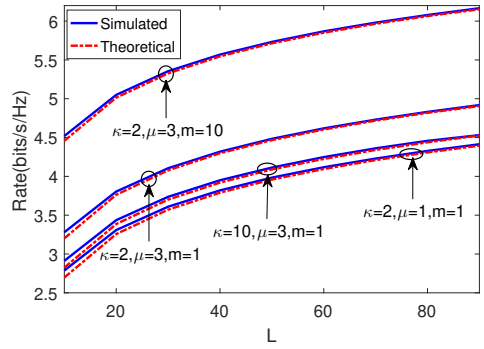


Figure 8.10: L vs ergodic rate for $\{\kappa_I = 2, \mu_I = 2, m_I = 3\}$, N =1.

Fig. 8.9 further shows variation in the rate for changes in the interferer fading parameters (here, interferers are assumed to be i.i.d.). Increase in μ_I, m_I results in stronger interferers and hence results in a decrease in the rate. Further, Fig. 8.10 shows the variation in rate for different values of source fading parameters. As discussed previously for the case of outage probability, increase in m, μ results in better coverage conditions and hence higher rates. Also, similar to the case of outage probability, the variation of the rate with respect to κ depends on the sign of $\mu - m$.

8.5.2 Upper bound for the rate in FAS (Full Antenna Selection) architecture in massive MIMO

In massive MIMO scenarios, there is often a restriction to the number of RF chains available for processing. Dedicated RF chains for each antenna in the massive MIMO antenna array is a cost prohibitive and power hungry design (Gao and Kaiser, 2016). Hence, a very common practice is to choose a subset of antennas from the array for further processing. This is known as the full array switching (FAS) architecture. Very recently, the authors of (Gao and Kaiser, 2016; Gao *et al.*, 2018) discussed the analysis of FAS systems in a noise-limited scenario using EVT, when the channel experiences Rayleigh fading. In this subsection, we derive upper bounds on the rate for FAS architecture in an interference-limited scenario for $\kappa - \mu$ shadowed fading channels. Note that our results will generalize the FAS results of (Gao *et al.*, 2018).

Consider a multi-antenna receiver with L antennas, out of which L_s antennas are selected for further processing. Upper bound on the rate in a FAS scenario would cor-

respond to the condition where the first L_s antennas in terms of SIR are selected. The corresponding bound is given by (Gao *et al.*, 2018) :

$$R_{ub,fas} = \mathbb{E}\left[\sum_{l=1}^{L_s} log_2(1+\gamma_{(l)}^L)\right], \tag{8.18}$$

where $\{\gamma_{(l)}^L\}_{l=1,2,\cdots,L}$ are the ordered SIR RVs; ie. $\gamma_{(1)}^L > \gamma_{(2)}^L > \cdots > \gamma_{(L)}^L > 0$. The convergence of the above moment to the first moment of the limiting distribution is guaranteed by extending the same claims used in the previous section to prove convergence of rate in MSC scheduling systems. Now, to compute the ergodic rate as in (8.18), we need the joint distribution of the first L_s SIR RVs. The exact expression for the joint pdf of L_s ordered i.i.d. random variables $\{x_{(1)} \geq x_{(2)} \geq \cdots \geq x_{(L_s)}\}$ is given by

$$f_{x_{(1)},x_{(2)},\cdots,x_{(L_s)}}(x_{(1)}, x_{(2)}, \cdots, x_{(L_s)}) = L_s! \prod_{l=1}^{L_s} f(x_{(l)}), \tag{8.19}$$

where $f(x)$ is the pdf of each of the RV $x_{(i)}; i = 1, \cdots, L$. The evaluation of (8.18) using the true joint distribution as given in (8.19) will result in a very complex expression due to the complicated nature of the L_s product terms and the L_s fold integration. In fact, even the evaluation of a single term of the product using (L.5) will take close to an hour in Mathematica. However, we make use of the following result from EVT to derive the asymptotic joint distribution of the first L_s RVs.

Lemma 8.3. *Given a sequence of i.i.d RVs $X_1, X_2, \cdots, X_L$ with a common CDF $F(x)$ that belongs to the MDA of one of the three EVD $G(x)$ such that $\frac{max(X_1,X_2,\cdots,X_L)-b_L}{a_L} \xrightarrow{D} G(x)$. Suppose that $X_{(1)} > X_{(2)} > \cdots > X_{(L)}$ is the ordered sequence of $X_1, X_2, \cdots, X_L$ then the L_s dimensional vector $\left(\frac{X_{(1)}-b_L}{a_L}, \frac{X_{(2)}-b_L}{a_L}, \cdots, \frac{X_{(L_s)}-b_L}{a_L}\right)$ has the following asymptotic joint distribution :*

$$g_{1,2,\cdots,L_s}(x_{(1)}, x_{(2)}, \cdots, x_{(L_s)}) = G(x_{(L_s)}) \prod_{l=1}^{L_s} \frac{g(x_{(l)})}{G(x_{(l)})}, \tag{8.20}$$

with $x_{(1)} > x_{(2)} > \cdots > x_{(L_s)}$ and $g(x)$ is the pdf of $max(X_1, X_2, \cdots, X_L)$.

Proof. Please refer to page 219 in (Arnold *et al.*, 1992) for the proof. □

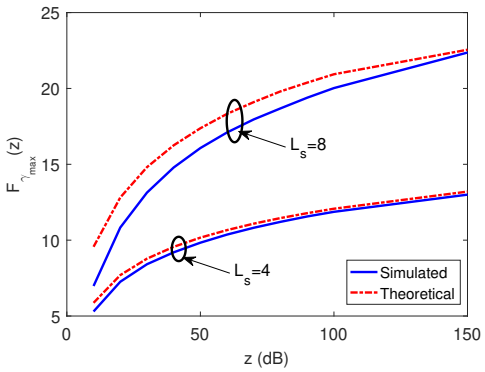


Figure 8.11: Asymptotic upper bound of rate and simulated rate of FAS systems in Rayleigh fading.

The corresponding joint distribution in our case is as follows :

$$f_{x_{(1)},x_{(2)},\cdots,x_{(L_s)}}(x_{(1)},x_{(2)},\cdots,x_{(L_s)}) = \left(\beta(a_L)^{\beta+1}\right)^{L_s} \; exp\left(-\left(\frac{x_{(L_s)}}{a_L}\right)^{-\beta}\right)\prod_{l=1}^{L_s}\left(x_{(l)}\right)^{-1-\beta}. \tag{8.21}$$

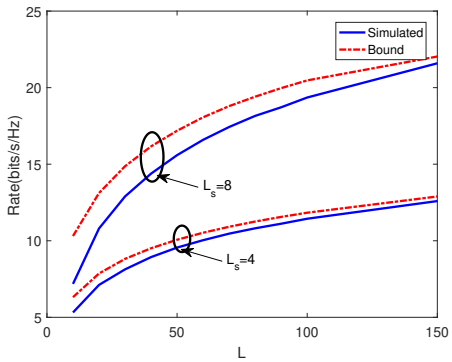


Figure 8.12: Asymptotic upper bound of rate and simulated rate of FAS systems in $\kappa-\mu$ shadowed fading with i.i.d. interferers.

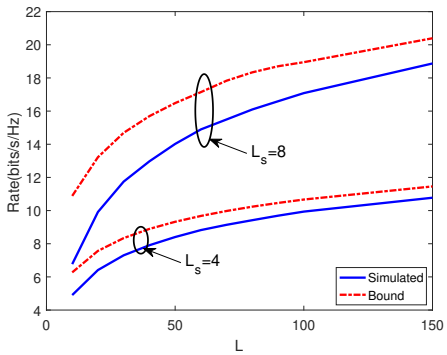


Figure 8.13: Asymptotic upper bound of rate and simulated rate of FAS systems in $\kappa-\mu$ shadowed fading with i.i.d. interferers.

Now, using the expression for the joint distribution of ordered RVs in (8.21) we can compute the upper bound for the rate easily. We show typical simulation results to validate the bounds for FAS systems. Here, the best 4 or 8 antennas are selected for further processing in each case. We can see that the bound is tight for the case of $L_s = 4$ and the bound gets looser for larger values of L_s. Here, the asymptotic statistics of the ordered RVs allows easy evaluation of the upper bound on the rate. Fig. 8.11

shows the asymptotic bound and the simulated rate with different antenna lengths, for the case of FAS architectures in Rayleigh fading environment. Further, Fig. 8.12 and Fig. 8.13 shows the simulated and asymptotic bound for the rate in FAS system in a $\kappa - \mu$ shadowed fading environment. The fading parameters chosen for the simulations in Fig. 12 and Fig. 13 are $\kappa = 2$, $\mu = 3$, $m = 1$, $\kappa_I = 2$, $\mu_I = 1$, $m_I = 1$ and $\kappa = 2$, $\mu = 3$, $m = 1$, $\kappa_1 = 2$, $\mu_1 = 1$, $m_1 = 1$ and $\kappa_2 = 2$, $\mu_2 = 2$, $m_2 = 1$ respectively.

8.5.3 Other applications:

Besides the performance analysis of MSC scheduling and FAS systems, the results derived in our work can also be used in other applications. One can derive the asymptotic outage probability and ergodic rate of the maximum SIR antenna (i.e., selection combining) in a massive MIMO scenario. These metrics describe the performance of the best antenna from the antenna array under a given channel condition and hence are useful for system characterization and resource allocation, especially when limited number of high-resolution RF chains are available at the receiver (Fan *et al.*, 2015; Srinivasan and Kalyani, 2019*a*). Also, note that the asymptotic distribution of the kth maximum SIR RV can be derived easily from the distribution of the maximum RV using (Arnold *et al.*, 1992, Eqn. (8.4.2)). This can be used for the performance analysis of a general selection-diversity (SD) scheme as in (Al-Badarneh *et al.*, 2018). One can also determine the joint distribution of the largest and second largest SIR RV and determine performance bounds for generalized selection combining (GSC) schemes in massive MIMO receivers (Kong and Milstein, 2000; Annamalai and Tellambura, 2001; Chen *et al.*, 2004). Also, our asymptotic distribution of the maximum SIR can be used for the analysis of receive antenna selection schemes in spectrum sharing systems (Hanif *et al.*, 2018).

8.6 Secondary user power control policy

In the underlay mode, the SU-Tx transmits over the same frequency as the PU-Tx, even when the PU-Tx is active. Simultaneous transmission occurs as long as the Quality of Service (QoS) degradation at the PU-Rx due to the interference imposed by the SU-Tx is tolerable. This QoS degradation in the primary network is quantified by means of

different constraints at the PU-Rxs. In this section, we will look at the outage constraint at the PU-Rxs. Here, the SU-Tx must transmit at a power that keeps the outage at each of the PU-Rx below a predetermined level. Thus, transmit power policy at the SU-Tx can be mathematically formulated as follows (Sibomana and Zepernick, 2016; Al-Badarneh *et al.*, 2018),

$$max\ P_s, \tag{8.22a}$$

$$\text{s.t. } \mathbb{P}\{\gamma_{m,p}(P_s) \leq \gamma_0\} \leq p_0, \quad \forall\, m = 1, ..., M \tag{8.22b}$$

$$P_s \leq P_{s,max}, \tag{8.22c}$$

where p_0 is the maximum tolerable outage at each of the PU-Rx and γ_0 is the minimum desired SIR at the PU-Rx for a fixed PU transmit power P_p. The outage constraint in (8.22b) is equivalent to the condition where PU-Rx$_m$ with the lowest SIR satisfy the outage constraint. Hence, the power policy of SU-Tx can be alternatively formulated as

$$max\ P_s \tag{8.23a}$$

$$\text{s.t. } \mathbb{P}\{\min_{1\leq m\leq M} \gamma_{m,p}(P_s) \leq \gamma_0\} \leq p_0 \tag{8.23b}$$

$$P_s \leq P_{s,max}. \tag{8.23c}$$

Substituting the fading coefficients into (8.23b), we obtain

$$max\ P_s \tag{8.24a}$$

$$\text{s.t. } \mathbb{P}\left\{\min_{1\leq m\leq M} \frac{h_m}{\alpha_m} \leq \gamma_0 \frac{P_s}{P_p}\right\} \leq p_0. \tag{8.24b}$$

$$P_s \leq P_{s,max}. \tag{8.24c}$$

Here, $\{h_m; m = 1, \cdots, M\}$ and $\{\alpha_m; m = 1, \cdots, M\}$ are sets of i.i.d. $\kappa - \mu$ shadowed RVs with fading parameters $(\kappa_p, \mu_p, m_p, \bar{h}_p)$ and $(\kappa_{p,s}, \mu_{p,s},\ m_{p,s}, \bar{\alpha}_{p,s})$ respectively.

To determine the optimum value of P_s that satisfies the outage constraint in (8.23b), we have to determine the CDF of the minimum of SIR RVs in a $\kappa - \mu$ shadowed fading environment. Note that, we can evaluate this using the CDF of the minimum of ratio of two $\kappa - \mu$ shadowed RVs as given in (8.24b). The exact distribution of the mini-

mum of any set of i.i.d. RVs $\gamma_{\min} = \min\{\gamma_1, \gamma_2, \cdots, \gamma_M\}$, where $\gamma_i \sim F_\gamma(z); \forall\, i \in \{1, \cdots, M\}$ is given by

$$F_{\gamma_{\min}}(z) = 1 - (1 - F_\gamma(z))^M . \tag{8.25}$$

Hence, to evaluate the CDF in (8.24b), we have to evaluate the Mth power of the CCDF of ratio of $\kappa - \mu$ shadowed RVs. The complex nature of the CDF $F_\gamma(z)$ makes the evaluation of the Mth power of the CDF difficult. Note that even if we compute the exact distribution for large values of M, it will not be possible to derive any meaningful inference from them owing to the complex nature of those expressions. On the other hand, if we use the simple limiting distribution for (8.25), which closely approximates the CDF values, we can obtain a closed-form expression for the optimum P_s that satisfies (8.24b).

Using the asymptotic distribution (8.6), we can now determine the optimum P_s, when the number of PU-Rxs M, is moderate to large. To evaluate the CDF of $\gamma_{\min,p} := \lim_{M\to\infty} \min_m \left\{ \frac{h_m}{\alpha_m}; m = 1, \cdots, M \right\}$ (to approximate (8.24b)), we now substitute $N = 1$, $K = M$, $(\kappa, \mu, m, \bar{x}) = (\kappa_p, \mu_p, m_p, \bar{h}_p)$, $(\kappa_1, \mu_1, m_1, \bar{x}_1) = (\kappa_{p,s}, \mu_{p,s}, m_{p,s}, \bar{\alpha}_{p,s})$, $a_K = a_M = F_\gamma^{-1}\left(\frac{1}{M}\right)$ and $\upsilon = \mu_p$ in Theorem 8.4[4]. Hence, we have (8.23b) is equivalent to the following,

$$1 - \exp\left(- \left(\frac{\gamma_0 P_s}{P_p a_M} \right)^{\mu_p} \right) \leq p_0. \tag{8.26}$$

Further rearrangement of (8.26) gives,

$$P_s \leq \frac{P_p a_M}{\gamma_0} \left[-ln(1 - p_0) \right]^{1/\mu_p} . \tag{8.27}$$

[4]Here, $F_\gamma(z)$ is evaluated using (J.1) for $N = 1$. Even if we consider multiple primary interferers, note that Theorem 1 gives the asymptotic distribution of the minimum SIR for a case where the receiver suffers from the interference of N other transmitters. Therefore, the theoretical framework developed is applicable for a much broader framework. However, when we consider $(N - 1)$ primary interferers having known transmit powers, the expression of the outage probability will be different and we will not have a closed form expression for the secondary user's power allocation. Furthermore, in cells having large cell radius, the interference arising from other primary transmitters can be neglected due to the associated high path loss.

The largest P_s that satisfies the above constraint is given by

$$P_s^+ = \frac{P_p a_M}{\gamma_0} \left[-ln(1-p_0)\right]^{1/\mu_p}. \tag{8.28}$$

Now, using (8.28) and (8.23c), the optimal P_s for the SU-Tx power policy is given by

$$\bar{P}_s = \min\{P_s^+, P_{s,max}\}. \tag{8.29}$$

Now that we have derived the optimal SU-Tx power, we will analyze the impact of fading parameters on this power policy. From (8.29), we can observe that the optimum power at the SU-Tx, $\bar{P}_s$, is dependent on P_s^+ given in (8.28). The variations in P_s^+ are in turn governed by P_P, γ_0, p_0, μ_p and a_M.

Observation 1: From (8.28), it is plausible that an increase in either P_p or p_0 or alternatively a decrease in γ_0 leads to an increase in P_s^+.

Note that, the variation in the SU transmit power P_s^+ with respect to the variations in the desired channel's fading conditions can be studied by analyzing the variations in a_M and μ_p. Since analyzing the effect of channel conditions on a_M would mean analyzing the inverse CDF of SIR, which is difficult, we proceed by approximating each of the $\kappa-\mu$ shadowed RV by a Gamma RV. Following analysis similar to limiting distribution of maximum, we can use stochastic ordering to make inferences about the approximate variation in $F_\gamma(z)$, with respect to the changes in κ_p, μ_p, m_p, $\kappa_{p,s}, \mu_{p,s}$ and $m_{p,s}$. The corresponding observations are presented below.

Observation 2 : P_s^+ increases upon increasing $\boldsymbol{\mu_p}$ or $\boldsymbol{m_p}$ or decreasing $\boldsymbol{\mu_{p,s}}$ or $\boldsymbol{m_{p,s}}$.

Observation 3 : P_s^+ increases upon increasing $\boldsymbol{\kappa_p}$ if $\boldsymbol{m_p - \mu_p \geq 0}$ and decreases otherwise. Alternatively, P_s^+ increases upon decreasing $\boldsymbol{\kappa_{p,s}}$ if $\boldsymbol{m_{p,s} - \mu_{p,s} \geq 0}$ and decreases otherwise.

The PU-Tx's target rate is chosen to be $R_0 = 0.03$ bps/Hz for all the simulations. This is to match the performance target for the operational long-term evolution (LTE) network, which requires the cell edge user throughput to be higher than 0.02 bps/Hz/cell/userSibomana and Zepernick (2016); Sesia *et al.* (2011). Similarly, all the results are generated for the choice of $P_{s,max} = 20\ dB$. Here, Fig. 8.14 shows the SU-Tx power allocation for various combinations of PU-Tx power P_p and

PU-Rx outage constraint p_0 computed using (8.29). Furthermore, we have chosen $(\kappa_p = 3, \mu_p = 2, m_p = 1)$, $(\kappa_{p,s} = 2, \mu_{p,s} = 2, m_{p,s} = 1)$, $(\kappa_s = 2, \mu_s = 2, m_s = 1)$, $(\kappa_{s,p} = 3, \mu_{s,p} = 3, m_{s,p} = 1)$, $M = 10$ and $L = 10$ for generating Figs. 8.14-8.16. The results indicate that the optimum SU-Tx power $\bar{P}_s$ increases upon increasing the PU-Tx power P_p. This is because, upon increasing P_p, the PU-Rxs become capable of handling a higher interference arriving from the SU-Tx at the same outage constraints. Furthermore, for constant P_p, $\bar{P}_s$ decreases with a reduction in p_0. This is because a reduction in p_0 results in stricter outage constraints at the PU-Rxs. In order to satisfy these stricter reliability conditions, the SU-Tx has to transmit at a lower power for reducing the interference at the PU-Rx. Note that the optimum transmit power $\bar{P}_s$ is always limited by $P_{s,max}$. For the power allocation considered in Fig. 8.14, we show the simulated values of outage probabilities of both the primary and of the secondary receiver having lowest SIR (P_p^{out} and P_s^{out}) in Fig. 8.15 and 8.16, respectively, for a threshold of $R_0 = 0.03$. Here, note that we are not constraining the outage probability of the secondary users in the allocation scheme and hence the probability of outage of the secondary users may change with the channel conditions or system model.

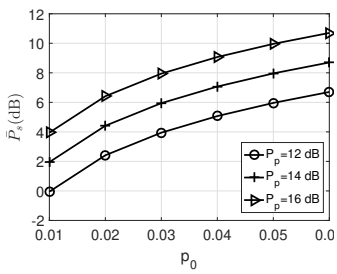


Figure 8.14: p_0 vs $\bar{P}_s$ for M=10 and L=10.

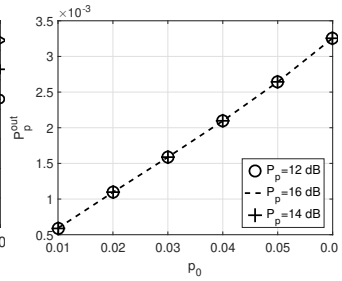


Figure 8.15: p_o vs P_p^{out}.

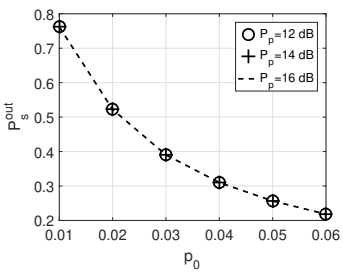


Figure 8.16: p_0 vs P_s^{out}.

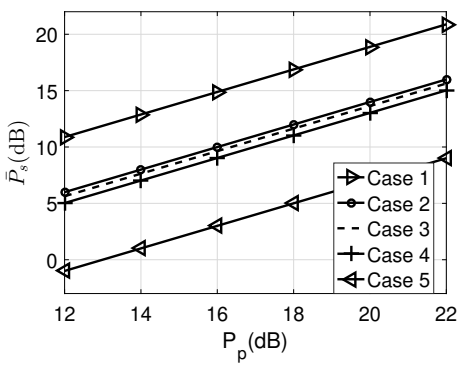


Figure 8.17: P_p vs $\bar{P}_s$ for M=20.

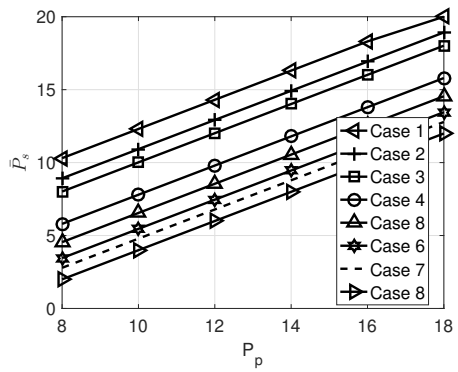


Figure 8.18: P_p vs $\bar{P}_s$ for M=20.

Fig. 8.17 shows plots of P_p versus $\bar{P}_s$ for different channel conditions to validate *Observation 2*. The channel parameters corresponding to the cases shown in the figure are given in Table 8.2. From Cases 1,2 and 5 we can observe an increase in $\bar{P}_s$ with an increase in μ_p and m_p. Similarly, we can observe an decrease in $\bar{P}_s$ with an increase in $\mu_{p,s}$ and $m_{p,s}$ from cases 2,3 and 4. Next, *Observation 3* is validated using simulations in Fig. 8.18. The fading channel parameters used for simulation are given in Table 8.3. According to *Observation 3*, variation in $\bar{P}_s$ with changes in κ_p or $\kappa_{p,s}$ depends upon the sign of $\mu_p - m_p$ and $\mu_{p,s} - m_{p,s}$ respectively. We verify all such variations possible using cases 1-8 in Fig. 8.18. The above figures validate the claim that the proposed asymptotic results can be readily used to derive inferences on the system performance. Without the proposed simple distribution for the minimum SIR RV, predicting the changes in the underlay CRN performance with respect to variations in channel fading conditions would have been non-trivial.

Case #	κ_p	μ_p	m_p	$\kappa_{p,s}$	$\mu_{p,s}$	$m_{p,s}$
1	3	2	1	2	1	1
2	3	1	1	2	1	1
3	3	1	1	2	20	1
4	3	1	1	2	1	10
5	3	1	0.1	2	1	1

Table 8.2: Channel parameters used for simulation of Fig.8.17.

Case #	κ_p	μ_p	m_p	$\kappa_{p,s}$	$\mu_{p,s}$	$m_{p,s}$
1	3	2	1	2	1	1
2	10	2	1	2	1	1
3	3	1	10	2	1	2
4	3	1	10	10	1	2
5	10	1	2	2	1	1
6	3	1	2	2	1	1
7	3	1	1	30	2	1
8	3	1	1	2	2	1

Table 8.3: Channel parameters used for simulation of Fig.8.18

8.7 Conclusions

We considered L wireless links in the presence of N co-channel interferers experiencing non-identical $\kappa-\mu$ shadowed fading conditions with $\{\kappa, \mu, m, \bar{x}\}$ and $\{\kappa_i, \mu_i, m_i, \bar{y}_i; i = 1, \cdots, N\}$ being the fading parameters of the source and interferers respectively. Let, $\gamma = \gamma_j, \ \forall\, j \ \in \ \{1, \cdots, L\}$ represent the SIR available at each user/antenna and γ_{max}^L is the maximum over all γ_j. Following are our key results and observations :

- The asymptotic distribution of γ_{max}^{L} is a Frechet distribution with scale parameter $a_L = F_{\gamma}^{-1}(1 - L^{-1})$ and shape parameter $\beta = \sum_{i=1}^{N} \mu_i$. The asymptotic distribution of γ_{min}^{L} is a Weibull distribution with scale parameter $a_L = F_{\gamma}^{-1}(L^{-1})$ and shape parameter μ.

- The νth moment of γ_{max}^{L} converges to the νth moment of the corresponding Frechet distribution for all $\nu < \sum_{i=1}^{N} \mu_i$.

- The rate of convergence of $F_{\gamma_{max}^{L}}(z)$ to the Frechet distribution is

$$O\left(L^{-\left(\sum_{i=1}^{N}\mu_i\right)^{-1}} + L^{-1}\right)$$

  . From this result, we can observe that the maximum deviation between the true distribution of the maximum and the asymptotic distribution of the maximum over all the points decreases with an increase in L. Also, in order to demonstrate the practical validity of the work, we studied the empirical KL divergence between the exact distribution of the maximum and the corresponding asymptotic distribution. The KL divergence results indicated the closeness between the asymptotic results and the exact results, even for finite L. The rate of convergence of $F_{\gamma_{min}^{L}}(z)$ to the Weibull distribution is $O\left(L^{-(\mu)^{-1}} + L^{-1}\right)$.

- Using results from stochastic ordering, the variation in the behaviour of the asymptotic outage probability and asymptotic ergodic rate with respect to variations in the source fading environment is studied.

- We analyze the utility of the derived asymptotic results in the following applications :
  1. Analysis of asymptotic outage probability and asymptotic ergodic rate of the user in each time slot of an MSC system.
  2. Derivation of the asymptotic upper bound on the rate in FAS architectures for antenna selection.
  3. Determining the secondary user power policy in cognitive radio scenario.

Further, simulations are provided to validate the above results and to confirm the utility of our results. As the $\kappa - \mu$ shadowed fading model is a generalized fading model encompassing most of the general fading scenarios as special cases, our results can be used in a number of problem scenarios involving maxima/minima statistics.

CHAPTER 9

Conclusions

In this thesis, we first explored three applications in the context of generalized fading channels. The common feature among all these applications was that either the actual performance metric expressions are mathematically complicated or expressions do not exist at all. For all these applications, we derived simple approximations that are easy to analyze and draw inferences. The first application is a classic three-node model, where the transmitter communicates with a receiver in the presence of an eavesdropper. Approximate SPSC expressions were derived for the case when all the channels were subject to $\kappa - \mu$ shadowed fading. From the approximate expression, which is in terms of a Gauss hypergeometric function, a number of interesting inferences on the behaviour of SPSC were drawn. The second application considered the uplink of a massive-MIMO network with low resolution analog-to-digital converter (ADC) architecture. By means of moment matching and second order bi-variate Taylor series expansion, the signal-to-interference and quantization noise ratio (SIQNR) was approximated by a gamma random variable (RV). Using this approximate pdf, tight approximations for metrics like OP and rate were derived. In this chapter, we considered receiver diversity system in the presence of multiple unequal power uncorrelated $\eta - \mu$ interferers. The third application considered a MRC system when the desired user and the interfering signals are subject to $\eta - \mu$ fading. In this chapter, we considered receiver diversity system in the presence of multiple unequal power uncorrelated $\eta - \mu$ interferers. Closed-form approximate outage probability and rate expressions were also derived.

Approximate symbol error rate (SER), outage probability and rate expressions have been derived for optimum combining (OC) for the case of Rician faded users and Rician faded interferers. SER is also derived for an interference-limited scenario and the expressions obtained are significantly simpler than the existing expressions. Nevertheless, these expressions were complicated, even when the eigenvalue statistics of non-central Wishart distribution were available. For the case of $\kappa - \mu$ or $\eta - \mu$ random variables, we developed approximate random matrix models in terms of complex Wishart matrix having the least KL-divergence from the original matrix distribution. The utility of

our result is shown by a) deriving approximate capacity expressions for $\kappa - \mu$ or $\eta - \mu$ MIMO models b) deriving approximate outage expressions of MIMO-MRC for $\kappa - \mu$ or $\eta - \mu$ channels c) deriving ergodic rate expressions for zero-forcing (ZF) receiver in an uplink single cell massive MIMO scenario with low resolution ADCs in the antennas.

We considered L wireless links in the presence of N co-channel interferers experiencing non-identical $\kappa - \mu$ shadowed fading conditions. We showed that the asymptotic distribution of the maximum of the L SIR random variables is a Frechet distribution and that of the minimum is a Weibull distribution. Using results from stochastic ordering, the variation in the behaviour of the asymptotic outage probability and asymptotic ergodic rate with respect to variations in the source fading environment is studied. We also analyzed the utility of the derived asymptotic results in the analysis of asymptotic outage probability and asymptotic ergodic rate of the user in each time slot of an MSC system, the derivation of the asymptotic upper bound on the rate in FAS architectures for antenna selection and determining the secondary user power policy in cognitive radio scenario.

9.1 Some possible extensions

1. Note that the first three applications discussed matching the first two moments of SINR/SIR RVs to that of gamma RV. An interesting future work will be determining the gamma RV that minimizes the KL-divergence and compare it against the moment-matched gamma RV. This may yield a better fit at the cost of increase in mathematical complexity.

2. In the case of the matrix approximation for $\kappa - \mu$ fading channels, determining the degree of freedom involves some complicated algebraic calculations. An algorithmic or machine learning approach to determining the degree of freedom will be an interesting extension. Also, another possibility will be exploring distributions other than Wishart for approximating the matrix distributions. Again, this will increase the mathematical complexity.

3. We also believe extending OC analysis to take into account receiver side correlation is an interesting future work.

APPENDIX A

Truncation error for 3.7

The error due to truncation of the infinite summations in (3.7) till $p = M - 1$ and $q = N - 1$ is given by

$$E(M,N) = D\Big[\sum_{p=0}^{\infty}\sum_{q=N}^{\infty} A_{pq}G_{pq} + \sum_{p=M}^{\infty}\sum_{q=0}^{N-1} A_{pq}G_{pq}\Big]. \tag{A.1}$$

We can identify G_{pq} to be the cdf of a beta-prime distribution given by (Cordeiro and Lemonte, 2012, Eq 2), evaluated at $\frac{y}{x}$. Since the cdf takes only values less than unity, we obtain

$$E(M,N) < D\Big[\sum_{p=0}^{\infty}\sum_{q=N}^{\infty} A_{pq} + \sum_{p=M}^{\infty}\sum_{q=0}^{N-1} A_{pq}\Big]. \tag{A.2}$$

The second term can again be upper bounded by $\sum_{p=M}^{\infty}\sum_{q=0}^{\infty} A_{pq}$. The truncation error now becomes

$$E(M,N) < D\Big[\sum_{p=0}^{\infty}\sum_{q=N}^{\infty} A_{pq} + \sum_{p=M}^{\infty}\sum_{q=0}^{\infty} A_{pq}\Big]. \tag{A.3}$$

This can be further simplified using identities (Abramowitz and Stegun, 1972, Eq. 15.1.1), (Gradshteyn and Ryzhik, 2007, Eq 1.110), and (Abramowitz and Stegun, 1972, 15.3.4) and by canceling redundant terms to obtain

$$\begin{aligned} E(M,N) < \Big[& \frac{{}_2F_1\left(N, N_E m_e + N, N+1, \frac{-Y}{1-Y}\right)(N_E m_e)_N}{\left(\frac{Y}{1-Y}\right)^{-N} N!} \\ & + \frac{{}_2F_1\left(M, N_R m_b + M, M+1, \frac{-X}{1-X}\right)(N_R m_b)_M}{\left(\frac{X}{1-X}\right)^{-M} M!}\Big]. \end{aligned} \tag{A.4}$$

Both the terms are cdf of beta-prime distributed random variables with parameters $(N, N_E m_e)$ and $(M, N_R m_b)$ and evaluated at $\frac{Y}{1-Y}$ and $\frac{X}{1-X}$ respectively (Cordeiro and Lemonte, 2012, Eq 2). From I2, we can observe that the cdf evaluated at $N+1$ is smaller than the cdf evaluated at N. Similarly, the cdf evaluated at $M+1$ is smaller than the cdf evaluated at M. This implies that the truncation error bound decreases with increasing M and N. With $M = 50$ and $N = 50$, we get an upper bound of 10^{-1} on the truncation error for $N_R = N_E = 2$, $\kappa_b = \kappa_e = 3$, $\mu_b = \mu_e = 5$ and $m_b = m_e = 6$.

As m_b or m_e increases, or κ_b, κ_e, μ_b or μ_e decreases, $\frac{Y}{1-Y}$ and $\frac{X}{1-X}$ decreases leading to a tighter bound.

APPENDIX B

Determining moments for gamma RV

To determine $\mathbb{E}[X_n]$, first observe that $|g_{mn}|^2 \sim \mathcal{G}\left(m_n, \frac{\beta_n}{m_n}\right)$. From the moment expressions of gamma RV in (Papoulis and Pillai, 2002, Eq 5.106) we can obtain $\mathbb{E}[||\mathbf{g}_n||^2] = Mm_n\frac{\beta_n}{m_n}$ and $Var[||\mathbf{g}_n||^2] = Mm_n\frac{\beta_n^2}{m_n^2}$. This implies that $\mathbb{E}[||\mathbf{g}_n||^4] = \frac{\beta_n^2}{m_n^2}\left(M^2m_n^2 + Mm_n\right)$. Therefore,

$$\mathbb{E}[X_n] = \alpha^2\frac{\beta_n^2}{m_n^2}(M^2m_n^2 + Mm_n). \tag{B.1}$$

Similarly, $\mathbb{E}[||\mathbf{g}_n||^8] = \frac{\beta_n^4\Gamma(4+Mm_n)}{m_n^4\Gamma(Mm_n)}$. This implies that

$$\mathbb{E}(X_n^2) = \alpha^4\frac{\beta_n^4}{m_n^4}\frac{\Gamma(4+Mm_n)}{\Gamma(Mm_n)}. \tag{B.2}$$

By expanding $|\mathbf{g}_n^H\mathbf{g}_i|^2$ and applying expectation term by term, we obtain $\mathbb{E}[|\mathbf{g}_n^H\mathbf{g}_i|^2] = \frac{\beta_n}{m_n}\beta_i Mm_n$. Therefore

$$\begin{aligned}\mathbb{E}[Y_n] = \alpha^2 \sum_{i=1,i\neq n}^{N} \frac{\beta_n}{m_n}\beta_i Mm_n + \alpha Mm_n\frac{\beta_n}{m_n} + \alpha(1-\alpha) \\ \times Mm_n\left(\frac{\beta_n}{m_n}\sum_{i=1,i\neq n}^{N}\beta_i + \frac{\beta_n^2}{m_n^2}(m_n+1)\right).\end{aligned} \tag{B.3}$$

Similarly, we can determine $\mathbb{E}[Y_n^2]$ by squaring Y_n and determining the expectation term by term. Hence,

$$\begin{aligned}\mathbb{E}[Y_n^2] &= \alpha^2(M^2m_n^2 + Mm_n)a \\ &+ (\alpha^2 - \alpha^4)\sum_{i\neq n} M(Mm_nm_i + m_n + m_i + 1)b(i) \\ &+ \alpha^4\sum_{i\neq n} M(2Mm_nm_i - m_nm_i + m_n + m_i + 1)b(i) \\ &+ \alpha^2(1-\alpha)^2M(m_n+1)[Mm_n^2 + Mm_n + 4m_n + 6]c \\ &+ 2\alpha^2(1-\alpha)M(m_n+1)(Mm_n+2)d,\end{aligned} \tag{B.4}$$

where $a = \frac{\beta_n^2}{m_n^2}\left(\sum_{i,j=1,i,j\neq n,j\neq i}^{N}\beta_i\beta_j + 2\sum_{i=1,i\neq n}^{N}\beta_i + 1\right)$, $b(i) = \frac{\beta_n^2\beta_i^2}{m_n m_i}$, $c = \frac{\beta_n^4}{m_n^3}$ and $d = \frac{\beta_n^3}{m_n^2}\left(\sum_{i=1,i\neq n}^{N}\beta_i + 1\right)$. Finally,

$$\begin{aligned}\mathbb{E}[X_nY_n] = \alpha^3 M(Mm_n+2)\big((Mm_n+1)d \\ + (1-\alpha)(m_n+1)(Mm_n+3)c\big).\end{aligned} \tag{B.5}$$

Using all the derived expressions, one can obtain finally compute $\mathbb{E}[\tilde{\gamma}_n]$ and $\mathbb{E}[\tilde{\gamma}_n^2]$ using (4.8) and (4.9).

APPENDIX C

Simplification of $N_{i,j}$

We first substitute the value of $h(t,x)$ from (6.6) for N_{ij} $j=1$ and then use the following identities Gradshteyn and Ryzhik (2007), for $p<q$ and Re$(s)>0$,

$$\int_0^\infty e^{-x}x^{s-1}\,{}_pF_q(a_1,..,a_p;b_1,..,b_q;ax)dx$$
$$= \Gamma(s)\,{}_{p+1}F_q(s,a_1,..,a_p;b_1,..,b_q;a), \quad \text{(C.1)}$$

$$\int_0^\infty e^{-x}x^{s-1}dx = \Gamma(s), \quad \text{(C.2)}$$

to solve the integrals in $N_{i,j}$ entries for $j=2,..,n_1$ in (6.9). For $j=1$, $i=1,...,L$ and $i=L+1,...,n_1$, the integrals to be solved are of the form,

$$I = \int_0^\infty \frac{p+x}{v-x}e^{-x}x^z({}_0F_1(q,w_ix))({}_1F_1(1;N_R;\frac{u}{x-v}))dx, \quad \text{(C.3)}$$

where $p=\sigma^2/E_I$, $q=n_2-n_1+1$, $u=aN_RsE_D/E_I$, $v=bsE_D/E_I-\sigma^2/E_I$ and z is a positive integer greater than zero. To obtain a solution for I, we substitute the series expansion for ${}_0F_1$ and ${}_1F_1$, and interchange summations and integration. The integral to be solved becomes,

$$I = \sum_{k=0}^\infty \frac{w_i^k}{(q)_k k!}\left[\sum_{l=0}^\infty \frac{u^l(1)_l}{(N_R)_l l!}\int_0^\infty \frac{p+x}{v-x}e^{-x}\frac{x^{k+z}}{(x-v)^l}dx\right]. \quad \text{(C.4)}$$

The justification for the interchange of summations and integration can be done in two steps. We first expand the ${}_0F_1$ term and apply Tonelli's theorem to justify the exchange of the first summation. The ${}_1F_1$ term is now expanded and the summation is interchanged with the integration using Lebesgue dominated convergence theorem. Let $A1=\int_0^\infty \frac{p}{v-x}e^{-x}\frac{x^{k+z}}{(x-v)^l}dx$ and $A2=\int_0^\infty \frac{x}{v-x}e^{-x}\frac{x^{k+z}}{(x-v)^l}dx$. The Tricomi function or confluent Hyper-geometric function of the second kind is given by Gradshteyn and Ryzhik (2007), $U(\alpha,\gamma,z)=\frac{1}{\Gamma(\alpha)}\int_0^\infty e^{-zt}t^{\alpha-1}(1+t)^{\gamma-\alpha-1}dt$ for Re$(\alpha)>0$, Re$(z)>$

0. In our case $-v > 0$ and $k + z > 0$. Hence using the above identity, A_1 and A_2 can be simplified as,

$$A1 = -p\Gamma(k+z+1)(-v)^{l-k-z}U(k+z+1, k+z+1-l, -v), \tag{C.5}$$

$$A2 = -\Gamma(k+z+2)(-v)^{l-k-z-1}U(k+z+2, k+z+2-l, -v). \tag{C.6}$$

Further, using the functional identity $U(a, b, z) = z^{1-b}U(a-b+1, 2-b, z)$ from Abramowitz and Stegun (1972), (C.5) and (C.6) are simplified and substituted back in (C.4) to obtain,

$$\begin{aligned} I = \sum_{k=0}^{\infty} \frac{w_i^k}{(q)_k k!} \Bigg[\sum_{l=0}^{\infty} \frac{u^l (1)_l}{(N_R)_l l!} \Bigg[&- \Gamma(k+z+2) \\ &\times U(l+1, l-k-z, -v) \\ &- p\Gamma(k+z+1)U(l+1, -k-z+l+1, -v) \Bigg] \Bigg]. \end{aligned} \tag{C.7}$$

To reduce the computation time, we can use the recurrence identity for Tricomi hypergeometric functions given in (Wolfram Research, 2020*d*). To prove the convergence of the above infinite summation, first consider the summation $I_1 = \sum_{k=0}^{\infty} \frac{w_i^k}{(q)_k k!} \Bigg[\sum_{l=0}^{\infty} \frac{|u|^l (1)_l}{(N_R)_l l!} \Gamma(k+z+2)U(l+1, l-k-z, -v) \Bigg]$. From Theorem 3 in C.M. Joshi (1981), we get the identity $U(a, b, x) < x^{-a}$ for $x > 0,\ a > 0$ and $a - b + 1 > 0$. In our case, we can see that $a = l + 1 > 0$ and $a - b + 1 = k + z + 2 > 0$ and $x = -v > 0$. Therefore,

$$\begin{aligned} I_1 &< \sum_{k=0}^{\infty} \frac{w_i^k}{(q)_k k!} \sum_{l=0}^{\infty} \frac{|u|^l (1)_l}{(N_R)_l l!} \Gamma(k+z+2)(-v)^{l+1} \\ &= (-v)\Gamma(z+2) \sum_{k=0}^{\infty} \frac{w_i^k (z+2)_k}{(q)_k k!} \sum_{l=0}^{\infty} \frac{|uv|^l (1)_l}{(N_R)_l l!} \\ &= (-v)\Gamma(z+2)\, {}_1F_1(z+2, q, w_i)\, {}_1F_1(1, N_R, |uv|). \end{aligned}$$

The last equality is obtained from the series expansion definition of ${}_1F_1$ Hypergeometric function Abramowitz and Stegun (1972). A similar argument can be used to prove the absolute convergence of the other infinite summation. Hence, I is convergent, which implies that we can truncate the double summation to T_1 and T_2 values such that $I - \sum_{l=0}^{T_1} \sum_{k=0}^{T_2} \frac{u^l}{(N_R)_l} \frac{w_i^k}{(q)_k} \Bigg[-(k+1)U(l+1, l-k, -v) - pU(l+1, -k+l+1, -v) \Bigg] \leq \epsilon$

for any $\epsilon > 0$. Hence, the simplified $N_{i,j}$ entry is given by (6.10).

Analysis of truncation error

An exact analysis of the truncation error is mathematically intractable. So, we upper bound the truncation error by an upper bound and determine how the bound varies with various parameters. The magnitude of the error in truncation is

$$\begin{aligned} E(T_1, T_2) = \Big| \sum_{k=0}^{\infty} \frac{w_i^k}{(q)_k k!} \sum_{l=0}^{\infty} \frac{u^l (1)_l}{(N_R)_l l!} \Gamma(k+z+2) \\ U(l+1, l-k-z, -v) - \sum_{k=0}^{T_1} \frac{w_i^k}{(q)_k k!} \sum_{l=0}^{T_2} \frac{u^l (1)_l}{(N_R)_l l!} \\ \Gamma(k+z+2) U(l+1, l-k-z, -v) \Big|. \end{aligned}$$

From Theorem 3 in C.M. Joshi (1981), we get the identity $U(a, b, x) < x^{-a}$ for $x > 0$, $a > 0$ and $a - b + 1 > 0$. In our case, we can see that $a = l + 1 > 0$ and $a - b + 1 = k + z + 2 > 0$ and $x = -v > 0$. Hence, the truncation error can be upper bounded by

$$\begin{aligned} E(T_1, T_2) < |v| \sum_{k=0}^{T_1} \frac{w_i^k}{(q)_k k!} \left[\sum_{l=T_2+1}^{\infty} \frac{|uv|^l (1)_l}{(N_R)_l l!} \Gamma(k+z+2) \right] \\ + |v| \sum_{k=T_1+1}^{\infty} \frac{w_i^k}{(q)_k k!} \left[\sum_{l=0}^{T_2} \frac{|uv|^l (1)_l}{(N_R)_l l!} \Gamma(k+z+2) \right] \\ + |v| \sum_{k=T_1+1}^{\infty} \frac{w_i^k}{(q)_k k!} \left[\sum_{l=T_2+1}^{\infty} \frac{|u|^l (1)_l}{(N_R)_l l!} \Gamma(k+z+2) \right]. \end{aligned}$$

Combining the last two terms, then upper-bounding the first term, and finally using hypergeometric expansion identity for ${}_1F_1$, we obtain

$$\begin{aligned} E(T_1, T_2) < |v| \Gamma(z+2) [{}_1F_1(z+2, q, w_i) \frac{|uv|^{T_2+1}}{(N_R)_{T_2+1}} \\ {}_1F_1(1, N_R + T_2 + 1, |uv|) + \frac{(z+2)_{T_1+1} w_i^{T_1+1}}{(q)_{T_1+1} (T_1+1)!} \\ {}_1F_1(1, N_R, |uv|) \\ {}_2F_2(z + T_1 + 3, 1, q + T_1 + 1, T_1 + 2, w_i)]. \end{aligned}$$

As the Rician factor κ_i at the interferers decreases, the eigenvalues of the centrality matrix $\mathbf{M}\mathbf{M}^H$, given by w_i, decreases. Therefore, $E(T_1, T_2)$ decreases and the bound becomes tighter. This implies that we need lesser terms in the infinite summation as κ_i increases. Also, observe that a decrease in $|u|$ or $|v|$ decreases $E(T_1, T_2)$. A decrease in $|u|$ or $|v|$ is true for an increase in E_I or σ^2.

APPENDIX D

Approximation for Rayleigh interferers

Consider the expression to be simplified, $M_\eta(s) = c\det\left(\mathbf{N}_{\sigma^2=0,L=0}\right)$, where

$$c = \frac{((n_2-n_1)!)^{-n_1}}{\prod_{i=1}^{n_1}(n_1-i)!}(-1)^{N_R}(\sigma^2/E_I)^{(N_R-n_1)}$$

and $\mathbf{N}_{\sigma^2=0,L=0}$ is from (6.10) for $L=0$. First, the common terms inside each column or row of the determinant are taken out of the determinant and canceled with the existing terms in the constant c. All columns $j=2,...,n_1$ are flipped and all rows $i=1,..,n_1$ are flipped. The term $\Gamma(n_2-n_1+i+j-1)$ is then removed from each row to obtain $\tilde{\mathbf{N}}$. Now, $M_\eta(s) = c\det\left(\tilde{\mathbf{N}}\right)$, where

$$c = \frac{\prod_{i=1}^{n_1}(n_2-n_1+i)!\left(\frac{\sigma^2}{E_I}\right)^{(N_R-n_1)}}{\prod_{i=1}^{n_1}(n_1-i)!\prod_{i=1}^{n_1}(n_2-i)!}(-1)^{N_R+1}, \tag{D.1}$$

$$\tilde{\mathbf{N}}_{i,j} = \begin{cases} A(i) & j=1,\ i=1,...,n_1, \\ 1, & j=2,\ i=1,...,n_1, \\ \prod_{k=1}^{j-2}(n_2-n_1+i+k), & j=3,...,n_1, 1\le i\le n_1, \end{cases} \tag{D.2}$$

where

$$A(i) = \frac{1}{(n_2-n_1+i)!}\Bigg[\sum_{l=0}^{T_1}\frac{(aN_RsE_D/E_I)^l}{(N_R)_l}\Bigg[-\Gamma(n_2-N_R+i+1)$$
$$U(l+1,l-n_2+N_R-i+1,-bsE_D/E_I)\Bigg] - \sum_{t=1}^{N_R-n_1}\frac{{}_1F_1(t;N_R;aN_RsE_D/(-bsE_D)}{(bsE_D/E_I)^t}$$
$$\times\Bigg[\Gamma(t+n_2-N_R+i)\Bigg]\Bigg]$$

From Normand (2004), shifted factorials are defined by,

$$(z)_{s;n} = \begin{cases} 1, & n=0, \\ z(z+s)....(z+(n-1)s), & n=1,2,... \end{cases} \tag{D.3}$$

A special case of this is the Pochhammer's symbols, when $s = 1$.

$$(z)_n = (z)_{1;n} = \begin{cases} 1, & n = 0, \\ z(z+1)....(z+(n-1)), & n = 1, 2, ... \end{cases} \tag{D.4}$$

In our case, in the $\tilde{\mathbf{N}}$ matrix, we have such Pochhammer's symbols in all columns except in the first. From (Normand, 2004, Lemma.1), we have the relation that determinant of a matrix with ij^{th} element for $0 \leq i, j \leq n-1$, being a shifted factorial $(z_j)_{s;i}$ is given by $\det\left((z_j)_{s;i}\right) = \Delta_n(\mathbf{z})$, where $\Delta_n(\mathbf{z}) = \prod_{0 \leq i < j \leq n-1}(z_j - z_i)$. Evaluating $\det\left(\tilde{\mathbf{N}}\right)$ by Laplace expansion along the first column and using the above relation from Normand (2004), we get

$$\det\left(\tilde{\mathbf{N}}\right) = \sum_{i=1}^{n_1}(-1)^{i+1}A(i)\Delta^i_{n_1-1}(\mathbf{z}), \tag{D.5}$$

where $\mathbf{z} = [n_2 - n_1 + 1 + 1, n_2 - n_1 + 1 + 2, ..., n_2 - n_1 + 1 + n_1]$ and $\Delta^i_{n_1-1}(z)$ is the Vandermonde determinant formed by all elements of the vector $\mathbf{z}$ except the ith element. Any Vandermonde determinant remains unchanged if from each element of the matrix, one subtracts the same constant, i.e., $\Delta_n(\mathbf{z} + c) = \prod_{0 \leq k < j \leq n-1}((z_j + c) - (z_k + c)) = \prod_{0 \leq k < j \leq n-1}(z_j - z_k)$. Hence, the constant $n_2 - n_1 + 1$ can be subtracted from each element of the vector $\mathbf{z}$. Hence,

$$\det\left(\tilde{\mathbf{N}}\right) = \sum_{i=1}^{n_1}(-1)^{i+1}A(i)\Delta^i_{n_1-1}(\mathbf{z}), \tag{D.6}$$

where $\mathbf{z} = [1, ..., n_1]$. The Vandermonde determinant $\Delta_n(\mathbf{z})$, whose nodes are given by first n_1 integers, i.e., $\mathbf{z} = [1, ..., n_1]$, is given by $\Delta_n(\mathbf{z}) = \prod_{1 \leq k < j \leq n_1}(j - k)$. For simplifying this expression, we expand the double product as follows:

$$\Delta_n(\mathbf{z}) = (n_1 - 1)!(n_1 - 2)!...(1)! = \prod_{j=1}^{n_1}(j-1)!. \tag{D.7}$$

However, we actually want to evaluate $\Delta^i_{n_1-1}(\mathbf{z})$ and not $\Delta_n(\mathbf{z})$. Note that the Vandermonde determinant $\Delta^i_{n_1-1}(\mathbf{z})$ in which the ith element is missing, is given by, $\Delta^i_{n_1-1}(\mathbf{z}) = \prod_{1 \leq k < j \leq n_1; k, j \neq i}(j - k)$. Note that the above expression is difficult to evaluate. Hence, to obtain a simplified expression we multiply and divide the expression for $\Delta^i_{n_1-1}(\mathbf{z})$

by the terms that are present in $\Delta_{n_1}(\mathbf{z})$, but are missing in $\Delta^i_{n_1-1}(\mathbf{z})$. We thus obtain,

$$\Delta^i_{n_1-1}(\mathbf{z}) = \frac{\prod_{1\leq k<j\leq n_1;}(j-k)}{(i-1)!(n_1-i)!.}$$

Substituting (D.7) in the above expression, we obtain, $\Delta^i_{n_1-1}(\mathbf{z})$ in terms of $\Delta_{n_1}(\mathbf{z})$ as,

$$\Delta^i_{n_1-1}(\mathbf{z}) = \frac{\Delta_{n_1}(\mathbf{z})}{(i-1)!(n_1-i)!} = \frac{\prod_{j=1}^{n_1}(j-1)!}{(i-1)!(n_1-i)!}.$$

Hence the final expression becomes (6.11).

APPENDIX E

Moments of SINR

For the case of $L = n_1$ we will derive the lth moment. The mgf equation for this case can be written as

$$M_\eta(s) = c\sum_{k=1}^{n_1}(-1)^{k+1}\rho_k \det\left(Y_k\right), \tag{E.1}$$

where $\rho_k = \int_0^\infty (\frac{\sigma^2}{E_I} + x)e^{-x}x^{n_2-N_R}\,{}_0F_1(n_2 - n_1 + 1; w_k x)\frac{{}_1F_1(1;N_R;\frac{aN_R s}{xE_I/E_D+\sigma^2/E_D-bs})}{(bsE_D/E_I-\sigma^2/E_I-x)}dx$ $- \sum_{t=1}^{N_R-n_1}\frac{{}_1F_1(t;N_R;aN_R sE_D/(\sigma^2-bsE_D)}{(bsE_D/E_I-\sigma^2/E_I)^t} \times \Bigg[\frac{\sigma^2}{E_I}\Gamma(t + n_2 - N_R)\,{}_1F_1(t + n_2 - N_R; n_2 - n_1 + 1; w_k)$ $+ \Gamma(t + n_2 - N_R + 1){}_1F_1(t + n_2 - N_R + 1; n_2 - n_1 + 1; w_i)\Bigg]$, $Y_{i,j} = \frac{\sigma^2}{E_I}\,{}_1F_1(n_2 - j + 1; n_2 - n_1 + 1; w_k)\Gamma(n_2 - j + 1) + \Gamma(n_2 - j + 2){}_1F_1(n_2 - j + 2; n_2 - n_1 + 1; w_i)$ and Y_k is the matrix Y with k^{th} row and first column removed. The lth moment is given by $\mu_l = \frac{d^l}{ds^l}M_\eta(s)|_{s=0}$. We need to evaluate $\frac{d^l}{ds^l}\rho_k$. We use the relations in (McKay *et al.*, 2009) to evaluate the differential and obtain, $\frac{d^l}{ds^l}\rho_k|_{s=0} = -\alpha_l^{Ric}d_l$ where $\alpha_l^{Ric} = b^l\sum_{k=0}^{l}\binom{l}{k}\frac{(aN_R/b)^k}{(N_R)_k}$ and

$$\begin{aligned} d_l =& l!(\frac{E_D}{E_I})^l\sum_{n=0}^{\infty}\frac{(w_k)^n\Gamma(n_2 - N_R + n + 1)}{(q)_n n!} \\ & U(l, l - n_2 + N_R - n, \frac{\sigma^2}{E_I}) \\ & + \sum_{t=1}^{N_R-n_1}(t)_l(-\frac{E_I}{\sigma^2})^t(\frac{E_D}{\sigma^2})^l\Bigg[\frac{\sigma^2}{E_I}\Gamma(t + n_2 - N_R) \\ & {}_1F_1(t + n_2 - N_R; q; w_k) \\ & + \Gamma(t + n_2 - N_R + 1){}_1F_1(t + n_2 - N_R + 1; q; w_i)\Bigg]. \end{aligned} \tag{E.2}$$

APPENDIX F

Correlated and/or Unequal power Interferers

The determinant evaluation of $\det(\mathbf{N})$ can be significantly simplified for $\sigma^2 \approx 0$. We first substitute $\sigma^2 = 0$ in $N_{i,j}$, to obtain

$$\mathbf{N}_{i,j} = \begin{cases} B(i), & j = 1,\, i = 1, ..., N_I, \\ r_i^{N_R - j + 2}\Gamma(N_R - j + 2), & j = 2, ..., N_I,\, i = 1, ..., N_I. \end{cases}$$

where $B(i) = \sum_{l=0}^{\infty} \frac{(aN_r s)^l}{(N_R)_l}(|bsE_D|^{-l+1}U(2, 2-l, r_i|bsE_D|))$
$-\sum_{t=1}^{N_R - N_I} \frac{{}_1F_1(t; N_R; \frac{aN_R s}{-bs})}{(bsE_D)^t}(r_i^{t+1}\Gamma(t+1))$. By taking $r_i^{N_r - N_I + 2}$ and common gamma terms outside the determinant term we obtain, $M_\eta(s) \approx c \det(\mathbf{N})$, where
$c = \frac{(-1)^{N_R}(\sigma^2)^{(N_R - N_I)}(-1)^{\frac{1}{2}N_I(N_I-1)}\det(\mathbf{\Psi})^{-N_R}}{\prod_{i<j}^{N_I}(\frac{1}{r_i} - \frac{1}{r_j})\prod_{k=1}^{N_I}(N_R - k)!}$
$\times \prod_{i=1}^{N_I} r_i^{N_R - N_I + 2} \prod_{j=2}^{N_I} \Gamma(N_R - j + 2)$,

$$\mathbf{N}_{i,j} = \begin{cases} B(i) r_i^{-N_R + N_I - 2}, & j = 1,\, i = 1, ..., N_I, \\ r_i^{N_I - j}, & j = 2, ..., N_I,\, i = 1, ..., N_I. \end{cases}$$

Expanding along the first column, we obtain an approximation for the Laplace transform for $\sigma^2 = 0$ as, $M_\eta(s) \approx c \sum_{i=1}^{N_I} (-1)^{i+1} B(i) r_i^{-N_R + N_I - 2} \det(V^i(\mathbf{r}))$, where $V^i(\mathbf{r})$ denotes the Vandermonde matrix formed from all elements of $\mathbf{r} = (r_1, r_2, .., r_{N_I})$ except the ith element. Note that, we do not substitute $\sigma^2 \approx 0$ in the c term but only in the $\det(\mathbf{N})$ term, to obtain the approximation.

APPENDIX G

Approximate mean of complex $\kappa - \mu$ random variables

The expectations to be approximated are,

$$\mathbb{E}[x_{ik}] = \int_{-\infty}^{\infty} x \frac{|x|^{\mu/2}}{2\sigma^2|p|^{\mu/2-1}} exp(-\frac{(x-p)^2}{2\sigma^2}) sech(\frac{px}{\sigma^2}) I_{\frac{\mu}{2}-1}(\frac{|px|}{\sigma^2}) dx \tag{G.1}$$

$$\mathbb{E}[y_{ik}] = \int_{-\infty}^{\infty} y \frac{|y|^{\mu/2}}{2\sigma^2|q|^{\mu/2-1}} exp(-\frac{(y-q)^2}{2\sigma^2}) sech(\frac{qy}{\sigma^2}) I_{\frac{\mu}{2}-1}(\frac{|qy|}{\sigma^2}) dy. \tag{G.2}$$

The expectation $\mathbb{E}[x_{ik}^r]$, for some odd integer $r > 0$, using the trigonometric identity $tanh(z) = 1 - e^{-z} sech(z)$, is given as,

$$\mathbb{E}[x_{ik}^r] = 2\int_0^{\infty} \frac{x^{\mu/2+r}}{2\sigma^2|p|^{\mu/2-1}} exp(-\frac{x^2}{2\sigma^2}) exp(-\frac{p^2}{2\sigma^2}) tanh(\frac{px}{\sigma^2}) I_{\frac{\mu}{2}-1}(\frac{px}{\sigma^2}) dx.$$

Since, the above integral cannot be solved to obtain a solution in closed form, we can approximate $tanh(\frac{px}{\sigma^2})$ by $erf(\frac{\sqrt{\pi}}{2}\frac{px}{\sigma^2})$ for $x \in [0, \infty)$ to obtain (Reyes and Arcos-Olalla, 2015),

$$\mathbb{E}[x_{ik}^r] \approx 2\int_0^{\infty} \frac{x^{\mu/2+r}}{2\sigma^2|p|^{\mu/2-1}} exp(-\frac{x^2}{2\sigma^2}) exp(-\frac{p^2}{2\sigma^2}) erf(\frac{\sqrt{\pi}}{2}\frac{px}{\sigma^2}) I_{\frac{\mu}{2}-1}(\frac{px}{\sigma^2}) dx.$$

Using the identity $I_v(z) = \frac{1}{\Gamma(v+1)}(\frac{z}{2})^v \, {}_0F_1(v+1, \frac{z^2}{4})$ from (Abramowitz and Stegun, 1972),

$$\begin{aligned}\mathbb{E}[x_{ik}^r] \approx 2\int_0^{\infty} & \frac{x^{\mu/2+r}}{2\sigma^2|p|^{\mu/2-1}} exp(-\frac{x^2}{2\sigma^2}) exp(-\frac{p^2}{2\sigma^2}) \\ & erf(\frac{\sqrt{\pi}}{2}\frac{px}{\sigma^2}) \frac{1}{\Gamma(\mu/2)} (\frac{px}{2\sigma^2})^{\mu/2-1} \, {}_0F_1(\frac{\mu}{2}, \frac{p^2x^2}{4\sigma^4}) dx.\end{aligned}$$

Now, expanding the hypergeometric series and interchanging the integration and summation, we obtain,

$$\mathbb{E}[x_{ik}^4] \approx 2\int_0^\infty \frac{x^{\mu/2+r}}{2\sigma^2|p|^{\mu/2-1}} exp(-\frac{x^2}{2\sigma^2}) exp(-\frac{p^2}{2\sigma^2}) erf(\frac{\sqrt{\pi}}{2}\frac{px}{\sigma^2}) \frac{1}{\Gamma(\mu/2)} (\frac{px}{2\sigma^2})^{\mu/2-1}$$
$$\sum_{n=0}^{\infty} \frac{1}{(\frac{\mu}{2})_n n!} (\frac{p^2x^2}{4\sigma^4})^n dx$$
$$= 2\sum_{n=0}^{\infty} \frac{p^{2n}}{(\frac{\mu}{2})_n n!} \frac{e^{-\frac{p^2}{2\sigma^2}}}{\Gamma(\mu/2)} \frac{1}{(2\sigma^2)^{\mu/2+2n}} \int_0^\infty x^{\mu+2n+r-1} exp(-\frac{x^2}{2\sigma^2}) erf(\frac{\sqrt{\pi}}{2}\frac{px}{\sigma^2}) dx$$

Applying the integration identity $\int_0^\infty erf(ax)e^{-b^2x^2}x^p dx = \frac{a}{\sqrt{\pi}} b^{-p-2}\Gamma(\frac{p}{2}+1)\, {}_2F_1(\frac{1}{2}, \frac{p}{2}+1, \frac{3}{2}, -\frac{a^2}{b^2})$ for $b^2 > 0$ and $p > -2$ from (Ng and Geller, 1969), we obtain,

$$\mathbb{E}[x_{ik}^r] = 2pe^{-\frac{p^2}{2\sigma^2}} \sum_{n=0}^{\infty} p^{2n} (\frac{1}{2\sigma^2})^{n-r/2+1/2} \frac{1}{(\frac{\mu}{2})_n n! \Gamma(\mu/2)} \Gamma(\mu/2+n+r/2-1/2+1)$$
$$\times\ {}_2F_1(\frac{1}{2}, \mu/2+n+r/2-1/2+1, \frac{3}{2}, -\frac{2p^2}{\sigma^2}\frac{\pi}{4}).$$

Using the transformation ${}_2F_1(a,b,c,z) = (1-z)^{-b}\, {}_2F_1(c-a,b,c,\frac{z}{z-1})$ for the Gauss Hypergeometric function from (Wolfram Research, 2020*b*), we obtain,

$$\mathbb{E}[x_{ik}^r] \approx 2pe^{-\frac{p^2}{2\sigma^2}} \sum_{n=0}^{\infty} p^{2n} (\frac{1}{2\sigma^2})^{n-r/2+1/2} \frac{\Gamma(\mu/2+n+r/2-1/2+1)}{(\frac{\mu}{2})_n n! \Gamma(\mu/2)}$$
$$(1+\frac{2p^2}{\sigma^2}\frac{\pi}{4})^{-\mu/2-n-1-r/2+1/2}\, {}_2F_1(1, \mu/2+n+r/2-1/2+1, \frac{3}{2}, \frac{2p^2\pi}{2p^2\pi+4\sigma^2})$$
$$= 2pe^{-\frac{p^2}{2\sigma^2}} (\frac{4\sigma^2}{4\sigma^2+2p^2\pi})^{\mu/2+1} 2^{r-1} \left(\frac{1}{4\sigma^2+2p^2\pi}\right)^{r/2-1/2}$$
$$\sum_{n=0}^{\infty} (\frac{2p^2}{4\sigma^2+2p^2\pi})^n \frac{\Gamma(\mu/2+n+r/2-1/2+1)}{(\frac{\mu}{2})_n n! \Gamma(\mu/2)}$$
$${}_2F_1(1, \mu/2+n+r/2-1/2+1, \frac{3}{2}, \frac{2p^2\pi}{2p^2\pi+4\sigma^2}). \tag{G.3}$$

Similarly,

$$\mathbb{E}[y_{ik}^r] = 2qe^{-\frac{q^2}{2\sigma^2}} (\frac{4\sigma^2}{4\sigma^2+2q^2\pi})^{\mu/2+1} 2^{r-1} \left(\frac{1}{4\sigma^2+2q^2\pi}\right)^{r/2-1/2}$$
$$\sum_{n=0}^{\infty} (\frac{2q^2}{4\sigma^2+2q^2\pi})^n \frac{\Gamma(\mu/2+n+r/2-1/2+1)}{(\frac{\mu}{2})_n n! \Gamma(\mu/2)}$$
$${}_2F_1(1, \mu/2+n+r/2-1/2+1, \frac{3}{2}, \frac{2q^2\pi}{2q^2\pi+4\sigma^2}). \tag{G.4}$$

We can use the above expression to compute $\mathbb{E}[x_{ik}^r]$ and $\mathbb{E}[y_{ik}^r]$ or the expression can be further simplified further in terms of the Appell function. For example, for $r = 1$, expanding the ${}_2F_1$ as series

$$\mathbb{E}[x_{ik}] = 2pe^{-\frac{p^2}{2\sigma^2}}(\frac{4\sigma^2}{4\sigma^2 + 2p^2\pi})^{\mu/2+1}\frac{\Gamma(\mu/2+1)}{\Gamma(\mu/2)}$$
$$\sum_{n=0}^{\infty}\sum_{k=0}^{\infty}\frac{(\mu/2+1)_{n+k}(1)_k}{(\frac{3}{2})_k(\frac{\mu}{2})_n}\frac{(\frac{2p^2}{4\sigma^2+2p^2\pi})^n}{n!}\frac{(\frac{2p^2\pi}{2p^2\pi+4\sigma^2})^k}{k!}.$$

Rewriting the above using confluent Appell function Ψ_1 (Brychkov and Saad, 2014),

$$\mathbb{E}[x_{ik}] \approx 2pe^{-\frac{p^2}{2\sigma^2}}(\frac{4\sigma^2}{4\sigma^2 + 2p^2\pi})^{\mu/2+1}\frac{\Gamma(\mu/2+1)}{\Gamma(\mu/2)}$$
$$\Psi_1(\mu/2+1, 1, 3/2, \mu/2, \frac{2p^2\pi}{2p^2\pi + 4\sigma^2}, \frac{2p^2}{4\sigma^2 + 2p^2\pi}). \tag{G.5}$$

Similarly,

$$\mathbb{E}[y_{ik}] \approx 2qe^{-\frac{q^2}{2\sigma^2}}(\frac{4\sigma^2}{4\sigma^2 + 2q^2\pi})^{\mu/2+1}\frac{\Gamma(\mu/2+1)}{\Gamma(\mu/2)}$$
$$\Psi_1(\mu/2+1, 1, 3/2, \mu/2, \frac{2q^2\pi}{2q^2\pi + 4\sigma^2}, \frac{2q^2}{4\sigma^2 + 2q^2\pi}). \tag{G.6}$$

We have compared (G.5) and (G.6) with numerical evaluation of the expectation integrals and also empirical average of simulated $\kappa - \mu$ variables for a wide range of parameters. In all cases, an excellent match has been observed.

APPENDIX H

Determining $tr\left[E_p\left(E_p[\mathbf{X}]^{-1}\mathbf{X}E_p[\mathbf{X}]^{-1}\mathbf{X}\right)\right]$ for $\mathbf{X}$

Assume $\mathbf{H}$ is of dimension $2 \times N_T$ and ijth element ($i = 1, 2$ and $j = 1, ..., N_T$) of $\mathbf{H}$ is $x_{ij} + y_{ij}$ and a complex $\kappa - \mu$ RV. $\mathbf{X} = \mathbf{H}\mathbf{H}^H$ is of dimension 2×2. Denote $\mathbf{X}$ by $\begin{bmatrix} a & b \\ \bar{b} & c \end{bmatrix}$. Now to evaluate $tr\left(E_p\left(E_p[\mathbf{X}]^{-1}\mathbf{X}E_p[\mathbf{X}]^{-1}\mathbf{X}\right)\right)$, we substitute $E_p[X] = Z = (z_{diag} - z_{off})\mathbf{I}_{n_1} + z_{off}\mathbf{1}\mathbf{1}^T$, where z_{diag} and z_{off} are given by (7.12) and (7.14) respectively. Applying this substitution and using Sherman-Morrison formula for Z^{-1}, we obtain

$$Z^{-1}X = (z_{diag} - z_{off})^{-1}X - \frac{z_{off}(z_{diag} - z_{off})^{-2}\mathbf{1}\mathbf{1}^T X}{1 + n_1 z_{off}(z_{diag} - z_{off})^{-1}} \tag{H.1}$$

Therefore,

$$\begin{aligned}(Z^{-1}X)^2 = (z_{diag} - z_{off})^{-2}X^2 &+ \frac{z_{off}^2(z_{diag} - z_{off})^{-4}(\mathbf{1}\mathbf{1}^T X)^2}{(1 + n_1 z_{off}(z_{diag} - z_{off})^{-1})^2} \\ &- \frac{z_{off}(z_{diag} - z_{off})^{-3}X\mathbf{1}\mathbf{1}^T X}{1 + n_1 z_{off}(z_{diag} - z_{off})^{-1}} - \frac{z_{off}(z_{diag} - z_{off})^{-3}\mathbf{1}\mathbf{1}^T X X}{1 + n_1 z_{off}(z_{diag} - z_{off})^{-1}}.\end{aligned} \tag{H.2}$$

Hence, to determine $tr\left[E_p\left((\mathbf{Z}^{-1}\mathbf{X})^2\right)\right]$, we need to evaluate $E_p[tr(\mathbf{X}^2)]$, $E_p[tr((\mathbf{1}\mathbf{1}^T\mathbf{X})^2)]$ and $E_p[tr(\mathbf{1}\mathbf{1}^T\mathbf{X}^2)]$. Substituting $\mathbf{X} = \begin{bmatrix} a & b \\ \bar{b} & c \end{bmatrix}$, we obtain:

$$E_p[tr(\mathbf{X}^2)] = E_p\left[a^2 + c^2 + 2b\bar{b}\right], \tag{H.3}$$

$$\begin{aligned}E_p[tr((\mathbf{1}\mathbf{1}^T\mathbf{X})^2)] &= E_p\left[(a + \bar{b})^2 + 2(a + \bar{b})(b + c) + (b + c)^2\right] \\ &= E_p\left[a^2 + \bar{b}^2 + 2a\bar{b} + 2ab + 2\bar{b}b + 2ac + 2\bar{b}c + b^2 + c^2 + 2bc\right]\end{aligned} \tag{H.4}$$

and

$$E_p[tr(\mathbf{1}\mathbf{1}^T\mathbf{X}^2)] = E_p\left[a^2 + 2b\bar{b} + c^2 + ab + a\bar{b} + bc + \bar{b}c\right]. \tag{H.5}$$

The terms in the above equations can be trivially evaluated by the substituting the terms of $\mathbf{X}$ in terms of the elements of $\mathbf{H}$ and simplifying. For a $\mathbf{H}$ of dimension $2 \times N_T$, we obtain

$$E_p[a^2] = E_p[c^2] = N_T(E_p[x_{ik}^4]+E_p[y_{ik}^4])+(N_T-1)(E_p[x_{ik}^2]+E_p[y_{ik}^2])^2+2(E_p[x_{ik}^2]E_p[y_{ik}^2]) \tag{H.6}$$

$$E_p[b\bar{b}] = N_T((E_p[x_{ik}^2] + E_p[y_{ik}^2])^2 + (N_T - 1)(E_p[x_{ik}]^2 + E_p[y_{ik}]^2)^2) \tag{H.7}$$

$$\begin{aligned} E_p[ab] = E_p[cb] = {} & N_T(E_p[x_{ik}^3]E_p[x_{ik}] + N_T E_p[x_{ik}^2]E_p[y_{ik}]^2 \\ & + (N_T - 1)E_p[x_{ik}^2]E_p[x_{ik}]^2 - jE_p[x_{ik}^3]E_p[y_{ik}] + jE_p[x_{ik}^2]E_p[x_{ik}]E_p[y_{ik}] \\ & + E_p[y_{ik}^3]E_p[y_{ik}] + N_T E_p[y_{ik}^2]E_p[x_{ik}]^2 \\ & + (N_T - 1)E_p[y_{ik}^2]E_p[y_{ik}]^2 + jE_p[y_{ik}^3]E_p[x_{ik}] - jE_p[y_{ik}^2]E_p[y_{ik}]E_p[x_{ik}] \end{aligned} \tag{H.8}$$

where $E_p[x_{ik}]$, $E_p[x_{ik}^2]$, $E_p[x_{ik}^3]$ and $E_p[x_{ik}^4]$ can be obtained using (G.3) and $E_p[y_{ik}]$, $E_p[y_{ik}^2]$, $E_p[y_{ik}^3]$ and $E_p[y_{ik}^4]$ can be obtained using (G.4).

APPENDIX I

Capacity and outage for $\kappa - \mu$

I.0.1 Capacity

We have to determine an approximation for $C = E_{\mathbf{\Lambda}}[\sum_{i=1}^{n_1} log_2(1 + \frac{\rho}{N_T}\lambda_i)]$, where λ_k for $k = 1, .., n_1$ are eigenvalues of a $n_1 \times n_1$ random matrix $\mathbf{R} = \mathbf{H}\mathbf{H}^H$, where $\mathbf{H}$ have i.i.d. $\kappa - \mu$ or $\eta - \mu$ elements. We approximate the matrix $\mathbf{R}$ by a $n_1 \times n_1$ central Wishart matrix $\mathbf{W} \sim \mathcal{CW}_{n_1}(n_2, \mathbf{\Sigma})$, such that $n_1 \leq n_2$ and $\mathbf{\Sigma}$ as in (7.20). The eigenvalue distribution of the unordered eigenvalues of $\mathbf{W}$ is given by,

$$f(\mathbf{\Lambda}) = (-1)^{\frac{1}{2}n_1(n_1-1)} \frac{1}{n_1!} \frac{det(\{e^{-\lambda_i w_j}\})}{\det(\mathbf{\Sigma})^{n_2}} \frac{\Delta(\mathbf{\Lambda})}{\Delta(\mathbf{\Sigma}^{-1})} \prod_{j=1}^{n_1} \frac{\lambda_j^{n_2-n_1}}{(n_2-j)!} \tag{I.1}$$

where $w_1 > w_2 > .... > w_{n_1}$ are the eigenvalues of $\mathbf{\Sigma}^{-1}$ and $\lambda_1, ..., \lambda_{n_1}$ are the eigenvalues of $\mathbf{W}$. But if some eigenvalues of $\mathbf{\Sigma}^{-1}$ are not distinct, then the above distribution cannot be used because $det(\{e^{-\lambda_i w_j}\}) = \Delta(\mathbf{\Sigma}^{-1}) = 0$ leading to an indeterminate form. Hence, we apply the following theorem from (Couillet and Debbah, 2011), to modify the distribution and account for non-distinct eigenvalues.

Theorem I.1. *Let $f_1, ..., f_N$ be a family of infinitely differentiable functions and let $x_1, ..., x_N \in \mathcal{R}$. Denote $R(x_1, .., x_N) \triangleq \frac{det\left(\{f_i(x_j)\}\right)}{\prod_{i<j}(x_j - x_i)}$. Then, for $N_1, ..., N_p$ such that $N_1 + ... + N_p = N$ and for $y_1, ..., y_p \in R$ distinct,*

$$\lim_{\substack{x_1,...,x_{N_1} \to y_1 \\ .... \\ x_{N-N_p+1},..,x_N \to y_p}} R(x_1, ..., x_N)$$
$$= \frac{det\left[f_i(y_1), f_i'(y_1), ..., f_i^{(N_1-1)}(y_1), ..., f_i(y_p), f_i'(y_p), ..., f_i^{(N_p-1)}(y_p)\right]}{\prod_{1\leq i<j\leq p}(y_j - y_i)^{N_i N_j} \prod_{l=1}^{p} \prod_{j=1}^{N_l-1} j!}.$$

In our case, $\mathbf{\Sigma}^{-1}$ has two eigenvalues w_1 and w_2 with multiplicity $n_1 - 1$ and 1 respectively. Hence, applying the above theorem to (I.1), we obtain, the eigenvalue

distribution as,

$$f(\mathbf{\Lambda}) = \frac{(-1)^{\frac{1}{2}n_1(n_1-1)}}{n_1!} \frac{det(\{e^{-\lambda_i w_1}\ (-\lambda_i)e^{-\lambda_i w_1}\ ...\ (-\lambda_i)^{n_1-2}e^{-\lambda_i w_1}\ e^{-\lambda_i w_2}\})}{(w_2-w_1)^{n_1-1}\prod_{j=1}^{n_1-2} j!}$$
$$\frac{\Delta(\mathbf{\Lambda})}{\det(\mathbf{\Sigma})^{n_2}} \prod_{j=1}^{n_1} \frac{\lambda_j^{n_2-n_1}}{(n_2-j)!}. \tag{I.2}$$

Hence

$$C \approx (-1)^{\frac{1}{2}n_1(n_1-1)} \frac{1}{n_1! \prod_{j=1}^{n_1}(n_2-j)!} \frac{1}{\det(\mathbf{\Sigma})^{n_2}\,(w_2-w_1)^{n_1-1}\prod_{j=1}^{n_1-2} j!}$$
$$\times \int_0^\infty [\sum_{i=1}^{n_1} log_2(1+\frac{\rho}{N_T}\lambda_i)]\lambda_k^{n_2-n_1}\Delta(\mathbf{\Lambda})$$
$$det(\{e^{-\lambda_i w_1}\ (-\lambda_i)e^{-\lambda_i w_1}\ ...\ (-\lambda_i)^{n_1-2}e^{-\lambda_i w_1}\ e^{-\lambda_i w_2}\})d\mathbf{\Lambda}.$$

From Theorem 3 in Appendix of (Chiani *et al.*, 2003*b*) and in (Andreev, 1884), it can be observed that, for two arbitrary $n_1 \times n_1$ matrices $\mathbf{\Phi}(\mathbf{y})$ and $\mathbf{\Psi}(\mathbf{y})$ with ij^{th} elements $\phi_i(y_j)$ and $\Psi_i(y_j)$, and two arbitrary functions $\xi(.)$ and $\xi'(.)$, where $\mathbf{y} = [y_1\, y_2 \ldots y_{n_1}]^T$, the following identity holds:

$$\int \cdots \int_{d \geq y_i \geq c} \det(\mathbf{\Phi}(\mathbf{y}))\det(\mathbf{\Psi}(\mathbf{y}))\prod_{n=1}^{N}\xi(y_n)\sum_{k=1}^{N}\xi'(y_k)dy_1..dy_{n_1}$$
$$= N!\sum_{k=1}^{N} det\left(\left\{\int_c^d \phi_i(y)\Psi_j(y)\xi(y)U_{k,j}(\xi'(y))dy\right\}_{1\leq i,j\leq n_1}\right), \tag{I.3}$$

where, $U_{j,k}(x) = x, \quad if\ k = j$ and $U_{j,k}(x) = 1, \quad if\ k \neq j$. Applying the above identity, we obtain,

$$C \approx (-1)^{\frac{1}{2}n_1(n_1-1)} \frac{1}{\prod_{j=1}^{n_1}(n_2-j)!} \frac{1}{\det(\mathbf{\Sigma})^{n_2}\,(w_2-w_1)^{n_1-1}\prod_{j=1}^{n_1-2} j!} \sum_{k=1}^{n_1} det(N^k]. \tag{I.4}$$

where

$$\mathbf{N}_{i,j}^{k}(n_1,n_2)=\begin{cases}\int_0^\infty \lambda^{n_2-n_1}\lambda^{i-1}(-\lambda)^{j-1}e^{-\lambda w_1}d\lambda; \quad 1\le i\le n_1, 1\le j\le n_1-1, j\neq k,\\ \int_0^\infty log_2(1+\frac{\rho}{N_T}\lambda)\lambda^{n_2-n_1}\lambda^{i-1}(-\lambda)^{j-1}e^{-\lambda w_1}d\lambda;\\ \quad 1\le i\le n_1, 1\le j\le n_1-1, j=k,\\ \int_0^\infty \lambda^{n_2-n_1}\lambda^{i-1}e^{-\lambda w_2}d\lambda; \quad 1\le i\le n_1, j=n_1, j\neq k,\\ \int_0^\infty log_2(1+\frac{\rho}{N_T}\lambda)\lambda^{n_2-n_1}\lambda^{i-1}e^{-\lambda w_2}d\lambda; \quad 1\le i\le n_1, j=n_1, j=k.\end{cases}$$

First writing the logarithm in terms of Meijer-G function using the identity $ln(1+x)=G_{2,2}^{1,2}\left(\begin{smallmatrix}1,1\\1,0\end{smallmatrix}\middle|\, x\right)$ (Wolfram Research, 2020*c*) and solving the integrals using identities $\int_0^\infty x^{v-1}e^{-\mu x}dx=\Gamma(v)\mu^{-v}$ and $\int_0^\infty x^{-\rho}e^{-\beta x}\, G_{p,q}^{m,n}\left(\begin{smallmatrix}a_1,\ldots,a_p\\b_1,\ldots,b_q\end{smallmatrix}\middle|\,\alpha x\right)dx=\beta^{\rho-1}G_{p+1,q}^{m,n+1}\left(\begin{smallmatrix}\rho,a_1,\ldots,a_p\\b_1,\ldots,b_q\end{smallmatrix}\middle|\,\frac{\alpha}{\beta}\right)$ from (Gradshteyn and Ryzhik, 2007), we obtain,

$$\mathbf{N}_{i,j}^{k}(n_1,n_2)=\begin{cases}(-1)^{j-1}\Gamma(n_2-n_1+i+j-1)w_1^{-n_2+n_1-i-j+1};\\ \quad 1\le i\le n_1, 1\le j\le n_1-1, j\neq k,\\ (-1)^{j-1}\frac{1}{ln2}G_{3,2}^{1,3}\left(\begin{smallmatrix}1-n_2+n_1-i-j+1,1,1\\1,0\end{smallmatrix}\middle|\,\frac{\rho}{N_T w_1}\right)w_1^{-n_2+n_1-i-j+1};\\ \qquad 1\le i\le n_1, 1\le j\le n_1-1, j=k,\\ \Gamma(n_2-n_1+i)w_2^{-n_2+n_1-i}; \quad 1\le i\le n_1, j=n_1, j\neq k,\\ \frac{1}{ln2}G_{3,2}^{1,3}\left(\begin{smallmatrix}1-n_2+n_1-i,1,1\\1,0\end{smallmatrix}\middle|\,\frac{\rho}{N_T w_2}\right)w_2^{-n_2+n_1-i};\\ \qquad 1\le i\le n_1, j=n_1, j=k.\end{cases} \tag{I.5}$$

I.0.2 Outage probability of MRC

For the outage probability of MRC, from (I.2),

$$P(\lambda_{max}(W)<x)\approx(-1)^{\frac{1}{2}n_1(n_1-1)}\frac{1}{n_1!\prod_{j=1}^{n_1}(n_2-j)!}\frac{1}{\det(\mathbf{\Sigma})^{n_2}\,(w_2-w_1)^{n_1-1}\prod_{j=1}^{n_1-2}j!}$$
$$\times\int_0^x \lambda_k^{n_2-n_1}\Delta(\mathbf{\Lambda})$$
$$det(\{e^{-\lambda_i w_1}\,(-\lambda_i)e^{-\lambda_i w_1}\,...\,(-\lambda_i)^{n_1-2}e^{-\lambda_i w_1}\,e^{-\lambda_i w_2}\})d\mathbf{\Lambda}.$$

By following a procedure similar to the one followed for deriving capacity expressions, we can simplify the outage to,

$$P(\lambda_{max}(W) < x) \approx \frac{(-1)^{\frac{1}{2}n_1(n_1-1)}}{\prod_{j=1}^{n_1}(n_2-j)!} \frac{1}{\det(\mathbf{\Sigma})^{n_2}(w_2-w_1)^{n_1-1}\prod_{j=1}^{n_1-2} j!} det(\mathbf{N}). \quad \text{(I.6)}$$

where

$$\mathbf{N}_{i,j}(n_1,n_2) = \begin{cases} \int_0^x \lambda^{n_2-n_1}\lambda^{i-1}(-\lambda)^{j-1}e^{-\lambda w_1}d\lambda; & 1 \le i \le n_1, 1 \le j \le n_1-1 \\ \int_0^x \lambda^{n_2-n_1}\lambda^{i-1}e^{-\lambda w_2}d\lambda; & 1 \le i \le n_1, j = n_1. \end{cases}$$

Using the identity $\int_0^x u^{v-1}e^{-\mu u}du = \mu^{-v}\gamma(v,\mu x)$, where $\gamma(.,.)$ is the incomplete gamma function, we obtain,

$$\mathbf{N}_{i,j}(n_1,n_2) = \begin{cases} (-1)^{j-1}w_1^{-n_2+n_1-i-j+1}\gamma(n_2-n_1+i+j-1, w_1x); \\ \quad 1 \le i \le n_1, 1 \le j \le n_1-1 \\ w_2^{-n_2+n_1-i}\gamma(n_2-n_1+i, w_2x); \quad 1 \le i \le n_1, j = n_1. \end{cases}$$

APPENDIX J

Proof for Theorem 8.2

$F_\gamma(z)$ belongs to the MDA of the Frechet distribution, if it satisfies (8.2). The CDF of the SIR RV γ is given by $F_\gamma(z) = \mathbb{P}(\gamma \leq z)$,, where $\mathbb{P}(.)$ represents the probability of an event. Note that, this is equivalent to the expression for outage probability with a threshold z. The expression for outage probability in a $\kappa - \mu$ shadowed interference-limited scenario is given in (Kumar and Kalyani, 2017*b*, Eqn.(6)) [1]. An equivalent expression for the above CDF in the form of an infinite sum of Lauricella functions of the fourth kind is given below (Kumar and Kalyani, 2017*b*, Eqn. (20)). We assume that the source and interferers undergo $\kappa - \mu$ shadowed fading with parameters $(\kappa, \mu, m, \bar{x})$ and $\{(\kappa_i, \mu_i, m_i, \bar{y}_i); i = 1, \cdots, N\}$ respectively, where N is the number of interferers and $\bar{x}, \{\bar{y}_i; i = 1, \cdots, N\}$ are the expectations of the corresponding RVs. Then the CDF of γ is given by

$$\begin{aligned} F_\gamma(z) =& 1 - K_1 \sum_{p=0}^{\infty} \frac{(m)_p \left(1 - \frac{\theta}{\lambda}\right)^p \Gamma\left[\sum_{i=1}^{N} \mu_i + \mu + p\right]}{(\mu)_p p!} \\ & F_D^{(2N)}(1 - p - \mu, \mu_1 - m_1, \cdots, \mu_N - m_N, \\ & m_1, \cdots, m_N; 1 + \sum_{i=1}^{N} \mu_i; \frac{\theta}{\theta + z\theta_1}, \cdots, \frac{\theta}{\theta + z\theta_N}, \frac{\theta}{\theta + z\lambda_1}, \cdots, \frac{\theta}{\theta + z\lambda_N}), \end{aligned} \tag{J.1}$$

where $K_1 = \dfrac{\prod_{i=1}^{N} \left(\left(\frac{\theta}{\theta + z\theta_i}\right)^{\mu_i - m_i} \left(\frac{\theta}{\theta + z\lambda_i}\right)^{m_i} \right) \theta^m}{\Gamma\left[\sum_{i=1}^{N} \mu_i + 1\right] \Gamma[\mu] \lambda^m}$, $(m)_p = \dfrac{\Gamma[m+p]}{\Gamma[m]}$ is the Pochhammer symbol and

$$\theta = \frac{\bar{x}}{\mu_(1 + \kappa)}, \ \theta_i = \frac{\bar{y}_i}{\mu_i(1 + \kappa_i)}, \ \lambda = \frac{(\mu\kappa + m)\bar{x}}{\mu_(1 + \kappa_)}, \lambda_i = \frac{(\mu_i\kappa_i + m_i)\bar{y}_i}{\mu_i(1 + \kappa_i)m_i}. \tag{J.2}$$

[1] Note that there was an typo in the equation in the original version of (Kumar and Kalyani, 2017*b*) and the correct expressions are used above. An errata for (Kumar and Kalyani, 2017*b*) has also been communicated.

To show that $F_\gamma(z)$ belongs to the *MDA* of the Frechet distribution, we first do some simplifications of the CDF in (J.1). This CDF expression has a Lauricella function of the fourth kind ($F_D^{(2N)}(.)$), which has the following series expansion (Exton, 1976):

$$F_D^{(N)}(a, b_1, \cdots, b_N; c; x_1, \cdots, x_N) = \sum_{p_1,\cdots,p_N=0}^{\infty} \frac{(a)_{p_1+\cdots,p_N}}{(c)_{p_1+\cdots+p_N}} \prod_{i=1}^{N} (b_i)_{p_i} \frac{x_i^{p_i}}{p_i!}. \tag{J.3}$$

Substituting (J.3) in (J.1), we get an expanded form for the CDF of γ as given in (J.4). Further, by rewriting the inner $2N$ fold summation in (J.4) as two separate terms one with $p_1 = \cdots = p_{2N} = 0$ and the second with rest of the terms,we get (J.5).

$$F_\gamma(z) = 1 - \frac{\prod_{i=1}^{N} \left(\frac{\theta}{\theta+z\theta_i}\right)^{\mu_i-m_i} \left(\frac{\theta}{\theta+z\lambda_i}\right)^{m_i} \theta^m}{\Gamma\left[\sum_{i=1}^{N}\mu_i+1\right]\Gamma[\mu]\lambda^m} \times \sum_{p=0}^{\infty} \frac{(m)_p \left(1-\frac{\theta}{\lambda}\right)^p}{(\mu)_p p!} \Gamma\left[\sum_{i=1}^{N}\mu_i + p + \mu\right] \times$$
$$\sum_{p_1,\cdots,p_{2N}=0}^{\infty} \frac{(1-p-\mu)_{p_1+\cdots+p_{2N}} \prod_{i=1}^{N} (\mu_i - m_i)_{p_i} (m_i)_{p_{i+N}}}{\left(1+\sum_{i=1}^{N}\mu_i\right)_{p_1+\cdots+p_{2N}}} \prod_{i=1}^{N} \frac{\left(\frac{\theta}{\theta+z\theta_i}\right)^{p_i} \left(\frac{\theta}{\theta+z\lambda_i}\right)^{p_{i+N}}}{p_i! p_{i+N}!}. \tag{J.4}$$

$$1 - F_\gamma(z) = K_2 \sum_{p=0}^{\infty} C_1 \left\{ C_2 z^{-\sum_{i=1}^{N}\mu_i} \prod_{i=1}^{N} \left(\frac{\theta}{z}+\theta_i\right)^{-(\mu_i-m_i)} \left(\frac{\theta}{z}+\lambda_i\right)^{-m_i} + \right.$$
$$\left. \sum_{\substack{p_1\cdots p_{2N}=0,\\ s.t\ \exists\ p_{i_1}\neq 0;\ i_1\in\{1,\cdots,2N\}}}^{\infty} C_2 z^{-\sum_{i=1}^{N}\mu_i} z^{-\sum_{i=1}^{N}(p_i+p_{i+N})} \prod_{i=1}^{N} \frac{\left(\frac{\theta}{z}+\theta_i\right)^{m_i-\mu_i-p_i} \left(\frac{\theta}{z}+\lambda_i\right)^{-m_i-p_{i+N}}}{p_i! p_{i+N}!} \right\}, \tag{J.5}$$

where $K_2 = \dfrac{(\theta/\lambda)^m}{\Gamma\left[1+\sum_{i=1}^{N}\mu_i\right]\Gamma[\mu]}$, $C_1 = \dfrac{(m)_p \left(1-\frac{\theta}{\lambda}\right)^p}{(\mu)_p p!} \Gamma\left[\sum_{i=1}^{N}\mu_i + p + \mu\right]$ and $C_2 = \dfrac{(1-p-\mu)_{p_1+\cdots+p_{2N}} \prod_{i=1}^{N} (\mu_i - m_i)_{p_i} (m_i)_{p_{i+N}}}{\left(1+\sum_{i=1}^{N}\mu_i\right)_{p_1+\cdots+p_{2N}}} \theta^{\sum_{i=1}^{N}\mu_i+p_i+p_{i+N}}$. After further rear-

rangement of the terms, we obtain,

$$1 - F_\gamma(z) = K_2 \sum_{p=0}^{\infty} C_1 \left\{ C_2 z^{-\sum_{i=1}^{N} \mu_i} \prod_{i=1}^{N} \left(\frac{\theta}{z} + \theta_i \right)^{-(\mu_i - m_i)} \left(\frac{\theta}{z} + \lambda_i \right)^{-(m_i)} + z^{-\sum_{i=1}^{N} \mu_i} h(z) \right\}, \tag{J.6}$$

where $h(z) = \sum_{\substack{p_1 \cdots p_{2N}=0, \\ s.t\ \exists\ p_{i_1} \neq 0;\ i_1 \in \{1, \cdots, 2N\}}}^{\infty} C_2 z^{-\sum_{i=1}^{N}(p_i + p_{i+N})} \prod_{i=1}^{N} \frac{\left(\frac{\theta}{z} + \theta_i\right)^{m_i - \mu_i - p_i} \left(\frac{\theta}{z} + \lambda_i\right)^{-m_i - p_{i+N}}}{p_i! p_{i+N}!}$.

Now, let us focus on the term $h(z)$. As $z \to \infty$, $z^{-\sum_{i=1}^{N}(p_i + p_{i+N})}$ tends to zero and the product $\prod_{i=1}^{N} \frac{\left(\frac{\theta}{z} + \theta_i\right)^{m_i - \mu_i - p_i} \left(\frac{\theta}{z} + \lambda_i\right)^{-m_i - p_{i+N}}}{p_i! p_{i+N}!}$ will tend to a finite and positive value, which is $\frac{\prod_{i=1}^{N} \theta_i^{m_i - \mu_i - p_i} \lambda_i^{-m_i - p_{i+N}}}{p_i! p_{i+N}!}$. Hence $\lim_{z \to \infty} h(z) = 0$. Recall from (8.2) that the condition for a CDF F to belong to the MDA for Frechet distribution is

$$\lim_{t \to \infty} \frac{1 - F(tz)}{1 - F(t)} = z^{-\beta}. \tag{J.7}$$

Substituting (J.6) in the left hand side (LHS) of the above relation, we have,

$$\lim_{t \to \infty} \frac{K_2 \sum_{p=0}^{\infty} C_1 \{ C_2 (tz)^{-\sum_{i=1}^{N} \mu_i} \prod_{i=1}^{N} (\frac{\theta}{tz} + \theta_i)^{-(\mu_i - m_i)} (\frac{\theta}{tz} + \lambda_i)^{-(m_i)} + (tz)^{-\sum_{i=1}^{N} \mu_i} h(tz) \}}{K_2 \sum_{p=0}^{\infty} C_1 \{ C_2 (t)^{-\sum_{i=1}^{N} \mu_i} \prod_{i=1}^{N} (\frac{\theta}{t} + \theta_i)^{-(\mu_i - m_i)} (\frac{\theta}{t} + \lambda_i)^{-(m_i)} + (t)^{-\sum_{i=1}^{N} \mu_i} h(t) \}}. \tag{J.8}$$

Cancelling the terms common to both numerator and denominator, we obtain,

$$\lim_{t \to \infty} \frac{z^{-\sum_{i=1}^{N} \mu_i} K_2 \sum_{p=0}^{\infty} C_1 \{ C_2 \prod_{i=1}^{N} (\frac{\theta}{tz} + \theta_i)^{-(\mu_i - m_i)} (\frac{\theta}{tz} + \lambda_i)^{-(m_i)} + h(tz) \}}{K_2 \sum_{p=0}^{\infty} C_1 \{ C_2 \prod_{i=1}^{N} (\frac{\theta}{t} + \theta_i)^{-(\mu_i - m_i)} (\frac{\theta}{t} + \lambda_i)^{-(m_i)} + h(t) \}}. \tag{J.9}$$

Since we have already proven that $\lim_{z \to \infty} h(z) = 0$, (J.9) evaluates to $z^{-\sum_{i=1}^{N} \mu_i}$.

APPENDIX K

Derivation of rate of convergence

To derive the result in Theorem 8.6, we first define the δ-neighborhood of GPD for a Frechet RV. Let the δ-neighbourhood be denoted by $Q_1(\delta)$ and the GPD for a Frechet RV be denoted by $W_{\{1,\beta\}}$. The Extreme Value Distributions (EVDs) lies in the δ neighbourhood of one of three GPD $W_{\{i,\beta\}};\ i = 1, 2, 3$ with $\delta = 1$.

Definition 2. δ-neighborhood $Q_1(\delta)$ of the GPD $W_{\{1,\beta\}}$ (Falk *et al.*, 2010) is defined as $Q_1(\delta)$:= {F : ω(F) = ∞} and F has a density f on $[z_0, \infty]$ for some $z_0 > 0$ such that for some shape parameter $\beta > 0$ and some scale parameter $a > 0$ on $[z_0, \infty]$, we have,

$$f(z) = \frac{1}{a} W'_{1,\beta}\left(\frac{z}{a}\right)(1 + O((1 - W_{1,\beta}(z))^{\delta})\}, \tag{K.1}$$

where $\omega(F) := sup\{z \in \mathbb{R} : F(z) < 1\}$. In fact the GPD for the Frechet distribution is defined in (Falk *et al.*, 2010) as $W_{1,\beta} = 1 - z^{-\beta}; z \geq 1$ and using this, (K.1) can be rewritten as

$$f(z) = \frac{\beta}{a}\left(\frac{z}{a}\right)^{-\beta-1}\left(1 + O((z^{-\beta})^{\delta})\right). \tag{K.2}$$

This definition says that, if a PDF f on $[z_0, \infty]$ for some $z_0 > 0$ can be written in the form of (K.2), then the corresponding CDF F belongs to the δ-neighborhood $Q_1(\delta)$ of the Frechet distribution[1]. The PDF of the SIR RV at the jth antenna is given by,

$$\begin{aligned} f_\gamma(z) = K_5 z^{-\left(1+\sum_{i=1}^{N}\mu_i\right)}\left(1 + \frac{\theta}{z\theta_1}\right)^{-\left(\mu+\sum_{i=1}^{N}\mu_i\right)} \\ \times {}^{(1)}_{(1)}E_D^{(2N)}\Bigg[\mu + \sum_{i=1}^{N}\mu_i, m, \mu_2 - m_2, \cdots, \\ \mu_N - m_N, m_1, \cdots, m_N; \mu, \sum_{i=1}^{N}\mu_i; \frac{z\theta_1(\lambda - \theta)}{\lambda(\theta + z\theta_1)}, \frac{\theta(\theta_2 - \theta_1)}{\theta_2(\theta + z\theta_1)}, \cdots, \\ \frac{\theta(\theta_N - \theta_1)}{\theta_N(\theta + z\theta_1)}, \frac{\theta(\lambda_1 - \theta_1)}{\lambda_1(\theta + z\theta_1)}, \cdots, \frac{\theta(\lambda_N - \theta_1)}{\lambda_N(\theta + z\theta_1)}\Bigg], \end{aligned} \tag{K.3}$$

[1]For a real or complex valued function $g_1(x)$ and a strictly positive real valued function $g_2(x)$ both defined on some unbounded subset of $\mathbb{R}^+$, we say $g_1(x) = O(g_2(x))$, iff $\exists\ M \in \mathbb{R}^+$ and $x_0 \in \mathbb{R}$ such that, $|g_1(x)| \leq M g_2(x)\ \forall x \geq x_0$.

where $K_5 = \dfrac{\theta^{(m+\sum_{i=1}^{N}\mu_i)}\Gamma\left[\mu+\sum_{i=1}^{N}\mu_i\right]}{\lambda^m\Gamma[\mu]\Gamma\left[\sum_{i=1}^{N}\mu_i\right]\prod_{i=1}^{N}\theta_i^{\mu_i-m_i}\lambda_i^{m_i}}$ (The PDF expression is not given in (Kumar and Kalyani, 2017*b*) and is a non-trivial derivation. Hence, we derive the PDF in Appendix L). Also, we get the following form for the PDF $f_\gamma(z)$ by following the simplification steps given in Appendix M:

$$f_\gamma(z) = K_6 z^{-(1+\sum_{i=1}^{N}\mu_i)}\left(1+\frac{(\mu+\sum_{i=1}^{N}\mu_i)m(\lambda-\theta)}{\mu\lambda}\right)(1+O(z^{-1})). \tag{K.4}$$

This is in the same form as that of (K.2). Comparing (K.4) with (K.2), we can identify that the CDF $F_\gamma(z)$ belongs to the δ neighborhood of $Q_1(\delta)$ with $\delta = \left(\sum_{i=1}^{N}\mu_i\right)^{-1}$ and $\beta = \sum_{i=1}^{N}\mu_i$. Now that we have identified the δ neighbourhood for $F_\gamma(z)$, we make use of the following lemma from (Falk *et al.*, 2010) to conclude the proof.

Lemma K.1. *Suppose that the CDF F (of i.i.d. RVs $z_1, \cdots, z_L$) is in the δ neighborhood $Q_1(\delta)$ of the GPD $W_{1,\beta}$ then there obviously exist constant $a > 0$ such that $f(z) = \frac{1}{a}W'_{1,\beta}(\frac{z}{a})(1+O((1-W_{1,\beta}(z))^\delta)$ for all z in the left neighborhood of $\omega(W_{1,\beta})$. Consequently we have,*

$$\sup_{B\in\mathbb{B}}\left|\mathbb{P}\left(\left(\left(\frac{M_L}{a}\right)/L^\beta\right)\in B\right) - G_\beta(B)\right| = O\left(\left(\frac{1}{L}\right)^\delta + \frac{1}{L}\right), \tag{K.5}$$

where $\mathbb{B}$ denotes the Borel σ algebra on $\mathbb{R}$ and $M_L = max\{z_1, \cdots, z_L\}$.

Since the CDF $F_\gamma(z)$ belongs to the δ neighborhood of $Q_1(\delta)$, by the previous lemma, the rate of convergence is $O\left(\left(\frac{1}{L}\right)^\delta + \frac{1}{L}\right)$ with $\delta = \left(\sum_{i=1}^{N}\mu_i\right)^{-1}$.

APPENDIX L

Derivation of PDF of SIR random variable

Let $Z = \frac{X}{Y}$, $Y = \sum_{i=1}^{N} Y_i$. Here, X and Y_i are $\kappa - \mu$ shadowed RVs with parameters $\{\kappa, \mu, m, \bar{x}\}$ and $\{\kappa_i, \mu_i, m_i, \bar{y}_i\}$ respectively. Then by the method of transformation of RVs, the PDF of Z can be expressed as $f_Z(z) = \int_0^\infty y f_X(yz) f_Y(y)\, dy$, where $f_X(x)$ and $f_Y(y)$ represent the PDFs of X and Y respectively. The expression for the PDF of X is given in ((Paris, 2014, Eqn. (4))). The PDF of sum of i.n.i.d. $\kappa - \mu$ shadowed RVs is given in (Paris, 2014) as follows,

$$f_Y(y) = \frac{y^{\sum_{i=1}^{N} \mu_i - 1}}{\Gamma\left[\sum_{i=1}^{N} \mu_i\right] \prod_{i=1}^{N} \left(\theta_i^{\mu_i - m_i} \lambda_i^{m_i}\right)} \phi_2^{(2N)} \left(\mu_1 - m_1, \cdots, \mu_N - m_N, m_1, \cdots, m_N; \sum_{i=1}^{N} \mu_i; -\frac{y}{\theta_1}, \cdots, -\frac{y}{\theta_N}, -\frac{y}{\lambda_1}, \cdots, -\frac{y}{\lambda_N}\right), \tag{L.1}$$

where $\theta_i = \dfrac{\bar{y}_i}{\mu_i(1+\kappa_i)}$, $\lambda_i = \dfrac{(\mu_i \kappa_i + m_i)\bar{y}_i}{\mu_i(1+\kappa_)m_i}$ for $i = 1, \cdots, N$ and $\phi_2^{(2N)}(.)$ is the confluent multivariate hypergeometric function of $2N$ variables. Substituting the pdfs of X and Y in the expression for $f_Z(z)$, we obtain,

$$\begin{aligned} f_Z(z) = \; & K_3 \int_0^\infty y^{\mu + \sum_{i=1}^{N} \mu_i - 1} e^{-\frac{yz}{\theta}} \times_1 F_1\left(m, \mu, \frac{yz}{\theta} - \frac{yz}{\lambda}\right) \times \\ & \phi_2^{(2N)} \left(\mu_1 - m_1, \cdots, \mu_N - m_N, m_1, \cdots, m_N; \sum_{i=1}^{N} \mu_i; \right. \\ & \left. -\frac{y}{\theta_1}, \cdots, -\frac{y}{\theta_N}, -\frac{y}{\lambda_1}, \cdots, -\frac{y}{\lambda_N}\right), \end{aligned} \tag{L.2}$$

where $K_3 = \dfrac{z^{\mu-1}}{\theta^{\mu-m} \lambda^m \Gamma[\mu] \Gamma\left[\sum_{i=1}^{N} \mu_i\right]} \dfrac{1}{\prod_{i=1}^{N} \theta_i^{\mu_i - m_i} \lambda_i^{m_i}}$. We use the following integral identity from (Exton, 1976) to simplify the integral expression in (L.2):

$$\Gamma[a]^{(1)}_{(1)}E_D^{(N)}[a,b_1,\cdots,b_N;c,c';x_1,\cdots,x_N] = \int_0^\infty e^{-t}t^{a-1}\phi_2^{(k)}[b_1,\cdots,b_k;c;x_1t,\cdots,x_kt]$$
$$\times\phi_2^{(N-k)}[b_{k+1},\cdots,b_N;c';x_{k+1}t,\cdots,x_Nt]\,dt. \tag{L.3}$$

Now, the PDF of Z is given by,

$$f_Z(z) = K_4 \times^{(1)}_{(1)} E_D^{(2N+1)}\left[\mu+\sum_{i=1}^N \mu_i, m, \mu_1-m_1,\cdots,\mu_N-m_N,m_1,\cdots,m_N;\right.$$
$$\left.\mu,\sum_{i=1}^N\mu_i; 1-\frac{\theta}{\lambda},-\frac{\theta}{z\theta_1},\cdots,-\frac{\theta}{z\theta_N},-\frac{\theta}{z\lambda_1},\cdots,-\frac{\theta}{z\lambda_N}\right], \tag{L.4}$$

where $K_4 = K_3\Gamma\left[\mu+\sum_{i=1}^N\mu_i\right]$. Now, we make use of the following transformation from (Exton, 1976) to get a converging form for the PDF:

$$^{(1)}_{(1)}E_D^{(N)}(a,b_1,\cdots,b_N;c,c';x_1,\cdots,x_N)$$
$$= (1-x_2)^{-a}\times^{(1)}_{(1)}E_D^{(N)}(a,b_1,c'-b_2-\cdots-b_N,b_3,\cdots,b_N;c,c';\tfrac{x_1}{1-x_2},\tfrac{x_2}{x_2-1},\tfrac{x_2-x_3}{x_3-1},\cdots,\tfrac{x_2-x_N}{x_2-1}).$$

Hence, we obtain the following expression:

$$f_Z(z) = K_5 z^{-(1+\sum_{i=1}^N\mu_i)}\left(1+\frac{\theta}{z\theta_1}\right)^{-\left(\mu+\sum_{i=1}^N\mu_i\right)}$$
$$\times^{(1)}_{(1)}E_D^{(2N)}\left[\mu+\sum_{i=1}^N\mu_i, m, \mu_2-m_2,\cdots,\right.$$
$$\mu_N-m_N,m_1,\cdots,m_N;\mu,\sum_{i=1}^N\mu_i;\frac{z\theta_1(\lambda-\theta)}{\lambda(\theta+z\theta_1)},\frac{\theta(\theta_2-\theta_1)}{\theta_2(\theta+z\theta_1)},\cdots,$$
$$\left.\frac{\theta(\theta_N-\theta_1)}{\theta_N(\theta+z\theta_1)},\frac{\theta(\lambda_1-\theta_1)}{\lambda_1(\theta+z\theta_1)},\cdots,\frac{\theta(\lambda_N-\theta_1)}{\lambda_N(\theta+z\theta_1)}\right], \tag{L.5}$$

where $K_5 = \dfrac{\theta^{(m+\sum_{i=1}^N\mu_i)}\Gamma\left[\mu+\sum_{i=1}^N\mu_i\right]}{\lambda^m\Gamma[\mu]\Gamma\left[\sum_{i=1}^N\mu_i\right]\prod_{i=1}^N\theta_i^{\mu_i-m_i}\lambda_i^{m_i}}$.

APPENDIX M

Simplification of PDF to identify δ neighbourhood

To begin with, the PDF $f_\gamma(z)$ given by (K.3) is rewritten as in Eqn. M.2, where $z_1 = \dfrac{(\lambda-\theta)z\theta_1}{\lambda(z\theta_1+\theta)}$, $z_i = \dfrac{\theta(\theta_i-\theta_1)}{\theta_i(\theta+z\theta_1)}$ for $i \in \{2,\cdots N\}$ and $z_i = \dfrac{\theta(\lambda_i-\theta_1)}{\lambda_i(\theta+z\theta_1)}$ for $i \in \{N+1,\cdots,2N\}$. The $E_D^{(2N)}(.)$ term in this expression has the following series expansion from (Exton, 1976):

$$ {}^{(1)}_{(1)}E_D^{(N)}[a,b_1,\cdots,b_N;c,c';x_1,\cdots,x_N] = \sum_{p_1,\cdots,p_N=0}^{\infty} \frac{(a)_{p_1+\cdots+p_N}\prod_{i=1}^{N}(b_i)_{p_i}\prod_{i=1}^{N}x_i^{p_i}}{(c)_{p_1}(c')_{p_2+\cdots+p_N}p_1!\cdots p_N!}. \tag{M.1}$$

$$ f_\gamma(z) = \frac{z^{-\left(1+\sum_{i=1}^{N}\mu_i\right)}\theta^{\left(m+\sum_{i=1}^{N}\mu_i\right)}\Gamma\left[\mu+\sum_{i=1}^{N}\mu_i\right]}{\lambda^m\Gamma[\mu]\Gamma\left[\sum_{i=1}^{N}\mu_i\right]\prod_{i=1}^{N}\theta_i^{\mu_i-m_i}\lambda_i^{m_i}}\left(1+\frac{\theta}{z\theta_1}\right)^{-\left(\mu+\sum_{i=1}^{N}\mu_i\right)} $$
$$ \times {}^{(1)}_{(1)}E_D^{(2N)}\left(\mu+\sum_{i=1}^{N}\mu_i, m, \mu_2-m_2,\cdots,\mu_N-m_N, m_1,\cdots,m_N;\mu,\sum_{i=1}^{N}\mu_i;z_1,\cdots,z_{2N}\right). \tag{M.2}$$

Using the above series expansion, we rewrite (M.2) as

$$ f_\gamma(z) = K_6 z^{-\left(1+\sum_{i=1}^{N}\mu_i\right)}\left(1+\frac{\theta}{z\theta_1}\right)^{-\left(\mu+\sum_{i=1}^{N}\mu_i\right)}\sum_{p_1,\cdots,p_{2N}=0}^{\infty}\frac{\left(\mu+\sum_{i=1}^{N}\mu_i\right)_{p_1+\cdots+p_{2N}}}{(\mu)_{p_1}} $$
$$ \times \frac{(m)_{p_1}\prod_{i=2}^{N}(\mu_i-m_i)_{p_i}\prod_{i=N+1}^{2N}(m_i)_{p_i}}{\left(\sum_{i=1}^{N}\mu_i\right)_{p_2+\cdots+p_{2N}}}\prod_{i=1}^{2N}\frac{z_i^{p_i}}{p_i!}, \tag{M.3}$$

where $K_6 := \dfrac{\theta^{\left(m+\sum_{i=1}^{N}\mu_i\right)}\Gamma\left[\mu+\sum_{i=1}^{N}\mu_i\right]}{\lambda^m\Gamma[\mu]\Gamma\left[\sum_{i=1}^{N}\mu_i\right]\prod_{i=1}^{N}\theta_i^{\mu_i-m_i}\lambda_i^{m_i}}$. We then expand the $2N$ fold summation in (M.3) into three terms: the first term with all the iterating variables $p_1, p_2, ..., p_{2N}$ taking the value zero, the second term with exactly one non-zero iterating variable and the third term with the rest. By expanding, (M.3) becomes the expression given in (M.7)

where $\rho = \mu + \sum_{i=1}^{N} \mu_i$. The term $\frac{1}{\theta + z\theta_1}$ present in $Term\ a$ and $Term\ b$ of (M.7) has the following converging series expansion:

$$\frac{1}{z\theta_1 + \theta} = \frac{1}{\theta}\left\{\frac{1}{z\theta_1/\theta} - \frac{1}{(z\theta_1/\theta)^2} + \frac{1}{(z\theta_1/\theta)^3} - \frac{1}{(z\theta_1/\theta)^4} + O(z^{-5})\right\}. \tag{M.4}$$

Using (M.4), $Term\ a$ can be represented as

$$\frac{(\mu + \sum_{i=1}^{N} \mu_i)m(\lambda - \theta)}{\mu\lambda}\left\{1 - \frac{1}{z\theta_1/\theta} + \frac{1}{(z\theta_1/\theta)^2} - \frac{1}{(z\theta_1/\theta)^3} + O(z^{-4})\right\}. \tag{M.5}$$

Similarly, $Term\ b$ can also be expanded to get a series expression, but the expansion will have only negative powers of z. $Term\ 3$ will also have only powers of z less than 1. Combining these series expansions, the SIR PDF can be finally expressed as,

$$f_\gamma(z) = K_6 z^{-(1+\sum_{i=1}^{N} \mu_i)}\left(1 + \frac{(\mu + \sum_{i=1}^{N} \mu_i)m(\lambda - \theta)}{\mu\lambda}\right)(1 + O(z^{-1})). \tag{M.6}$$

$$\begin{aligned} f_\gamma(z) = \left(K_6 z^{-\left(1+\sum_{i=1}^{N} \mu_i\right)}\left(1 + \frac{\theta}{z\theta_1}\right)^{-\rho}\right)\Bigg\{&\underbrace{1}_{\textit{Term 1}} + \\ &\underbrace{\underbrace{\frac{\rho m(\lambda - \theta)z\theta_1}{\mu\lambda(z\theta_1 + \theta)}}_{\textit{Term a}} + \sum_{k=2}^{N} \underbrace{\frac{\rho(\mu_k - m_k)\theta(\theta_k - \theta_1)}{\sum_{i=1}^{N} \mu_i\theta_k(\theta + z\theta_1)} + \sum_{k=N+1}^{2N} \frac{\rho(m_k)\theta(\lambda_k - \theta_1)}{\sum_{i=1}^{N} \mu_i\lambda_k(\theta + z\theta_1)}}_{\textit{Term b}}}_{\textit{Term 2}} \\ &+ \underbrace{\sum_{\substack{p_1,\cdots,p_{2N}=0;\\ \exists\ i_1,i_2 s.t\ p_{i_1}p_{i_2}\neq 0\ \forall\ i_1\neq i_2}}^{\infty} \frac{(\rho)_{p_1+\cdots+p_{2N}}}{(\mu)_{p_1}} \frac{(m)_{p_1}\prod_{i=2}^{N}(\mu_i - m_i)_{p_i}\prod_{i=N+1}^{2N}(m_i)_{p_i}\prod_{i=1}^{2N}\frac{z_i^{p_i}}{p_i!}}{\left(\sum_{i=1}^{N}\mu_i\right)_{p_2+\cdots+p_{2N}}}}_{\textit{Term 3}}\Bigg\}. \end{aligned} \tag{M.7}$$

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LIST OF PAPERS BASED ON THESIS

Publications

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I Journal

1. M. Srinivasan and S. Kalyani, "Secrecy Capacity of $\kappa-\mu$ Shadowed Fading Channels," in IEEE Communications Letters, vol. 22, no. 8, pp. 1728-1731, Aug. 2018.
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ized Fading", to appear in *IEEE Transactions on Communications* (Early Access).

II Conference

1. M. Srinivasan, A. Subhash and S. Kalyani, "Joint power and resource allocation of D2D communication with low-resolution ADC," in Asilomar 2019.

Under review

1. M. Srinivasan and S. Kalyani, "Approximate Random Matrix Models for Generalized Fading MIMO Channels" in *IEEE Transactions on Wireless Communications*, preprint arXiv:1707.09734